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6TSL02

Shaimaa

Test begins on the next page.

Shaimaa

Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

Former astronaut Ellen Ochoa says that although she doesn't have a definite idea of when it might happen, she _____ that humans will someday need to be able to live in other environments than those found on Earth. This conjecture informs her interest in future research missions to the moon.

Which choice completes the text with the most logical and precise word or phrase?

A) demands
B) speculates
C) doubts
D) establishes

2

Beginning in the 1950s, Navajo Nation legislator Annie Dodge Wauneka continuously worked to promote public health; this _____ effort involved traveling throughout the vast Navajo homeland and writing a medical dictionary for speakers of *Diné bizaad*, the Navajo language.

Which choice completes the text with the most logical and precise word or phrase?

A) impartial
B) offhand
C) persistent
D) mandatory

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CONTINUE

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3

Following the principles of community-based participatory research, tribal nations and research institutions are equal partners in health studies conducted on reservations. A collaboration between the Crow Tribe and Montana State University _____ this model: tribal citizens worked alongside scientists to design the methodology and continue to assist in data collection.

Which choice completes the text with the most logical and precise word or phrase?

A) circumvents
B) eclipses
C) fabricates
D) exemplifies

4

The parasitic dodder plant increases its reproductive success by flowering at the same time as the host plant it has latched onto. In 2020, Jianqiang Wu and his colleagues determined that the tiny dodder achieves this _____ with its host by absorbing and utilizing a protein the host produces when it is about to flower.

Which choice completes the text with the most logical and precise word or phrase?

A) synchronization
B) hibernation
C) prediction
D) moderation

5

Given that the conditions in binary star systems should make planetary formation nearly impossible, it's not surprising that the existence of planets in such systems has lacked _____ explanation. Roman Rafikov and Kedron Silsbee shed light on the subject when they used modeling to determine a complex set of factors that could support planets' development.

Which choice completes the text with the most logical and precise word or phrase?

A) a discernible
B) a straightforward
C) an inconclusive
D) an unbiased

6

Seminole/Muscogee director Sterlin Harjo _____ television's tendency to situate Native characters in the distant past: this rejection is evident in his series *Reservation Dogs*, which revolves around teenagers who dress in contemporary styles and whose dialogue is laced with current slang.

Which choice completes the text with the most logical and precise word or phrase?

A) repudiates
B) proclaims
C) foretells
D) recants

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7

In 2007, computer scientist Luis von Ahn was working on converting printed books into a digital format. He found that some words were distorted enough that digital scanners couldn't recognize them, but most humans could easily read them. Based on that finding, von Ahn invented a simple security test to keep automated "bots" out of websites. The first version of the reCAPTCHA test asked users to type one known word and one of the many words scanners couldn't recognize. Correct answers proved the users were humans and added data to the book-digitizing project.

Which choice best states the main purpose of the text?

A) To discuss von Ahn's invention of reCAPTCHA

B) To explain how digital scanners work

C) To call attention to von Ahn's book-digitizing project

D) To indicate how popular reCAPTCHA is

8

The following text is from Edith Wharton's 1905 novel *The House of Mirth*. Lily Bart and a companion are walking through a park.

> Lily had no real intimacy with nature, but she had a passion for the appropriate and could be keenly sensitive to a scene which was the fitting background of her own sensations. The landscape outspread below her seemed an enlargement of her present mood, and she found something of herself in its calmness, its breadth, its long free reaches. On the nearer slopes the sugar-maples wavered like pyres of light; lower down was a massing of grey orchards, and here and there the lingering green of an oak-grove.

Which choice best describes the function of the underlined sentence in the text as a whole?

A) It creates a detailed image of the physical setting of the scene.

B) It establishes that a character is experiencing an internal conflict.

C) It makes an assertion that the next sentence then expands on.

D) It illustrates an idea that is introduced in the previous sentence.

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9

A study by a team including finance professor Madhu Veeraraghavan suggests that exposure to sunshine during the workday can lead to overly optimistic behavior. Using data spanning from 1994 to 2010 for a set of US companies, the team compared over 29,000 annual earnings forecasts to the actual earnings later reported by those companies. The team found that the greater the exposure to sunshine at work in the two weeks before a manager submitted an earnings forecast, the more the manager's forecast exceeded what the company actually earned that year.

Which choice best states the function of the underlined sentence in the overall structure of the text?

A) To summarize the results of the team's analysis

B) To present a specific example that illustrates the study's findings

C) To explain part of the methodology used in the team's study

D) To call out a challenge the team faced in conducting its analysis

10

The following text is adapted from Edith Nesbit's 1906 novel *The Railway Children*.

> Mother did not spend all her time in paying dull [visits] to dull ladies, and sitting dully at home waiting for dull ladies to pay [visits] to her. She was almost always there, ready to play with the children, and read to them, and help them to do their home-lessons. Besides this she used to write stories for them while they were at school, and read them aloud after tea, and she always made up funny pieces of poetry for their birthdays and for other great occasions.

According to the text, what is true about Mother?

A) She wishes that more ladies would visit her.

B) Birthdays are her favorite special occasion.

C) She creates stories and poems for her children.

D) Reading to her children is her favorite activity.

11

The following text is from Maggie Pogue Johnson's 1910 poem "Poet of Our Race." In this poem, the speaker is addressing Paul Laurence Dunbar, a Black author.

> Thou, with stroke of mighty pen,
> Hast told of joy and mirth,
> And read the hearts and souls of men
> As cradled from their birth.
> The language of the flowers,
> Thou hast read them all,
> And e'en the little brook
> Responded to thy call.

Which choice best states the main purpose of the text?

A) To praise a certain writer for being especially perceptive regarding people and nature

B) To establish that a certain writer has read extensively about a variety of topics

C) To call attention to a certain writer's careful and elaborately detailed writing process

D) To recount fond memories of an afternoon spent in nature with a certain writer

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12

"To You" is an 1856 poem by Walt Whitman. In the poem, Whitman suggests that readers, whom he addresses directly, have not fully understood themselves, writing, _____

Which quotation from "To You" most effectively illustrates the claim?

A) "You have not known what you are, you have slumber'd upon yourself / all your life, / Your eyelids have been the same as closed most of the time."

B) "These immense meadows, these interminable rivers, you are immense / and interminable as they."

C) "I should have made my way straight to you long ago, / I should have blabb'd nothing but you, I should have chanted nothing / but you."

D) "I will leave all and come and make the hymns of you, / None has understood you, but I understand you."

13

Born in 1891 to a Quechua-speaking family in the Andes Mountains of Peru, Martín Chambi is today considered to be one of the most renowned figures of Latin American photography. In a paper for an art history class, a student claims that Chambi's photographs have considerable ethnographic value—in his work, Chambi was able to capture diverse elements of Peruvian society, representing his subjects with both dignity and authenticity.

Which finding, if true, would most directly support the student's claim?

A) Chambi took many commissioned portraits of wealthy Peruvians, but he also produced hundreds of images carefully documenting the peoples, sites, and customs of Indigenous communities of the Andes.

B) Chambi's photographs demonstrate a high level of technical skill, as seen in his strategic use of illumination to create dramatic light and shadow contrasts.

C) During his lifetime, Chambi was known and celebrated both within and outside his native Peru, as his work was published in places like Argentina, Spain, and Mexico.

D) Some of the peoples and places Chambi photographed had long been popular subjects for Peruvian photographers.

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14

Credited Film Output of James Young Deer, Dark Cloud, Edwin Carewe, and Lillian St. Cyr

Individual	Years active	Number of films known and commonly credited
James Young Deer	1909–1924	33 (actor), 35 (director), 10 (writer)
Dark Cloud	1910–1920	35 (actor), 1 (writer)
Edwin Carewe	1912–1934	47 (actor), 58 (director), 20 (producer), 4 (writer)
Lillian St. Cyr (Red Wing)	1908–1921	66 (actor)

Some researchers studying Indigenous actors and filmmakers in the United States have turned their attention to the early days of cinema, particularly the 1910s and 1920s, when people like James Young Deer, Dark Cloud, Edwin Carewe, and Lillian St. Cyr (known professionally as Red Wing) were involved in one way or another with numerous films. In fact, so many films and associated records for this era have been lost that counts of those four figures' output should be taken as bare minimums rather than totals; it's entirely possible, for example, that _____

Which choice most effectively uses data from the table to complete the example?

A) Dark Cloud acted in significantly fewer films than did Lillian St. Cyr, who is credited with 66 performances.

B) Edwin Carewe's 47 credited acting roles includes only films made after 1934.

C) Lillian St. Cyr acted in far more than 66 films and Edwin Carewe directed more than 58.

D) James Young Deer actually directed 33 films and acted in only 10.

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15

Juvenile Plants Found Growing on Bare Ground and in Patches of Vegetation for Five Species

Species	Bare ground	Patches of vegetation	Total	Percent found in patches of vegetation
T. moroderi	9	13	22	59.1%
T. libanitis	83	120	203	59.1%
H. syriacim	95	106	201	52.7%
H. squamatum	218	321	539	59.6%
H. stoechas	11	12	23	52.2%

Alicia Montesinos-Navarro, Isabelle Storer, and Rocío Perez-Barrales recently examined several plots within a diverse plant community in southeast Spain. The researchers calculated that if individual plants were randomly distributed on this particular landscape, only about 15% would be with other plants in patches of vegetation. They counted the number of juvenile plants of five species growing in patches of vegetation and the number growing alone on bare ground and compared those numbers to what would be expected if the plants were randomly distributed. Based on these results, they claim that plants of these species that grow in close proximity to other plants gain an advantage at an early developmental stage.

Which choice best describes data from the table that support the researchers' claim?

A) For all five species, less than 75% of juvenile plants were growing in patches of vegetation.

B) The species with the greatest number of juvenile plants growing in patches of vegetation was *H. stoechas*.

C) For *T. libanitis* and *T. moroderi*, the percentage of juvenile plants growing in patches of vegetation was less than what would be expected if plants were randomly distributed.

D) For each species, the percentage of juvenile plants growing in patches of vegetation was substantially higher than what would be expected if plants were randomly distributed.

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16

In the mountains of Brazil, *Barbacenia tomentosa* and *Barbacenia macrantha*—two plants in the Velloziaceae family—establish themselves on soilless, nutrient-poor patches of quartzite rock. Plant ecologists Anna Abrahão and Patricia de Britto Costa used microscopic analysis to determine that the roots of *B. tomentosa* and *B. macrantha*, which grow directly into the quartzite, have clusters of fine hairs near the root tip; further analysis indicated that these hairs secrete both malic and citric acids. The researchers hypothesize that the plants depend on dissolving underlying rock with these acids, as the process not only creates channels for continued growth but also releases phosphates that provide the vital nutrient phosphorus.

Which finding, if true, would most directly support the researchers' hypothesis?

A) Other species in the Velloziaceae family are found in terrains with more soil but have root structures similar to those of *B. tomentosa* and *B. macrantha*.

B) Though *B. tomentosa* and *B. macrantha* both secrete citric and malic acids, each species produces the acids in different proportions.

C) The roots of *B. tomentosa* and *B. macrantha* carve new entry points into rocks even when cracks in the surface are readily available.

D) *B. tomentosa* and *B. macrantha* thrive even when transferred to the surfaces of rocks that do not contain phosphates.

17

Herbivorous sauropod dinosaurs could grow more than 100 feet long and weigh up to 80 tons, and some researchers have attributed the evolution of sauropods to such massive sizes to increased plant production resulting from high levels of atmospheric carbon dioxide during the Mesozoic era. However, there is no evidence of significant spikes in carbon dioxide levels coinciding with relevant periods in sauropod evolution, such as when the first large sauropods appeared, when several sauropod lineages underwent further evolution toward gigantism, or when sauropods reached their maximum known sizes, suggesting that _____

Which choice most logically completes the text?

A) fluctuations in atmospheric carbon dioxide affected different sauropod lineages differently.

B) the evolution of larger body sizes in sauropods did not depend on increased atmospheric carbon dioxide.

C) atmospheric carbon dioxide was higher when the largest known sauropods lived than it was when the first sauropods appeared.

D) sauropods probably would not have evolved to such immense sizes if atmospheric carbon dioxide had been even slightly higher.

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18

In documents called judicial opinions, judges explain the reasoning behind their legal rulings, and in those explanations they sometimes cite and discuss historical and contemporary philosophers. Legal scholar and philosopher Anita L. Allen argues that while judges are naturally inclined to mention philosophers whose views align with their own positions, the strongest judicial opinions consider and rebut potential objections; discussing philosophers whose views conflict with judges' views could therefore _____

Which choice most logically completes the text?

A) allow judges to craft judicial opinions without needing to consult philosophical works.

B) help judges improve the arguments they put forward in their judicial opinions.

C) make judicial opinions more comprehensible to readers without legal or philosophical training.

D) bring judicial opinions in line with views that are broadly held among philosophers.

19

Public-awareness campaigns about the need to reduce single-use plastics can be successful, says researcher Kim Borg of Monash University in Australia, when these campaigns give consumers a choice: for example, Japan achieved a 40 percent reduction in plastic-bag use after cashiers were instructed to ask customers whether _____ wanted a bag.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) they

B) one

C) you

D) it

20

In ancient Greece, an Epicurean was a follower of Epicurus, a philosopher whose beliefs revolved around the pursuit of pleasure. Epicurus defined pleasure as "the absence of pain in the body and of trouble in the _____ that all life's virtues derived from this absence.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) soul," positing

B) soul": positing

C) soul"; positing

D) soul." Positing

21

British scientists James Watson and Francis Crick won the Nobel Prize in part for their 1953 paper announcing the double helix structure of DNA, but it is misleading to say that Watson and Crick discovered the double helix. _____ findings were based on a famous X-ray image of DNA fibers, "Photo 51," developed by X-ray crystallographer Rosalind Franklin and her graduate student Raymond Gosling.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) They're

B) It's

C) Their

D) Its

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22

In 1937, Chinese American screen actor Anna May Wong, who had portrayed numerous villains and secondary characters but never a heroine, finally got a starring role in Paramount Pictures' *Daughter of Shanghai*, a film that _____ "expanded the range of possibilities for Asian images on screen."

Which choice completes the text so that it conforms to the conventions of Standard English?

A) critic, Stina Chyn, claims
B) critic, Stina Chyn, claims,
C) critic Stina Chyn claims
D) critic Stina Chyn, claims,

23

In 1637, the price of tulips skyrocketed in Amsterdam, with single bulbs of rare varieties selling for up to the equivalent of $200,000 in today's US dollars. Some historians _____ that this "tulip mania" was the first historical instance of an asset bubble, which occurs when investors drive prices to highs not supported by actual demand.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) claiming
B) claim
C) having claimed
D) to claim

24

Researchers studying magnetosensation have determined why some soil-dwelling roundworms in the Southern Hemisphere move in the opposite direction of Earth's magnetic field when searching for _____ in the Northern Hemisphere, the magnetic field points down, into the ground, but in the Southern Hemisphere, it points up, toward the surface and away from worms' food sources.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) food:
B) food,
C) food while
D) food

25

Scientists believe that, unlike most other species of barnacle, turtle barnacles (*Chelonibia testudinari*) can dissolve the cement-like secretions they use to attach _____ to a sea turtle shell, enabling the barnacles to move short distances across the shell's surface.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) it
B) themselves
C) them
D) itself

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26

The classic children's board game Chutes and Ladders is a version of an ancient Nepalese game, Paramapada Sopanapata. In both games, players encounter "good" or "bad" spaces while traveling along a path; landing on one of the good spaces _____ a player to skip ahead and arrive closer to the end goal.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) allows
B) are allowing
C) have allowed
D) allow

27

In 1943, in the midst of World War II, mathematics professor Grace Hopper was recruited by the US military to help the war effort by solving complex equations. Hopper's subsequent career would involve more than just _____ as a pioneering computer programmer, Hopper would help usher in the digital age.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) equations, though:
B) equations, though,
C) equations. Though,
D) equations though

28

In 1453, English King Henry VI became unfit to rule after falling gravely ill. As a result, Parliament appointed Richard, Third Duke of York, who had a strong claim to the English throne, to rule as Lord Protector. Upon recovering two years later, _____ forcing an angered Richard from the royal court and precipitating a series of battles later known as the Wars of the Roses.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) Henry resumed his reign,
B) the reign of Henry resumed,
C) Henry's reign resumed,
D) it was Henry who resumed his reign,

29

Although novels and poems are considered distinct literary forms, many authors have created hybrid works that incorporate elements of both. Bernardine Evaristo's *The Emperor's Babe*, _____ is a verse novel, a book-length narrative complete with characters and a plot but conveyed in short, crisp lines of poetry rather than prose.

Which choice completes the text with the most logical transition?

A) by contrast,
B) consequently,
C) secondly,
D) for example,

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30

At two weeks old, the time their critical socialization period begins, wolves can smell but cannot yet see or hear. Domesticated dogs, _____ can see, hear, and smell by the end of two weeks. This relative lack of sensory input may help explain why wolves behave so differently around humans than dogs do: from a very young age, wolves are more wary and less exploratory.

Which choice completes the text with the most logical transition?

A) in other words,
B) for instance,
C) by contrast,
D) accordingly,

31

Researchers Helena Mihaljević-Brandt, Lucía Santamaría, and Marco Tullney report that while mathematicians may have traditionally worked alone, evidence points to a shift in the opposite direction. _____ mathematicians are choosing to collaborate with their peers—a trend illustrated by a rise in the number of mathematics publications credited to multiple authors.

Which choice completes the text with the most logical transition?

A) Similarly,
B) For this reason,
C) Furthermore,
D) Increasingly,

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32

While researching a topic, a student has taken the following notes:

- Pterosaurs were flying reptiles that existed millions of years ago.
- In a 2021 study, Anusuya Chinsamy-Turan analyzed fragments of pterosaur jawbones located in the Sahara Desert.
- She was initially unsure if the bones belonged to juvenile or adult pterosaurs.
- She used advanced microscope techniques to determine that the bones had few growth lines relative to the bones of fully grown pterosaurs.
- She concluded that the bones belonged to juveniles.

The student wants to present the study and its findings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) In 2021, Chinsamy-Turan studied pterosaur jawbones and was initially unsure if the bones belonged to juveniles or adults.

B) Pterosaur jawbones located in the Sahara Desert were the focus of a 2021 study.

C) In a 2021 study, Chinsamy-Turan used advanced microscope techniques to analyze the jawbones of pterosaurs, flying reptiles that existed millions of years ago.

D) In a 2021 study, Chinsamy-Turan determined that pterosaur jawbones located in the Sahara Desert had few growth lines relative to the bones of fully grown pterosaurs and thus belonged to juveniles.

33

While researching a topic, a student has taken the following notes:

- African American women played prominent roles in the Civil Rights Movement, including at the famous 1963 March on Washington.
- Civil rights activist Anna Hedgeman, one of the march's organizers, was a political adviser who had worked for President Truman.
- Civil rights activist Daisy Bates was a well-known journalist and advocate for school desegregation.
- Hedgeman worked behind the scenes to make sure a woman was included in the lineup of speakers at the march.
- Bates was the sole woman to speak, delivering a brief but memorable address to the cheering crowd.

The student wants to compare the two women's contributions to the March on Washington. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Hedgeman and Bates contributed to the march in different ways; Bates, for example, delivered a brief but memorable address.

B) Hedgeman worked in politics and helped organize the march, while Bates was a journalist and school desegregation advocate.

C) Although Hedgeman worked behind the scenes to make sure a woman speaker was included, Bates was the sole woman to speak at the march.

D) Many African American women, including Bates and Hedgeman, fought for civil rights, but only one spoke at the march.

STOP

If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.

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Reading and Writing

33 QUESTIONS

DIRECTIONS

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All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

For painter Jacob Lawrence, being _____ was an important part of the artistic process. Because he paid close attention to all the details of his Harlem neighborhood, Lawrence's artwork captured nuances in the beauty and vitality of the Black experience during the Harlem Renaissance and the Great Migration.

Which choice completes the text with the most logical and precise word or phrase?

A) skeptical
B) observant
C) critical
D) confident

2

Mônica Lopes-Ferreira and others at Brazil's Butantan Institute are studying the freshwater stingray species *Potamotrygon rex* to determine whether biological characteristics such as the rays' age and sex have _____ effect on the toxicity of their venom—that is, to see if differences in these traits are associated with considerable variations in venom potency.

Which choice completes the text with the most logical and precise word or phrase?

A) a disconcerting
B) an acceptable
C) an imperceptible
D) a substantial

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3

Researchers have struggled to pinpoint specific causes for hiccups, which happen when a person's diaphragm contracts _____ . However, neuroscientist Kimberley Whitehead has found that these uncontrollable contractions may play an important role in helping infants regulate their breathing.

Which choice completes the text with the most logical and precise word or phrase?

A) involuntarily
B) beneficially
C) strenuously
D) smoothly

4

Critics have asserted that fine art and fashion rarely _____ in a world where artists create timeless works for exhibition and designers periodically produce new styles for the public to buy. Luiseño/Shoshone-Bannock beadwork artist and designer Jamie Okuma challenges this view: her work can be seen in the Metropolitan Museum of Art and purchased through her online boutique.

Which choice completes the text with the most logical and precise word or phrase?

A) prevail
B) succumb
C) diverge
D) intersect

5

Scholarly discussions of gender in Shakespeare's comedies often celebrate the rebellion of the playwright's characters against the rigid expectations _____ by Elizabethan society. Most of the comedies end in marriage, with characters returning to their socially dictated gender roles after previously defying them, but there are some notable exceptions.

Which choice completes the text with the most logical and precise word or phrase?

A) interjected
B) committed
C) illustrated
D) prescribed

6

In studying the use of external stimuli to reduce the itching sensation caused by an allergic histamine response, Louise Ward and colleagues found that while harmless applications of vibration or warming can provide a temporary distraction, such _____ stimuli actually offer less relief than a stimulus that seems less benign, like a mild electric shock.

Which choice completes the text with the most logical and precise word or phrase?

A) deceptive
B) innocuous
C) novel
D) impractical

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7

The province of Xoconochco was situated on the Pacific coast, hundreds of kilometers southeast of Tenochtitlan, the capital of the Aztec Empire. Because Xoconochco's location within the empire was so _____ , cacao and other trade goods produced there could reach the capital only after a long overland journey.

Which choice completes the text with the most logical and precise word or phrase?

A) unobtrusive

B) concealed

C) approximate

D) peripheral

8

The following text is from Charlotte Brontë's 1847 novel *Jane Eyre*. Jane works as a governess at Thornfield Hall.

> I went on with my day's business tranquilly; but ever and anon vague suggestions kept wandering across my brain of reasons why I should quit Thornfield; and I kept involuntarily framing advertisements and pondering conjectures about new situations: these thoughts I did not think to check; they might germinate and bear fruit if they could.

Which choice best states the main purpose of the text?

A) To convey a contrast between Jane's outward calmness and internal restlessness

B) To emphasize Jane's loyalty to the people she works for at Thornfield Hall

C) To demonstrate that Jane finds her situation both challenging and deeply fulfilling

D) To describe Jane's determination to secure employment outside of Thornfield Hall

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9

Text 1
Most animals can regenerate some parts of their bodies, such as skin. But when a three-banded panther worm is cut into three pieces, each piece grows into a new worm. Researchers are investigating this feat partly to learn more about humans' comparatively limited abilities to regenerate, and they're making exciting progress. An especially promising discovery is that both humans and panther worms have a gene for early growth response (EGR) linked to regeneration.

Text 2
When Mansi Srivastava and her team reported that panther worms, like humans, possess a gene for EGR, it caused excitement. However, as the team pointed out, the gene likely functions very differently in humans than it does in panther worms. Srivastava has likened EGR to a switch that activates other genes involved in regeneration in panther worms, but how this switch operates in humans remains unclear.

Based on the texts, what would the author of Text 2 most likely say about Text 1's characterization of the discovery involving EGR?

A) It is reasonable given that Srivastava and her team have identified how EGR functions in both humans and panther worms.

B) It is overly optimistic given additional observations from Srivastava and her team.

C) It is unexpected given that Srivastava and her team's findings were generally met with enthusiasm.

D) It is unfairly dismissive given the progress that Srivastava and her team have reported.

10

The following text is adapted from William Shakespeare's 1609 poem "Sonnet 27." The poem is addressed to a close friend as if he were physically present.

> Weary with toil, I [hurry] to my bed,
> The dear repose for limbs with travel tired;
> But then begins a journey in my head
> To work my mind, when body's work's expired:
> For then my thoughts—from far where I abide—
> [Begin] a zealous pilgrimage to thee,
> And keep my drooping eyelids open wide,

What is the main idea of the text?

A) The speaker is asleep and dreaming about traveling to see the friend.

B) The speaker is planning an upcoming trip to the friend's house.

C) The speaker is too fatigued to continue a discussion with the friend.

D) The speaker is thinking about the friend instead of immediately falling asleep.

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11

The following text is adapted from Lewis Carroll's 1889 satirical novel *Sylvie and Bruno*. A crowd has gathered outside a room belonging to the Warden, an official who reports to the Lord Chancellor.

> One man, who was more excited than the rest, flung his hat high into the air, and shouted (as well as I could make out) "Who roar for the Sub-Warden?" Everybody roared, but whether it was for the Sub-Warden, or not, did not clearly appear: some were shouting "Bread!" and some "Taxes!", but no one seemed to know what it was they really wanted.
>
> All this I saw from the open window of the Warden's breakfast-saloon, looking across the shoulder of the Lord Chancellor.
>
> "What can it all mean?" he kept repeating to himself. "I never heard such shouting before—and at this time of the morning, too! And with such unanimity!"

Based on the text, how does the Lord Chancellor respond to the crowd?

A) He asks about the meaning of the crowd's shouting, even though he claims to know what the crowd wants.

B) He indicates a desire to speak to the crowd, even though the crowd has asked to speak to the Sub-Warden.

C) He expresses sympathy for the crowd's demands, even though the crowd's shouting annoys him.

D) He describes the crowd as being united, even though the crowd clearly appears otherwise.

12

O Pioneers! is a 1913 novel by Willa Cather. In the novel, Cather portrays Alexandra Bergson as having a deep emotional connection to her natural surroundings: _____

Which quotation from *O Pioneers!* most effectively illustrates the claim?

A) "She had never known before how much the country meant to her. The chirping of the insects down in the long grass had been like the sweetest music. She had felt as if her heart were hiding down there, somewhere, with the quail and the plover and all the little wild things that crooned or buzzed in the sun. Under the long shaggy ridges, she felt the future stirring."

B) "Alexandra talked to the men about their crops and to the women about their poultry. She spent a whole day with one young farmer who had been away at school, and who was experimenting with a new kind of clover hay. She learned a great deal."

C) "Alexandra drove off alone. The rattle of her wagon was lost in the howling of the wind, but her lantern, held firmly between her feet, made a moving point of light along the highway, going deeper and deeper into the dark country."

D) "It was Alexandra who read the papers and followed the markets, and who learned by the mistakes of their neighbors. It was Alexandra who could always tell about what it had cost to fatten each steer, and who could guess the weight of a hog before it went on the scales closer than John Bergson [her father] himself."

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13

Approximate Rates of Speech and Information Conveyed for Five Languages

Language	Rate of speech (syllables per second)	Rate of information conveyed (bits per second)
Serbian	7.2	39.1
Spanish	7.7	42.0
Vietnamese	5.3	42.5
Thai	4.7	33.8
Hungarian	5.9	34.6

A group of researchers working in Europe, Asia, and Oceania conducted a study to determine how quickly different Eurasian languages are typically spoken (in syllables per second) and how much information they can effectively convey (in bits per second). They found that, although languages vary widely in the speed at which they are spoken, the amount of information languages can effectively convey tends to vary much less. Thus, they claim that two languages with very different spoken rates can nonetheless convey the same amount of information in a given amount of time.

Which choice best describes data from the table that support the researchers' claim?

A) Among the five languages in the table, Thai and Hungarian have the lowest rates of speech and the lowest rates of information conveyed.

B) Vietnamese conveys information at approximately the same rate as Spanish despite being spoken at a slower rate.

C) Among the five languages in the table, the language that is spoken the fastest is also the language that conveys information the fastest.

D) Serbian and Spanish are spoken at approximately the same rate, but Serbian conveys information faster than Spanish does.

14

Psychologists Dacher Keltner and Jonathan Haidt have argued that experiencing awe—a sensation of reverence and wonder typically brought on by perceiving something grand or powerful—can enable us to feel more connected to others and thereby inspire us to act more altruistically. Keltner, along with Paul K. Piff, Pia Dietze, and colleagues, claims to have found evidence for this effect in a recent study where participants were asked to either gaze up at exceptionally tall trees in a nearby grove (reported to be a universally awe-inspiring experience) or stare at the exterior of a nearby, nondescript building. After one minute, an experimenter deliberately spilled a box of pens nearby.

Which finding from the researchers' study, if true, would most strongly support their claim?

A) Participants who had been looking at the trees helped the experimenter pick up significantly more pens than did participants who had been looking at the building.

B) Participants who helped the experimenter pick up the pens used a greater number of positive words to describe the trees and the building in a postexperiment survey than did participants who did not help the experimenter.

C) Participants who did not help the experimenter pick up the pens were significantly more likely to report having experienced a feeling of awe, regardless of whether they looked at the building or the trees.

D) Participants who had been looking at the building were significantly more likely to notice that the experimenter had dropped the pens than were participants who had been looking at the trees.

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15

Employment by Sector in France and the United States, 1800–2012
(% of total employment)

Year	Agriculture in France	Manufacturing in France	Services in France	Agriculture in US	Manufacturing in US	Services in US
1800	64	22	14	68	18	13
1900	43	29	28	41	28	31
1950	32	33	35	14	33	53
2012	3	21	76	2	18	80

Rows in table may not add up to 100 due to rounding.

Over the past two hundred years, the percentage of the population employed in the agricultural sector has declined in both France and the United States, while employment in the service sector (which includes jobs in retail, consulting, real estate, etc.) has risen. However, this transition happened at very different rates in the two countries. This can be seen most clearly by comparing the employment by sector in both countries in _____

Which choice most effectively uses data from the table to complete the statement?

A) 1900 with the employment by sector in 1950.

B) 1800 with the employment by sector in 2012.

C) 1900 with the employment by sector in 2012.

D) 1800 with the employment by sector in 1900.

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16

Many archaeologists will tell you that categorizing excavated fragments of pottery by style, period, and what objects they belong to relies not only on standard criteria, but also on instinct developed over years of practice. In a recent study, however, researchers trained a deep-learning computer model on thousands of images of pottery fragments and found that it could categorize them as accurately as a team of expert archaeologists. Some archaeologists have expressed concern that they might be replaced by such computer models, but the researchers claim that outcome is highly unlikely.

Which finding, if true, would most directly support the researchers' claim?

A) In the researchers' study, the model was able to categorize the pottery fragments much more quickly than the archaeologists could.

B) In the researchers' study, neither the model nor the archaeologists were able to accurately categorize all the pottery fragments that were presented.

C) A survey of archaeologists showed that categorizing pottery fragments limits the amount of time they can dedicate to other important tasks that only human experts can do.

D) A survey of archaeologists showed that few of them received dedicated training in how to properly categorize pottery fragments.

17

Although military veterans make up a small proportion of the total population of the United States, they occupy a significantly higher proportion of the jobs in the civilian government. One possible explanation for this disproportionate representation is that military service familiarizes people with certain organizational structures that are also reflected in the civilian government bureaucracy, and this familiarity thus _____

Which choice most logically completes the text?

A) makes civilian government jobs especially appealing to military veterans.

B) alters the typical relationship between military service and subsequent career preferences.

C) encourages nonveterans applying for civilian government jobs to consider military service instead.

D) increases the number of civilian government jobs that require some amount of military experience to perform.

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18

Birds of many species ingest foods containing carotenoids, pigmented molecules that are converted into feather coloration. Coloration tends to be especially saturated in male birds' feathers, and because carotenoids also confer health benefits, the deeply saturated colors generally serve to communicate what is known as an honest signal of a bird's overall fitness to potential mates. However, ornithologist Allison J. Shultz and others have found that males in several species of the tanager genus *Ramphocelus* use microstructures in their feathers to manipulate light, creating the appearance of deeper saturation without the birds necessarily having to maintain a carotenoid-rich diet. These findings suggest that _____

Which choice most logically completes the text?

A) individual male tanagers can engage in honest signaling without relying on carotenoid consumption.

B) feather microstructures may be less effective than deeply saturated feathers for signaling overall fitness.

C) scientists have yet to determine why tanagers have a preference for mates with colorful appearances.

D) a male tanager's appearance may function as a dishonest signal of the individual's overall fitness.

19

When writing *The Other Black Girl* (2021), novelist Zakiya Dalila Harris drew on her own experiences working at a publishing office. The award-winning book is Harris's first novel, but her writing _____ honored before. At the age of twelve, she entered a contest to have a story published in *American Girl* magazine—and won.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) were

B) have been

C) has been

D) are

20

The Alvarez theory, developed in 1980 by physicist Luis Walter Alvarez and his geologist son Walter Alvarez, maintained that the secondary effects of an asteroid impact caused many dinosaurs and other animals to die _____ it left unexplored the question of whether unrelated volcanic activity might have also contributed to the mass extinctions.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) out but

B) out, but

C) out

D) out,

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21

In winter, the diets of Japanese macaques, also known as snow monkeys, are influenced more by food availability than by food preference. Although the monkeys prefer to eat vegetation and land-dwelling invertebrates, those food sources may become unavailable because of extensive snow and ice cover, _____ the monkeys to hunt for marine animals in any streams that have not frozen over.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) forces
B) to force
C) forcing
D) forced

22

Lucía Michel of the University of Chile observed that alkaline soils contain an insoluble form of iron that blueberry plants cannot absorb, thus inhibiting blueberry growth. If these plants were grown in alkaline soil alongside grasses that aid in iron solubilization, _____ Michel was determined to find out.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) could the blueberries thrive.
B) the blueberries could thrive.
C) the blueberries could thrive?
D) could the blueberries thrive?

23

In his 1963 exhibition *Exposition of Music—Electronic Television*, Korean American artist Nam June Paik showed how television images could be manipulated to express an artist's perspective. Today, Paik _____ considered the first video artist.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) will be
B) had been
C) was
D) is

24

The first computerized spreadsheet, Dan Bricklin's *VisiCalc*, improved financial recordkeeping not only by providing users with an easy means of adjusting data in spreadsheets but also by automatically updating all calculations that were dependent on these _____ to VisiCalc's release, changing a paper spreadsheet often required redoing the entire sheet by hand, a process that could take days.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) adjustments prior
B) adjustments, prior
C) adjustments. Prior
D) adjustments and prior

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25

In order to prevent nonnative fish species from moving freely between the Mediterranean and Red Seas, marine biologist Bella Galil has proposed that a saline lock system be installed along the Suez Canal in Egypt's Great Bitter Lakes. The lock would increase the salinity of the lakes and _____ a natural barrier of water most marine creatures would be unable to cross.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) creates
B) create
C) creating
D) created

26

Despite being cheap, versatile, and easy to produce, _____ they are made from nonrenewable petroleum, and most do not biodegrade in landfills.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) there are two problems associated with commercial plastics:
B) two problems are associated with commercial plastics:
C) commercial plastics' two associated problems are that
D) commercial plastics have two associated problems:

27

Stomata, tiny pore structures in a leaf that absorb gases needed for plant growth, open when guard cells surrounding each pore swell with water. In a pivotal 2007 article, plant cell _____ showed that lipid molecules called phosphatidylinositol phosphates are responsible for signaling guard cells to open stomata.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) biologist, Yuree Lee
B) biologist Yuree Lee,
C) biologist Yuree Lee
D) biologist, Yuree Lee,

28

Small, flat structures called spatulae are found at the tips of the hairs on a spider's leg. These spatulae temporarily bond with the atoms of whatever they touch. _____ spiders are able to cling to and climb almost any surface.

Which choice completes the text with the most logical transition?

A) For instance,
B) However,
C) Similarly,
D) As a result,

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29

In November 1934, Amrita Sher-Gil was living in what must have seemed like the ideal city for a young artist: Paris. She was studying firsthand the color-saturated style of France's modernist masters and beginning to make a name for herself as a painter. _____ Sher-Gil longed to return to her childhood home of India; only there, she believed, could her art truly flourish.

Which choice completes the text with the most logical transition?

A) Still,

B) Therefore,

C) Indeed,

D) Furthermore,

30

Before California's 1911 election to approve a proposition granting women the right to vote, activists across the state sold tea to promote the cause of suffrage. In San Francisco, the Woman's Suffrage Party sold Equality Tea at local fairs. _____ in Los Angeles, activist Nancy Tuttle Craig, who ran one of California's largest grocery store firms, distributed Votes for Women Tea.

Which choice completes the text with the most logical transition?

A) For example,

B) To conclude,

C) Similarly,

D) In other words,

31

While researching a topic, a student has taken the following notes:

- The Seikan Tunnel is a rail tunnel in Japan.
- It connects the island of Honshu to the island of Hokkaido.
- It is roughly 33 miles long.
- The Channel Tunnel is a rail tunnel in Europe.
- It connects Folkestone, England, to Coquelles, France.
- It is about 31 miles long.

The student wants to compare the lengths of the two rail tunnels. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Some of the world's rail tunnels, including one tunnel that extends from Folkestone, England, to Coquelles, France, are longer than 30 miles.

B) The Seikan Tunnel is roughly 33 miles long, while the slightly shorter Channel Tunnel is about 31 miles long.

C) The Seikan Tunnel, which is roughly 33 miles long, connects the Japanese islands of Honshu and Hokkaido.

D) Both the Seikan Tunnel, which is located in Japan, and the Channel Tunnel, which is located in Europe, are examples of rail tunnels.

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32

While researching a topic, a student has taken the following notes:

- Jon Ching is a Los Angeles-based painter.
- He uses the term "flauna" to describe the plant-animal hybrids that he depicts in his surreal paintings.
- "Flauna" is a combination of the words "flora" and "fauna."
- His painting *Nectar* depicts a parrot with leaves for feathers.
- His painting *Primaveral* depicts a snow leopard whose fur sprouts flowers.

The student wants to provide an explanation and example of "flauna." Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The term "flauna," used by Los Angeles-based painter Jon Ching, is a combination of the words "flora" and "fauna."

B) Jon Ching uses the term "flauna," a combination of the words "flora" and "fauna," to describe the subjects of his surreal paintings: plant-animal hybrids such as a parrot with leaves for feathers.

C) Jon Ching, who created *Nectar*, refers to the subjects of his paintings as "flauna."

D) The subjects of *Nectar* and *Primaveral* are types of "flauna," a term that the paintings' creator, Jon Ching, uses when describing his surreal artworks.

33

While researching a topic, a student has taken the following notes:

- In the midst of the US Civil War, Susie Taylor escaped slavery and fled to Union-army-occupied St. Simons Island off the Georgia coast.
- She began working for an all-Black army regiment as a nurse and teacher.
- In 1902, she published a book about the time she spent with the regiment.
- Her book was the only Civil War memoir to be published by a Black woman.
- It is still available to readers in print and online.

The student wants to emphasize the uniqueness of Taylor's accomplishment. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Taylor fled to St. Simons Island, which was then occupied by the Union army, for whom she began working.

B) After escaping slavery, Taylor began working for an all-Black army regiment as a nurse and teacher.

C) The book Taylor wrote about the time she spent with the regiment is still available to readers in print and online.

D) Taylor was the only Black woman to publish a Civil War memoir.

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

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No Test Material On This Page

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Math

27 QUESTIONS

DIRECTIONS

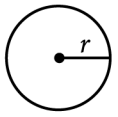
The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

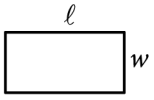
Unless otherwise indicated:

- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

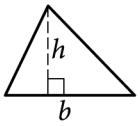
REFERENCE



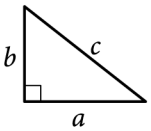
$A = \pi r^2$
$C = 2\pi r$



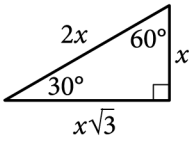
$A = \ell w$

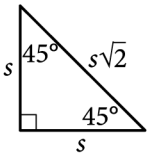


$A = \frac{1}{2}bh$

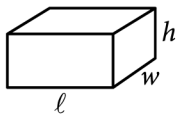


$c^2 = a^2 + b^2$

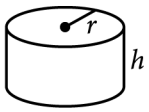




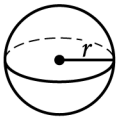
Special Right Triangles



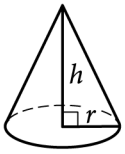
$V = \ell wh$



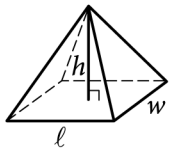
$V = \pi r^2 h$



$V = \frac{4}{3}\pi r^3$



$V = \frac{1}{3}\pi r^2 h$



$V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π.

The sum of the measures in degrees of the angles of a triangle is 180.

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For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find **more than one correct answer**, write and circle only one answer.
- Your answer can be up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer, but no more.
- If your answer is a **fraction** that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a **decimal** that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

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1

What is 10% of 470?

A) 37

B) 47

C) 423

D) 460

2

$$4x + 6 = 18$$

Which equation has the same solution as the given equation?

A) $4x = 108$

B) $4x = 24$

C) $4x = 12$

D) $4x = 3$

3

The total cost, in dollars, to rent a surfboard consists of a \$25 service fee and a \$10 per hour rental fee. A person rents a surfboard for t hours and intends to spend a maximum of \$75 to rent the surfboard. Which inequality represents this situation?

A) $10t \leq 75$

B) $10 + 25t \leq 75$

C) $25t \leq 75$

D) $25 + 10t \leq 75$

4

The function g is defined by $g(x) = x^2 + 9$. For which value of x is $g(x) = 25$?

A) 4

B) 5

C) 9

D) 13

5

Each face of a fair 14-sided die is labeled with a number from 1 through 14, with a different number appearing on each face. If the die is rolled one time, what is the probability of rolling a 2?

A) $\frac{1}{14}$

B) $\frac{2}{14}$

C) $\frac{12}{14}$

D) $\frac{13}{14}$

6

A printer produces posters at a constant rate of 42 posters per minute. At what rate, in posters per hour, does the printer produce the posters?

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7

The function f is defined by the equation $f(x) = 7x + 2$. What is the value of $f(x)$ when $x = 4$?

8

A teacher is creating an assignment worth 70 points. The assignment will consist of questions worth 1 point and questions worth 3 points. Which equation represents this situation, where x represents the number of 1-point questions and y represents the number of 3-point questions?

A) $4xy = 70$

B) $4(x + y) = 70$

C) $3x + y = 70$

D) $x + 3y = 70$

9

Right triangles LMN and PQR are similar, where L and M correspond to P and Q, respectively. Angle M has a measure of 53°. What is the measure of angle Q ?

A) 37°

B) 53°

C) 127°

D) 143°

10

$$y = -3x$$
$$4x + y = 15$$

The solution to the given system of equations is (x, y). What is the value of x ?

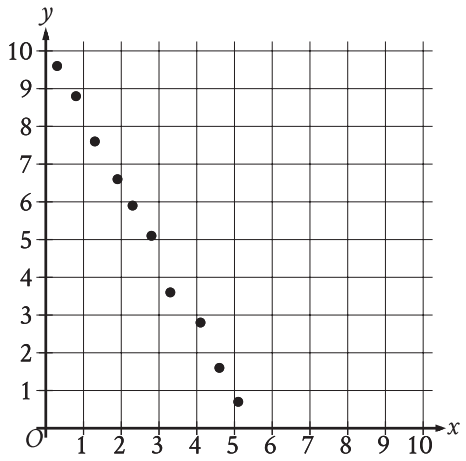
A) 1

B) 5

C) 15

D) 45

11



Which of the following equations is the most appropriate linear model for the data shown in the scatterplot?

A) $y = -1.9x - 10.1$

B) $y = -1.9x + 10.1$

C) $y = 1.9x - 10.1$

D) $y = 1.9x + 10.1$

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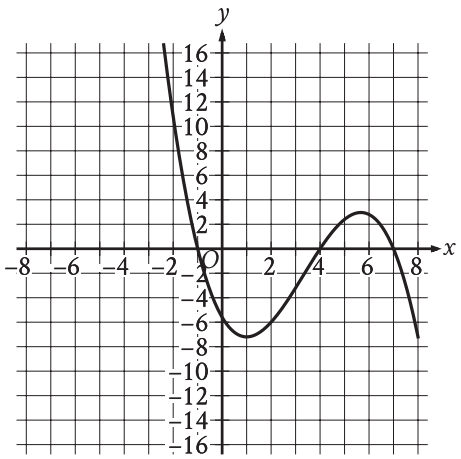
12



The graph of $y = f(x)$ is shown, where the function f is defined by $f(x) = ax^3 + bx^2 + cx + d$ and a, b, c, and d are constants. For how many values of x does $f(x) = 0$?

A) One

B) Two

C) Three

D) Four

13

Vivian bought party hats and cupcakes for $71. Each package of party hats cost $3, and each cupcake cost $1. If Vivian bought 10 packages of party hats, how many cupcakes did she buy?

14

$$z^2 + 10z - 24 = 0$$

What is one of the solutions to the given equation?

15

Bacteria are growing in a liquid growth medium. There were 300,000 cells per milliliter during an initial observation. The number of cells per milliliter doubles every 3 hours. How many cells per milliliter will there be 15 hours after the initial observation?

A) 1,500,000

B) 2,400,000

C) 4,500,000

D) 9,600,000

16

Which expression is equivalent to $6x^8y^2 + 12x^2y^2$?

A) $6x^2y^2(2x^6)$

B) $6x^2y^2(x^4)$

C) $6x^2y^2(x^6 + 2)$

D) $6x^2y^2(x^4 + 2)$

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CONTINUE

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17

A neighborhood consists of a 2-hectare park and a 35-hectare residential area. The total number of trees in the neighborhood is 3,934. The equation $2x + 35y = 3{,}934$ represents this situation. Which of the following is the best interpretation of x in this context?

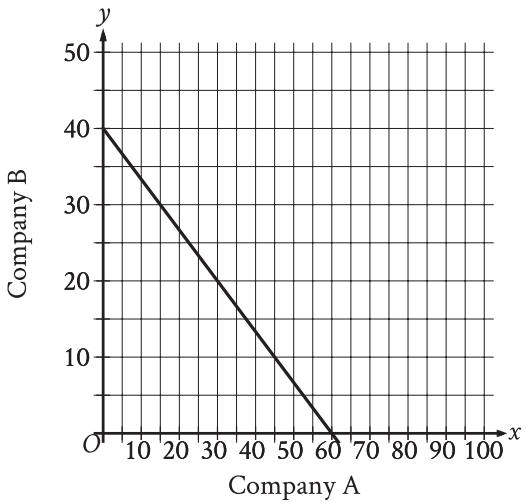
A) The average number of trees per hectare in the park

B) The average number of trees per hectare in the residential area

C) The total number of trees in the park

D) The total number of trees in the residential area

18



The graph shows the relationship between the number of shares of stock from Company A, x, and the number of shares of stock from Company B, y, that Simone can purchase. Which equation could represent this relationship?

A) $y = 8x + 12$

B) $8x + 12y = 480$

C) $y = 12x + 8$

D) $12x + 8y = 480$

19

Circle A has a radius of $3n$ and circle B has a radius of $129n$, where n is a positive constant. The area of circle B is how many times the area of circle A?

A) 43

B) 86

C) 129

D) 1,849

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CONTINUE

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20

Data value	Frequency
6	3
7	3
8	8
9	8
10	9
11	11
12	9
13	0
14	6

The frequency table summarizes the 57 data values in a data set. What is the maximum data value in the data set?

21

A circle in the xy-plane has a diameter with endpoints $(2, 4)$ and $(2, 14)$. An equation of this circle is $(x-2)^2 + (y-9)^2 = r^2$, where r is a positive constant. What is the value of r ?

22

The measure of angle R is $\frac{2\pi}{3}$ radians. The measure of angle T is $\frac{5\pi}{12}$ radians greater than the measure of angle R. What is the measure of angle T, in degrees?

A) 75
B) 120
C) 195
D) 390

23

A certain town has an area of 4.36 square miles. What is the area, in square yards, of this town? (1 mile = 1,760 yards)

A) 404
B) 7,674
C) 710,459
D) 13,505,536

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24

x	y
18	130
23	160
26	178

For line h, the table shows three values of x and their corresponding values of y. Line k is the result of translating line h down 5 units in the xy-plane. What is the x-intercept of line k?

A) $\left(-\frac{26}{3}, 0\right)$

B) $\left(-\frac{9}{2}, 0\right)$

C) $\left(-\frac{11}{3}, 0\right)$

D) $\left(-\frac{17}{6}, 0\right)$

25

In the xy-plane, the graph of the equation $y = -x^2 + 9x - 100$ intersects the line $y = c$ at exactly one point. What is the value of c?

A) $-\frac{481}{4}$

B) -100

C) $-\frac{319}{4}$

D) $-\frac{9}{2}$

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26

$$2x + 3y = 7$$
$$10x + 15y = 35$$

For each real number r, which of the following points lies on the graph of each equation in the xy-plane for the given system?

A) $\left(\frac{r}{5} + 7, -\frac{r}{5} + 35\right)$

B) $\left(-\frac{3r}{2} + \frac{7}{2}, r\right)$

C) $\left(r, \frac{2r}{3} + \frac{7}{3}\right)$

D) $\left(r, -\frac{3r}{2} + \frac{7}{2}\right)$

27

The perimeter of an equilateral triangle is 624 centimeters. The height of this triangle is $k\sqrt{3}$ centimeters, where k is a constant. What is the value of k?

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

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No Test Material On This Page

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Math

27 QUESTIONS

DIRECTIONS

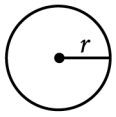
The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

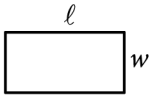
Unless otherwise indicated:

- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

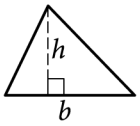
REFERENCE



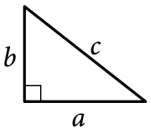
$A = \pi r^2$
$C = 2\pi r$



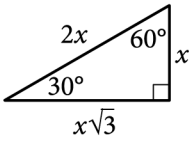
$A = \ell w$

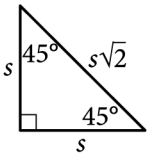


$A = \frac{1}{2}bh$

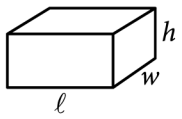


$c^2 = a^2 + b^2$

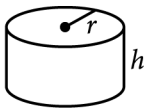




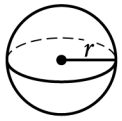
Special Right Triangles



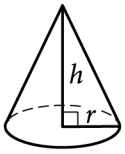
$V = \ell wh$



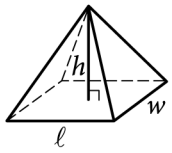
$V = \pi r^2 h$



$V = \frac{4}{3}\pi r^3$



$V = \frac{1}{3}\pi r^2 h$



$V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π.

The sum of the measures in degrees of the angles of a triangle is 180.

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CONTINUE

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For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find **more than one correct answer**, write and circle only one answer.
- Your answer can be up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer, but no more.
- If your answer is a **fraction** that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a **decimal** that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

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CONTINUE
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1

Tilly earns p dollars for every w hours of work. Which expression represents the amount of money, in dollars, Tilly earns for $39w$ hours of work?

A) $39p$

B) $\frac{p}{39}$

C) $p + 39$

D) $p - 39$

2

For a training program, Juan rides his bike at an average rate of 5.7 minutes per mile. Which function m models the number of minutes it will take Juan to ride x miles at this rate?

A) $m(x) = \frac{x}{5.7}$

B) $m(x) = x + 5.7$

C) $m(x) = x - 5.7$

D) $m(x) = 5.7x$

3

$$3x = 12$$
$$-3x + y = -6$$

The solution to the given system of equations is (x, y). What is the value of y ?

A) -3

B) 6

C) 18

D) 30

4

$$s = 40 + 3t$$

The equation gives the speed s, in miles per hour, of a certain car t seconds after it began to accelerate. What is the speed, in miles per hour, of the car 5 seconds after it began to accelerate?

A) 40

B) 43

C) 45

D) 55

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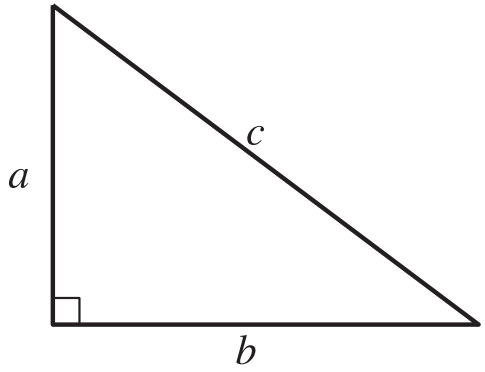
5



Note: Figure not drawn to scale.

For the right triangle shown, $a = 4$ and $b = 5$. Which expression represents the value of c ?

A) $4 + 5$

B) $\sqrt{(4)(5)}$

C) $\sqrt{4 + 5}$

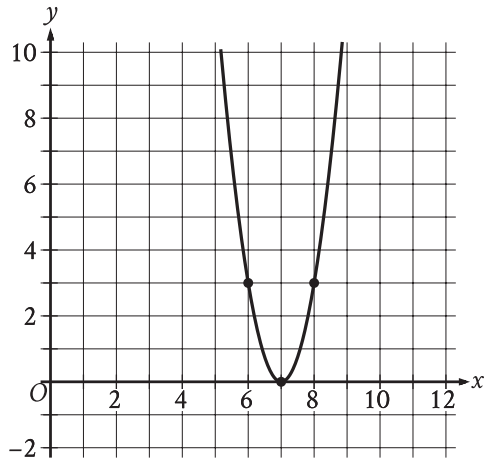
D) $\sqrt{4^2 + 5^2}$

6

$$4x + 5 = 165$$

What is the solution to the given equation?

7



The x-intercept of the graph shown is $(x, 0)$. What is the value of x ?

8

The function f is defined by $f(x) = \frac{1}{10}x - 2$. What is the y-intercept of the graph of $y = f(x)$ in the xy-plane?

A) $(-2, 0)$

B) $(0, -2)$

C) $\left(0, \frac{1}{10}\right)$

D) $\left(\frac{1}{10}, 0\right)$

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9

The function f is defined by $f(x) = 7x^3$. In the xy-plane, the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ down 2 units. Which equation defines function g ?

A) $g(x) = \frac{7}{2}x^3$

B) $g(x) = 7x^{\frac{3}{2}}$

C) $g(x) = 7x^3 + 2$

D) $g(x) = 7x^3 - 2$

10

$$x + 7 = 10$$
$$(x + 7)^2 = y$$

Which ordered pair (x, y) is a solution to the given system of equations?

A) $(3, 100)$

B) $(3, 3)$

C) $(3, 10)$

D) $(3, 70)$

11

Which expression is equivalent to $(7x^3 + 7x) - (6x^3 - 3x)$?

A) $x^3 + 10x$

B) $-13x^3 + 10x$

C) $-13x^3 + 4x$

D) $x^3 + 4x$

12

The function p is defined by $p(n) = 7n^3$. What is the value of n when $p(n)$ is equal to 56?

A) 2

B) $\frac{8}{3}$

C) 7

D) 8

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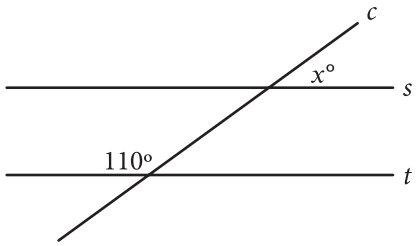
13



Note: Figure not drawn to scale.

In the figure shown, line c intersects parallel lines s and t. What is the value of x ?

14

A list of 10 data values is shown.

6, 8, 16, 4, 17, 26, 8, 5, 5, 5

What is the mean of these data?

15

The equation $E(t) = 5(1.8)^t$ gives the estimated number of employees at a restaurant, where t is the number of years since the restaurant opened. Which of the following is the best interpretation of the number 5 in this context?

A) The estimated number of employees when the restaurant opened

B) The increase in the estimated number of employees each year

C) The number of years the restaurant has been open

D) The percent increase in the estimated number of employees each year

16

$$g(x) = x^2 + 55$$

What is the minimum value of the given function?

A) 0

B) 55

C) 110

D) 3,025

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17

Each year, the value of an investment increases by 0.49% of its value the previous year. Which of the following functions best models how the value of the investment changes over time?

A) Decreasing exponential

B) Decreasing linear

C) Increasing exponential

D) Increasing linear

18

The population of Greenville increased by 7% from 2015 to 2016. If the 2016 population is k times the 2015 population, what is the value of k ?

A) 0.07

B) 0.7

C) 1.07

D) 1.7

19

Which expression is equivalent to $a^{\frac{11}{12}}$,

where $a > 0$?

A) $\sqrt[12]{a^{132}}$

B) $\sqrt[144]{a^{132}}$

C) $\sqrt[121]{a^{132}}$

D) $\sqrt[11]{a^{132}}$

20

An event planner is planning a party. It costs the event planner a onetime fee of \$35 to rent the venue and \$10.25 per attendee. The event planner has a budget of \$200. What is the greatest number of attendees possible without exceeding the budget?

21

If $|4x - 4| = 112$, what is the positive value of $x - 1$?

22

A cube has an edge length of 68 inches. A solid sphere with a radius of 34 inches is inside the cube, such that the sphere touches the center of each face of the cube. To the nearest cubic inch, what is the volume of the space in the cube not taken up by the sphere?

A) 149,796

B) 164,500

C) 190,955

D) 310,800

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CONTINUE

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23

What is the diameter of the circle in the xy-plane with equation $(x - 5)^2 + (y - 3)^2 = 16$?

A) 4

B) 8

C) 16

D) 32

24

For the exponential function f, the value of $f(1)$ is k, where k is a constant. Which of the following equivalent forms of the function f shows the value of k as the coefficient or the base?

A) $f(x) = 50(1.6)^{x+1}$

B) $f(x) = 80(1.6)^{x}$

C) $f(x) = 128(1.6)^{x-1}$

D) $f(x) = 204.8(1.6)^{x-2}$

25

A model estimates that at the end of each year from 2015 to 2020, the number of squirrels in a population was 150% more than the number of squirrels in the population at the end of the previous year. The model estimates that at the end of 2016, there were 180 squirrels in the population. Which of the following equations represents this model, where n is the estimated number of squirrels in the population t years after the end of 2015 and $t \leq 5$?

A) $n = 72(1.5)^{t}$

B) $n = 72(2.5)^{t}$

C) $n = 180(1.5)^{t}$

D) $n = 180(2.5)^{t}$

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CONTINUE

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26

$$5x + 7y = 1$$

$$ax + by = 1$$

In the given pair of equations, a and b are constants. The graph of this pair of equations in the xy-plane is a pair of perpendicular lines. Which of the following pairs of equations also represents a pair of perpendicular lines?

A) $10x + 7y = 1$
$ax - 2by = 1$

B) $10x + 7y = 1$
$ax + 2by = 1$

C) $10x + 7y = 1$
$2ax + by = 1$

D) $5x - 7y = 1$
$ax + by = 1$

27

$$x^2 - 34x + c = 0$$

In the given equation, c is a constant. The equation has no real solutions if $c > n$. What is the least possible value of n?

STOP

If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.

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GENERAL DIRECTIONS

- You may work on only one module at a time.
- If you finish a module before time is called, check your work on that module only. You may NOT turn to any other module.

MARKING YOUR ANSWERS

- Be sure to answer your questions properly in this book.
- Circle only one answer to each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

USING YOUR TEST BOOK

- You may use the test book for scratch work.
- You may not fold or remove pages or portions of a page from this book, or take the book from the testing room.

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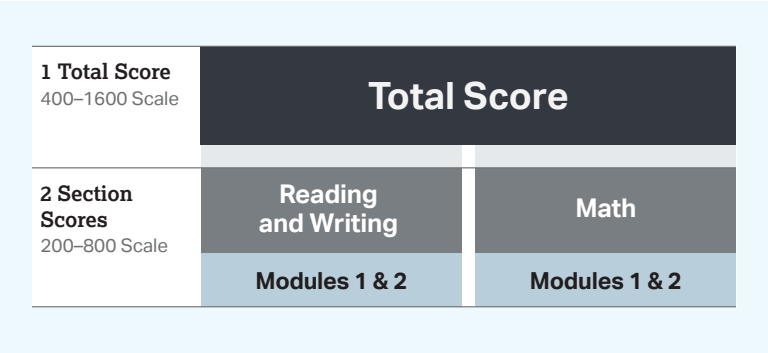
Scoring Your Paper SAT Practice Test #1

IMPORTANT: *This guide is for students who completed a paper version of the digital SAT practice test. The scoring method used for this nondigital version isn't exactly the same as the digital version.*



Congratulations on completing an SAT® practice test. To score your test, follow the instructions in this guide.

Scores Overview

Each assessment in the SAT Suite (SAT, PSAT/NMSQT®, PSAT™ 10, and PSAT™ 8/9) reports test scores on a common scale.

For more details about scores, visit **sat.org/scores**.

The College Board Assessment Design & Development team developed the practice tests using the same processes and review standards they used when developing the actual SAT.

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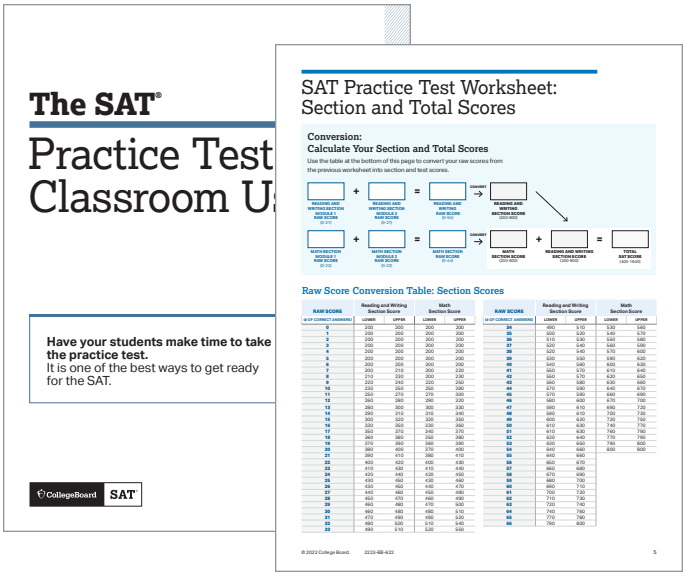
How to Calculate Your Practice Test Scores

The worksheets on pages 4 and 5 help you calculate your test scores.

GET SET UP

1. In addition to your practice test, you'll need the conversion tables and answer key at the end of this guide.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Practice Test
Classroom U

Have your students make time to take the practice test.
It is one of the best ways to get ready for the SAT.

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SAT Practice Test Worksheet:
Section and Total Scores

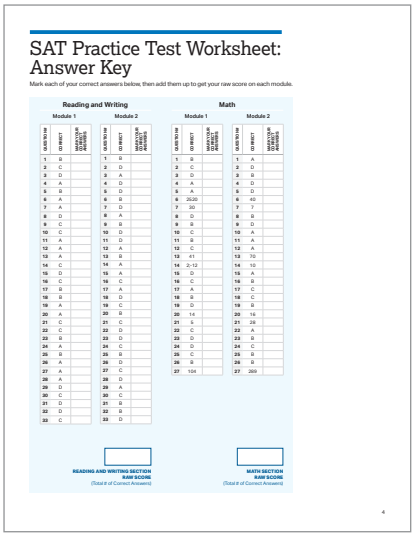
Conversion:
Calculate Your Section and Total Scores

Use the table at the bottom of this page to convert your raw scores from the previous worksheet into section and test scores.

Raw Score Conversion Table: Section Scores

SCORE YOUR PRACTICE TEST

2. Compare your answers to the answer key on page 4, and count your total correct answers for each section. Write the number of correct answers for each section in the answer key at the bottom of that section.


SAT Practice Test Worksheet:
Answer Key

Mark each of your correct answers below, then add them up to get your raw score on each module.

Reading and Writing

Module 1

Module 2

Math

Module 1

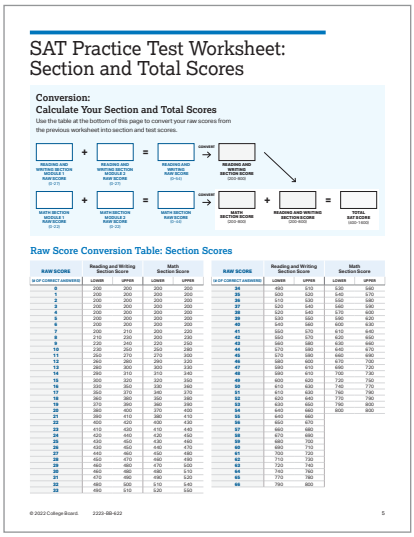
Module 2

READING AND WRITING SECTION RAW SCORE
(Total # of Correct Answers)

MATH SECTION RAW SCORE
(Total # of Correct Answers)

CALCULATE YOUR SCORES

3. Using your marked-up answer key and the conversion tables, follow the directions on page 5 to get your section and test scores.


SAT Practice Test Worksheet:
Section and Total Scores

Conversion:
Calculate Your Section and Total Scores

Use the table at the bottom of this page to convert your raw scores from the previous worksheet into section and test scores.

Raw Score Conversion Table: Section Scores

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Get Section and Total Scores

Your total score on an SAT practice test is the sum of your scores in the modules for both the Reading and Writing and Math sections. To get your total score, you'll convert what we call the "raw score" for each section—the number of questions you got right in that section—into the "scaled score" for that section, and then calculate the total score.

GET YOUR READING AND WRITING SECTION SCORE

Calculate your SAT Reading and Writing section score (it's on a scale of 200–800).

1. Use the answer key on page 4 to count the number of correct answers you got on module 1 and module 2.
2. To determine your Reading and Writing raw score, add the number of correct answers you got on module 1 and module 2.
3. Use the Raw Score Conversion Table: Section Scores on page 5 to turn your raw score into your Reading and Writing section score.

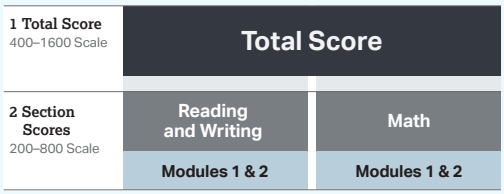
GET YOUR MATH SECTION SCORE

Calculate your SAT Math section score (it's on a scale of 200–800).

1. Use the answer key on page 4 to count the number of correct answers you got on module 1 and module 2.
2. To determine your Math raw score, add the number of correct answers you got on module 1 and module 2.
3. Use the Raw Score Conversion Table: Section Scores on page 5 to turn your raw score into your Math section score.

GET YOUR TOTAL SCORE

Add your Reading and Writing section score to your Math section score. The result is your total score on the SAT practice test, on a scale of 400–1600.



Your total score on the SAT practice test is the sum of your Reading and Writing section score and your Math section score.

Use the worksheets on pages 4 and 5 to calculate your section and total scores.

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SAT Practice Test Worksheet: Answer Key

Mark each of your correct answers below, then add them up to get your raw score on each module.

Reading and Writing

Module 1

QUESTION #	CORRECT	MARK YOUR CORRECT ANSWERS
1	B	
2	C	
3	D	
4	A	
5	B	
6	A	
7	A	
8	D	
9	C	
10	C	
11	A	
12	A	
13	A	
14	C	
15	D	
16	C	
17	B	
18	B	
19	A	
20	A	
21	C	
22	C	
23	B	
24	A	
25	B	
26	A	
27	A	
28	A	
29	D	
30	C	
31	D	
32	D	
33	C	

Module 2

QUESTION #	CORRECT	MARK YOUR CORRECT ANSWERS
1	B	
2	D	
3	A	
4	D	
5	D	
6	B	
7	D	
8	A	
9	B	
10	D	
11	D	
12	A	
13	B	
14	A	
15	A	
16	C	
17	A	
18	D	
19	C	
20	B	
21	C	
22	D	
23	D	
24	C	
25	B	
26	D	
27	C	
28	D	
29	A	
30	C	
31	B	
32	B	
33	D	

READING AND WRITING SECTION RAW SCORE
(Total # of Correct Answers)

Math

Module 1

QUESTION #	CORRECT	MARK YOUR CORRECT ANSWERS
1	B	
2	C	
3	D	
4	A	
5	A	
6	2520	
7	30	
8	D	
9	B	
10	C	
11	B	
12	C	
13	41	
14	2; -12	
15	D	
16	C	
17	A	
18	B	
19	D	
20	14	
21	5	
22	C	
23	D	
24	D	
25	C	
26	B	
27	104	

Module 2

QUESTION #	CORRECT	MARK YOUR CORRECT ANSWERS
1	A	
2	D	
3	B	
4	D	
5	D	
6	40	
7	7	
8	B	
9	D	
10	A	
11	A	
12	A	
13	70	
14	10	
15	A	
16	B	
17	C	
18	C	
19	B	
20	16	
21	28	
22	A	
23	B	
24	C	
25	B	
26	B	
27	289	

MATH SECTION RAW SCORE
(Total # of Correct Answers)

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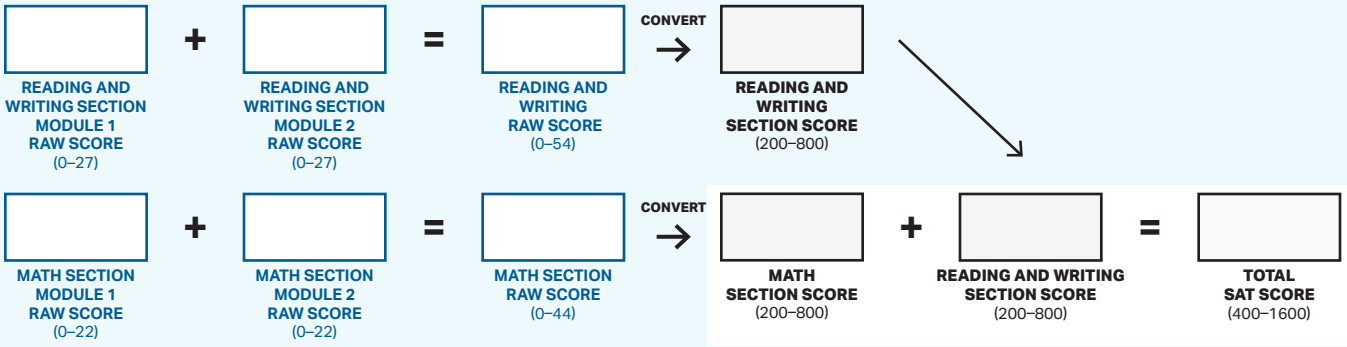
SAT Practice Test Worksheet: Section and Total Scores

Conversion: Calculate Your Section and Total Scores

Use the table at the bottom of this page to convert your raw scores from the previous worksheet into section and test scores.



Raw Score Conversion Table: Section Scores

RAW SCORE	Reading and Writing Section Score		Math Section Score	
(# OF CORRECT ANSWERS)	LOWER	UPPER	LOWER	UPPER
0	200	200	200	200
1	200	200	200	200
2	200	200	200	200
3	200	200	200	200
4	200	200	200	200
5	200	200	200	200
6	200	200	200	200
7	200	210	200	220
8	210	230	200	230
9	220	240	220	250
10	230	250	250	280
11	250	270	270	300
12	260	280	290	320
13	280	300	300	330
14	290	310	310	340
15	300	320	320	350
16	330	350	330	360
17	350	370	340	370
18	360	380	350	380
19	370	390	360	390
20	380	400	370	400
21	390	410	380	410
22	400	420	400	430
23	410	430	410	440
24	420	440	420	450
25	430	450	430	460
26	430	450	440	470
27	440	460	450	480
28	450	470	460	490
29	460	480	470	500
30	460	480	480	510
31	470	490	490	520
32	480	500	510	540
33	490	510	520	550

RAW SCORE	Reading and Writing Section Score		Math Section Score	
(# OF CORRECT ANSWERS)	LOWER	UPPER	LOWER	UPPER
34	490	510	530	560
35	500	520	540	570
36	510	530	550	580
37	520	540	560	590
38	520	540	570	600
39	530	550	590	620
40	540	560	600	630
41	550	570	610	640
42	550	570	620	650
43	560	580	630	660
44	570	590	640	670
45	570	590	660	690
46	580	600	670	700
47	590	610	690	720
48	590	610	700	730
49	600	620	720	750
50	610	630	740	770
51	610	630	760	790
52	620	640	770	790
53	630	650	790	800
54	640	660	800	800
55	640	660		
56	650	670		
57	660	680		
58	670	690		
59	680	700		
60	690	710		
61	700	720		
62	710	730		
63	720	740		
64	740	760		
65	770	780		
66	790	800		

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The SAT®

Practice Test #1

ANSWER EXPLANATIONS

These answer explanations are for students taking the digital SAT in nondigital format.

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Reading and Writing

Module 1
(33 questions)

QUESTION 1

Choice B is the best answer because it most logically completes the text's discussion of Ochoa's prediction that humans will one day need to live in places other than Earth. As used in this context, "speculates" would mean puts forward an idea without firm evidence. The text states that Ochoa "doesn't have a definite idea" about when humans might need to live in other environments and characterizes Ochoa's prediction as a "conjecture," or a conclusion presented without convincing evidence. This context indicates that Ochoa speculates when she makes this prediction.

Choice A is incorrect because saying that Ochoa "demands," or insists or requires, that humans will one day need to live in other environments than Earth's wouldn't make sense in context. The text indicates that she's unsure about the timing but hypothesizes that it will someday happen. *Choice C* is incorrect because saying that Ochoa "doubts," or questions or disbelieves, that humans will one day need to live in other environments than Earth's wouldn't make sense in context. The text indicates that although Ochoa is unsure about the timing, she hypothesizes that humans will need to live in places other than Earth and encourages research into future travel to the moon. *Choice D* is incorrect because saying that Ochoa "establishes," or proves, that humans will one day need to live in other environments than Earth's wouldn't make sense in context. Rather than stating that Ochoa discusses her idea with certainty and supports it with evidence, the text indicates that Ochoa is unsure about when humans might need to live in other environments.

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QUESTION 2

Choice C is the best answer because it most logically completes the text's discussion of Annie Dodge Wauneka's work as a Navajo Nation legislator. As used in this context, "persistent" means existing continuously. The text states that Wauneka "continuously worked to promote public health," traveling extensively and authoring a medical dictionary; this indicates that Wauneka's effort was persistent.

Choice A is incorrect because describing Wauneka's effort related to public health as "impartial," or not partial or biased and treating all things equally, wouldn't make sense in context. The text suggests that Wauneka's continuous work was partial in one way, as she focused specifically on promoting public health throughout the Navajo homeland and to speakers of the Navajo language. *Choice B* is incorrect because the text emphasizes that Wauneka's effort to promote public health as a Navajo Nation legislator was continuous and extensive, involving wide travels and the authoring of a medical dictionary. Because this work clearly involved care and dedication, it wouldn't make sense to describe it as "offhand," or casual and informal. *Choice D* is incorrect because nothing in the text suggests that Wauneka's effort to promote public health was "mandatory," or required by law or rule, even though Wauneka was a Navajo Nation legislator. Rather than suggesting that Wauneka's effort was required for any reason, the text emphasizes the continuous and extensive nature of her work.

QUESTION 3

Choice D is the best answer because it most logically completes the text's discussion of the collaboration between the Crow Tribe and Montana State University. As used in this context, "exemplifies" means demonstrates. The text conveys how the Crow Tribe–Montana State University collaboration serves to illustrate the model of community-based participatory research introduced earlier in the text and expanded on later in the text.

Choice A is incorrect because referring to "circumvents," or avoids, wouldn't make sense in context. The text suggests that the Crow Tribe–Montana State University collaboration serves as an example of the principles of community-based participatory research, not that the collaboration evades this model. *Choice B* is incorrect because referring to "eclipses," or overshadows, wouldn't make sense in context. The text describes the Crow Tribe–Montana State University collaboration as an equal partnership, which indicates that it's an example of the community-based participatory research model, not that it overshadows the model. *Choice C* is incorrect because saying that the collaboration "fabricates," or creates, the model wouldn't make sense in context. The text indicates that the Crow Tribe–Montana State University collaboration serves as an example of the model, not that it created the model.

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QUESTION 4

Choice A is the best answer because it most logically completes the text's discussion of a relationship between the dodder plant and its host plant. As used in this context, "synchronization" means the act of things happening at the same time. The text indicates that the dodder and its host plant flower in unison and that this synchronization occurs because the dodder makes use of a protein produced by the host shortly before flowering.

Choice B is incorrect because referring to "hibernation," or the state of being dormant or inactive, wouldn't make sense in context. The text focuses on something the dodder plant actively engages in—making use of a protein and producing flowers. *Choice C* is incorrect because stating that the dodder plant and its host engage together in "prediction," or the act of declaring or indicating something in advance, wouldn't make sense in context. Rather than indicating that the dodder plant and its host plant make a prediction about flowering activity, the text suggests that the host produces a protein as part of its regular flowering process and that the dodder then absorbs and uses that protein to flower at the same time. *Choice D* is incorrect because referring to "moderation," or the act of causing something to become less intense or extreme, wouldn't make sense in context. Although the text states that the dodder plant absorbs and uses a protein made by its host plant, it doesn't suggest that the dodder lessens the host plant's flowering activity; the two plants simply flower in unison.

QUESTION 5

Choice B is the best answer because it most logically completes the text's description of efforts to explain the existence of planets in binary star systems. As used in this context, describing an explanation as "a straightforward" one would mean that the explanation is direct and uncomplicated. The text asserts that since it should be "nearly impossible" for planets to form in binary star systems, it's "not surprising" that there isn't a straightforward explanation for the existence of planets in such systems; the fact that one potential approach involves "complex" factors offers further contextual support for this idea.

Choice A is incorrect because it would not make sense in context to say that there isn't "a discernible" explanation—meaning an explanation capable of being perceived—for the existence of planets in binary star systems. The text discusses just such an explanation offered by Roman Rafikov and Kedron Silsbee, which indicates that their explanation can be discerned. *Choice C* is incorrect because the text emphasizes how difficult it is to explain the existence of planets in binary star systems, suggesting that the situation isn't marked by the lack of "an inconclusive" explanation—an explanation that does not resolve the issue—but rather that if any explanations have been offered, they've likely been inconclusive ones. *Choice D* is incorrect because nothing in the text suggests that there is a lack of "an unbiased," or impartial and unprejudiced, explanation for the existence of planets in binary star systems. The text indicates that it's difficult to explain the existence of planets in such systems and it describes one attempt to do so, but there is no evidence that explanations from Roman Rafikov and Kedron Silsbee or others are biased.

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QUESTION 6

Choice A is the best answer because it most logically completes the text's discussion of Sterlin Harjo's approach to representing Native characters on television. As used in this context, "repudiates" means rejects or refuses to have anything to do with. The text indicates that television shows tend to depict Native characters as living long ago, but that Harjo's series *Reservation Dogs* focuses on Native teenagers in the present day, representing a "rejection" of the typical approach to depicting Native characters. This context thus indicates that Harjo repudiates television's general tendency regarding Native characters.

Choice B is incorrect because the text describes Harjo's "rejection" of the typical approach to representing Native characters on television, so it wouldn't make sense to say that Harjo "proclaims," or declares or affirms, television's general tendency regarding Native characters. Harjo is described as refusing to follow the pattern of depicting Native characters in the distant past, not as proclaiming that pattern. *Choice C* is incorrect because the text describes television's tendency to represent Native characters in the distant past as something that is already occurring, not as something that Harjo "foretells," or predicts will happen in the future. The text is focused on Harjo's "rejection" of this pattern, not on any predictions he may have about it. *Choice D* is incorrect because saying that Harjo "recants" something would mean that he withdraws a previously held belief, and it wouldn't make sense to say that Harjo recants television's tendency to represent Native characters as living in the past. No beliefs previously held by Harjo are mentioned. Additionally, a tendency isn't a belief and thus isn't something that can be recanted.

QUESTION 7

Choice A is the best answer because it most accurately states the main purpose of the text. After providing a brief introduction to computer scientist Luis von Ahn, the text focuses on discussing how von Ahn's digitization work led to the invention of a digital security test known as reCAPTCHA.

Choice B is incorrect because the text doesn't address how digital scanners work. *Choice C* is incorrect. Although the text mentions von Ahn's book-digitizing project, that information is provided as a detail, not as the main purpose of the text. *Choice D* is incorrect because the text doesn't provide any indication of reCAPTCHA's popularity; instead, it describes reCAPTCHA's origin.

QUESTION 8

Choice D is the best answer because it best describes how the underlined sentence functions in the text as a whole. The first sentence of the text establishes that Lily can be "keenly sensitive to" scenes that serve as a "fitting background" for her feelings—that is, she's very aware of when a setting seems to reflect her mood. The next sentence, which is underlined, then demonstrates this awareness: Lily views the landscape she's in as a large-scale reflection of her current mood, identifying with elements such as its calmness. Thus, the function of the underlined sentence is to illustrate an idea introduced in the previous sentence.

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Choice A is incorrect because the underlined sentence describes the scene only in very general terms, referring to its calmness, breadth, and long stretches of land. It's the next sentence that adds specific details about colors, light, and various trees nearby. *Choice B* is incorrect because nothing in the underlined sentence suggests that Lily is experiencing an internal conflict. In fact, the sentence indicates that Lily thinks the landscape reflects her own feeling of calmness. *Choice C* is incorrect because the only assertion in the underlined sentence is that Lily feels that broad aspects of the landscape, such as its calmness, reflect her current mood, and that assertion isn't expanded on in the next sentence. Instead, the next sentence describes specific details of the scene without connecting them to Lily's feelings.

QUESTION 9

Choice C is the best answer because it best describes how the underlined sentence functions in the text as a whole. The first sentence presents the implications of Veeraraghavan's team's study: sunshine exposure during work hours can cause overly optimistic behavior. The underlined sentence then describes the data the team consulted and how they were used (comparing predictions about earnings to what the companies actually earned), and the final sentence presents what the team found in their examination of the data. Thus, the underlined sentence mainly functions to explain part of the methodology used in the team's study.

Choice A is incorrect because the underlined sentence explains in part how the team conducted their analysis of the effect of sunshine but doesn't address what the team found; a broad summary is instead given in the other two sentences. *Choice B* is incorrect because the underlined sentence doesn't present any specific examples from the team's comparisons of 29,000 earnings predictions to actual earnings; it simply explains in part how the team conducted their analysis. *Choice D* is incorrect because the underlined sentence simply explains in part how the team conducted their analysis; the text never mentions any challenges that the team encountered in their study.

QUESTION 10

Choice C is the best answer because it describes something that is true of Mother, as presented in the text. The text indicates that in addition to other activities, Mother writes stories for her children while they are at school and makes up "funny pieces of poetry" for certain occasions.

Choice A is incorrect because the text suggests that Mother prefers to spend her time with her children and doesn't sit at home hoping that ladies will visit her. *Choice B* is incorrect because the text says only that Mother makes up poetry for the children's birthdays, not that she likes birthdays more than other special occasions. *Choice D* is incorrect because the text doesn't suggest that Mother prefers reading to her children over the other activities she does with them, such as playing with them and writing stories and poems for them.

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QUESTION 11

Choice A is the best answer because it most accurately states the main purpose of the text. In the first part of the text, the speaker addresses Paul Laurence Dunbar's ability to understand people (he has "read the hearts and souls of men" and written of their "joy and mirth"). In the second part of the text, the speaker describes Dunbar's thorough understanding of the natural world (he has read "the language of the flowers" and engaged with "the little brook"). Thus, the text mainly praises Dunbar for being especially perceptive about people and nature.

Choice B is incorrect because the speaker describes Dunbar as having read the "hearts and souls of men" and the "language of flowers" to convey Dunbar's ability to comprehend people and nature, not to suggest that Dunbar has literally read any of these things or has read a great deal about them. *Choice C* is incorrect because the text notes how well Dunbar has made sense of the topics he's written about but doesn't address any specific parts of Dunbar's writing process beyond the suggestion that he used a pen. *Choice D* is incorrect because the text focuses on Dunbar's understanding of people and nature as expressed in his writing. Nothing in the text suggests that the speaker is recalling a particular afternoon actually spent in nature with Dunbar; even if there had been a shared experience, the text isn't focused on reminiscing.

QUESTION 12

Choice A is the best answer because it presents the quotation that most directly illustrates the claim that Whitman's poem suggests that its readers haven't fully understood themselves. This quotation makes that point directly by saying to readers, "You have not known what you are." The quotation goes on to reinforce this point using a metaphor of sleep, saying that readers have "slumber'd" and that their "eyelids have been the same as closed most of the time."

Choice B is incorrect because this quotation doesn't suggest that readers haven't fully understood themselves but instead characterizes readers as "immense" and "interminable." Although immense and interminable things can be difficult to understand, this quotation doesn't make that point. *Choice C* is incorrect because this quotation doesn't suggest that readers haven't fully understood themselves but instead conveys the speaker's regret over not having celebrated readers sooner. In fact, this quotation says nothing at all about readers themselves—it's focused solely on the speaker's feelings about readers. *Choice D* is incorrect because this quotation doesn't suggest that readers haven't fully understood themselves; instead, this quotation makes the point that the speaker has understood readers and is determined to create "hymns" about them.

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QUESTION 13

Choice A is the best answer because it presents a finding that, if true, would support the claim about Chambi's photographs. The text describes a student advancing the claim that Chambi's photographs "have considerable ethnographic value"—meaning that they are valuable as records of cultures—and that they "capture diverse elements of Peruvian society" in a respectful way. If it's true that Chambi carefully photographed people from a range of different communities in Peru as well as photographed the customs and sites of different communities, that would lend support to the claim that the photographs have ethnographic value as depictions of diverse elements of society in Peru.

Choice B is incorrect because the student's claim is that Chambi's photographs have considerable ethnographic value because they depict diverse elements of Peruvian society; the student doesn't claim anything about the technical skill demonstrated in the photographs. *Choice C* is incorrect because neither Chambi's reputation nor the locations where his photographs may have been published would be relevant to the student's claim that his photographs are valuable as an ethnographic record of Peru's diverse society. *Choice D* is incorrect because the popularity among other photographers of the people and places that Chambi photographed would be irrelevant to the student's claim that Chambi's photographs are valuable as an ethnographic record of Peru's diverse society.

QUESTION 14

Choice C is the best answer because it uses data from the table to effectively exemplify the idea that the film outputs of the four individuals included in the table should be considered bare minimums—that is, that we should assume that the individuals actually had higher outputs than those recorded. The table presents the years during which the individuals were active and the number of known films the individuals are credited in. The table indicates that Lillian St. Cyr has 66 film credits as an actor and that Edwin Carewe has 58 film credits as a director; it follows that if some films and records for the era were lost, it's possible that Lillian St. Cyr acted in far more than 66 films and that Edwin Carewe directed more than 58 films.

Choice A is incorrect because it doesn't effectively exemplify the idea that the film outputs of the four individuals included in the table should be considered bare minimums. Rather than addressing the idea that the individuals likely had higher outputs than those presented in the table, this choice simply compares data from the table to make the point that Dark Cloud has fewer credited acting roles than Lillian St. Cyr (35 and 66, respectively). *Choice B* is incorrect because it misrepresents data from the table, even though it may exemplify the idea that the film outputs of the four individuals included in the table should be considered bare minimums by implying that Edwin Carewe acted in more than 47 films. The table indicates that Edwin Carewe was active from 1912 to 1934, meaning that his 47 credited acting roles were in films made before or during 1934, not after that time. *Choice D* is incorrect because it doesn't effectively exemplify the idea that the film outputs of the four individuals included in the table should be considered

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bare minimums. Instead of addressing the idea that the individuals likely had higher outputs than those recorded, this choice suggests that James Young Deer actually acted in and directed fewer films than presented in the table (only 33 known films as a director instead of 35, and only 10 known films as an actor instead of 33).

QUESTION 15

Choice D is the best answer because it provides the most direct support from the table for the claim that the plants growing in close proximity to other plants gained an advantage at an early developmental stage. The table shows the total number of juvenile plants from five species that were found growing on bare ground and in patches of vegetation as well as the percentage of the total number of each species that were growing in patches of vegetation. For each of the five species, more than 50% of the juvenile plants were growing in patches of vegetation. The text notes, however, that a random distribution of plants across the landscape should result in only about 15% of the plants being found in patches of vegetation. In other words, for each of the five species, the percentage of juvenile plants found growing in patches of vegetation was substantially higher than could be explained by chance alone. This finding supports the claim in the text: if plants growing in patches are overrepresented among plants that have survived to the juvenile stage, as the data show they are, then it suggests that it's advantageous for plants at an early stage of development to grow in patches of vegetation.

Choice A is incorrect because the statement that less than 75% of juvenile plants were found growing in patches of vegetation, while true, doesn't clearly support the claim that the plants growing in close proximity to other plants gained an advantage at an early developmental stage. Saying that less than 75% of plants were found in patches doesn't indicate how the percentage growing in patches compares with the percentage that would be expected to grow in patches on the basis of chance alone, which is the information necessary to evaluate whether the claim in the text has support in the table. Put another way, if the percentage of plants found growing in patches was 15% or less, it would be true that less than 75% were found in patches, but the data would in fact weaken the claim in the text, not strengthen it, since the data would show that growing in patches wasn't advantageous. *Choice B* is incorrect because only 12 plants of this species were found growing in patches, which was the lowest number of any species, not the greatest number. Additionally, even if it were true that this species had the greatest number of plants growing in patches, the finding would be irrelevant to the claim that plants of all five species gained an advantage by growing in close proximity to other plants. *Choice C* is incorrect because 59.1% of the plants of these species were found growing in patches, which is a far greater percentage, not a lower percentage, than what would be expected if plants were randomly distributed (around 15%). Additionally, if it were true that the percentage of plants growing in patches was lower for these species than what would be expected from chance alone, that finding would weaken, not strengthen, the claim that growing in patches is advantageous.

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QUESTION 16

Choice C is the best answer because it presents a finding that, if true, would support the researchers' hypothesis about the plants' dependence on dissolving rock. The text indicates that the roots of the two plant species grow directly into quartzite rock, where hairs on the roots secrete acids that dissolve the rock. The researchers hypothesize that the plants depend on this process because dissolving rock opens spaces for the roots to grow and releases phosphates that provide the plants with phosphorous, a vital nutrient. If the plants carry out this process of dissolving rock even when the rock already has spaces into which the roots could grow, that would support the researchers' hypothesis because it suggests that the plants are getting some advantage—such as access to phosphorous—from the action of dissolving rock. If the plants don't benefit from dissolving rock, they would be expected to grow in the cracks that already exist, as doing so would mean that the plants don't have to spend energy creating and secreting acids; if, however, the plants create new entry points by dissolving rock even when cracks already exist, that would support the hypothesis that they depend on dissolving rock for some benefit.

Choice A is incorrect because the existence of soil-inhabiting members of the Velloziaceae family with similar root structures to those of the two species discussed in the text wouldn't support the researchers' hypothesis that the species discussed in the text depend on dissolving rock. If other such members exist, that might suggest that the root structures can serve more functions than secreting acids to dissolve rock (since dissolving rock may not be necessary for plants living in soil), but that wouldn't suggest anything about whether the species discussed in the text benefit from dissolving rock. *Choice B* is incorrect because differences in the proportions of citric and malic acid secreted by the two species would be irrelevant to the hypothesis that the plants depend on dissolving rock. There's no information in the text to suggest that the proportion of each acid has any bearing on the process of dissolving rock or on any benefits the plants might receive from that process. *Choice D* is incorrect because if the two species thrive on rocks without phosphates, that would weaken the researchers' hypothesis that the plants depend on dissolving rock partly because dissolving rock gives them access to phosphates. If the plants can survive on rocks without getting a vital nutrient by dissolving those rocks, then either the nutrient isn't actually vital for those plants or they can get the nutrient in some way other than by dissolving rocks.

QUESTION 17

Choice B is the best answer because it presents the conclusion that most logically follows from the text's discussion of the relationship between atmospheric carbon dioxide and sauropod body size. The text establishes that sauropods evolved to reach enormous sizes, and it notes that some scientists have asserted that the cause of this phenomenon was increased plant production that resulted from increased atmospheric carbon dioxide. The text goes on to

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state, however, that atmospheric carbon dioxide levels didn't increase around the time of important periods in sauropods' evolution of larger body sizes. If significant periods of sauropod evolution toward larger sizes occurred without increased atmospheric carbon dioxide levels, that suggests that the evolution of larger sizes didn't depend on increased carbon dioxide in the atmosphere.

Choice A is incorrect because the text doesn't describe any fluctuations in atmospheric carbon dioxide, so there's no evidence in the text to support the conclusion that such fluctuations had different effects on different sauropod lineages. All that the text says about atmospheric carbon dioxide levels is that there weren't increases at particular points that correspond with key moments in sauropod evolution. *Choice C* is incorrect because the text indicates that there weren't significant increases in atmospheric carbon dioxide around the time of important periods in sauropods' evolution toward larger body sizes, not that atmospheric carbon dioxide was higher when the largest sauropods lived than when sauropods first appeared. *Choice D* is incorrect because the text indicates that atmospheric carbon dioxide levels didn't increase at important periods in sauropod evolution, not that higher levels would have affected that evolution. The text provides no information about how higher levels of atmospheric carbon dioxide might have affected sauropods.

QUESTION 18

Choice B is the best answer because it most logically completes the text's discussion of Anita Allen's argument about judges citing philosophers in their judicial opinions. The text indicates that judges sometimes cite philosophers when writing their judicial opinions and that, according to Allen, judges tend to cite philosophers whose views are in agreement with those of the judges themselves. Allen claims, however, that the best judicial opinions consider potential objections and rebut them, which suggests that judges may be able to strengthen their opinions by including discussions of philosophers with views contrary to their own.

Choice A is incorrect because Allen's claim is that judges could improve their judicial opinions by citing philosophers who disagree with the views expressed in the opinions, which would necessarily require judges to consult philosophical works. *Choice C* is incorrect because there's no discussion in the text about making judicial opinions more easily understood by any particular group of readers. The focus of the text is on Allen's claim that judicial opinions could be strengthened by the inclusion of discussions of philosophers whose views disagree with those of the judges authoring the opinions. *Choice D* is incorrect because the text presents Allen's argument that discussing philosophers whose views judges disagree with could strengthen judicial opinions, not that doing so could bring those opinions into line with views that are popular among philosophers.

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QUESTION 19

Choice A is the best answer. The convention being tested is pronoun-antecedent agreement. The plural pronoun "they" agrees in number with the plural antecedent "customers."

Choice B is incorrect because the singular pronoun "one" doesn't agree in number with the plural antecedent "customers." *Choice C* is incorrect because the second person pronoun "you" isn't conventional as a substitute for "customers." It suggests that the audience ("you") is the customer. *Choice D* is incorrect because the singular pronoun "it" doesn't agree in number with the plural antecedent "customers."

QUESTION 20

Choice A is the best answer. The convention being tested is punctuation use between a main clause and a participial phrase. This choice correctly uses a comma to mark the boundary between the main clause ("Epicurus…'soul'") and the participial phrase ("positing…absence") that provides additional information about how Epicurus defined pleasure.

Choice B is incorrect because a colon can't be used in this way to join a main clause and a participial phrase. *Choice C* is incorrect because a semicolon can't be used in this way to join a main clause and a participial phrase. *Choice D* is incorrect because it results in a rhetorically unacceptable sentence fragment beginning with "positing."

QUESTION 21

Choice C is the best answer. The convention being tested is the use of possessive determiners. The plural possessive determiner "their" agrees in number with the plural conjoined noun phrase "Watson and Crick" and thus indicates that the findings were those of Watson and Crick.

Choice A is incorrect because "they're" is the contraction for "they are," not a possessive determiner. *Choice B* is incorrect because "it's" is the contraction for "it is" or "it has," not a possessive determiner. *Choice D* is incorrect because the singular possessive determiner "its" doesn't agree in number with the plural conjoined noun phrase "Watson and Crick."

QUESTION 22

Choice C is the best answer. The conventions being tested are punctuation use between titles and proper nouns and between verbs and integrated quotations. No punctuation is needed to set off the proper noun "Stina Chyn" from the title that describes Chyn, "critic." Because "Stina Chyn" is essential information identifying the "critic," no punctuation is necessary. Further, no punctuation is needed between the verb "claims" and the following quotation because the quotation is integrated into the structure of the sentence.

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Choice A is incorrect because no punctuation is needed before or after the proper noun "Stina Chyn." Setting the critic's name off with commas suggests that it could be removed without affecting the coherence of the sentence, which isn't the case. *Choice B* is incorrect because no punctuation is needed before or after the proper noun "Stina Chyn." Setting the critic's name off with commas suggests that it could be removed without affecting the coherence of the sentence, which isn't the case. Additionally, no punctuation is needed between "claims" and the integrated quotation. *Choice D* is incorrect because no punctuation is needed between the verb "claims" and its subject, "critic Stina Chyn." Additionally, no punctuation is needed between the verb "claims" and the integrated quotation.

QUESTION 23

Choice B is the best answer. The convention being tested is the use of finite and nonfinite verb forms within a sentence. A main clause requires a finite verb to perform the action of the subject (in this case, "some historians"), and this choice supplies the finite present tense verb "claim" to indicate what some historians do.

Choice A is incorrect because the nonfinite participle "claiming" doesn't supply the main clause with a finite verb. *Choice C* is incorrect because the nonfinite participle "having claimed" doesn't supply the main clause with a finite verb. *Choice D* is incorrect because the nonfinite to-infinitive "to claim" doesn't supply the main clause with a finite verb.

QUESTION 24

Choice A is the best answer. The convention being tested is colon use within a sentence. A colon used in this way introduces information that illustrates or explains information that has come before it. In this case, the colon introduces the following explanation of why some roundworms in the Southern Hemisphere move in the opposite direction of Earth's magnetic field.

Choice B is incorrect because it results in a comma splice. A comma can't be used in this way to join two long independent clauses ("Researchers…food" and "in…sources") such as these. *Choice C* is incorrect because it results in a run-on sentence. The two clauses ("Researchers…food" and "in…sources") are fused without punctuation. Furthermore, the conjunction "while" fails to indicate that what follows is an explanation of why some roundworms in the Southern Hemisphere move in the opposite direction of Earth's magnetic field. *Choice D* is incorrect because it results in a run-on sentence. The two clauses ("Researchers…food" and "in…sources") are fused without punctuation and/or a conjunction.

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QUESTION 25

Choice B is the best answer. The convention being tested is pronoun-antecedent agreement. The plural reflexive pronoun "themselves" agrees in number with the plural antecedent "turtle barnacles," correctly indicating what is attached to a sea turtle shell.

Choice A is incorrect because the singular pronoun "it" doesn't agree in number with the plural antecedent "turtle barnacles." *Choice C* is incorrect because it results in an unclear and confusing sentence. In this context, it's unclear what the plural pronoun "them" refers to. *Choice D* is incorrect because the singular reflexive pronoun "itself" doesn't agree in number with the plural antecedent "turtle barnacles."

QUESTION 26

Choice A is the best answer. The convention being tested is subject–verb agreement. The singular verb "allows" agrees in number with the singular subject "landing."

Choice B is incorrect because the plural verb "are allowing" doesn't agree in number with the singular subject "landing." *Choice C* is incorrect because the plural verb "have allowed" doesn't agree in number with the singular subject "landing." *Choice D* is incorrect because the plural verb "allow" doesn't agree in number with the singular subject "landing."

QUESTION 27

Choice A is the best answer. The convention being tested is the use of punctuation to mark boundaries between supplements and clauses. The comma after "equations" is used to separate the independent clause ("Hopper's… equation") from the supplementary adverb phrase "though." The colon after "though" is used to mark the boundary between the clause ending with "though" and the following clause ("as…age"). A colon used in this way introduces information that illustrates or explains information that has come before it. In this case, the colon after "though" introduces the following explanation of how Hopper's subsequent career would involve more than just solving equations: she would become a pioneering computer programmer.

Choice B is incorrect because it results in a comma splice. A comma can't be used in this way to join two independent clauses ("Hopper's…though" and "as…age") such as these. *Choice C* is incorrect because it results in an illogical sequence of sentences. Placing the period after "equations" and beginning the next sentence with "Though" illogically suggests that the following information (that Hopper would help usher in the digital age) is contrary to the information in the previous sentence (Hopper's subsequent career would involve more than just solving equations). Instead, the information that follows supports the information from the previous sentence by explaining how her work and influence extended beyond solely solving equations. *Choice D* is incorrect because it results in a run-on sentence. The two independent clauses ("Hopper's…though" and "as…age") are fused without punctuation.

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QUESTION 28

Choice A is the best answer. The convention being tested is subject-modifier placement. This choice ensures that the introductory phrase "upon recovering two years later" appears immediately before the noun it modifies ("Henry"), clearly establishing that Henry recovered two years later.

Choice B is incorrect because it results in a dangling modifier. The placement of the noun phrase "the reign of Henry" immediately after the introductory phrase illogically suggests that the reign of Henry recovered two years later. *Choice C* is incorrect because it results in a dangling modifier. The placement of the noun phrase "Henry's reign" immediately after the introductory phrase illogically suggests that Henry's reign recovered two years later. *Choice D* is incorrect because it results in a dangling modifier. The placement of the function word "it" immediately after the introductory phrase illogically suggests that "it" recovered two years later.

QUESTION 29

Choice D is the best answer. "For example" logically signals that the information in this sentence—that *The Emperor's Babe* is a novel conveyed in lines of poetry—exemplifies the claim in the previous sentence about hybrid works that incorporate elements of both novels and poems.

Choice A is incorrect because "by contrast" illogically signals that the information in this sentence contrasts with the claim about hybrid works in the previous sentence. Instead, the information demonstrates that Evaristo's novel is an example of a hybrid work. *Choice B* is incorrect because "consequently" illogically signals that the information in this sentence is a consequence, or result, of the claim about hybrid works in the previous sentence. Instead, the information demonstrates that Evaristo's novel is an example of a hybrid work. *Choice C* is incorrect because "secondly" illogically signals that the information in this sentence is a second, separate claim from the previous sentence's claim about hybrid works. Instead, the information demonstrates that Evaristo's novel is an example of a hybrid work.

QUESTION 30

Choice C is the best answer. "By contrast" logically signals that the information in this sentence—that dogs can see, hear, and smell by the end of two weeks—contrasts with the preceding information (that wolves can smell but not see or hear at the same age).

Choice A is incorrect because "in other words" illogically signals that the information about domesticated dogs in this sentence paraphrases the information about wolves in the previous sentence. Instead, the information about dogs contrasts with what came before. *Choice B* is incorrect because "for instance" illogically signals that the information about domesticated dogs in this sentence exemplifies the information about wolves in the previous sentence. Instead, the information about dogs contrasts with what came before. *Choice D* is incorrect because "accordingly" illogically signals that the information about domesticated dogs in this sentence is in accordance with, or results from, the information about wolves in the previous sentence. Instead, the information about dogs contrasts with what came before.

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QUESTION 31

Choice D is the best answer. "Increasingly" logically signals that the claim in this sentence—that mathematicians are collaborating with their peers—marks a change relative to what was traditionally done. As the previous sentence explains, while mathematicians may have traditionally worked alone, evidence points to a shift in the opposite direction. The claim describes the shift: a rise in collaboration.

Choice A is incorrect because "similarly" illogically signals that the claim in this sentence is similar to, but separate from, the previous claim about the shift away from mathematicians working alone. Instead, the claim about the rise in collaboration elaborates on the previous claim, describing the shift. *Choice B* is incorrect because "for this reason" illogically signals that the claim in this sentence is caused by the previous claim about the shift away from mathematicians working alone. Instead, the claim about the rise in collaboration elaborates on the previous claim, describing the shift. *Choice C* is incorrect because "furthermore" illogically signals that the claim in this sentence is in addition to the previous claim about the shift away from mathematicians working alone. Instead, the claim about the rise in collaboration elaborates on the previous claim, describing the shift.

QUESTION 32

Choice D is the best answer. The sentence presents both the study and its findings, noting the study's date and the researcher's name as well as describing what the researcher determined about the jawbones and how she determined it.

Choice A is incorrect. While the sentence describes the study and the researcher's initial assessment, it doesn't present the study's findings. *Choice B* is incorrect. While the sentence describes the study and its focus, it doesn't present the study's findings or the name of the researcher who conducted it. *Choice C* is incorrect. While the sentence mentions the study's methodology and provides information about pterosaurs, it doesn't present the study's findings.

QUESTION 33

Choice C is the best answer. The sentence compares the two women's contributions to the march: Hedgeman worked behind the scenes to make sure a woman speaker was included, whereas Bates actually spoke at the event.

Choice A is incorrect. While it acknowledges that the two women both contributed to the march, it doesn't indicate what Hedgeman did, so no comparison is made. *Choice B* is incorrect. While the sentence provides information about the two women, it doesn't mention anything about Bates's contribution to the march. *Choice D* is incorrect. While the sentence indicates that the two women both fought for civil rights, it doesn't compare their individual contributions to the march.

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Reading and Writing

Module 2

(33 questions)

QUESTION 1

Choice B is the best answer because it most logically completes the text's discussion of Jacob Lawrence's artistic process. In this context, "observant" means watchful and perceptive. The text emphasizes that the "close attention" Lawrence paid to "all the details" of his neighborhood allowed him to reflect subtle elements of "the beauty and vitality of the Black experience" in his artwork. This context indicates that being observant of his surroundings was an important part of Lawrence's work as an artist.

Choice A is incorrect because the text gives no indication that Lawrence was "skeptical," or had an attitude of doubt in general or about particular things, let alone that skepticism was important to him as an artist. Rather than indicating that he was skeptical, the text focuses on how Lawrence paid careful attention to everything around him and reflected his observations in his artwork. *Choice C* is incorrect because the text gives no indication that Lawrence was "critical," which in this context would mean inclined to criticize harshly or unfairly. Rather than indicating that Lawrence found fault in things, the text suggests that he paid careful attention to everything around him and that his artwork reflects this careful attention. *Choice D* is incorrect because the text doesn't suggest that Lawrence was "confident," or self-assured. Rather than addressing how Lawrence felt about himself and how that feeling affected his artistic process, the text emphasizes the careful attention Lawrence paid to everything around him—attention that allowed him to capture subtle elements of a particular place and time in his artwork.

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QUESTION 2

Choice D is the best answer because it most logically completes the text's discussion of the research that Lopes-Ferreira and her colleagues are conducting on the stingray species *Potamotrygon rex*. As used in this context, "a substantial" effect means an effect that is sizable or noteworthy. The text indicates that the researchers are seeking to determine whether there are "considerable variations" in the potency of stingray venom that are associated with variation in the stingrays' age and sex. This context suggests that the researchers want to find out whether stingray age and sex have a substantial effect on venom toxicity.

Choice A is incorrect because there's nothing in the text that suggests that the researchers have been studying whether the stingrays' age and sex have "a disconcerting," or an unsettling and disturbing, effect on the stingrays' venom. The text indicates that the researchers wish to determine if stingray age and sex cause large variations in the toxicity of stingray venom, not if the effect of age and sex is disconcerting. *Choice B* is incorrect because the text indicates that researchers want to find out whether differences in stingray age and sex produce differences in stingray venom, not that the researchers want to find out whether age and sex have "an acceptable," or a satisfactory, effect on venom. The text makes no mention of what would make an effect on venom toxicity acceptable and gives no indication that the researchers are interested in that question. *Choice C* is incorrect because it wouldn't make sense in context for the researchers to be looking for "an imperceptible," or an unnoticeable, effect of age and sex on stingray venom. The text says that the researchers are trying to determine if there are "considerable variations" in venom toxicity linked to age and sex, not that the researchers are trying to find effects that they can't perceive.

QUESTION 3

Choice A is the best answer because it most logically completes the text's discussion of diaphragm contractions and hiccups. In this context, "involuntarily" means done without any control, or by reflex. The text explains that when a person's diaphragm repeatedly contracts and results in hiccups (which may be beneficial for infants), those muscle contractions are "uncontrollable." This context indicates that the diaphragm contractions occur without the person's control.

Choice B is incorrect because it wouldn't support the logical relationship established in the text's discussion of diaphragm contractions and hiccups. The text indicates that although specific causes for hiccups haven't been identified, it may be the case that the muscle contractions that occur have an important purpose in infants. It wouldn't make sense to say that even though the contractions occur "beneficially," or with a good or helpful effect, they might play a positive role in infants' breathing regulation. *Choice C* is incorrect because the text indicates that the diaphragm contractions that result in hiccups are "uncontrollable." Because those muscle contractions are described as happening automatically and without the person's control, it wouldn't make sense to

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describe them as occurring "strenuously," or in a way that requires great effort or energy. *Choice D* is incorrect because the text doesn't describe the quality of the diaphragm contractions that result in hiccups beyond stating that they are "uncontrollable." Nothing in the text indicates that those muscle contractions occur "smoothly," or evenly and continuously.

QUESTION 4

Choice D is the best answer because it most logically completes the text's discussion about the relationship between fine art and fashion. As used in this context, "intersect" means to connect or overlap. The text indicates that Jamie Okuma challenges the position held by critics because her work can be seen at an art museum and can be bought by the public from her online boutique. The text also presents the critics' view as being influenced by a perception that fine artists create works that are "timeless" and meant for exhibition, whereas fashion designers periodically produce new styles that are meant for purchase. This context suggests that the critics believe that fine art and fashion tend not to overlap—in other words, that they rarely intersect.

Choice A is incorrect because it wouldn't make sense in context to say that critics contend that fine art and fashion rarely "prevail," or prove to be triumphant or widespread. The text indicates that Okuma is an example of an artist who demonstrates that it's possible to make fine art that is also available to the public as fashion. *Choice B* is incorrect because it wouldn't make sense in context to say that fine art and fashion rarely "succumb," or surrender. The text establishes that unlike what critics believe, Okuma creates works that are in art museums and available for the public to purchase, suggesting that critics believe fine art and fashion rarely overlap, not that they rarely succumb. *Choice C* is incorrect because saying that critics believe that fine art and fashion rarely "diverge," or disagree or move in different directions, wouldn't make sense in context. The text presents Okuma's work as both fine art and fashion, thereby undermining what the critics assert. This suggests that the critics believe that fine art and fashion rarely intersect rather than that the two rarely diverge.

QUESTION 5

Choice D is the best answer because it most logically completes the text's discussion of gender roles in Shakespeare's comedies. As used in this context, "prescribed" would mean laid down as rules. The text indicates that the characters in the comedies often defy gender roles that are "socially dictated" (even if most characters do return to those roles eventually) and that scholars have been very interested in these acts of defiance. This context indicates that what the characters are rebelling against are standards of behavior prescribed by the society of the time.

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Choice A is incorrect because saying that expectations about gender were "interjected," or suddenly inserted between other things, wouldn't make sense in context. There's no suggestion in the text that the issue of gender roles was inserted between other things or was an interruption in a larger discussion. *Choice B* is incorrect because the text indicates that Shakespeare depicts characters rebelling against expectations about gender that have been "socially dictated," not expectations that society has "committed," or carried out, entrusted, or promised. *Choice C* is incorrect because the text indicates that Shakespeare depicts characters rebelling against expectations about gender that have been "socially dictated," not expectations that have been "illustrated," or clarified with examples. Although it's possible for expectations about gender roles to be illustrated, there's nothing in the text to indicate that characters in Shakespeare's comedies rebel against illustrations of gender expectations.

QUESTION 6

Choice B is the best answer because it most logically completes the text's discussion of Ward and colleagues' findings. As used in this context, "innocuous" means mild or unharmful. The text describes the vibration and warming that Ward and colleagues used to alleviate itching as "harmless applications" and goes on to contrast these applications with another stimulus that actually offers more relief even though it seems to be stronger and "less benign." This context conveys the idea that vibration and warming were innocuous stimuli.

Choice A is incorrect because the text focuses on a distinction between harmless stimuli and those that seem to be less benign. Nothing in the text suggests that any of the treatments are "deceptive," or misleading; indeed, even the less effective ones are described as offering some relief. *Choice C* is incorrect because the text focuses on the amount of relief from itching offered by harmless stimuli and those that seem to be less benign. The text doesn't suggest that any of these stimuli are "novel," or original and new; heat, vibration, and electricity aren't new inventions. *Choice D* is incorrect because it wouldn't make sense to describe an application of vibration or warming as "impractical," or not suitable for use. The text indicates that these harmless applications are useful in that they offer at least some temporary relief.

QUESTION 7

Choice D is the best answer because it most logically completes the text's discussion of the location of the province of Xoconochco within the Aztec Empire. As used in this context, "peripheral" means situated toward the outer bounds rather than the center. The text indicates that Xoconochco was located on a coast, hundreds of kilometers away from the capital of the Aztec Empire. The text also states that trade between the province and the capital required "a long overland journey." This context suggests that Xoconochco was situated toward an edge of the empire's territory rather than near its center.

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Choice A is incorrect because it wouldn't make sense in context to refer to Xoconochco's location within the Aztec Empire as "unobtrusive," or not blatant or undesirably prominent; it's not clear how a province's physical location would or wouldn't be blatant. Instead of focusing on how noticeable Xoconochco's location was, the text emphasizes the province's distance from the capital of the empire, pointing out that because of this distance trade between the two required "a long overland journey." *Choice B* is incorrect because the text indicates that the province of Xoconochco was located on a coast far from the capital of the Aztec Empire, not that it was "concealed," or kept out of sight or hidden from view. Nothing in the text suggests that Xoconochco was actually hidden such that people couldn't see it, and being hidden wouldn't necessarily result in trade between the province and the capital requiring "a long overland journey." *Choice C* is incorrect because to say that Xoconochco's location within the Aztec Empire was "approximate" would mean that the location either wasn't precisely correct or was close to some other location. Neither of these meanings would make sense in context because the text indicates that Xoconochco's location is known and that it was far from the empire's capital, so there's no reason to characterize the location as either not precisely correct or close to another location.

QUESTION 8

Choice A is the best answer because it most accurately describes the main purpose of the text, which is to show that while Jane calmly goes about her daily tasks, she is experiencing internal agitation about possibly seeking a new job. At the start of the text, Jane says, "I went on with my day's business tranquilly," indicating that she is outwardly calm. This outward calmness is then contrasted with her intense internal restlessness, as Jane says that thoughts of leaving her job keep running through her mind, that she is "involuntarily framing advertisements" (meaning that she can't stop herself from thinking up potential listings for jobs), and that she often wonders what new "situations" (or jobs) would be like.

Choice B is incorrect because the text gives no indication of Jane's feelings, either positive or negative, about the people she works for at Thornfield Hall. And rather than emphasizing that Jane feels particularly loyal to her employers, the text focuses on her constant consideration of leaving her job. *Choice C* is incorrect because the text gives no indication that Jane finds her current situation fulfilling, or satisfying. Given that much of the text is focused on Jane's thoughts about possibly leaving her job for a new one, it might be the case that she finds her situation challenging, but there is no evidence in the text that Jane also finds that situation satisfying—she says nothing positive about her current job at all, in fact. *Choice D* is incorrect because the text describes Jane as wondering about getting a new job, not as determined to definitely do so. Jane keeps thinking about reasons why she "should" quit her current job (indicating that she hasn't yet decided to) and imagining possible new situations she could find, but she says at the end of the text that these thoughts "might germinate and bear fruit if they could," meaning that the thoughts haven't yet led to a decision—that Jane isn't yet determined to get a new job somewhere else.

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QUESTION 9

Choice B is the best answer because it reflects how the author of Text 2 would most likely respond to Text 1 based on the information provided. Text 1 discusses the discovery of a regeneration-linked gene, EGR, in both three-banded panther worms (which are capable of full regeneration) and humans (who have relatively limited regeneration abilities). Text 1 characterizes this discovery as "especially promising" and a sign of "exciting progress" in understanding human regeneration. The author of Text 2, on the other hand, focuses on the fact that the team that reported the EGR finding pointed out that while EGR's function in humans isn't yet known, it's likely very different from its function in panther worms. Therefore, the author of Text 2 would most likely say that Text 1's enthusiasm about the EGR discovery is overly optimistic given Srivastava's team's observations about EGR in humans.

Choice A is incorrect because the author of Text 2 explains that Srivastava and her team explicitly reported that they haven't yet identified how EGR functions in humans; therefore, the author of Text 2 wouldn't say that Text 1's excitement is reasonable for the stated reason. Instead, the author of Text 2 would likely characterize Text 1's excitement as premature and overly optimistic. *Choice C* is incorrect because Text 1 does treat Srivastava's team's findings with enthusiasm; it describes the discovery of EGR in both three-banded panther worms and humans as promising and exciting. It would be illogical for the author of Text 2 to say that because most others treat the discovery with enthusiasm, Text 1's enthusiastic characterization of the discovery is unexpected. *Choice D* is incorrect because Text 1 isn't at all dismissive of Srivastava's team's findings; instead, Text 1 is optimistic about the EGR discovery, characterizing it as promising and exciting. There's nothing in Text 2 to suggest that the author of Text 2 would say that Text 1's praise for the discovery is dismissive, or disdainful.

QUESTION 10

Choice D is the best answer because it most accurately states the main idea of the text. The speaker describes the experience of being "weary" and "tired" and going to bed to seek "dear repose" (that is, sleep), but instead of sleeping, the speaker is kept awake ("keep my drooping eyelids open wide") by thoughts of a friend ("my thoughts… [Begin] a zealous pilgrimage to thee").

Choice A is incorrect because the text makes it clear that the speaker isn't asleep; thoughts about the friend are keeping the speaker awake ("keep my drooping eyelids open wide"). *Choice B* is incorrect because the speaker isn't talking about taking a literal trip when referring to "a zealous pilgrimage." Rather, the speaker is referring to the experience of thinking about the friend, of taking "a journey in my head." *Choice C* is incorrect because the text indicates that the speaker and the friend aren't in the same place and having a conversation. Rather, the speaker is at home and thinking of the friend, who is somewhere else ("from far where I abide").

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QUESTION 11

Choice D is the best answer because it presents a statement about how the Lord Chancellor responds to the crowd that is supported by the text. The text indicates that the people in the crowd are roaring and shouting "Bread!" or "Taxes!" and presents them as not knowing what they really want. The Lord Chancellor's response is to ask what their shouting means but also to observe that they're shouting with "unanimity," or total agreement. Clearly, this isn't the case, which supports the statement that the Lord Chancellor describes the crowd as being united even though it's not.

Choice A is incorrect because it isn't supported by the text. Although the text indicates that the Lord Chancellor asks about the meaning of the crowd's shouting, it doesn't suggest that he knows what the crowd really wants. *Choice B* is incorrect because the text doesn't suggest that the Lord Chancellor wants to speak to the crowd. Furthermore, the text doesn't indicate that the crowd wants to hear from the Sub-Warden. Although the crowd roars when asked "Who roar for the Sub-Warden?" it's unclear what the roaring means. *Choice C* is incorrect because the text doesn't suggest that the Lord Chancellor knows of or sympathizes with the crowd's demands. In addition, the text doesn't indicate that the crowd's shouting annoys the Lord Chancellor, just that it causes him to keep repeating "What can it all mean?"

QUESTION 12

Choice A is the best answer because it presents the quotation that most directly illustrates the claim that Cather portrays Alexandra as having a deep emotional connection to her natural surroundings. This quotation states that the country meant a great deal to Alexandra and then goes on to detail several ways in which her natural surroundings affect her emotionally: the insects sound like "the sweetest music," she feels as though "her heart were hiding" in the grass "with the quail and the plover," and near the ridges she feels "the future stirring."

Choice B is incorrect because the quotation doesn't suggest that Alexandra had a deep emotional connection to her natural surroundings but instead describes how she interacts with the people around her to learn more about crops, poultry, and experiments with clover hay. *Choice C* is incorrect because the quotation doesn't suggest that Alexandra has a deep emotional connection to her natural surroundings but instead describes her nighttime departure in a wagon. The quotation says nothing about Alexandra's emotional state. *Choice D* is incorrect because the quotation doesn't convey Alexandra's deep emotional connection to her natural surroundings; instead, this quotation describes how well she understands the markets and livestock.

QUESTION 13

Choice B is the best answer because it provides the most direct support from the table for the claim that two languages can convey similar amounts of information even if they're spoken at different rates. The table shows the approximate rates at which five languages are spoken and the rates at which those five languages convey information. Vietnamese is spoken at around 5.3 syllables per second,

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whereas Spanish is spoken at around 7.7 syllables per second, but the two languages convey information at very similar rates: Vietnamese at a rate of around 42.5 bits per second and Spanish at a rate of around 42.0 bits per second. Thus, the description of Vietnamese conveying information at around the same rate that Spanish does despite being spoken more slowly supports the claim in the text that languages can convey the same amount of information even if spoken at different rates.

Choice A is incorrect because it isn't true that Thai and Hungarian have the lowest rates of speech of the five languages shown. According to the table, Hungarian is spoken at around 5.9 syllables per second, which is faster than Vietnamese (5.3 syllables per second). Additionally, even if this statement were true, the assertion that two languages are spoken the slowest and convey information the slowest wouldn't support the claim that languages can convey the same amount of information even if they're spoken at different rates. *Choice C* is incorrect because it isn't true that the fastest-spoken language (Spanish, at 7.7 syllables per second) also conveys information the fastest: Spanish conveys information at 42.0 bits per second, which is slower than the 42.5 bits-per-second rate at which Vietnamese conveys information. Additionally, even if this statement were true, the assertion that the language spoken the fastest also conveys information the fastest has no bearing on the claim that languages can convey the same amount of information even if they're spoken at different rates. *Choice D* is incorrect because it isn't true that Serbian conveys information faster than Spanish does. According to the table, Serbian conveys information at a rate of around 39.1 bits per second, which is slower than the 42.0 bits-per-second rate at which Spanish conveys information.

QUESTION 14

Choice A is the best answer because it presents a finding that, if true, would most strongly support the researchers' claim that they found evidence that experiencing awe can make people feel more connected to others and thus more likely to behave altruistically (with beneficial and unselfish concern for others). According to the text, the researchers tested for this effect by first having participants look at either something known to be awe-inspiring (very tall trees) or something ordinary (a plain building) and then purposely spilling pens near the participants. The finding that participants who had looked at the trees helped pick up significantly more pens than did participants who had looked at the building would support the researchers' claim by demonstrating that the people who had experienced awe behaved more altruistically when the experimenter needed help than the other participants did.

Choice B is incorrect because a finding about helpful participants using positive words to describe the trees and the building after the experiment was over wouldn't have any bearing on the researchers' claim that experiencing awe increases altruistic behavior. The text doesn't address the use of positive words to describe things or suggest any connection between using such words and having experienced awe, so that behavior wouldn't serve as evidence that experiencing awe played a role in promoting helpful behavior. *Choice C* is incorrect because a finding that participants who didn't help the experimenter were significantly more

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likely than others to report having experienced awe whether they had looked at the building or the trees would weaken the researchers' claim that experiencing awe increases altruistic behavior by suggesting that the opposite might be true—that experiencing awe is in fact linked to choosing not to act in a way that benefits someone else. *Choice D* is incorrect because a finding about participants noticing that the experimenter had dropped the pens wouldn't have any bearing on the researchers' claim about people behaving altruistically. Being aware of a challenge or problem isn't necessarily beneficial on its own and isn't the same as offering help, so the finding wouldn't support the idea that experiencing awe increases altruistic behavior.

QUESTION 15

Choice A is the best answer because it presents data from the table that most effectively complete the statement about the rates at which employment shifted in France and the United States. The text states that over the last two hundred years employment in the agricultural sector has declined while employment in the service sector has risen in both France and the US, and the data from the table reflect these trends. The text asserts, however, that the transition from agriculture to services "happened at very different rates in the two countries." This assertion is best supported by a comparison of data from 1900 and 1950: the table shows that in those years, employment in agriculture went from 43% to 32% in France (a decline of 11 percentage points) and from 41% to 14% in the US (a decline of 27 percentage points), and that employment in services went from 28% to 35% in France (an increase of 7 percentage points) and from 31% to 53% in the US (an increase of 22 percentage points). In other words, the rate of change was greater in the US than in France for both sectors.

Choice B is incorrect because comparing the data for 1800 and 2012 would suggest a similar rate of change in the two countries, not very different rates: employment in agriculture went from 64% in 1800 to 3% in 2012 in France, which is close to the change from 68% in 1800 to 2% in 2012 in the US, while employment in services went from 14% in 1800 to 76% in 2012 in France, which is close to the change from 13% in 1800 to 80% in 2012 in the US. *Choice C* is incorrect because comparing the data for 1900 and 2012 would suggest a similar rate of change in the two countries rather than very different rates: employment in agriculture went from 43% in 1900 to 3% in 2012 in France, which is close to the change from 41% in 1900 to 2% in 2012 in the US, while employment in services went from 28% in 1900 to 76% in 2012 in France, which is close to the change from 31% in 1900 to 80% in 2012 in the US. *Choice D* is incorrect because comparing the data for 1800 and 1900 would suggest a similar rate of change in the two countries, not very different rates: employment in agriculture went from 64% in 1800 to 43% in 1900 in France, which is fairly close to the change from 68% in 1800 to 41% in 1900 in the US, while employment in services went from 14% in 1800 to 28% in 1900 in France, which is close to the change from 13% in 1800 to 31% in 1900 in the US.

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QUESTION 16

Choice C is the best answer because it presents a finding that, if true, would support the researchers' claim that archaeologists are unlikely to be replaced by certain computer models. The text explains that although archaeologists hold that categorizing pottery fragments relies on both objective criteria and instinct developed through direct experience, researchers have found that a computer model can categorize the fragments with the same degree of accuracy as the humans can—a finding that has caused some archaeologists to worry that their own work won't be needed any longer. If survey results indicate that categorizing pottery fragments limits the amount of time archaeologists can dedicate to other important tasks that only human experts can do, that would mean that computer models aren't able to do all of the important things archaeologists do, thus supporting the researchers' claim that computer models are unlikely to replace human archaeologists.

Choice A is incorrect because if it were true that the computer model could categorize the pottery fragments much more quickly than the archaeologists could, that would weaken the researchers' claim that archaeologists are unlikely to be replaced by certain computer models, since it would demonstrate that the models could conduct the archaeologists' work not only with equal accuracy but also at a faster pace. *Choice B* is incorrect because the inability of both the computer model and the archaeologists to accurately categorize all of the pottery fragments presented wouldn't support the researchers' claim that archaeologists are unlikely to be replaced by certain computer models. The text indicates that some archaeologists are worried because the computer model's accuracy is equal to their own, and that could be the case whether both were perfectly accurate or were unable to achieve complete accuracy. *Choice D* is incorrect because survey results showing that few archaeologists received special training in properly categorizing pottery fragments wouldn't support the researchers' claim that archaeologists are unlikely to be replaced by certain computer models. The amount of special training in categorizing pottery fragments that archaeologists have received has no direct bearing on whether computer models' success at categorizing fragments will lead to the models replacing the archaeologists.

QUESTION 17

Choice A is the best answer because it presents the conclusion that most logically follows from the text's discussion of military veterans working in civilian government jobs in the United States. The text indicates that the proportion of military veterans working in civilian government jobs is considerably higher than the proportion of military veterans in the population as a whole. The text also notes that the unusually high representation of military veterans in these jobs may be a result of the organizational structures shared by civilian government entities and the military. Hence, it's reasonable to infer that it's the familiarity of the structures of civilian government that makes jobs there particularly attractive to military veterans.

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Choice B is incorrect because the text doesn't address what a typical relationship between military service and later career preferences would be, and there's no indication that it's atypical for veterans to work in civilian government jobs after they've left the military. On the contrary, the text suggests that many military veterans are drawn to such jobs. *Choice C* is incorrect because the text is focused on the high representation of military veterans in civilian government jobs and doesn't address nonveterans or their possible interest in military service. *Choice D* is incorrect because the text conveys that military veterans may be particularly interested in civilian government jobs due to the familiarity of organizational structures that are already in place, but there's no reason to think that this interest would mean that more civilian government jobs will start to require military experience.

QUESTION 18

Choice D is the best answer because it most logically completes the text's discussion of Shultz's finding about male tanagers. The text explains that because carotenoids both contribute to deeply saturated feathers and offer health benefits, having deeply saturated feathers is usually "an honest signal" (a true indication) that a bird is generally fit. However, Shultz and others have found that certain male tanagers can appear to have deeply saturated feathers even if they haven't consumed a diet rich in carotenoids, thanks to microstructures in their feathers that manipulate light. If those birds aren't necessarily eating carotenoid-rich diets, they may actually be less fit than other birds that appear to have similarly saturated feathers; this suggests that a male tanager's appearance may function as a dishonest signal, or a false indication, of the bird's overall fitness.

Choice A is incorrect because Shultz's finding suggests that some tanagers can signal fitness without consuming the carotenoids that contribute to fitness, thereby making those signals dishonest, not that tanagers can give honest signals of their fitness without consuming carotenoids. *Choice B* is incorrect because Shultz's finding suggests that the microstructures in certain tanagers' feathers can give a dishonest signal of fitness, not that the microstructures are less effective than actual pigmentation for signaling fitness. Whether the signal of fitness is honest or dishonest has no bearing on how effective the signal is: a signal is effective if potential mates behave as though it's true, regardless of whether it's actually true. Since there's no information in the text about how potential mates respond to the dishonest signals of some tanagers, there's no support for the idea that the dishonest signals are less effective than the honest signals. *Choice C* is incorrect because Shultz's finding suggests that certain male tanagers may appear to be fitter than they actually are, not that scientists haven't determined why tanagers prefer mates with colorful appearances.

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QUESTION 19

Choice C is the best answer. The convention being tested is subject-verb agreement. The singular verb "has been" agrees in number with the singular subject "writing."

Choice A is incorrect because the plural verb "were" doesn't agree in number with the singular subject "writing." *Choice B* is incorrect because the plural verb "have been" doesn't agree in number with the singular subject "writing." *Choice D* is incorrect because the plural verb "are" doesn't agree in number with the singular subject "writing."

QUESTION 20

Choice B is the best answer. The convention being tested is the coordination of main clauses within a sentence. This choice correctly uses a comma and the coordinating conjunction "but" to join the first main clause ("the Alvarez…out") and the second main clause ("it left … extinctions").

Choice A is incorrect because when coordinating two longer main clauses such as these, it's conventional to use a comma before the coordinating conjunction. *Choice C* is incorrect because it results in a run-on sentence. The two main clauses are fused without punctuation and/or a conjunction. *Choice D* is incorrect because it results in a comma splice. Without a conjunction following it, a comma can't be used in this way to join two main clauses.

QUESTION 21

Choice C is the best answer. The convention being tested is the use of finite and nonfinite verb forms within a sentence. The nonfinite present participle "forcing" is correctly used to form a participial phrase that supplements the main clause "those...cover," describing the effects on monkeys of the lack of food sources.

Choice A is incorrect because the finite present tense verb "forces" can't be used in this way to supplement the main clause ("those...cover"). *Choice B* is incorrect. While the nonfinite to-infinitive "to force" could be used to form a subordinate clause that supplements the main clause ("those...cover"), to-infinitives conventionally express purpose, and nothing in the sentence suggests that the food sources become unavailable for the purpose of forcing monkeys to hunt marine animals. *Choice D* is incorrect because the finite past tense verb "forced" can't be used in this way to supplement the main clause ("those...cover").

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QUESTION 22

Choice D is the best answer. The convention being tested is end-of-sentence punctuation. This choice correctly uses a question mark to punctuate the interrogative clause "could the blueberries thrive," which asks a direct question at the end of the sentence.

Choice A is incorrect because a period can't be used in this way to punctuate an interrogative clause, such as "could the blueberries thrive," at the end of a sentence. *Choice B* is incorrect because the context requires an interrogative clause. The declarative clause "the blueberries could thrive" incorrectly indicates that it was known that the blueberries could thrive in alkaline soil, whereas Michel had yet to find this out. *Choice C* is incorrect because a question mark can't be used in this way to punctuate a declarative clause, such as "the blueberries could thrive," at the end of a sentence.

QUESTION 23

Choice D is the best answer. The convention being tested is the use of verbs to express tense. In this choice, the present tense verb "is," used in conjunction with the word "today," correctly indicates that Paik is currently considered the first video artist.

Choice A is incorrect because the future-indicating verb "will be" doesn't indicate that Paik is currently considered the first video artist. *Choice B* is incorrect because the past perfect tense verb "had been" doesn't indicate that Paik is currently considered the first video artist. *Choice C* is incorrect because the past tense verb "was" doesn't indicate that Paik is currently considered the first video artist.

QUESTION 24

Choice C is the best answer. The convention being tested is punctuation use between sentences. In this choice, the period is used correctly to mark the boundary between the first sentence ("The...adjustments") and the second sentence ("Prior...days"). Because the adverbial phrase beginning with "prior" indicates when changing a spreadsheet required redoing the sheet by hand, that phrase belongs with the second sentence.

Choice A is incorrect because it results in a run-on sentence. Two sentences are fused without punctuation and/or a conjunction. *Choice B* is incorrect because it results in a comma splice. A comma can't be used in this way to mark the boundary between sentences. *Choice D* is incorrect. Without a comma preceding it, the conjunction "and" can't be used in this way to join the sentences.

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QUESTION 25

Choice B is the best answer. The convention being tested is the use of finite and nonfinite verb forms within a sentence. The modal "would," which indicates the future from a perspective in the past, should be accompanied by a nonfinite base form verb. In this choice, the nonfinite base form verb "create" is used correctly in conjunction with the nonfinite base form verb "increase" to describe what the lock would do.

Choice A is incorrect because the finite present tense verb "creates" can't be used in this way with the modal "would" to describe what the lock would do. *Choice C* is incorrect because the present participle "creating" can't be used in this way with the modal "would" to describe what the lock would do. *Choice D* is incorrect because the finite past tense verb "created" can't be used in this way with the modal "would" to describe what the lock would do.

QUESTION 26

Choice D is the best answer. The convention being tested is subject-modifier placement. This choice ensures that the modifying phrase "despite being cheap, versatile, and easy to produce" appears immediately before the noun it modifies, "commercial plastics," clearly establishing that the commercial plastics—and not another noun in the sentence—are being described as cheap, versatile, and easy to produce.

Choice A is incorrect because it results in a dangling modifier. The placement of the function word "there" immediately after the modifying phrase illogically and confusingly suggests that "there" is cheap, versatile, and easy to produce. *Choice B* is incorrect because it results in a dangling modifier. The placement of the noun "two problems" immediately after the modifying phrase illogically suggests that the "problems" are cheap, versatile, and easy to produce. *Choice C* is incorrect because it results in a dangling modifier. The placement of the noun phrase "commercial plastics' two associated problems" immediately after the modifying phrase illogically suggests that the "problems" are cheap, versatile, and easy to produce.

QUESTION 27

Choice C is the best answer. The convention being tested is the use of punctuation between titles and proper nouns. No punctuation is needed to set off the proper noun "Yuree Lee" from the title that describes Lee, "plant cell biologist." Because "Yuree Lee" is essential information identifying the "plant cell biologist," no punctuation is necessary.

Choice A is incorrect because no punctuation is needed. *Choice B* is incorrect because no punctuation is needed. *Choice D* is incorrect because no punctuation is needed before or after the proper noun "Yuree Lee." Setting the biologist's name off with commas suggests that it could be removed without affecting the coherence of the sentence, which isn't the case.

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QUESTION 28

Choice D is the best answer. "As a result" logically signals that the claim in this sentence—that spiders can cling to and climb almost any surface—is because of the previous information about the bonding properties of spiders' spatulae.

Choice A is incorrect because "for instance" illogically signals that the claim in this sentence exemplifies the information in the previous sentences. Instead, the claim is because of the previous information about the bonding properties of spiders' spatulae. *Choice B* is incorrect because "however" illogically signals that the claim in this sentence contrasts with the information in the previous sentences. Instead, the claim is because of the previous information about the bonding properties of spiders' spatulae. *Choice C* is incorrect because "similarly" illogically signals that the claim in this sentence is similar to, but separate from, the information in the previous sentences. Instead, the claim is because of the previous information about the bonding properties of spiders' spatulae.

QUESTION 29

Choice A is the best answer. "Still" logically signals that the information about Sher-Gil in this sentence— that she longed to leave Paris and return to India— contrasts with what one would expect after reading about Sher-Gil's experiences in Paris in the previous sentences.

Choice B is incorrect because "therefore" illogically signals that the information about Sher-Gil in this sentence is a result or consequence of the descriptions in the previous sentences. Instead, this information contrasts with what one would expect after reading about Sher-Gil's experiences in Paris. *Choice C* is incorrect because "indeed" illogically signals that the information about Sher-Gil in this sentence offers additional emphasis in support of the descriptions in the previous sentences. Instead, this information contrasts with what one would expect after reading about Sher-Gil's experiences in Paris. *Choice D* is incorrect because "furthermore" illogically signals that the information about Sher-Gil in this sentence offers additional support for or confirmation of the descriptions in the previous sentences. Instead, this information contrasts with what one would expect after reading about Sher-Gil's experiences in Paris.

QUESTION 30

Choice C is the best answer. "Similarly" logically signals that the activity described in this sentence (Nancy Tuttle Craig distributing Votes for Women Tea in her Los Angeles grocery stores) is like the activity described in the previous sentence (the Woman's Suffrage Party selling Equality Tea at fairs in San Francisco). Together, the two examples support the preceding claim that "activists across the state sold tea to promote the cause of suffrage."

Choice A is incorrect because "for example" illogically signals that the activity described in this sentence exemplifies the activity described in the previous sentence. Instead, the two activities are similar, and both support the preceding claim about selling tea to promote women's right to vote. *Choice B* is incorrect because "to conclude" illogically signals that the activity described in this

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sentence concludes or summarizes the information in the previous sentences. Instead, the activity is similar to the one described in the previous sentence, and both support the preceding claim about selling tea to promote women's right to vote. *Choice D* is incorrect because "in other words" illogically signals that the activity described in this sentence paraphrases the activity described in the previous sentence. Instead, the two activities are similar, and both support the preceding claim about selling tea to promote women's right to vote.

QUESTION 31

Choice B is the best answer. The sentence compares the lengths of the two rail tunnels, noting that the Channel Tunnel (about 31 miles long) is slightly shorter than the Seikan Tunnel (roughly 33 miles long).

Choice A is incorrect. The sentence makes a generalization about the length of some rail tunnels; it doesn't compare the lengths of the two rail tunnels. *Choice C* is incorrect. The sentence describes a single rail tunnel; it doesn't compare the lengths of the two rail tunnels. *Choice D* is incorrect. While the sentence mentions the two rail tunnels, it doesn't compare their lengths.

QUESTION 32

Choice B is the best answer because it provides both an explanation and an example of "flauna." The sentence explains that flauna, a combination of the words "flora" and "fauna," is a term used by Jon Ching to describe the plant-animal hybrids in his paintings. The sentence also mentions an example of Ching's flauna: a parrot with leaves for feathers.

Choice A is incorrect. While the sentence partially explains what "flauna" is, it doesn't provide a full explanation or specific example of Ching's flauna. *Choice C* is incorrect. While the sentence partially explains what "flauna" is and includes a title of a Ching painting, it doesn't provide a full explanation or specific example of Ching's flauna. *Choice D* is incorrect. While the sentence partially explains what "flauna" is and includes the titles of two Ching paintings, it doesn't provide a full explanation of Ching's flauna.

QUESTION 33

Choice D is the best answer. By indicating that Taylor's book was the only Civil War memoir published by a Black woman, this sentence emphasizes the uniqueness, or one-of-a-kind nature, of Taylor's accomplishment.

Choice A is incorrect. While the sentence describes some of Taylor's accomplishments, it doesn't emphasize the uniqueness of them. *Choice B* is incorrect. While the sentence describes some of Taylor's accomplishments, it doesn't emphasize that they were unique. *Choice C* is incorrect. While the sentence provides information about Taylor's book, it doesn't emphasize what made the book unique.

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Math

Module 1
(27 questions)

QUESTION 1

Choice B is correct. 10% of a quantity means $\frac{10}{100}$ times the quantity. Therefore, 10% of 470 can be represented as $\frac{10}{100}(470)$, which is equivalent to $0.10(470)$, or 47. Therefore, 10% of 470 is 47.

Choice A is incorrect. This is 10% of 370, not 10% of 470. *Choice C* is incorrect. This is 90% of 470, not 10% of 470. *Choice D* is incorrect. This is $470-10$, not 10% of 470.

QUESTION 2

Choice C is correct. Subtracting 6 from both sides of the given equation yields $4x=12$, which is the equation given in choice C. Since this equation is equivalent to the given equation, it has the same solution as the given equation.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice B* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 3

Choice D is correct. The cost of the rental fee depends on the number of hours the surfboard is rented. Multiplying t hours by 10 dollars per hour yields a rental fee of $10t$ dollars. The total cost of the rental consists of the rental fee plus the 25 dollar service fee, which yields a total cost of $25+10t$ dollars. Since the person intends to spend a maximum of 75 dollars to rent the surfboard, the total cost must be at most 75 dollars. Therefore, the inequality $25+10t\leq 75$ represents this situation.

Choice A is incorrect. This represents a situation where the rental fee, not the total cost, is at most 75 dollars. *Choice B* is incorrect and may result from conceptual

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or calculation errors. *Choice C* is incorrect and may result from conceptual or calculation errors.

QUESTION 4

Choice A is correct. It's given that $g(x)=x^2+9$. Substituting 25 for $g(x)$ in this equation yields $25=x^2+9$. Subtracting 9 from both sides of this equation yields $16=x^2$. Taking the square root of each side of this equation yields $x=\pm 4$. It follows that $g(x)=25$ when the value of x is 4 or -4. Only 4 is listed among the choices.

Choice B is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 5

Choice A is correct. The total number of possible outcomes for rolling a fair 14-sided die is 14. The number of possible outcomes for rolling a 2 is 1. The probability of rolling a 2 is the number of possible outcomes for rolling a 2 divided by the total number of possible outcomes, or $\frac{1}{14}$.

Choice B is incorrect. This is the probability of rolling a number no greater than 2. *Choice C* is incorrect. This is the probability of rolling a number greater than 2. *Choice D* is incorrect. This is the probability of rolling a number other than 2.

QUESTION 6

The correct answer is $2{,}520$. There are 60 minutes in one hour. At a rate of 42 posters per minute, the number of posters produced in one hour can be determined by $\left(\frac{42 \text{ posters}}{1 \text{ minute}}\right)\left(\frac{60 \text{ minutes}}{1 \text{ hour}}\right)$, which is $2{,}520$ posters per hour.

QUESTION 7

The correct answer is 30. The value of $f(x)$ when $x=4$ can be found by substituting 4 for x in the given equation $f(x)=7x+2$. This yields $f(4)=7(4)+2$, or $f(4)=30$. Therefore, when $x=4$, the value of $f(x)$ is 30.

QUESTION 8

Choice D is correct. Since x represents the number of 1-point questions and y represents the number of 3-point questions, the assignment is worth a total of $1 \cdot x+3 \cdot y$, or $x+3y$, points. Since the assignment is worth 70 points, the equation $x+3y=70$ represents this situation.

Choice A is incorrect and may result from conceptual errors. *Choice B* is incorrect and may result from conceptual errors. *Choice C* is incorrect and may result from conceptual errors.

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QUESTION 9

Choice B is correct. It's given that triangle LMN is similar to triangle PQR. Corresponding angles of similar triangles are congruent. Since angle M and angle Q correspond to each other, they must be congruent. Therefore, if the measure of angle M is 53°, then the measure of angle Q is also 53°.

Choice A is incorrect and may result from concluding that angle M and angle Q are complementary rather than congruent. *Choice C* is incorrect and may result from concluding that angle M and angle Q are supplementary rather than congruent. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 10

Choice C is correct. The given system of linear equations can be solved by the substitution method. Substituting $-3x$ for y from the first equation in the given system into the second equation yields $4x+\left(-3x\right)=15$, or $x=15$.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice B* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect. This is the absolute value of y, not the value of x.

QUESTION 11

Choice B is correct. The equation representing a linear model can be written in the form $y=a+bx$, or $y=bx+a$, where b is the slope of the graph of the model and $\left(0,a\right)$ is the *y*-intercept of the graph of the model. The scatterplot shows that as the *x*-values of the data points increase, the *y*-values of the data points decrease, which means the graph of an appropriate linear model has a negative slope. Therefore, $b<0$. The scatterplot also shows that the data points are close to the *y*-axis at a positive value of y. Therefore, the *y*-intercept of the graph of an appropriate linear model has a positive *y*-coordinate, which means $a>0$. Of the given choices, only choice B, $y=-1.9x+10.1$, has a negative value for b, the slope, and a positive value for a, the *y*-coordinate of the *y*-intercept.

Choice A is incorrect. The graph of this model has a *y*-intercept with a negative *y*-coordinate, not a positive *y*-coordinate. *Choice C* is incorrect. The graph of this model has a positive slope, not a negative slope, and a *y*-intercept with a negative *y*-coordinate, not a positive *y*-coordinate. *Choice D* is incorrect. The graph of this model has a positive slope, not a negative slope.

QUESTION 12

Choice C is correct. If a value of x satisfies $f\left(x\right)=0$, the graph of $y=f\left(x\right)$ will contain a point $\left(x,0\right)$ and thus touch the *x*-axis. Since there are 3 points at which this graph touches the *x*-axis, there are 3 values of x for which $f\left(x\right)=0$.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice B* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect and may result from conceptual or calculation errors.

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QUESTION 13

The correct answer is 41. The number of cupcakes Vivian bought can be found by first finding the amount Vivian spent on cupcakes. The amount Vivian spent on cupcakes can be found by subtracting the amount Vivian spent on party hats from the total amount Vivian spent. The amount Vivian spent on party hats can be found by multiplying the cost per package of party hats by the number of packages of party hats, which yields $\$3 \cdot 10$, or $\$30$. Subtracting the amount Vivian spent on party hats, $\$30$, from the total amount Vivian spent, $\$71$, yields $\$71 - \$30$, or $\$41$. Since the amount Vivian spent on cupcakes was $\$41$ and each cupcake cost $\$1$, it follows that Vivian bought 41 cupcakes.

QUESTION 14

The correct answer is either 2 or -12. The left-hand side of the given equation can be rewritten by factoring. The two values that multiply to -24 and add to 10 are 12 and -2. It follows that the given equation can be rewritten as $(z+12)(z-2)=0$. Setting each factor equal to 0 yields two equations: $z+12=0$ and $z-2=0$. Subtracting 12 from both sides of the equation $z+12=0$ results in $z=-12$. Adding 2 to both sides of the equation $z-2=0$ results in $z=2$. Note that 2 and -12 are examples of ways to enter a correct answer.

QUESTION 15

Choice D is correct. Let y represent the number of cells per milliliter x hours after the initial observation. Since the number of cells per milliliter doubles every 3 hours, the relationship between x and y can be represented by an exponential equation of the form $y=a(b)^{\frac{x}{k}}$, where a is the number of cells per milliliter during the initial observation and the number of cells per milliliter increases by a factor of b every k hours. It's given that there were 300,000 cells per milliliter during the initial observation. Therefore, $a=300{,}000$. It's also given that the number of cells per milliliter doubles, or increases by a factor of 2, every 3 hours. Therefore, $b=2$ and $k=3$. Substituting 300,000 for a, 2 for b, and 3 for k in the equation $y=a(b)^{\frac{x}{k}}$ yields $y=300{,}000(2)^{\frac{x}{3}}$. The number of cells per milliliter there will be 15 hours after the initial observation is the value of y in this equation when $x=15$. Substituting 15 for x in the equation $y=300{,}000(2)^{\frac{x}{3}}$ yields $y=300{,}000(2)^{\frac{15}{3}}$, or $y=300{,}000(2)^5$. This is equivalent to $y=300{,}000(32)$, or $y=9{,}600{,}000$. Therefore, 15 hours after the initial observation, there will be 9,600,000 cells per milliliter.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice B is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.

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QUESTION 16

Choice C is correct. Since each term of the given expression has a common factor of $6x^2y^2$, it may be rewritten as $6x^2y^2\left(x^6\right)+6x^2y^2\left(2\right)$, or $6x^2y^2\left(x^6+2\right)$.

Choice A is incorrect. This expression is equivalent to $12x^8y^2$, not $6x^8y^2+12x^2y^2$. *Choice B* is incorrect. This expression is equivalent to $6x^6y^2$, not $6x^8y^2+12x^2y^2$. *Choice D* is incorrect. This expression is equivalent to $6x^6y^2+12x^2y^2$, not $6x^8y^2+12x^2y^2$.

QUESTION 17

Choice A is correct. It's given that a neighborhood consists of a 2-hectare park and a 35-hectare residential area and that the total number of trees in the neighborhood is 3,934. It's also given that the equation $2x+35y=3{,}934$ represents this situation. Since the total number of trees for a given area can be determined by taking the number of hectares times the average number of trees per hectare, this must mean that the terms $2x$ and $35y$ correspond to the number of trees in the park and in the residential area, respectively. Since $2x$ corresponds to the number of trees in the park, and 2 is the size of the park, in hectares, x must represent the average number of trees per hectare in the park.

Choice B is incorrect and may result from conceptual errors. *Choice C* is incorrect and may result from conceptual errors. *Choice D* is incorrect and may result from conceptual errors.

QUESTION 18

Choice B is correct. The graph shown is a line passing through the points $\left(0,40\right)$ and $\left(60,0\right)$. Since the relationship between x and y is linear, if two points on the graph make a linear equation true, then the equation represents the relationship. Substituting 0 for x and 40 for y in the equation in choice B, $8x+12y=480$, yields $8\left(0\right)+12\left(40\right)=480$, or $480=480$, which is true. Substituting 60 for x and 0 for y in the equation $8x+12y=480$ yields $8\left(60\right)+12\left(0\right)=480$, or $480=480$, which is true. Therefore, the equation $8x+12y=480$ represents the relationship between x and y.

Choice A is incorrect. The point $\left(0,40\right)$ is not on the graph of this equation, since $40=8\left(0\right)+12$, or $40=12$, is not true. *Choice C* is incorrect. The point $\left(0,40\right)$ is not on the graph of this equation, since $40=12\left(0\right)+8$, or $40=8$, is not true. *Choice D* is incorrect. The point $\left(0,40\right)$ is not on the graph of this equation, since $12\left(0\right)+8\left(40\right)=480$, or $320=480$, is not true.

QUESTION 19

Choice D is correct. The area of a circle can be found by using the formula $A=\pi r^2$, where A is the area and r is the radius of the circle. It's given that the radius of circle A is $3n$. Substituting this value for r into the formula $A=\pi r^2$ gives $A=\pi\left(3n\right)^2$, or $9\pi n^2$. It's also given that the radius of circle B is $129n$. Substituting this value for r into the formula $A=\pi r^2$ gives $A=\pi\left(129n\right)^2$, or

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$16{,}641\pi n^2$. Dividing the area of circle *B* by the area of circle *A* gives $\frac{16{,}641\pi n^2}{9\pi n^2}$, which simplifies to $1{,}849$. Therefore, the area of circle *B* is $1{,}849$ times the area of circle *A*.

Choice A is incorrect. This is how many times greater the radius of circle *B* is than the radius of circle *A*. *Choice B* is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect. This is the coefficient on the term that describes the radius of circle *B*.

QUESTION 20

The correct answer is 14. The maximum value is the largest value in the data set. The frequency refers to the number of times a data value occurs. The given frequency table shows that for this data set, the data value 6 occurs three times, the data value 7 occurs three times, the data value 8 occurs eight times, the data value 9 occurs eight times, the data value 10 occurs nine times, the data value 11 occurs eleven times, the data value 12 occurs nine times, the data value 13 occurs zero times, and the data value 14 occurs six times. Therefore, the maximum data value in the data set is 14.

QUESTION 21

The correct answer is 5. The standard form of an equation of a circle in the *xy*-plane is $(x-h)^2+(y-k)^2=r^2$, where h, k, and r are constants, the coordinates of the center of the circle are (h,k), and the length of the radius of the circle is r. It's given that an equation of the circle is $(x-2)^2+(y-9)^2=r^2$. Therefore, the center of this circle is $(2,9)$. It's given that the endpoints of a diameter of the circle are $(2,4)$ and $(2,14)$. The length of the radius is the distance from the center of the circle to an endpoint of a diameter of the circle, which can be found using the distance formula, $\sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$. Substituting the center of the circle $(2,9)$ and one endpoint of the diameter $(2,4)$ in this formula gives a distance of $\sqrt{(2-2)^2+(9-4)^2}$, or $\sqrt{0^2+5^2}$, which is equivalent to 5. Since the distance from the center of the circle to an endpoint of a diameter is 5, the value of r is 5.

QUESTION 22

Choice C is correct. It's given that the measure of angle R is $\frac{2\pi}{3}$ radians, and the measure of angle T is $\frac{5\pi}{12}$ radians greater than the measure of angle R. Therefore, the measure of angle T is equal to $\frac{2\pi}{3}+\frac{5\pi}{12}$ radians. Multiplying $\frac{2\pi}{3}$ by $\frac{4}{4}$ to get a common denominator with $\frac{5\pi}{12}$ yields $\frac{8\pi}{12}$. Therefore, $\frac{2\pi}{3}+\frac{5\pi}{12}$ is equivalent to $\frac{8\pi}{12}+\frac{5\pi}{12}$, or $\frac{13\pi}{12}$. Therefore, the measure of angle T is $\frac{13\pi}{12}$ radians. The measure of angle T, in degrees, can be found by multiplying its measure, in

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radians, by $\frac{180}{\pi}$. This yields $\frac{13\pi}{12}\times\frac{180}{\pi}$, which is equivalent to 195 degrees. Therefore, the measure of angle T is 195 degrees.

Choice A is incorrect. This is the number of degrees that the measure of angle T is greater than the measure of angle R. *Choice B* is incorrect. This is the measure of angle R, in degrees. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 23

Choice D is correct. Since the number of yards in 1 mile is $1{,}760$, the number of square yards in 1 square mile is $(1{,}760)(1{,}760)=3{,}097{,}600$. Therefore, if the area of the town is 4.36 square miles, it is $4.36(3{,}097{,}600)=13{,}505{,}536$, in square yards.

Choice A is incorrect and may result from dividing the number of yards in a mile by the square mileage of the town. *Choice B* is incorrect and may result from multiplying the number of yards in a mile by the square mileage of the town. *Choice C* is incorrect and may result from dividing the number of square yards in a square mile by the square mileage of the town.

QUESTION 24

Choice D is correct. The equation of line h can be written in slope-intercept form $y=mx+b$, where m is the slope of the line and $(0,b)$ is the y-intercept of the line. It's given that line h contains the points $(18,130)$, $(23,160)$, and $(26,178)$. Therefore, its slope m can be found as $\frac{160-130}{23-18}$, or 6. Substituting 6 for m in the equation $y=mx+b$ yields $y=6x+b$. Substituting 130 for y and 18 for x in this equation yields $130=6(18)+b$, or $130=108+b$. Subtracting 108 from both sides of this equation yields $22=b$. Substituting 22 for b in $y=6x+b$ yields $y=6x+22$. Since line k is the result of translating line h down 5 units, an equation of line k is $y=6x+22-5$, or $y=6x+17$. Substituting 0 for y in this equation yields $0=6x+17$. Solving this equation for x yields $x=-\frac{17}{6}$. Therefore, the x-intercept of line k is $\left(-\frac{17}{6},0\right)$.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice B* is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect and may result from conceptual or calculation errors.

QUESTION 25

Choice C is correct. In the xy-plane, the graph of the line $y=c$ is a horizontal line that crosses the y-axis at $y=c$ and the graph of the quadratic equation $y=-x^2+9x-100$ is a parabola. A parabola can intersect a horizontal line at exactly one point only at its vertex. Therefore, the value of c should be equal to the y-coordinate of the vertex of the graph of the given equation. For a quadratic equation in vertex form, $y=a(x-h)^2+k$, the vertex of its graph in the xy-plane is (h,k). The given quadratic equation, $y=-x^2+9x-100$, can be rewritten as

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$y=-\left(x^2-2\left(\frac{9}{2}\right)x+\left(\frac{9}{2}\right)^2\right)+\left(\frac{9}{2}\right)^2-100$, or $y=-\left(x-\frac{9}{2}\right)^2+\left(-\frac{319}{4}\right)$. Thus, the value of c is equal to $-\frac{319}{4}$.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice B is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 26

Choice B is correct. The two given equations are equivalent because the second equation can be obtained from the first equation by multiplying each side of the equation by 5. Thus, the graphs of the equations are coincident, so if a point lies on the graph of one of the equations, it also lies on the graph of the other equation. A point (x,y) lies on the graph of an equation in the *xy*-plane if and only if this point represents a solution to the equation. It is sufficient, therefore, to find the point that represents a solution to the first given equation. Substituting the *x*- and *y*-coordinates of choice B, $-\frac{3r}{2}+\frac{7}{2}$ and r, for x and y, respectively, in the first equation yields $2\left(-\frac{3r}{2}+\frac{7}{2}\right)+3r=7$, which is equivalent to $-3r+7+3r=7$, or $7=7$. Therefore, the point $\left(-\frac{3r}{2}+\frac{7}{2},r\right)$ represents a solution to the first equation and thus lies on the graph of each equation in the *xy*-plane for the given system.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 27

The correct answer is 104. An equilateral triangle is a triangle in which all three sides have the same length and all three angles have a measure of $60°$. The height of the triangle, $k\sqrt{3}$, is the length of the altitude from one vertex. The altitude divides the equilateral triangle into two congruent 30-60-90 right triangles, where the altitude is the side across from the $60°$ angle in each 30-60-90 right triangle. Since the altitude has a length of $k\sqrt{3}$, it follows from the properties of 30-60-90 right triangles that the side across from each $30°$ angle has a length of k and each hypotenuse has a length of $2k$. In this case, the hypotenuse of each 30-60-90 right triangle is a side of the equilateral triangle; therefore, each side length of the equilateral triangle is $2k$. The perimeter of a triangle is the sum of the lengths of each side. It's given that the perimeter of the equilateral triangle is 624; therefore, $2k+2k+2k=624$, or $6k=624$. Dividing both sides of this equation by 6 yields $k=104$.

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Math

Module 2

(27 questions)

QUESTION 1

Choice A is correct. It's given that Tilly earns p dollars for every w hours of work. This can be represented by the proportion $\frac{p}{w}$. The amount of money, x, Tilly earns for $39w$ hours of work can be found by setting up the proportion $\frac{p}{w}=\frac{x}{39w}$. This can be rewritten as $39pw=xw$. Dividing both sides by w results in $x=39p$.

Choice B is incorrect. This is the amount of money Tilly earns in dollars per hour, not the amount of money Tilly earns for $39w$ hours of work. *Choice C* is incorrect. This is the amount of money Tilly earns for w hours of work plus 39, not the amount of money Tilly earns for $39w$ hours of work. *Choice D* is incorrect. This is the amount of money Tilly earns for w hours of work minus 39, not the amount of money Tilly earns for $39w$ hours of work.

QUESTION 2

Choice D is correct. It's given that Juan rides his bike at an average rate of 5.7 minutes per mile. The number of minutes it will take Juan to ride x miles can be determined by multiplying his average rate by the number of miles, x, which yields $5.7x$. Therefore, the function $m(x)=5.7x$ models the number of minutes it will take Juan to ride x miles.

Choice A is incorrect and may result from conceptual errors. *Choice B* is incorrect and may result from conceptual errors. *Choice C* is incorrect and may result from conceptual errors.

QUESTION 3

Choice B is correct. Adding the second equation in the given system to the first equation in the given system yields $3x+(-3x+y)=12+(-6)$, which is equivalent to $y=6$.

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Choice A is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 4

Choice D is correct. In the given equation, s is the speed, in miles per hour, of a certain car t seconds after it began to accelerate. Therefore, the speed of the car, in miles per hour, 5 seconds after it began to accelerate can be found by substituting 5 for t in the given equation, which yields $s=40+3(5)$, or $s=55$. Thus, the speed of the car 5 seconds after it began to accelerate is 55 miles per hour.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice B is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.

QUESTION 5

Choice D is correct. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b, then $c^2=a^2+b^2$. In the right triangle shown, the hypotenuse has length c and the legs have lengths a and b. It's given that $a=4$ and $b=5$. Substituting 4 for a and 5 for b in the Pythagorean theorem yields $c^2=4^2+5^2$. Taking the square root of both sides of this equation yields $c=\pm\sqrt{4^2+5^2}$. Since the length of a side of a triangle must be positive, the value of c is $\sqrt{4^2+5^2}$.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice B is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.

QUESTION 6

The correct answer is 40. Subtracting 5 from both sides of the given equation yields $4x=160$. Dividing both sides of this equation by 4 yields $x=40$. Therefore, the solution to the given equation is 40.

QUESTION 7

The correct answer is 7. It's given that the x-intercept of the graph shown is $(x,0)$. The graph passes through the point $(7,0)$. Therefore, the value of x is 7.

QUESTION 8

Choice B is correct. The y-intercept of the graph of a function in the xy-plane is the point on the graph where $x=0$. It's given that $f(x)=\frac{1}{10}x-2$. Substituting 0 for x in this equation yields $f(0)=\frac{1}{10}(0)-2$, or $f(0)=-2$. Since it's given that $y=f(x)$, it follows that $y=-2$ when $x=0$. Therefore, the y-intercept of the graph of $y=f(x)$ in the xy-plane is $(0,-2)$.

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Choice A is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 9

Choice D is correct. If the graph of $y=g(x)$ is the result of shifting the graph of $y=f(x)$ down k units in the *xy*-plane, the function g can be defined by an equation of the form $g(x)=f(x)-k$. It's given that $f(x)=7x^3$ and the graph of $y=g(x)$ is the result of shifting the graph of $y=f(x)$ down 2 units. Substituting $7x^3$ for $f(x)$ and 2 for k in the equation $g(x)=f(x)-k$ yields $g(x)=7x^3-2$.

Choice A is incorrect and may result from conceptual errors. *Choice B* is incorrect and may result from conceptual errors. *Choice C* is incorrect. This equation defines a function g for which the graph of $y=g(x)$ is the result of shifting the graph of $y=f(x)$ up, not down, 2 units.

QUESTION 10

Choice A is correct. The solution to a system of equations is the ordered pair (x,y) that satisfies all equations in the system. It's given by the first equation in the system that $x+7=10$. Substituting 10 for $x+7$ into the second equation yields $10^2=y$, or $y=100$. The *x*-coordinate of the solution to the system of equations can be found by subtracting 7 from both sides of the equation $x+7=10$, which yields $x=3$. Therefore, the ordered pair $(3, 100)$ is a solution to the given system of equations.

Choice B is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 11

Choice A is correct. Applying the distributive property, the given expression can be written as $7x^3+7x-6x^3+3x$. Grouping like terms in this expression yields $(7x^3-6x^3)+(7x+3x)$. Combining like terms in this expression yields x^3+10x.

Choice B is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 12

Choice A is correct. It's given that $p(n)=7n^3$. Substituting 56 for $p(n)$ in this equation yields $56=7n^3$. Dividing each side of this equation by 7 yields $8=n^3$. Taking the cube root of each side of this equation yields $2=n$. Therefore, when $p(n)$ is equal to 56, the value of n is 2.

Choice B is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

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QUESTION 13

The correct answer is 70. Based on the figure, the angle with measure 110° and the angle vertical to the angle with measure x° are same side interior angles. Since vertical angles are congruent, the angle vertical to the angle with measure x° also has measure x°. It's given that lines s and t are parallel. Therefore, same side interior angles between lines s and t are supplementary. It follows that $x+110=180$. Subtracting 110 from both sides of this equation yields $x=70$.

QUESTION 14

The correct answer is 10. The mean of a data set is calculated by dividing the sum of the data values by the number of data values in the data set. For this data set, the mean can be calculated as $\frac{6+8+16+4+17+26+8+5+5+5}{10}$, which is equivalent to $\frac{100}{10}$, or 10.

QUESTION 15

Choice A is correct. For an exponential function of the form $E(t)=a(b)^t$, where a and b are constants, the initial value of the function—that is, the value of the function when $t=0$—is a and the value of the function increases by a factor of b each time t increases by 1. Since the function $E(t)=5(1.8)^t$ gives the estimated number of employees at a restaurant and t is the number of years since the restaurant opened, the best interpretation of the number 5 in this context is the estimated number of employees when $t=0$, or when the restaurant opened.

Choice B is incorrect and may result from conceptual errors. *Choice C* is incorrect and may result from conceptual errors. *Choice D* is incorrect and may result from conceptual errors.

QUESTION 16

Choice B is correct. For a quadratic function defined by an equation of the form $g(x)=a(x-h)^2+k$, where a, h, and k are constants and $a>0$, the minimum value of the function is k. In the given function, $a=1$, $h=0$, and $k=55$. Therefore, the minimum value of the given function is 55.

Choice A is incorrect. This is the value of x for which the given function reaches its minimum value, not the minimum value of the function. *Choice C* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 17

Choice C is correct. Because the value of the investment increases each year, the function that best models how the value of the investment changes over time is an increasing function. It's given that each year, the value of the investment increases by 0.49% of its value the previous year. Since the value of the investment changes by a fixed percentage each year, the function that best

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models how the value of the investment changes over time is an exponential function. Therefore, the function that best models how the value of the investment changes over time is an increasing exponential function.

Choice A is incorrect and may result from conceptual errors. *Choice B* is incorrect and may result from conceptual errors. *Choice D* is incorrect and may result from conceptual errors.

QUESTION 18

Choice C is correct. Let x be the 2015 population of Greenville. It's given that the population increased by 7% from 2015 to 2016. The increase in population can be written as $(0.07)x$. The 2016 population of Greenville is given as the sum of the 2015 population of Greenville and the increase in population from 2015 to 2016. This can be rewritten as $x+(0.07)x$, or $1.07x$. Therefore, the value of k is 1.07.

Choice A is incorrect. This is the percent, represented as a decimal, that the population increased from 2015 to 2016, not the value of k. *Choice B* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect. This is the value of k if the population increased by 70%, not 7%, from 2015 to 2016.

QUESTION 19

Choice B is correct. Since $\frac{12}{12}=1$, multiplying the exponent of the given expression by $\frac{12}{12}$ yields an equivalent expression: $a^{\left(\frac{11}{12}\right)\left(\frac{12}{12}\right)}=a^{\left(\frac{132}{144}\right)}$. Since $\frac{132}{144}=132\left(\frac{1}{144}\right)$, the expression $a^{\frac{132}{144}}$ can be rewritten as $a^{(132)\left(\frac{1}{144}\right)}$. Applying properties of exponents, this expression can be rewritten as $\left(a^{132}\right)^{\frac{1}{144}}$. An expression of the form $(m)^{\frac{1}{k}}$, where $m>0$ and $k>0$, is equivalent to $\sqrt[k]{m}$. Therefore, $\left(a^{132}\right)^{\frac{1}{144}}$ is equivalent to $\sqrt[144]{a^{132}}$.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 20

The correct answer is 16. The total cost of the party is found by adding the onetime fee of the venue to the cost per attendee times the number of attendees. Let x be the number of attendees. The expression $35+10.25x$ thus represents the total cost of the party. It's given that the budget is $\$200$, so this situation can be represented by the inequality $35+10.25x\le 200$. The greatest number of attendees can be found by solving this inequality for x. Subtracting 35 from both sides of this inequality gives $10.25x\le 165$. Dividing both sides of this inequality by 10.25 results in approximately $x\le 16.098$. Since the question is stated in terms of attendees, rounding x down to the nearest whole number, 16, gives the greatest number of attendees possible.

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QUESTION 21

The correct answer is 28. The given absolute value equation can be rewritten as two linear equations: $4x-4=112$ and $-(4x-4)=112$, or $4x-4=-112$. Adding 4 to both sides of the equation $4x-4=112$ results in $4x=116$. Dividing both sides of this equation by 4 results in $x=29$. Adding 4 to both sides of the equation $4x-4=-112$ results in $4x=-108$. Dividing both sides of this equation by 4 results in $x=-27$. Therefore, the two values of $x-1$ are $29-1$, or 28, and $-27-1$, or -28. Thus, the positive value of $x-1$ is 28.

Alternate approach: The given equation can be rewritten as $|4(x-1)|=112$, which is equivalent to $4|x-1|=112$. Dividing both sides of this equation by 4 yields $|x-1|=28$. This equation can be rewritten as two linear equations: $x-1=28$ and $-(x-1)=28$, or $x-1=-28$. Therefore, the positive value of $x-1$ is 28.

QUESTION 22

Choice A is correct. The volume of a cube can be found by using the formula $V=s^3$, where V is the volume and s is the edge length of the cube. Therefore, the volume of the given cube is $V=68^3$, or 314,432 cubic inches. The volume of a sphere can be found by using the formula $V=\frac{4}{3}\pi r^3$, where V is the volume and r is the radius of the sphere. Therefore, the volume of the given sphere is $V=\frac{4}{3}\pi(34)^3$, or approximately 164,636 cubic inches. The volume of the space in the cube not taken up by the sphere is the difference between the volume of the cube and volume of the sphere. Subtracting the approximate volume of the sphere from the volume of the cube gives $314,432-164,636=149,796$ cubic inches.

Choice B is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 23

Choice B is correct. The standard form of an equation of a circle in the *xy*-plane is $(x-h)^2+(y-k)^2=r^2$, where the coordinates of the center of the circle are (h,k) and the length of the radius of the circle is r. For the circle in the *xy*-plane with equation $(x-5)^2+(y-3)^2=16$, it follows that $r^2=16$. Taking the square root of both sides of this equation yields $r=4$ or $r=-4$. Because r represents the length of the radius of the circle and this length must be positive, $r=4$. Therefore, the radius of the circle is 4. The diameter of a circle is twice the length of the radius of the circle. Thus, $2(4)$ yields 8. Therefore, the diameter of the circle is 8.

Choice A is incorrect. This is the radius of the circle. *Choice C* is incorrect. This is the square of the radius of the circle. *Choice D* is incorrect and may result from conceptual or calculation errors.

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QUESTION 24

Choice C is correct. For the form of the function in choice C, $f(x)=128(1.6)^{x-1}$, the value of $f(1)$ can be found as $128(1.6)^{1-1}$, which is equivalent to $128(1.6)^{0}$, or 128. Therefore, $k=128$, which is shown in $f(x)=128(1.6)^{x-1}$ as the coefficient.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice B is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 25

Choice B is correct. Since the model estimates that the number of squirrels in the population increased by a fixed percentage, 150%, each year, the model can be represented by an exponential equation of the form $n=a\left(1+\frac{p}{100}\right)^{t}$, where a is the estimated number of squirrels in the population at the end of 2015, and the model estimates that at the end of each year, the number is p% more than the number at the end of the previous year. Since the model estimates that at the end of each year, the number was 150% more than the number at the end of the previous year, $p=150$. Substituting 150 for p in the equation $n=a\left(1+\frac{p}{100}\right)^{t}$ yields $n=a\left(1+\frac{150}{100}\right)^{t}$, which is equivalent to $n=a(1+1.5)^{t}$, or $n=a(2.5)^{t}$. It's given that the estimated number of squirrels at the end of 2016 was 180. This means that when $t=1$, $n=180$. Substituting 1 for t and 180 for n in the equation $n=a(2.5)^{t}$ yields $180=a(2.5)^{1}$, or $180=2.5a$. Dividing each side of this equation by 2.5 yields $72=a$. Substituting 72 for a in the equation $n=a(2.5)^{t}$ yields $n=72(2.5)^{t}$.

Choice A is incorrect. This equation represents a model where at the end of each year, the estimated number of squirrels was 150% of, not 150% more than, the estimated number at the end of the previous year. *Choice C* is incorrect. This equation represents a model where at the end of each year, the estimated number of squirrels was 150% of, not 150% more than, the estimated number at the end of the previous year, and the estimated number of squirrels at the end of 2015, not the end of 2016, was 180. *Choice D* is incorrect. This equation represents a model where the estimated number of squirrels at the end of 2015, not the end of 2016, was 180.

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QUESTION 26

Choice B is correct. Two lines are perpendicular if their slopes are negative reciprocals, meaning that the slope of the first line is equal to -1 divided by the slope of the second line. Each equation in the given pair of equations can be written in slope-intercept form, $y=mx+b$, where m is the slope of the graph of the equation in the xy-plane and $(0,b)$ is the y-intercept. For the first equation, $5x+7y=1$, subtracting $5x$ from both sides gives $7y=-5x+1$, and dividing both sides of this equation by 7 gives $y=-\frac{5}{7}x+\frac{1}{7}$. Therefore, the slope of the graph of this equation is $-\frac{5}{7}$. For the second equation, $ax+by=1$, subtracting ax from both sides gives $by=-ax+1$, and dividing both sides of this equation by b gives $y=-\frac{a}{b}x+\frac{1}{b}$. Therefore, the slope of the graph of this equation is $-\frac{a}{b}$. Since the graph of the given pair of equations is a pair of perpendicular lines, the slope of the graph of the second equation, $-\frac{a}{b}$, must be the negative reciprocal of the slope of the graph of the first equation, $-\frac{5}{7}$. The negative reciprocal of $-\frac{5}{7}$ is $\frac{-1}{\left(-\frac{5}{7}\right)}$, or $\frac{7}{5}$. Therefore, $-\frac{a}{b}=\frac{7}{5}$, or $\frac{a}{b}=-\frac{7}{5}$. Similarly, rewriting the equations in choice B in slope-intercept form yields $y=-\frac{10}{7}x+\frac{1}{7}$ and $y=-\frac{a}{2b}x+\frac{1}{2b}$. It follows that the slope of the graph of the first equation in choice B is $-\frac{10}{7}$ and the slope of the graph of the second equation in choice B is $-\frac{a}{2b}$. Since $\frac{a}{b}=-\frac{7}{5}$, $-\frac{a}{2b}$ is equal to $-\left(\frac{1}{2}\right)\left(-\frac{7}{5}\right)$, or $\frac{7}{10}$. Since $\frac{7}{10}$ is the negative reciprocal of $-\frac{10}{7}$, the pair of equations in choice B represents a pair of perpendicular lines.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 27

The correct answer is 289. A quadratic equation of the form $ax^2+bx+c=0$, where a, b, and c are constants, has no real solutions when the value of the discriminant, b^2-4ac, is less than 0. In the given equation, $x^2-34x+c=0$, $a=1$ and $b=-34$. Therefore, the discriminant of the given equation can be expressed as $(-34)^2-4(1)(c)$, or $1{,}156-4c$. It follows that the given equation has no real solutions when $1{,}156-4c<0$. Adding $4c$ to both sides of this inequality yields $1{,}156<4c$. Dividing both sides of this inequality by 4 yields $289<c$, or $c>289$. It's given that the equation $x^2-34x+c=0$ has no real solutions when $c>n$. Therefore, the least possible value of n is 289.

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6TSL03

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Test begins on the next page.

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Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

As Mexico's first president from an Indigenous community, Benito Juarez became one of the most _____ figures in his country's history: among the many significant accomplishments of his long tenure in office (1858–1872), Juarez consolidated the authority of the national government and advanced the rights of Indigenous peoples.

Which choice completes the text with the most logical and precise word or phrase?

A) unpredictable
B) important
C) secretive
D) ordinary

2

Due to their often strange images, highly experimental syntax, and opaque subject matter, many of John Ashbery's poems can be quite difficult to _____ and thus are the object of heated debate among scholars.

Which choice completes the text with the most logical and precise word or phrase?

A) delegate
B) compose
C) interpret
D) renounce

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3

The Cambrian explosion gets its name from the sudden appearance and rapid diversification of animal remains in the fossil record about 541 million years ago, during the Cambrian period. Some scientists argue that this _____ change in the fossil record might be because of a shift in many organisms to body types that were more likely to be preserved.

Which choice completes the text with the most logical and precise word or phrase?

A) catastrophic
B) elusive
C) abrupt
D) imminent

4

During a 2014 archaeological dig in Spain, Vicente Lull and his team uncovered the skeleton of a woman from El Algar, an Early Bronze Age society, buried with valuable objects signaling a high position of power. This finding may persuade researchers who have argued that Bronze Age societies were ruled by men to _____ that women may have also held leadership roles.

Which choice completes the text with the most logical and precise word or phrase?

A) waive
B) concede
C) refute
D) require

5

Within baleen whale species, some individuals develop an accessory spleen—a seemingly functionless formation of splenetic tissue outside the normal spleen. Given the formation's greater prevalence among whales known to make deeper dives, some researchers hypothesize that its role isn't _____; rather, the accessory spleen may actively support diving mechanisms.

Which choice completes the text with the most logical and precise word or phrase?

A) replicable
B) predetermined
C) operative
D) latent

6

According to a US tax policy expert, state taxes are _____ other factors when considering an interstate move. Even significant differences in state taxation have almost no effect on most people's decisions, while differences in employment opportunities, housing availability, and climate are strong influences.

Which choice completes the text with the most logical and precise word or phrase?

A) consistent with
B) representative of
C) overshadowed by
D) irrelevant to

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7

The author's claim about the relationship between Neanderthals and *Homo sapiens* is _____, as it fails to account for several recent archaeological discoveries. To be convincing, his argument would need to address recent finds of additional hominid fossils, such as the latest Denisovan specimens and *Homo longi*.

Which choice completes the text with the most logical and precise word or phrase?

A) disorienting
B) tenuous
C) nuanced
D) unoriginal

8

The following text is from Georgia Douglas Johnson's 1922 poem "Benediction."

> Go forth, my son,
> Winged by my heart's desire!
> Great reaches, yet unknown,
> Await
> For your possession.
> I may not, if I would,
> Retrace the way with you,
> My pilgrimage is through,
> But life is calling you!

Which choice best states the main purpose of the text?

A) To express hope that a child will have the same accomplishments as his parent did
B) To suggest that raising a child involves many struggles
C) To warn a child that he will face many challenges throughout his life
D) To encourage a child to embrace the experiences life will offer

9

The following text is adapted from *Indian Boyhood*, a 1902 memoir by Ohiyesa (Charles A. Eastman), a Santee Dakota writer. In the text, Ohiyesa recalls how the women in his tribe harvested maple syrup during his childhood.

> Now the women began to test the trees—moving leisurely among them, axe in hand, and striking a single quick blow, to see if the sap would appear. The trees, like people, have their individual characters; some were ready to yield up their life-blood, while others were more reluctant. Now one of the birchen basins was set under each tree, and a hardwood chip driven deep into the cut which the axe had made. From the corners of this chip—at first drop by drop, then more freely—the sap trickled into the little dishes.

Which choice best describes the function of the underlined sentence in the text as a whole?

A) It portrays the range of personality traits displayed by the women as they work.
B) It foregrounds the beneficial relationship between humans and maple trees.
C) It demonstrates how human behavior can be influenced by the natural environment.
D) It elaborates on an aspect of the maple trees that the women evaluate.

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10

Text 1
Ecologists have long wondered how thousands of microscopic phytoplankton species can live together near ocean surfaces competing for the same resources. According to conventional wisdom, one species should emerge after outcompeting the rest. So why do so many species remain? Ecologists' many efforts to explain this phenomenon still haven't uncovered a satisfactory explanation.

Text 2
Ecologist Michael Behrenfeld and colleagues have connected phytoplankton's diversity to their microscopic size. Because these organisms are so tiny, they are spaced relatively far apart from each other in ocean water and, moreover, experience that water as a relatively dense substance. This in turn makes it hard for them to move around and interact with one another. Therefore, says Behrenfeld's team, direct competition among phytoplankton probably happens much less than previously thought.

Based on the texts, how would Behrenfeld and colleagues (Text 2) most likely respond to the "conventional wisdom" discussed in Text 1?

A) By arguing that it is based on a misconception about phytoplankton species competing with one another

B) By asserting that it fails to recognize that routine replenishment of ocean nutrients prevents competition between phytoplankton species

C) By suggesting that their own findings help clarify how phytoplankton species are able to compete with larger organisms

D) By recommending that more ecologists focus their research on how competition among phytoplankton species is increased with water density

11

In 2014, Amelia Quon and her team at NASA set out to build a helicopter capable of flying on Mars. Because Mars's atmosphere is only one percent as dense as Earth's, the air of Mars would not provide enough resistance to the rotating blades of a standard helicopter for the aircraft to stay aloft. For five years, Quon's team tested designs in a lab that mimicked Mars's atmospheric conditions. The craft the team ultimately designed can fly on Mars because its blades are longer and rotate faster than those of a helicopter of the same size built for Earth.

According to the text, why would a helicopter built for Earth be unable to fly on Mars?

A) Because Mars and Earth have different atmospheric conditions

B) Because the blades of helicopters built for Earth are too large to work on Mars

C) Because the gravity of Mars is much weaker than the gravity of Earth

D) Because helicopters built for Earth are too small to handle the conditions on Mars

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12

In West Africa, jalis have traditionally been keepers of information about family histories and records of important events. They have often served as teachers and advisers, too. New technologies may have changed some aspects of the role today, but jalis continue to be valued for knowing and protecting their peoples' stories.

Which choice best states the main idea of the text?

A) Even though there have been some changes in their role, jalis continue to preserve their communities' histories.

B) Although jalis have many roles, many of them like teaching best.

C) Jalis have been entertaining the people within their communities for centuries.

D) Technology can now do some of the things jalis used to be responsible for.

13

In 1934 physicist Eugene Wigner posited the existence of a crystal consisting entirely of electrons in a honeycomb-like structure. The so-called Wigner crystal remained largely conjecture, however, until Feng Wang and colleagues announced in 2021 that they had captured an image of one. The researchers trapped electrons between two semiconductors and then cooled the apparatus, causing the electrons to settle into a crystalline structure. By inserting an ultrathin sheet of graphene above the crystal, the researchers obtained an impression—the first visual confirmation of the Wigner crystal.

Which choice best states the main idea of the text?

A) Researchers have obtained the most definitive evidence to date of the existence of the Wigner crystal.

B) Researchers have identified an innovative new method for working with unusual crystalline structures.

C) Graphene is the most important of the components required to capture an image of a Wigner crystal.

D) It's difficult to acquire an image of a Wigner crystal because of the crystal's honeycomb structure.

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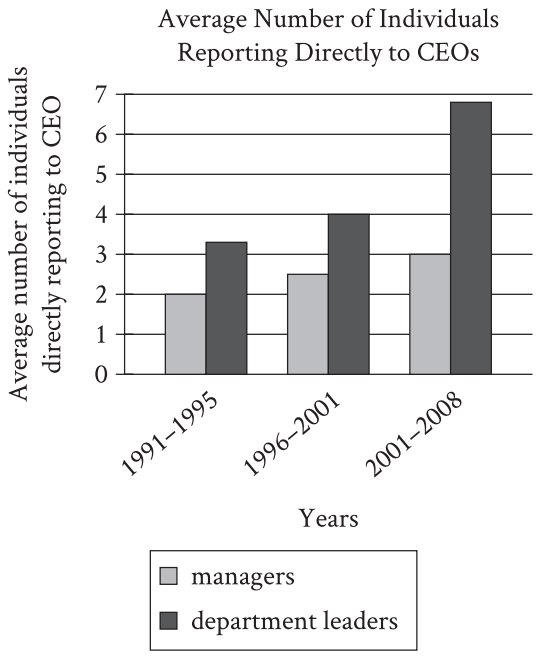
14



Considering a large sample of companies, economics experts Maria Guadalupe, Julie Wulf, and Raghuram Rajan assessed the number of managers and leaders from different departments who reported directly to a chief executive officer (CEO). According to the researchers, the findings suggest that across the years analyzed, there was a growing interest among CEOs in connecting with more departments in their companies.

Which choice best describes data from the graph that support the researchers' conclusion?

A) The average numbers of managers and department leaders reporting directly to their CEO didn't fluctuate from the 1991–1995 period to the 2001–2008 period.

B) The average number of managers reporting directly to their CEO was highest in the 1996–2001 period.

C) The average number of department leaders reporting directly to their CEO was greater than the average number of managers reporting directly to their CEO in each of the three periods studied.

D) The average number of department leaders reporting directly to their CEO rose over the three periods studied.

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15

When digging for clams, their primary food, sea otters damage the roots of eelgrass plants growing on the seafloor. Near Vancouver Island in Canada, the otter population is large and well established, yet the eelgrass meadows are healthier than those found elsewhere off Canada's coast. To explain this, conservation scientist Erin Foster and colleagues compared the Vancouver Island meadows to meadows where otters are absent or were reintroduced only recently. Finding that the Vancouver Island meadows have a more diverse gene pool than the others do, Foster hypothesized that damage to eelgrass roots increases the plant's rate of sexual reproduction; this, in turn, boosts genetic diversity, which benefits the meadow's health overall.

Which finding, if true, would most directly undermine Foster's hypothesis?

A) At some sites in the study, eelgrass meadows are found near otter populations that are small and have only recently been reintroduced.

B) At several sites not included in the study, there are large, well-established sea otter populations but no eelgrass meadows.

C) At several sites not included in the study, eelgrass meadows' health correlates negatively with the length of residence and size of otter populations.

D) At some sites in the study, the health of plants unrelated to eelgrass correlates negatively with the length of residence and size of otter populations.

16

Scholars have noted that F. Scott Fitzgerald's writings were likely influenced in part by his marriage to Zelda Fitzgerald, but many don't recognize Zelda as a writer in her own right. Indeed, Zelda authored several works herself, such as the novel *Save Me the Waltz* and numerous short stories. Thus, those who primarily view Zelda as an inspiration for F. Scott's writings _____

Which choice most logically completes the text?

A) overlook the many other factors that motivated F. Scott to write.

B) risk misrepresenting the full range of Zelda's contributions to literature.

C) may draw inaccurate conclusions about how F. Scott and Zelda viewed each other's works.

D) tend to read the works of F. Scott and Zelda in an overly autobiographical light.

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17

Among social animals that care for their young, such as chickens, macaque monkeys, and humans, newborns appear to show an innate attraction to faces and face-like stimuli. Elisabetta Versace and her colleagues used an image of three black dots arranged in the shape of eyes and a nose or mouth to test whether this trait also occurs in *Testudo* tortoises, which live alone and do not engage in parental care. They found that tortoise hatchlings showed a significant preference for the image, suggesting that _____

Which choice most logically completes the text?

A) face-like stimuli are likely perceived as harmless by newborns of social species that practice parental care but as threatening by newborns of solitary species without parental care.

B) researchers should not assume that an innate attraction to face-like stimuli is necessarily an adaptation related to social interaction or parental care.

C) researchers can assume that the attraction to face-like stimuli that is seen in social species that practice parental care is learned rather than innate.

D) newly hatched *Testudo* tortoises show a stronger preference for face-like stimuli than adult *Testudo* tortoises do.

18

Compiled in the late 1500s largely through the efforts of Indigenous scribes, *Cantares Mexicanos* is the most important collection of poetry in Classical Nahuatl, the principal language of the Aztec Empire. The poems portray Aztec society before the occupation of the empire by the army of Spain, and marginal notes in *Cantares Mexicanos* indicate that much of the collection's content predates the initial invasion. Nonetheless, some of the poems contain inarguable references to beliefs and customs common in Spain during this era. Thus, some scholars have concluded that _____

Which choice most logically completes the text?

A) while its content largely predates the invasion, *Cantares Mexicanos* also contains additions made after the invasion.

B) although those who compiled *Cantares Mexicanos* were fluent in Nahuatl, they had limited knowledge of the Spanish language.

C) before the invasion by Spain, the poets of the Aztec Empire borrowed from the literary traditions of other societies.

D) the references to beliefs and customs in Spain should be attributed to a coincidental resemblance between the societies of Spain and the Aztec Empire.

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19

In a study of the cognitive abilities of white-faced capuchin monkeys (*Cebus imitator*), researchers neglected to control for the physical difficulty of the tasks they used to evaluate the monkeys. The cognitive abilities of monkeys given problems requiring little dexterity, such as sliding a panel to retrieve food, were judged by the same criteria as were those of monkeys given physically demanding problems, such as unscrewing a bottle and inserting a straw. The results of the study, therefore, _____

Which choice most logically completes the text?

A) could suggest that there are differences in cognitive ability among the monkeys even though such differences may not actually exist.

B) are useful for identifying tasks that the monkeys lack the cognitive capacity to perform but not for identifying tasks that the monkeys can perform.

C) should not be taken as indicative of the cognitive abilities of any monkey species other than *C. imitator*.

D) reveal more about the monkeys' cognitive abilities when solving artificial problems than when solving problems encountered in the wild.

20

To survive when water is scarce, embryos inside African turquoise killifish eggs _____ a dormant state known as diapause. In this state, embryonic development is paused for as long as two years—longer than the life span of an adult killifish.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) enter

B) to enter

C) having entered

D) entering

21

Formed in 1967 to foster political and economic stability within the Asia-Pacific region, the Association of Southeast Asian Nations was originally made up of five members: Thailand, the Philippines, Singapore, Malaysia, and Indonesia. By the end of the 1990s, the organization _____ its initial membership.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) has doubled

B) had doubled

C) doubles

D) will double

22

The intense pressure found in the deep ocean can affect the structure of proteins in fish's cells, distorting the proteins' shape. The chemical trimethylamine N-oxide (TMAO) counters this effect, ensuring that proteins retain their original _____ is found in high concentrations in the cells of the deepest-dwelling fish.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) configurations. TMAO

B) configurations TMAO

C) configurations, TMAO

D) configurations and TMAO

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23

Food and the sensation of taste are central to Monique Truong's novels. In *The Book of Salt*, for example, the exiled character of Bình connects to his native Saigon through the food he prepares, while in *Bitter in the Mouth*, the character of Linda _____ a form of synesthesia whereby the words she hears evoke tastes.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) experienced
B) had experienced
C) experiences
D) will be experiencing

24

Inventor John Friedman created a prototype of the first flexible straw by inserting a screw into a paper straw and, using dental floss, binding the straw tightly around the _____ When the floss and screw were removed, the resulting corrugations in the paper allowed the straw to bend easily over the edge of a glass.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) screw's thread's.
B) screws' threads.
C) screw's threads.
D) screws threads'.

25

In her analysis of Edith Wharton's *The House of Mirth* (1905), scholar Candace Waid observes that the novel depicts the upper classes of New York society as "consumed by the appetite of a soulless _____ an apt assessment given that *The House of Mirth* is set during the Gilded Age, a period marked by rapid industrialization, economic greed, and widening wealth disparities.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) materialism"; and
B) materialism" and
C) materialism,"
D) materialism"

26

To humans, it does not appear that the golden orb-weaver spider uses camouflage to capture its _____ the brightly colored arachnid seems to wait conspicuously in the center of its large circular web for insects to approach. Researcher Po Peng of the University of Melbourne has explained that the spider's distinctive coloration may in fact be part of its appeal.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) prey, rather,
B) prey rather,
C) prey, rather;
D) prey; rather,

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27

In Death Valley National Park's Racetrack Playa, a flat, dry lakebed, are 162 rocks—some weighing less than a pound but others almost 700 pounds—that move periodically from place to place, seemingly of their own volition. Racetrack-like trails in the _____ mysterious migration.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) playas sediment mark the rock's
B) playa's sediment mark the rocks
C) playa's sediment mark the rocks'
D) playas' sediment mark the rocks'

28

In crafting her fantasy fiction, Nigerian-born British author Helen Oyeyemi has drawn inspiration from the classic nineteenth-century fairy tales of the Brothers Grimm. Her 2014 novel *Boy, Snow, Bird*, for instance, is a complex retelling of the story of Snow White, while her 2019 novel _____ offers a delicious twist on the classic tale of Hansel and Gretel.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) *Gingerbread*—
B) *Gingerbread*,
C) *Gingerbread*
D) *Gingerbread*:

29

While researching a topic, a student has taken the following notes:

- NASA uses rovers, large remote vehicles with wheels, to explore the surface of Mars.
- NASA's rovers can't explore regions inaccessible to wheeled vehicles.
- Rovers are also heavy, making them difficult to land on the planet's surface.
- Microprobes, robotic probes that weigh as little as 50 milligrams, could be deployed virtually anywhere on the surface of Mars.
- Microprobes have been proposed as an alternative to rovers.

The student wants to explain an advantage of microprobes. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Despite being heavy, NASA's rovers can land successfully on the surface of Mars.
B) Microprobes, which weigh as little as 50 milligrams, could explore areas of Mars that are inaccessible to NASA's heavy, wheeled rovers.
C) NASA currently uses its rovers on Mars, but microprobes have been proposed as an alternative.
D) Though they are different sizes, both microprobes and rovers can be used to explore the surface of Mars.

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30

While researching a topic, a student has taken the following notes:

- Abdulrazak Gurnah was awarded the 2021 Nobel Prize in Literature.
- Gurnah was born in Zanzibar in East Africa and currently lives in the United Kingdom.
- Many readers have singled out Gurnah's 1994 book *Paradise* for praise.
- *Paradise* is a historical novel about events that occurred in colonial East Africa.

The student wants to introduce *Paradise* to an audience unfamiliar with the novel and its author. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Abdulrazak Gurnah, who wrote *Paradise* and later was awarded the Nobel Prize in Literature, was born in Zanzibar in East Africa and currently lives in the United Kingdom.

B) Many readers have singled out Abdulrazak Gurnah's 1994 book *Paradise,* a historical novel about colonial East Africa, for praise.

C) A much-praised historical novel about colonial East Africa, *Paradise* (1994) was written by Abdulrazak Gurnah, winner of the 2021 Nobel Prize in Literature.

D) *Paradise* is a historical novel about events that occurred in colonial East Africa, Abdulrazak Gurnah's homeland.

31

While researching a topic, a student has taken the following notes:

- Ulaanbaatar is the capital of Mongolia.
- The city's population is 907,802.
- Ulaanbaatar contains 31.98 percent of Mongolia's population.
- Hanoi is the capital of Vietnam.
- The city's population is 7,781,631.
- Hanoi contains 8.14 percent of Vietnam's population.

Which choice most effectively uses information from the given sentences to emphasize the relative sizes of the two capitals' populations?

A) Mongolia's capital is Ulaanbaatar, which has 907,802 people, and Vietnam's capital is Hanoi, which has 7,781,631 people.

B) The populations of the capitals of Mongolia and Vietnam are 907,802 (Ulaanbaatar) and 7,781,631 (Hanoi), respectively.

C) Even though Hanoi (population 7,781,631) is larger than Ulaanbaatar (population 907,802), Ulaanbaatar accounts for more of its country's population.

D) Comparing Vietnam and Mongolia, 7,781,631 is 8.14 percent of Vietnam's population, and 907,802 is 31.98 percent of Mongolia's.

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32

While researching a topic, a student has taken the following notes:

- One of history's greatest libraries was the House of Wisdom in Baghdad, Iraq.
- It was founded in the eighth century with the goal of preserving all the world's knowledge.
- Scholars at the House of Wisdom collected ancient and contemporary texts from Greece, India, and elsewhere and translated them into Arabic.
- Writings included those of the Greek philosopher Aristotle and the Indian mathematician Aryabhata.
- The House of Wisdom used Chinese papermaking technology to create paper versions to be studied and shared.

The student wants to explain how the House of Wisdom preserved the world's knowledge. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The House of Wisdom was known for bringing together knowledge from around the world, including from Greece, India, and China.

B) Founded in Iraq in the eighth century, the House of Wisdom employed many scholars as translators.

C) Writings from the Greek philosopher Aristotle and the Indian mathematician Aryabhata were preserved at the House of Wisdom.

D) The House of Wisdom collected writings from different countries and created paper versions in Arabic to be studied and shared.

33

While researching a topic, a student has taken the following notes:

- British musicians John Lennon and Paul McCartney shared writing credit for numerous Beatles songs.
- Many Lennon-McCartney songs were actually written by either Lennon or McCartney, not by both.
- The exact authorship of specific parts of many Beatles songs, such as the verse for "In My Life," is disputed.
- Mark Glickman, Jason Brown, and Ryan Song used statistical methods to analyze the musical content of Beatles songs.
- They concluded that there is 18.9% probability that McCartney wrote the verse for "In My Life," stating that the verse is "consistent with Lennon's songwriting style."

The student wants to make a generalization about the kind of study conducted by Glickman, Brown, and Song. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Based on statistical analysis, Glickman, Brown, and Song claim that John Lennon wrote the verse of "In My Life."

B) There is only an 18.9% probability that Paul McCartney wrote the verse for "In My Life"; John Lennon is the more likely author.

C) It is likely that John Lennon, not Paul McCartney, wrote the verse for "In My Life."

D) Researchers have used statistical methods to address questions of authorship within the field of music.

STOP

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Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

The Mule Bone, a 1930 play written by Zora Neale Hurston and Langston Hughes, is perhaps the best-known of the few examples of _____ in literature. Most writers prefer working alone, and given that working together cost Hurston and Hughes their friendship, it is not hard to see why.

Which choice completes the text with the most logical and precise word or phrase?

A) characterization
B) interpretation
C) collaboration
D) commercialization

2

The process of mechanically recycling plastics is often considered _____ because of the environmental impact and the loss of material quality that often occurs. But chemist Takunda Chazovachii has helped develop a cleaner process of chemical recycling that converts superabsorbent polymers from diapers into a desirable reusable adhesive.

Which choice completes the text with the most logical and precise word or phrase?

A) resilient
B) inadequate
C) dynamic
D) satisfactory

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3

Interruptions in the supply chain for microchips used in personal electronics have challenged an economist's assertion that retailers can expect robust growth in sales of those devices in the coming months. The delays are unlikely to _____ her projection entirely but will almost certainly extend its time frame.

Which choice completes the text with the most logical and precise word or phrase?

A) dispute
B) withdraw
C) underscore
D) invalidate

4

For her 2021 art installation *Anthem*, Wu Tsang joined forces with singer and composer Beverly Glenn-Copeland to produce a piece that critics found truly _____: they praised Tsang for creatively transforming a museum rotunda into a dynamic exhibit by projecting filmed images of Glenn-Copeland onto a massive 84-foot curtain and filling the space with the sounds of his and other voices singing.

Which choice completes the text with the most logical and precise word or phrase?

A) restrained
B) inventive
C) inexplicable
D) mystifying

5

Some scientists have suggested that mammals in the Mesozoic era were not a very _____ group, but paleontologist Zhe-Xi Luo's research suggests that early mammals living in the shadow of dinosaurs weren't all ground-dwelling insectivores. Fossils of various plant-eating mammals have been found in China, including species like *Vilevolodon diplomylos*, which Luo says could glide like a flying squirrel.

Which choice completes the text with the most logical and precise word or phrase?

A) predatory
B) obscure
C) diverse
D) localized

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6

The following text is adapted from Gwendolyn Bennett's 1926 poem "Street Lamps in Early Spring."

> Night wears a garment
> All velvet soft, all violet blue...
> And over her face she draws a veil
> As shimmering fine as floating dew...
> And here and there
> In the black of her hair
> The subtle hands of Night
> Move slowly with their gem-starred light.

Which choice best describes the overall structure of the text?

A) It presents alternating descriptions of night in a rural area and in a city.

B) It sketches an image of nightfall, then an image of sunrise.

C) It makes an extended comparison of night to a human being.

D) It portrays how night changes from one season of the year to the next.

7

According to historian Vicki L. Ruiz, Mexican American women made crucial contributions to the labor movement during World War II. At the time, food processing companies entered into contracts to supply United States armed forces with canned goods. Increased production quotas conferred greater bargaining power on the companies' employees, many of whom were Mexican American women: employees insisted on more favorable benefits, and employers, who were anxious to fulfill the contracts, complied. Thus, labor activism became a platform for Mexican American women to assert their agency.

Which choice best describes the function of the underlined portion in the text as a whole?

A) It elaborates on a claim about labor relations in a particular industry made earlier in the text.

B) It offers an example of a trend in the World War II–era economy discussed earlier in the text.

C) It notes a possible exception to the historical narrative of labor activism sketched earlier in the text.

D) It provides further details about the identities of the workers discussed earlier in the text.

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8

The following text is adapted from Zora Neale Hurston's 1921 short story "John Redding Goes to Sea." John is a child who lives in a town in the woods.

> Perhaps ten-year-old John was puzzling to the folk there in the Florida woods for he was an imaginative child and fond of day-dreams. The St. John River flowed a scarce three hundred feet from his back door. On its banks at this point grow numerous palms, luxuriant magnolias and bay trees. On the bosom of the stream float millions of delicately colored hyacinths. [John Redding] loved to wander down to the water's edge, and, casting in dry twigs, watch them sail away down stream to Jacksonville, the sea, the wide world and [he] wanted to follow them.

Which choice best describes the function of the underlined sentence in the text as a whole?

A) It provides an extended description of a location that John likes to visit.

B) It reveals that some residents of John's town are confused by his behavior.

C) It illustrates the uniqueness of John's imagination compared to the imaginations of other children.

D) It suggests that John longs to experience a larger life outside the Florida woods.

9

The following text is adapted from Oscar Wilde's 1891 novel *The Picture of Dorian Gray*. Dorian Gray is taking his first look at a portrait that Hallward has painted of him.

> Dorian passed listlessly in front of his picture and turned towards it. When he saw it he drew back, and his cheeks flushed for a moment with pleasure. A look of joy came into his eyes, as if he had recognized himself for the first time. He stood there motionless and in wonder, dimly conscious that Hallward was speaking to him, but not catching the meaning of his words. The sense of his own beauty came on him like a revelation. He had never felt it before.

According to the text, what is true about Dorian?

A) He wants to know Hallward's opinion of the portrait.

B) He is delighted by what he sees in the portrait.

C) He prefers portraits to other types of paintings.

D) He is uncertain of Hallward's talent as an artist.

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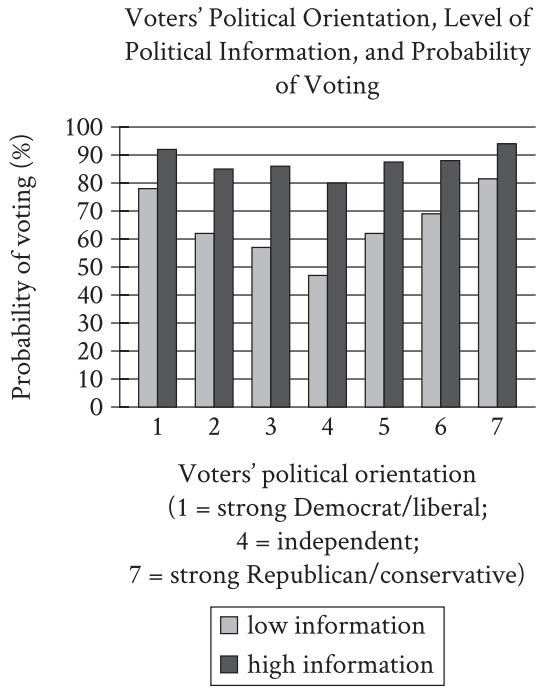
10



Economists Kerwin Kofi Charles and Melvin Stephens Jr. investigated a variety of factors that influence voter turnout in the United States. Using survey data that revealed whether respondents voted in national elections and how knowledgeable respondents are about politics, Charles and Stephens claim that the likelihood of voting is driven in part by potential voters' confidence in their assessments of candidates—essentially, the more informed voters are about politics, the more confident they are at evaluating whether candidates share their views, and thus the more likely they are to vote.

Which choice best describes data in the graph that support Charles and Stephens's claim?

A) At each point on the political orientation scale, high-information voters were more likely than low-information voters to vote.

B) Only low-information voters who identify as independents had a voting probability below 50%.

C) The closer that low-information voters are to the ends of the political orientation scale, the more likely they were to vote.

D) High-information voters were more likely to identify as strong Democrats or strong Republicans than low-information voters were.

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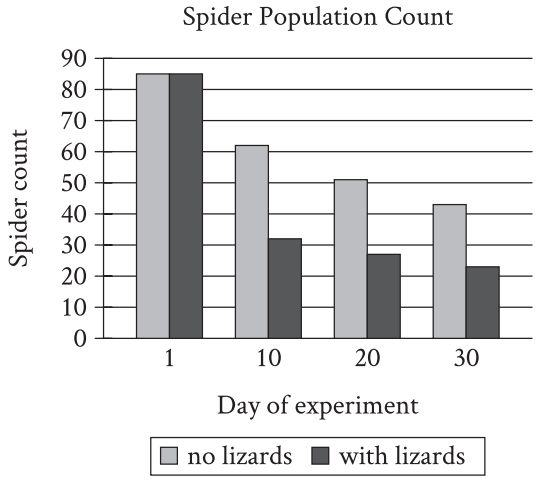
11



To investigate the effect of lizard predation on spider populations, a student in a biology class placed spiders in two enclosures, one with lizards and one without, and tracked the number of spiders in the enclosures for 30 days. The student concluded that the reduction in the spider population count in the enclosure with lizards by day 30 was entirely attributable to the presence of the lizards.

Which choice best describes data from the graph that weaken the student's conclusion?

A) The spider population count was the same in both enclosures on day 1.

B) The spider population count also substantially declined by day 30 in the enclosure without lizards.

C) The largest decline in spider population count in the enclosure with lizards occurred from day 1 to day 10.

D) The spider population count on day 30 was lower in the enclosure with lizards than in the enclosure without lizards.

12

Archaeologist Petra Vaiglova, anthropologist Xinyi Liu, and their colleagues investigated the domestication of farm animals in China during the Bronze Age (approximately 2000 to 1000 BCE). By analyzing the chemical composition of the bones of sheep, goats, and cattle from this era, the team determined that wild plants made up the bulk of sheep's and goats' diets, while the cattle's diet consisted largely of millet, a crop cultivated by humans. The team concluded that cattle were likely raised closer to human settlements, whereas sheep and goats were allowed to roam farther away.

Which finding, if true, would most strongly support the team's conclusion?

A) Analysis of the animal bones showed that the cattle's diet also consisted of wheat, which humans widely cultivated in China during the Bronze Age.

B) Further investigation of sheep and goat bones revealed that their diets consisted of small portions of millet as well.

C) Cattle's diets generally require larger amounts of food and a greater variety of nutrients than do sheep's and goats' diets.

D) The diets of sheep, goats, and cattle were found to vary based on what the farmers in each Bronze Age settlement could grow.

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13

Mosasaurs were large marine reptiles that lived in the Late Cretaceous period, approximately 100 million to 66 million years ago. Celina Suarez, Alberto Pérez-Huerta, and T. Lynn Harrell Jr. examined oxygen-18 isotopes in mosasaur tooth enamel in order to calculate likely mosasaur body temperatures and determined that mosasaurs were endothermic—that is, they used internal metabolic processes to maintain a stable body temperature in a variety of ambient temperatures. Suarez, Pérez-Huerta, and Harrell claim that endothermy would have enabled mosasaurs to include relatively cold polar waters in their range.

Which finding, if true, would most directly support Suarez, Pérez-Huerta, and Harrell's claim?

A) Mosasaurs' likely body temperatures are easier to determine from tooth enamel oxygen-18 isotope data than the body temperatures of nonendothermic Late Cretaceous marine reptiles are.

B) Fossils of both mosasaurs and nonendothermic marine reptiles have been found in roughly equal numbers in regions known to be near the poles during the Late Cretaceous, though in lower concentrations than elsewhere.

C) Several mosasaur fossils have been found in regions known to be near the poles during the Late Cretaceous, while relatively few fossils of nonendothermic marine reptiles have been found in those locations.

D) During the Late Cretaceous, seawater temperatures were likely higher throughout mosasaurs' range, including near the poles, than seawater temperatures at those same latitudes are today.

14

Researchers hypothesized that a decline in the population of dusky sharks near the mid-Atlantic coast of North America led to a decline in the population of eastern oysters in the region. Dusky sharks do not typically consume eastern oysters but do consume cownose rays, which are the main predators of the oysters.

Which finding, if true, would most directly support the researchers' hypothesis?

A) Declines in the regional abundance of dusky sharks' prey other than cownose rays are associated with regional declines in dusky shark abundance.

B) Eastern oyster abundance tends to be greater in areas with both dusky sharks and cownose rays than in areas with only dusky sharks.

C) Consumption of eastern oysters by cownose rays in the region substantially increased before the regional decline in dusky shark abundance began.

D) Cownose rays have increased in regional abundance as dusky sharks have decreased in regional abundance.

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15

Political scientists who favor the traditional view of voter behavior claim that voting in an election does not change a voter's attitude toward the candidates in that election. Focusing on each US presidential election from 1976 to 1996, Ebonya Washington and Sendhil Mullainathan tested this claim by distinguishing between subjects who had just become old enough to vote (around half of whom actually voted) and otherwise similar subjects who were slightly too young to vote (and thus none of whom voted). Washington and Mullainathan compared the attitudes of the groups of subjects toward the winning candidate two years after each election.

Which finding from Washington and Mullainathan's study, if true, would most directly weaken the claim made by people who favor the traditional view of voter behavior?

A) Subjects' attitudes toward the winning candidate two years after a given election were strongly predicted by subjects' general political orientation, regardless of whether subjects were old enough to vote at the time of the election.

B) Subjects who were not old enough to vote in a given election held significantly more positive attitudes towards the winning candidate two years later than they held at the time of the election.

C) Subjects who voted in a given election held significantly more polarized attitudes toward the winning candidate two years later than did subjects who were not old enough to vote in that election.

D) Two years after a given election, subjects who voted and subjects who were not old enough to vote were significantly more likely to express negative attitudes than positive attitudes toward the winning candidate in that election.

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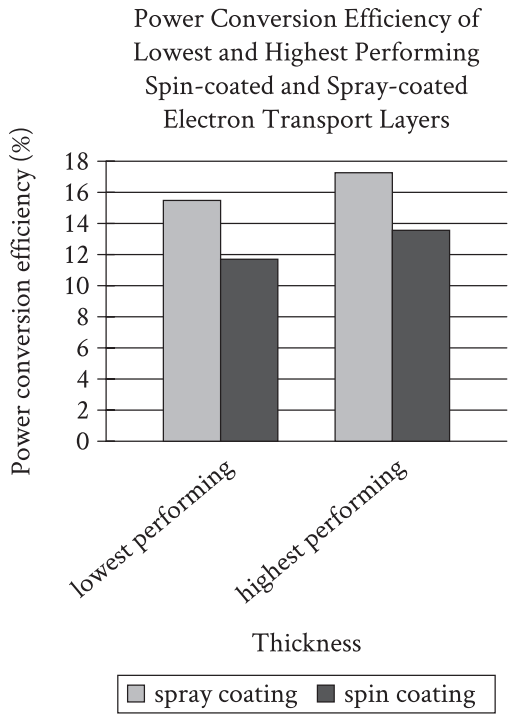
16



Perovskite solar cells convert light into electricity more efficiently than earlier kinds of solar cells, and manufacturing advances have recently made them commercially attractive. One limitation of the cells, however, has to do with their electron transport layer (ETL), through which absorbed electrons must pass. Often the ETL is applied through a process called spin coating, but such ETLs are fairly inefficient at converting input power to output power. André Taylor and colleagues tested a novel spray coating method for applying the ETL. The team produced ETLs of various thicknesses and concluded that spray coating holds promise for improving the power conversion efficiency of ETLs in perovskite solar cells.

Which choice best describes data from the graph that support Taylor and colleagues' conclusion?

A) Both the ETL applied through spin coating and the ETL applied through spray coating showed a power conversion efficiency greater than 10% at their lowest performing thickness.

B) The lowest performing ETL applied through spray coating had a higher power conversion efficiency than the highest performing ETL applied through spin coating.

C) The highest performing ETL applied through spray coating showed a power conversion efficiency of approximately 13%, while the highest performing ETL applied through spin coating showed a power conversion efficiency of approximately 11%.

D) There was a substantial difference in power conversion efficiency between the lowest and highest performing ETLs applied through spray coating.

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17

While attending school in New York City in the 1980s, Okwui Enwezor encountered few works by African artists in exhibitions, despite New York's reputation as one of the best places to view contemporary art from around the world. According to an arts journalist, later in his career as a renowned curator and art historian, Enwezor sought to remedy this deficiency, not by focusing solely on modern African artists, but by showing how their work fits into the larger context of global modern art and art history.

Which finding, if true, would most directly support the journalist's claim?

A) As curator of the Haus der Kunst in Munich, Germany, Enwezor organized a retrospective of Ghanaian sculptor El Anatsui's work entitled *El Anatsui: Triumphant Scale*, one of the largest art exhibitions devoted to a Black artist in Europe's history.

B) In the exhibition *Postwar: Art Between the Pacific and the Atlantic, 1945–1965*, Enwezor and cocurator Katy Siegel brought works by African artists such as Malangatana Ngwenya together with pieces by major figures from other countries, like US artist Andy Warhol and Mexico's David Siqueiros.

C) Enwezor's work as curator of the 2001 exhibition *The Short Century: Independence and Liberation Movements in Africa, 1945–1994* showed how African movements for independence from European colonial powers following the Second World War profoundly influenced work by African artists of the period, such as Kamala Ibrahim Ishaq and Thomas Mukarobgwa.

D) Enwezor organized the exhibition *In/sight: African Photographers, 1940 to the Present* not to emphasize a particular aesthetic trend but to demonstrate the broad range of ways in which African artists have approached the medium of photography.

18

For thousands of years, people in the Americas _____ the bottle gourd, a large bitter fruit with a thick rind, to make bottles, other types of containers, and even musical instruments. Oddly, there is no evidence that any type of bottle gourd is native to the Western Hemisphere; either the fruit or its seeds must have somehow been carried from Asia or Africa.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) to use

B) have used

C) having used

D) using

19

While many video game creators strive to make their graphics ever more _____ others look to the past, developing titles with visuals inspired by the "8-bit" games of the 1980s and 1990s. (The term "8-bit" refers to a console whose processor could only handle eight bits of data at once.)

Which choice completes the text so that it conforms to the conventions of Standard English?

A) lifelike but

B) lifelike

C) lifelike,

D) lifelike, but

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20

In the 1950s, a man named Joseph McVicker was struggling to keep his business afloat when his sister-in-law Kay Zufall advised him to repurpose the company's product, a nontoxic, clay-like substance for removing soot from wallpaper, as a modeling putty for kids. In addition, Zufall _____ selling the product under a child-friendly name: Play-Doh.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) suggested
B) suggests
C) had suggested
D) was suggesting

21

Beatrix Potter is perhaps best known for writing and illustrating children's books such as *The Tale of Peter Rabbit* (1902), but she also dedicated herself to mycology, the study of _____ more than 350 paintings of the fungal species she observed in nature and submitting her research on spore germination to the Linnean Society of London.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) fungi; producing
B) fungi. Producing
C) fungi producing
D) fungi, producing

22

In assessing the films of Japanese director Akira Kurosawa, _____ have missed his equally deep engagement with Japanese artistic traditions such as Noh theater.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) many critics have focused on Kurosawa's use of Western literary sources but
B) Kurosawa's use of Western literary sources has been the focus of many critics, who
C) there are many critics who have focused on Kurosawa's use of Western literary sources, but they
D) the focus of many critics has been on Kurosawa's use of Western literary sources; they

23

Joshua Hinson, director of the language revitalization program of the Chickasaw Nation in Oklahoma, helped produce the world's first Indigenous-language instructional app, Chickasaw _____ Chickasaw TV, in 2010; and a Rosetta Stone language course in Chickasaw, in 2015.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) Basic; in 2009, an online television network;
B) Basic; in 2009, an online television network,
C) Basic, in 2009; an online television network,
D) Basic, in 2009, an online television network,

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24

The forty-seven geothermal springs of Arkansas' Hot Springs National Park are sourced via a process known as natural groundwater recharge, in which rainwater percolates downward through the earth—in this case, the porous rocks of the hills around Hot _____ collect in a subterranean basin.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) Springs to
B) Springs: to
C) Springs—to
D) Springs, to

25

Over twenty years ago, in a landmark experiment in the psychology of choice, professor Sheena Iyengar set up a jam-tasting booth at a grocery store. The number of jams available for tasting _____ some shoppers had twenty-four different options, others only six. Interestingly, the shoppers with fewer jams to choose from purchased more jam.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) varied:
B) varied,
C) varied, while
D) varied while

26

Nigerian author Buchi Emecheta's celebrated literary oeuvre includes *The Joys of Motherhood*, a novel about the changing roles of women in 1950s _____ a television play about the private struggles of a newlywed couple in Nigeria; and *Head Above Water*, her autobiography.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) Lagos, *A Kind of Marriage*,
B) Lagos; *A Kind of Marriage*,
C) Lagos, *A Kind of Marriage*:
D) Lagos; *A Kind of Marriage*

27

Chimamanda Ngozi Adichie's 2013 novel *Americanah* chronicles the divergent experiences of Ifemelu and Obinze, a young Nigerian couple, after high school. Ifemelu moves to the United States to attend a prestigious university. _____ Obinze travels to London, hoping to start a career there. However, frustrated with the lack of opportunities, he soon returns to Nigeria.

Which choice completes the text with the most logical transition?

A) Meanwhile,
B) Nevertheless,
C) Secondly,
D) In fact,

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28

Organisms have evolved a number of surprising adaptations to ensure their survival in adverse conditions. Tadpole shrimp (*Triops longicaudatus*) embryos, _____ can pause development for over ten years during extended periods of drought.

Which choice completes the text with the most logical transition?

A) in contrast,
B) for example,
C) meanwhile,
D) consequently,

29

In 1933, the Twentieth Amendment to the US Constitution was ratified. The amendment mandates that presidential inaugurations be held on January 20, approximately ten weeks after the November election. _____ this amendment requires newly elected US senators and representatives to be sworn into their respective offices on January 3.

Which choice completes the text with the most logical transition?

A) Instead,
B) For instance,
C) Specifically,
D) In addition,

30

In her poetry collection *Thomas and Beulah*, Rita Dove interweaves the titular characters' personal stories with broader historical narratives. She places Thomas's journey from the American South to the Midwest in the early 1900s within the larger context of the Great Migration. _____ Dove sets events from Beulah's personal life against the backdrop of the US Civil Rights Movement.

Which choice completes the text with the most logical transition?

A) Specifically,
B) Thus,
C) Regardless,
D) Similarly,

31

While researching a topic, a student has taken the following notes:

- The Philadelphia and Lancaster Turnpike was a road built between 1792 and 1794.
- It was the first private turnpike in the United States.
- It connected the cities of Philadelphia and Lancaster in the state of Pennsylvania.
- It was sixty-two miles long.

The student wants to emphasize the distance covered by the Philadelphia and Lancaster Turnpike. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The sixty-two-mile-long Philadelphia and Lancaster Turnpike connected the Pennsylvania cities of Philadelphia and Lancaster.
B) The Philadelphia and Lancaster Turnpike was the first private turnpike in the United States.
C) The Philadelphia and Lancaster Turnpike, which connected two Pennsylvania cities, was built between 1792 and 1794.
D) A historic Pennsylvania road, the Philadelphia and Lancaster Turnpike was completed in 1794.

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32

While researching a topic, a student has taken the following notes:

- Most, but not all, of the Moon's oxygen comes from the Sun, via solar wind.
- Cosmochemist Kentaro Terada from Osaka University wondered if some of the unaccounted-for oxygen could be coming from Earth.
- In 2008, he analyzed data from the Japanese satellite Kaguya.
- Kaguya gathered data about gases and particles it encountered while orbiting the Moon.
- Based on the Kaguya data, Terada confirmed his suspicion that Earth is sending oxygen to the Moon.

The student wants to emphasize the aim of the research study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) As it orbited the Moon, the Kaguya satellite collected data that was later analyzed by cosmochemist Kentaro Terada.

B) Before 2008, Kentaro Terada wondered if the Moon was receiving some of its oxygen from Earth.

C) Cosmochemist Kentaro Terada set out to determine whether some of the Moon's oxygen was coming from Earth.

D) Kentaro Terada's study determined that Earth is sending a small amount of oxygen to the Moon.

33

While researching a topic, a student has taken the following notes:

- Ducklings expend up to 62.8% less energy when swimming in a line behind their mother than when swimming alone.
- The physics behind this energy savings hasn't always been well understood.
- Naval architect Zhiming Yuan used computer simulations to study the effect of the mother duck's wake.
- The study revealed that ducklings are pushed in a forward direction by the wake's waves.
- Yuan determined this push reduces the effect of wave drag on the ducklings by 158%.

The student wants to present the study and its methodology. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) A study revealed that ducklings, which expend up to 62.8% less energy when swimming in a line behind their mother, also experience 158% less drag.

B) Seeking to understand how ducklings swimming in a line behind their mother save energy, Zhiming Yuan used computer simulations to study the effect of the mother duck's wake.

C) Zhiming Yuan studied the physics behind the fact that by being pushed in a forward direction by waves, ducklings save energy.

D) Naval architect Zhiming Yuan discovered that ducklings are pushed in a forward direction by the waves of their mother's wake, reducing the effect of drag by 158%.

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

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Math

27 QUESTIONS

DIRECTIONS

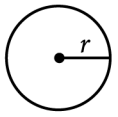
The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

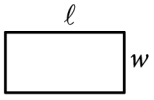
Unless otherwise indicated:

- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

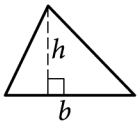
REFERENCE



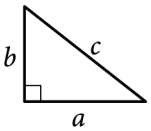
$A = \pi r^2$
$C = 2\pi r$



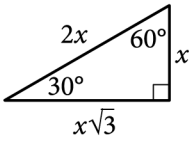
$A = \ell w$

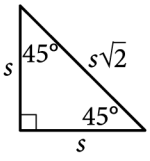


$A = \frac{1}{2}bh$

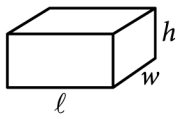


$c^2 = a^2 + b^2$

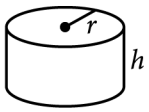




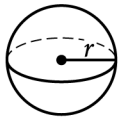
Special Right Triangles



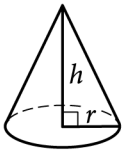
$V = \ell wh$



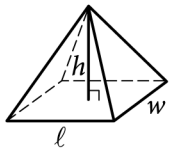
$V = \pi r^2 h$



$V = \frac{4}{3}\pi r^3$



$V = \frac{1}{3}\pi r^2 h$



$V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π.

The sum of the measures in degrees of the angles of a triangle is 180.

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CONTINUE

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For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find **more than one correct answer**, write and circle only one answer.
- Your answer can be up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer, but no more.
- If your answer is a **fraction** that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a **decimal** that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

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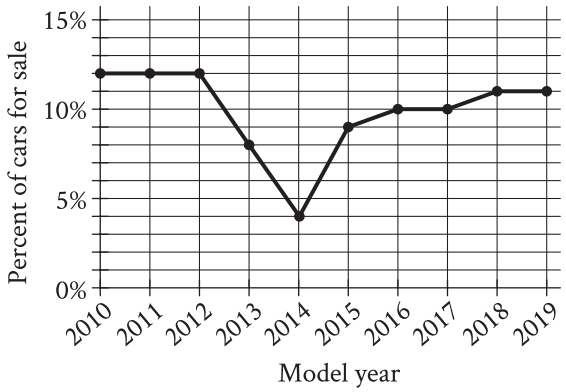
1

The line graph shows the percent of cars for sale at a used car lot on a given day by model year.



For what model year is the percent of cars for sale the smallest?

A) 2012

B) 2013

C) 2014

D) 2015

2

For a particular machine that produces beads, 29 out of every 100 beads it produces have a defect. A bead produced by the machine will be selected at random. What is the probability of selecting a bead that has a defect?

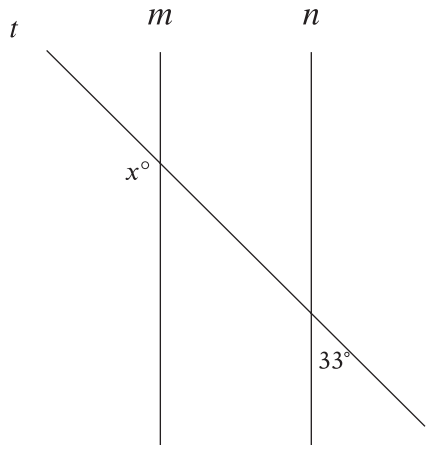
A) $\frac{1}{2{,}900}$

B) $\frac{1}{29}$

C) $\frac{29}{100}$

D) $\frac{29}{10}$

3



Note: Figure not drawn to scale.

In the figure, line m is parallel to line n, and line t intersects both lines. What is the value of x ?

A) 33

B) 57

C) 123

D) 147

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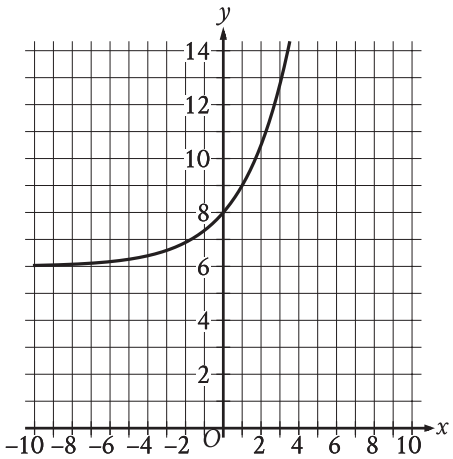
4



What is the y-intercept of the graph shown?

A) $(-8, 0)$

B) $(-6, 0)$

C) $(0, 6)$

D) $(0, 8)$

5

The total cost $f(x)$, in dollars, to lease a car for 36 months from a particular car dealership is given by $f(x) = 36x + 1{,}000$, where x is the monthly payment, in dollars. What is the total cost to lease a car when the monthly payment is $400?

A) $13,400

B) $13,000

C) $15,400

D) $37,400

6

Each side of a square has a length of 45. What is the perimeter of this square?

7

$$\frac{55}{x + 6} = x$$

What is the positive solution to the given equation?

8

An object travels at a constant speed of 12 centimeters per second. At this speed, what is the time, in seconds, that it would take for the object to travel 108 centimeters?

A) 9

B) 96

C) 120

D) 972

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9

Data set X: 5, 9, 9, 13
Data set Y: 5, 9, 9, 13, 27

The lists give the values in data sets X and Y. Which statement correctly compares the mean of data set X and the mean of data set Y?

A) The mean of data set X is greater than the mean of data set Y.
B) The mean of data set X is less than the mean of data set Y.
C) The means of data set X and data set Y are equal.
D) There is not enough information to compare the means.

10

A rocket contained 467,000 kilograms (kg) of propellant before launch. Exactly 21 seconds after launch, 362,105 kg of this propellant remained. On average, approximately how much propellant, in kg, did the rocket burn each second after launch?

A) 4,995
B) 17,243
C) 39,481
D) 104,895

11

If $4x + 2 = 12$, what is the value of $16x + 8$?

A) 40
B) 48
C) 56
D) 60

12

An object is kicked from a platform. The equation $h = -4.9t^2 + 7t + 9$ represents this situation, where h is the height of the object above the ground, in meters, t seconds after it is kicked. Which number represents the height, in meters, from which the object was kicked?

A) 0
B) 4.9
C) 7
D) 9

13

$$f(x) = 4x^2 - 50x + 126$$

The given equation defines the function f. For what value of x does $f(x)$ reach its minimum?

14

A small business owner budgets $2,200 to purchase candles. The owner must purchase a minimum of 200 candles to maintain the discounted pricing. If the owner pays $4.90 per candle to purchase small candles and $11.60 per candle to purchase large candles, what is the maximum number of large candles the owner can purchase to stay within the budget and maintain the discounted pricing?

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15

In the linear function f, $f(0) = 8$ and $f(1) = 12$. Which equation defines f ?

A) $f(x) = 12x + 8$

B) $f(x) = 4x$

C) $f(x) = 4x + 12$

D) $f(x) = 4x + 8$

16

The function $f(w) = 6w^2$ gives the area of a rectangle, in square feet (ft^2), if its width is w ft and its length is 6 times its width. Which of the following is the best interpretation of $f(14) = 1{,}176$?

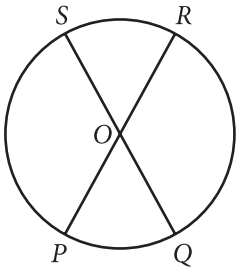
A) If the width of the rectangle is 14 ft, then the area of the rectangle is 1,176 ft^2.

B) If the width of the rectangle is 14 ft, then the length of the rectangle is 1,176 ft.

C) If the width of the rectangle is 1,176 ft, then the length of the rectangle is 14 ft.

D) If the width of the rectangle is 1,176 ft, then the area of the rectangle is 14 ft^2.

17



Note: Figure not drawn to scale.

The circle shown has center O, circumference 144π, and diameters $\overline{PR}$ and $\overline{QS}$. The length of arc PS is twice the length of arc PQ. What is the length of arc QR ?

A) 24π

B) 48π

C) 72π

D) 96π

18

A company that provides whale-watching tours takes groups of 21 people at a time. The company's revenue is 80 dollars per adult and 60 dollars per child. If the company's revenue for one group consisting of adults and children was 1,440 dollars, how many people in the group were children?

A) 3

B) 9

C) 12

D) 18

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19

The function h is defined by $h(x) = 4x + 28$. The graph of $y = h(x)$ in the xy-plane has an x-intercept at $(a, 0)$ and a y-intercept at $(0, b)$, where a and b are constants. What is the value of $a + b$?

A) 21
B) 28
C) 32
D) 35

20

One of the factors of $2x^3 + 42x^2 + 208x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b ?

21

$$y = -1.5$$
$$y = x^2 + 8x + a$$

In the given system of equations, a is a positive constant. The system has exactly one distinct real solution. What is the value of a ?

22

$$f(x) = (x + 6)(x + 5)(x - 4)$$

The function f is given. Which table of values represents $y = f(x) - 3$?

A)

x	y
−6	−9
−5	−8
4	1

B)

x	y
−6	−3
−5	−3
4	−3

C)

x	y
−6	−3
−5	−2
4	7

D)

x	y
−6	3
−5	3
4	3

23

For the function q, the value of $q(x)$ decreases by 45% for every increase in the value of x by 1. If $q(0) = 14$, which equation defines q ?

A) $q(x) = 0.55(14)^x$
B) $q(x) = 1.45(14)^x$
C) $q(x) = 14(0.55)^x$
D) $q(x) = 14(1.45)^x$

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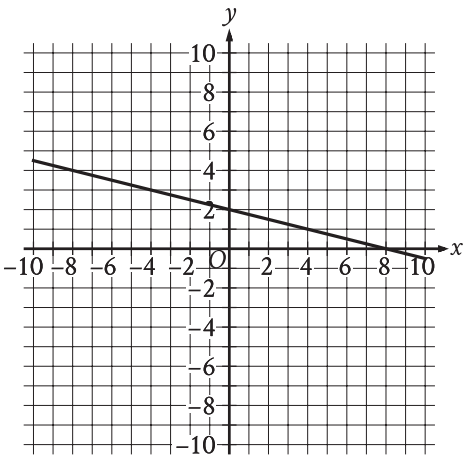
24



The graph of $y = f(x) + 14$ is shown. Which equation defines function f?

A) $f(x) = -\frac{1}{4}x - 12$

B) $f(x) = -\frac{1}{4}x + 16$

C) $f(x) = -\frac{1}{4}x + 2$

D) $f(x) = -\frac{1}{4}x - 14$

25

$RS = 20$
$ST = 48$
$TR = 52$

The side lengths of right triangle RST are given. Triangle RST is similar to triangle UVW, where S corresponds to V and T corresponds to W. What is the value of $\tan W$?

A) $\frac{5}{13}$

B) $\frac{5}{12}$

C) $\frac{12}{13}$

D) $\frac{12}{5}$

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26

One gallon of paint will cover 220 square feet of a surface. A room has a total wall area of w square feet. Which equation represents the total amount of paint P, in gallons, needed to paint the walls of the room twice?

A) $P = \frac{w}{110}$

B) $P = 440w$

C) $P = \frac{w}{220}$

D) $P = 220w$

27

The number a is 110% greater than the number b. The number b is 90% less than 47. What is the value of a ?

STOP

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Do not turn to any other module in the test.

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Math

27 QUESTIONS

DIRECTIONS

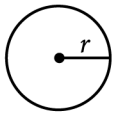
The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

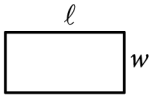
Unless otherwise indicated:

- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

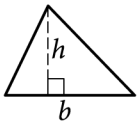
REFERENCE



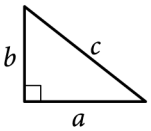
$A = \pi r^2$
$C = 2\pi r$



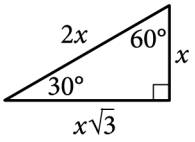
$A = \ell w$

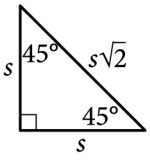


$A = \frac{1}{2}bh$

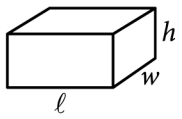


$c^2 = a^2 + b^2$

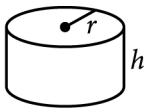




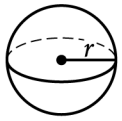
Special Right Triangles



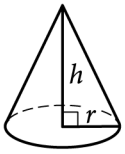
$V = \ell wh$



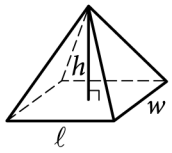
$V = \pi r^2 h$



$V = \frac{4}{3}\pi r^3$



$V = \frac{1}{3}\pi r^2 h$



$V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π.

The sum of the measures in degrees of the angles of a triangle is 180.

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For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find **more than one correct answer**, write and circle only one answer.
- Your answer can be up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer, but no more.
- If your answer is a **fraction** that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a **decimal** that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

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1

There are 55 students in Spanish club. A sample of the Spanish club students was selected at random and asked whether they intend to enroll in a new study program. Of those surveyed, 20% responded that they intend to enroll in the study program. Based on this survey, which of the following is the best estimate of the total number of Spanish club students who intend to enroll in the study program?

A) 11

B) 20

C) 44

D) 55

2

Jay walks at a speed of 3 miles per hour and runs at a speed of 5 miles per hour. He walks for w hours and runs for r hours for a combined total of 14 miles. Which equation represents this situation?

A) $3w + 5r = 14$

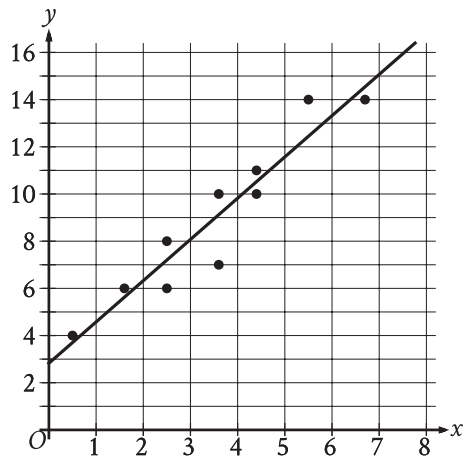
B) $\frac{1}{3}w + \frac{1}{5}r = 14$

C) $\frac{1}{3}w + \frac{1}{5}r = 112$

D) $3w + 5r = 112$

3

The scatterplot shows the relationship between two variables, x and y. A line of best fit is also shown.



Which of the following equations best represents the line of best fit shown?

A) $y = 2.8 + 1.7x$

B) $y = 2.8 - 1.7x$

C) $y = -2.8 + 1.7x$

D) $y = -2.8 - 1.7x$

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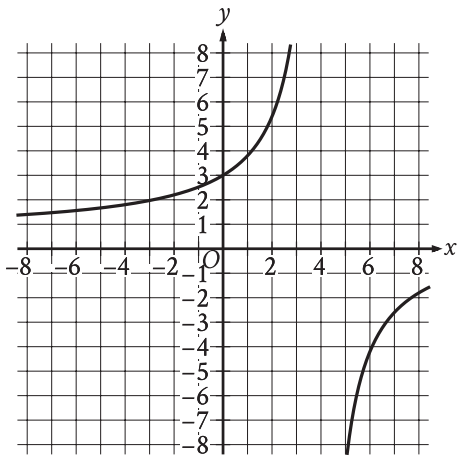
4



The graph of $y = f(x)$ is shown in the xy-plane. What is the value of $f(0)$?

A) -3

B) 0

C) $\frac{3}{5}$

D) 3

5

Which expression is equivalent to $(m^4q^4z^{-1})(mq^5z^3)$, where m, q, and z are positive?

A) $m^4q^{20}z^{-3}$

B) $m^5q^9z^2$

C) $m^6q^8z^{-1}$

D) $m^{20}q^{12}z^{-2}$

6

73, 74, 75, 77, 79, 82, 84, 85, 91

What is the median of the data shown?

7

$$x + 40 = 95$$

What value of x is the solution to the given equation?

8

$$5x = 15$$
$$-4x + y = -2$$

The solution to the given system of equations is (x, y). What is the value of $x + y$?

A) -17

B) -13

C) 13

D) 17

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9

$$g(m) = -0.05m + 12.1$$

The given function g models the number of gallons of gasoline that remains from a full gas tank in a car after driving m miles. According to the model, about how many gallons of gasoline are used to drive each mile?

A) 0.05

B) 12.1

C) 20

D) 242.0

10

$$\frac{1}{7b} = \frac{11x}{y}$$

The given equation relates the positive numbers b, x, and y. Which equation correctly expresses x in terms of b and y ?

A) $x = \frac{7by}{11}$

B) $x = y - 77b$

C) $x = \frac{y}{77b}$

D) $x = 77by$

11

$$y = 76$$

$$y = x^2 - 5$$

The graphs of the given equations in the xy-plane intersect at the point (x, y). What is a possible value of x ?

A) $-\frac{76}{5}$

B) -9

C) 5

D) 76

12

$$y > 14$$

$$4x + y < 18$$

The point $(x, 53)$ is a solution to the system of inequalities in the xy-plane. Which of the following could be the value of x ?

A) -9

B) -5

C) 5

D) 9

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13

Out of 300 seeds that were planted, 80% sprouted. How many of these seeds sprouted?

14

The function f is defined by $f(x) = 4x$. For what value of x does $f(x) = 8$?

15

Which expression is equivalent to $\frac{8x(x-7)-3(x-7)}{2x-14}$, where $x > 7$?

A) $\frac{x-7}{5}$

B) $\frac{8x-3}{2}$

C) $\frac{8x^2-3x-14}{2x-14}$

D) $\frac{8x^2-3x-77}{2x-14}$

16

Line p is defined by $2y + 18x = 9$. Line r is perpendicular to line p in the xy-plane. What is the slope of line r ?

A) -9

B) $-\frac{1}{9}$

C) $\frac{1}{9}$

D) 9

17

$$f(t) = 8{,}000(0.65)^t$$

The given function f models the number of coupons a company sent to their customers at the end of each year, where t represents the number of years since the end of 1998, and $0 \le t \le 5$. If $y = f(t)$ is graphed in the ty-plane, which of the following is the best interpretation of the y-intercept of the graph in this context?

A) The minimum estimated number of coupons the company sent to their customers during the 5 years was 1,428.

B) The minimum estimated number of coupons the company sent to their customers during the 5 years was 8,000.

C) The estimated number of coupons the company sent to their customers at the end of 1998 was 1,428.

D) The estimated number of coupons the company sent to their customers at the end of 1998 was 8,000.

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18

Triangle XYZ is similar to triangle RST such that X, Y, and Z correspond to R, S, and T, respectively. The measure of $\angle Z$ is 20° and $2XY = RS$. What is the measure of $\angle T$?

A) 2°

B) 10°

C) 20°

D) 40°

19

$$y = 6x + 18$$

One of the equations in a system of two linear equations is given. The system has no solution. Which equation could be the second equation in the system?

A) $-6x + y = 18$

B) $-6x + y = 22$

C) $-12x + y = 36$

D) $-12x + y = 18$

20

What is the area, in square centimeters, of a rectangle with a length of 34 centimeters (cm) and a width of 29 cm?

21

$$y = 4x + 1$$
$$4y = 15x - 8$$

The solution to the given system of equations is (x, y). What is the value of $x - y$?

22

$$5x^2 + 10x + 16 = 0$$

How many distinct real solutions does the given equation have?

A) Exactly one

B) Exactly two

C) Infinitely many

D) Zero

23

A certain park has an area of 11,863,808 square yards. What is the area, in square miles, of this park? (1 mile = 1,760 yards)

A) 1.96

B) 3.83

C) 3,444.39

D) 6,740.8

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24

Which of the following equations represents a circle in the xy-plane that intersects the y-axis at exactly one point?

A) $(x-8)^2+(y-8)^2=16$

B) $(x-8)^2+(y-4)^2=16$

C) $(x-4)^2+(y-9)^2=16$

D) $x^2+(y-9)^2=16$

25

In triangles ABC and DEF, angles B and E each have measure 27° and angles C and F each have measure 41°. Which additional piece of information is sufficient to determine whether triangle ABC is congruent to triangle DEF ?

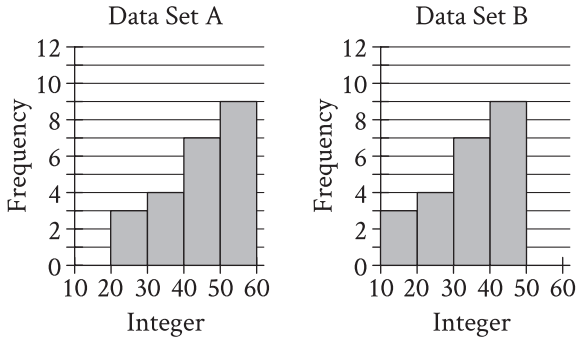
A) The measure of angle A

B) The length of side AB

C) The lengths of sides BC and EF

D) No additional information is necessary.

26



Two data sets of 23 integers each are summarized in the histograms shown. For each of the histograms, the first interval represents the frequency of integers greater than or equal to 10, but less than 20. The second interval represents the frequency of integers greater than or equal to 20, but less than 30, and so on. What is the smallest possible difference between the mean of data set A and the mean of data set B?

A) 0

B) 1

C) 10

D) 23

27

A right triangle has legs with lengths of 24 centimeters and 21 centimeters. If the length of this triangle's hypotenuse, in centimeters, can be written in the form $3\sqrt{d}$, where d is an integer, what is the value of d ?

STOP

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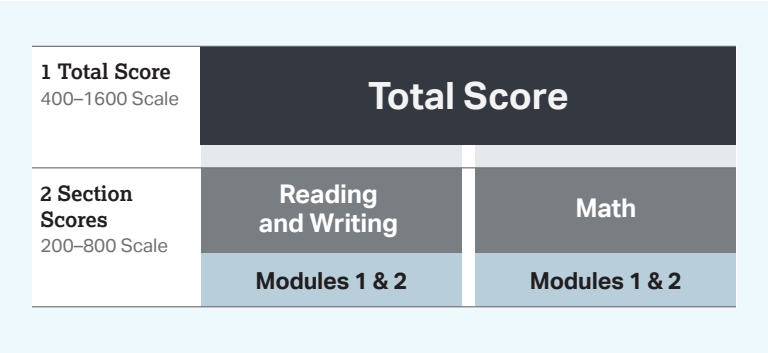
Scoring Your Paper SAT Practice Test #2

IMPORTANT: *This guide is for students who completed a paper version of the digital SAT practice test. The scoring method used for this nondigital version isn't exactly the same as the digital version.*



Congratulations on completing an SAT® practice test. To score your test, follow the instructions in this guide.

Scores Overview

Each assessment in the SAT Suite (SAT, PSAT/NMSQT®, PSAT™ 10, and PSAT™ 8/9) reports test scores on a common scale.

For more details about scores, visit **sat.org/scores**.

The College Board Assessment Design & Development team developed the practice tests using the same processes and review standards they used when developing the actual SAT.

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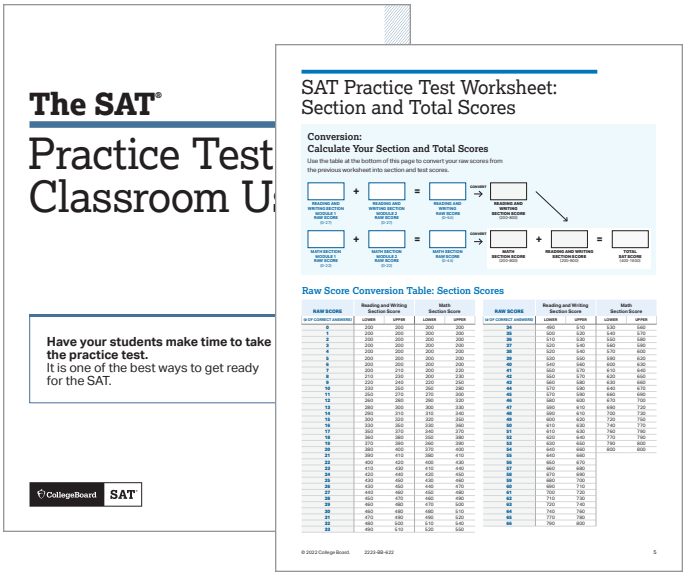
How to Calculate Your Practice Test Scores

The worksheets on pages 4 and 5 help you calculate your test scores.

GET SET UP

1. In addition to your practice test, you'll need the conversion tables and answer key at the end of this guide.


The SAT®

Practice Test
Classroom U

Have your students make time to take the practice test.
It is one of the best ways to get ready for the SAT.

CollegeBoard SAT

SAT Practice Test Worksheet:
Section and Total Scores

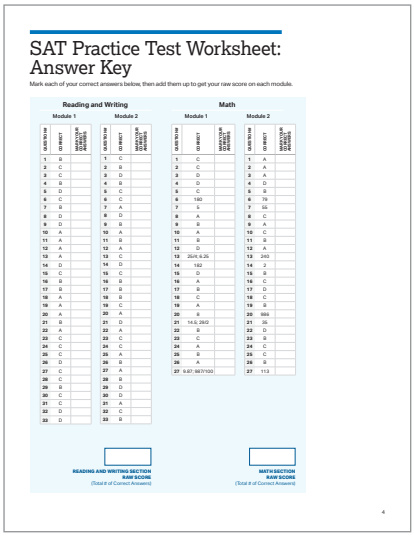
Conversion:
Calculate Your Section and Total Scores

Use the table at the bottom of this page to convert your raw scores from the previous worksheet into section and test scores.

Raw Score Conversion Table: Section Scores

SCORE YOUR PRACTICE TEST

2. Compare your answers to the answer key on page 4, and count your total correct answers for each section. Write the number of correct answers for each section in the answer key at the bottom of that section.


SAT Practice Test Worksheet:
Answer Key

Mark each of your correct answers below, then add them up to get your raw score on each module.

Reading and Writing

Module 1

QUESTION NO.	CORRECT ANSWER	MARK FOR REVIEW (IF ANSWERED)
1	B	
2	C	
3	C	
4	B	
5	D	
6	B	
7	B	
8	D	
9	D	
10	A	
11	A	
12	A	
13	A	
14	D	
15	C	
16	B	
17	B	
18	A	
19	A	
20	A	
21	B	
22	A	
23	C	
24	C	
25	C	
26	D	
27	C	
28	C	
29	B	
30	C	
31	C	
32	D	
33	D	

Module 2

QUESTION NO.	CORRECT ANSWER	MARK FOR REVIEW (IF ANSWERED)
1	C	
2	B	
3	D	
4	B	
5	C	
6	C	
7	A	
8	D	
9	B	
10	A	
11	B	
12	A	
13	C	
14	D	
15	C	
16	B	
17	B	
18	B	
19	C	
20	A	
21	D	
22	A	
23	C	
24	C	
25	A	
26	B	
27	A	
28	B	
29	D	
30	D	
31	A	
32	C	
33	B	

READING AND WRITING SECTION RAW SCORE (Total # of Correct Answers)

Math

Module 1

QUESTION NO.	CORRECT ANSWER	MARK FOR REVIEW (IF ANSWERED)
1	C	
2	C	
3	D	
4	D	
5	C	
6	162	
7	5	
8	A	
9	B	
10	A	
11	B	
12	D	
13	25/4, 6.25	
14	182	
15	D	
16	A	
17	B	
18	C	
19	A	
20	E	
21	14.8, 29/2	
22	B	
23	C	
24	A	
25	B	
26	A	
27	6.87, 687/100	

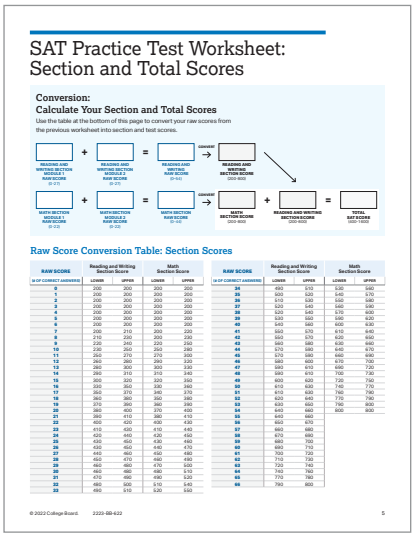
Module 2

QUESTION NO.	CORRECT ANSWER	MARK FOR REVIEW (IF ANSWERED)
1	A	
2	A	
3	A	
4	D	
5	B	
6	79	
7	55	
8	C	
9	A	
10	C	
11	A	
12	A	
13	240	
14	2	
15	B	
16	C	
17	D	
18	C	
19	B	
20	666	
21	35	
22	D	
23	B	
24	C	
25	C	
26	B	
27	113	

MATH SECTION RAW SCORE (Total # of Correct Answers)

CALCULATE YOUR SCORES

3. Using your marked-up answer key and the conversion tables, follow the directions on page 5 to get your section and test scores.


SAT Practice Test Worksheet:
Section and Total Scores

Conversion:
Calculate Your Section and Total Scores

Use the table at the bottom of this page to convert your raw scores from the previous worksheet into section and test scores.

Raw Score Conversion Table: Section Scores

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Get Section and Total Scores

Your total score on an SAT practice test is the sum of your scores in the modules for both the Reading and Writing and Math sections. To get your total score, you'll convert what we call the "raw score" for each section—the number of questions you got right in that section—into the "scaled score" for that section, and then calculate the total score.

GET YOUR READING AND WRITING SECTION SCORE

Calculate your SAT Reading and Writing section score (it's on a scale of 200–800).

1. Use the answer key on page 4 to count the number of correct answers you got on module 1 and module 2.
2. To determine your Reading and Writing raw score, add the number of correct answers you got on module 1 and module 2.
3. Use the Raw Score Conversion Table: Section Scores on page 5 to turn your raw score into your Reading and Writing section score.

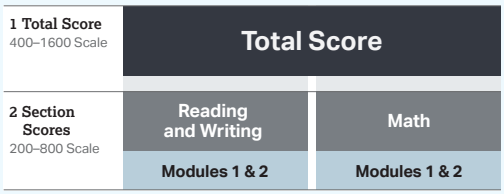
GET YOUR MATH SECTION SCORE

Calculate your SAT Math section score (it's on a scale of 200–800).

1. Use the answer key on page 4 to count the number of correct answers you got on module 1 and module 2.
2. To determine your Math raw score, add the number of correct answers you got on module 1 and module 2.
3. Use the Raw Score Conversion Table: Section Scores on page 5 to turn your raw score into your Math section score.

GET YOUR TOTAL SCORE

Add your Reading and Writing section score to your Math section score. The result is your total score on the SAT practice test, on a scale of 400–1600.



Your total score on the SAT practice test is the sum of your Reading and Writing section score and your Math section score.

Use the worksheets on pages 4 and 5 to calculate your section and total scores.

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SAT Practice Test Worksheet: Answer Key

Mark each of your correct answers below, then add them up to get your raw score on each module.

Reading and Writing

Module 1

QUESTION #	CORRECT	MARK YOUR CORRECT ANSWERS
1	B	
2	C	
3	C	
4	B	
5	D	
6	C	
7	B	
8	D	
9	D	
10	A	
11	A	
12	A	
13	A	
14	D	
15	C	
16	B	
17	B	
18	A	
19	A	
20	A	
21	B	
22	A	
23	C	
24	C	
25	C	
26	D	
27	C	
28	C	
29	B	
30	C	
31	C	
32	D	
33	D	

Module 2

QUESTION #	CORRECT	MARK YOUR CORRECT ANSWERS
1	C	
2	B	
3	D	
4	B	
5	C	
6	C	
7	A	
8	D	
9	B	
10	A	
11	B	
12	A	
13	C	
14	D	
15	C	
16	B	
17	B	
18	B	
19	C	
20	A	
21	D	
22	A	
23	C	
24	C	
25	A	
26	B	
27	A	
28	B	
29	D	
30	D	
31	A	
32	C	
33	B	

READING AND WRITING SECTION RAW SCORE
(Total # of Correct Answers)

Math

Module 1

QUESTION #	CORRECT	MARK YOUR CORRECT ANSWERS
1	C	
2	C	
3	D	
4	D	
5	C	
6	180	
7	5	
8	A	
9	B	
10	A	
11	B	
12	D	
13	25/4; 6.25	
14	182	
15	D	
16	A	
17	B	
18	C	
19	A	
20	8	
21	14.5; 29/2	
22	B	
23	C	
24	A	
25	B	
26	A	
27	9.87; 987/100	

Module 2

QUESTION #	CORRECT	MARK YOUR CORRECT ANSWERS
1	A	
2	A	
3	A	
4	D	
5	B	
6	79	
7	55	
8	C	
9	A	
10	C	
11	B	
12	A	
13	240	
14	2	
15	B	
16	C	
17	D	
18	C	
19	B	
20	986	
21	35	
22	D	
23	B	
24	C	
25	C	
26	B	
27	113	

MATH SECTION RAW SCORE
(Total # of Correct Answers)

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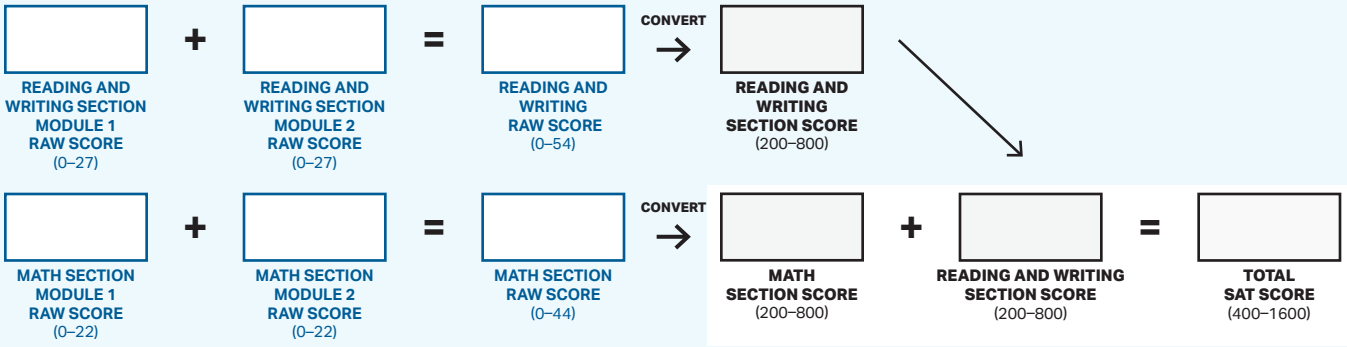
SAT Practice Test Worksheet: Section and Total Scores

Conversion: Calculate Your Section and Total Scores

Use the table at the bottom of this page to convert your raw scores from the previous worksheet into section and test scores.



Raw Score Conversion Table: Section Scores

RAW SCORE	Reading and Writing Section Score		Math Section Score	
(# OF CORRECT ANSWERS)	LOWER	UPPER	LOWER	UPPER
0	200	200	200	200
1	200	200	200	200
2	200	200	200	200
3	200	200	200	200
4	200	200	200	200
5	200	200	200	200
6	200	200	200	200
7	200	210	200	220
8	210	230	200	230
9	220	240	220	250
10	230	250	250	280
11	250	270	270	300
12	260	280	290	320
13	280	300	300	330
14	290	310	310	340
15	300	320	320	350
16	330	350	330	360
17	350	370	340	370
18	360	380	350	380
19	370	390	360	390
20	380	400	370	400
21	390	410	380	410
22	400	420	400	430
23	410	430	410	440
24	420	440	420	450
25	430	450	430	460
26	430	450	440	470
27	440	460	450	480
28	450	470	460	490
29	460	480	470	500
30	460	480	480	510
31	470	490	490	520
32	480	500	510	540
33	490	510	520	550

RAW SCORE	Reading and Writing Section Score		Math Section Score	
(# OF CORRECT ANSWERS)	LOWER	UPPER	LOWER	UPPER
34	490	510	530	560
35	500	520	540	570
36	510	530	550	580
37	520	540	560	590
38	520	540	570	600
39	530	550	590	620
40	540	560	600	630
41	550	570	610	640
42	550	570	620	650
43	560	580	630	660
44	570	590	640	670
45	570	590	660	690
46	580	600	670	700
47	590	610	690	720
48	590	610	700	730
49	600	620	720	750
50	610	630	740	770
51	610	630	760	790
52	620	640	770	790
53	630	650	790	800
54	640	660	800	800
55	640	660		
56	650	670		
57	660	680		
58	670	690		
59	680	700		
60	690	710		
61	700	720		
62	710	730		
63	720	740		
64	740	760		
65	770	780		
66	790	800		

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The SAT®

Practice Test #2

ANSWER EXPLANATIONS

These answer explanations are for students taking the digital SAT in nondigital format.

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Reading and Writing

Module 1
(33 questions)

QUESTION 1

Choice B is the best answer because it most logically completes the text's discussion of Juarez. In this context, "important" means marked by significant work or consequence. The text indicates that Juarez, who was the first president of Mexico from an Indigenous community, became a certain kind of figure in Mexico's history. It then supports that claim by describing some of the "many significant accomplishments" from Juarez's long tenure in office. This context conveys that Juarez is a significant and consequential figure in Mexico's history.

Choice A is incorrect because the text focuses on Juarez's role as the first president of Mexico from an Indigenous community and on his many major accomplishments during his lengthy time in office; nothing in the text suggests that Juarez was "unpredictable," or tended to behave in ways that couldn't be predicted. *Choice C* is incorrect because nothing in the text suggests that Juarez was a particularly "secretive" figure, or that he tended to keep things private or hidden from others. Instead, the text focuses on things that are known about Juarez: that he was the first president of Mexico from an Indigenous community, that he had a lengthy tenure, and that his many major accomplishments included consolidating the national government's authority and advancing Indigenous rights. *Choice D* is incorrect because the text focuses on the idea that Juarez, who was the first president of Mexico from an Indigenous community, had many major accomplishments during his lengthy time in office. Rather than suggesting that Juarez was an "ordinary," or common and typical, figure in Mexico's history, this context conveys that Juarez was instead a notable figure.

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QUESTION 2

Choice C is the best answer because it most logically completes the text's discussion of John Ashbery's poems. As used in this context, "interpret" would mean decipher the meaning of. The text indicates that Ashbery's poems have many unusual features, that it's difficult to tell what exactly the poems' subject matter is, and that scholars strongly disagree about the poems. This context conveys the idea that it's difficult to interpret Ashbery's poems.

Choice A is incorrect because "delegate" means to assign someone as a representative of another person or to entrust something to someone else, neither of which would make sense in context. The text is focused only on the difficulty that readers have interpreting Ashbery's poems due to their many unusual features; it doesn't suggest anything about the poems being difficult to delegate. *Choice B* is incorrect because describing Ashbery's poems as difficult to "compose," or put together or produce, would make sense only if the text were about Ashbery's experience of writing the poems. It could be true that it was difficult for Ashbery to compose his poems, but the text doesn't address this; it instead discusses how readers interpret and engage with the poems. *Choice D* is incorrect because describing Ashbery's poems as being difficult to "renounce," or give up or refuse, wouldn't make sense in context. The text focuses on the idea that features of Ashbery's poems are odd or unclear and have caused heated scholarly debate. This context suggests that the poems are difficult to interpret, not that the poems are difficult to renounce.

QUESTION 3

Choice C is the best answer because it most logically and precisely completes the text's discussion of the fossil record from the Cambrian period. In this context, "abrupt" means sudden. The text explains that the fossil record reflects the unexpected appearance and rapid diversification, or increase in variety, of animal remains during the Cambrian period. This context establishes that these remains' entry into the fossil record was sudden.

Choice A is incorrect. Although the word "explosion" appears in the name of the event marked by the fossil record change, the text never suggests that the change was "catastrophic," or disastrous. In context, "explosion" refers to the rapid diversification, or the swift increase in variety, of animal remains in the fossil record—a phenomenon that the text presents in a relatively neutral manner, without commenting on whether it was negative or positive. *Choice B* is incorrect because the text never suggests that the change toward greater diversification is "elusive," or difficult to locate, in the fossil record. Rather, the text notes that the change occurred about 541 million years ago, suggesting that scientists have indeed been able to locate it. *Choice D* is incorrect because it wouldn't make sense in context to describe the change in the fossil record as "imminent," or about to occur, since the text indicates that the change already occurred millions of years ago.

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QUESTION 4

Choice B is the best answer because it most logically completes the text's discussion of the significance of the 2014 archaeological finding at El Algar. In this context, "concede" means to admit something is true after first resisting that admission. The text indicates that some researchers believe "Bronze Age societies were ruled by men." But the Bronze Age burial of a woman at El Algar included "valuable objects signaling a high position of power," which would raise the possibility that "women may have also held leadership roles." Thus, the text is calling into question the notion that only men were leaders in these societies and speculating that people holding this view may reconsider their opinion.

Choice A is incorrect because "waive" means to refrain from insisting that something, such as a right or a requirement, be observed; the word isn't used, however, in contexts where someone acknowledges that an opinion they hold may be invalid, as is the case in the text. *Choice C* is incorrect. According to the text, the finding from the El Algar burial site undermines the view that Bronze Age societies were exclusively ruled by men. However, "refute" means to demonstrate that something is false and would not make sense in context. Lull and team's finding supports the view that women may have also held leadership roles, not that they did not participate in such roles. *Choice D* is incorrect because in this context, "require" means to demand or specify as mandatory. However, it would not make sense for contemporary researchers to demand that Bronze Age "women may have also held leadership roles."

QUESTION 5

Choice D is the best answer because it most logically completes the text's discussion of baleen whale accessory spleens. In this context, "latent" means dormant or functionless. The text sets up a contrast between the idea that baleen whale accessory spleens appear not to have a function and the research indicating that the accessory spleen may actually have a role in supporting the whales' diving mechanisms. This context therefore conveys the idea that the assumption that baleen whale accessory spleens are latent may be incorrect.

Choice A is incorrect because it wouldn't make sense to say that the role of the accessory spleen is "replicable," or capable of being reproduced. The text indicates that the role of the accessory spleen seems to have no function, but some researchers think it does have a role; the text doesn't address whether the role of the accessory spleen could or couldn't be reproduced. *Choice B* is incorrect because suggesting that the role of the accessory spleen is "predetermined," or decided in advance, wouldn't make sense in context. Although the researchers may agree that the role of the accessory spleen or any other organ hasn't been determined in advance, the text focuses on the idea that the accessory spleen was thought to have been functionless but may in fact serve an active role for baleen whales. *Choice C* is incorrect because it's the opposite of what the context of the text is conveying. The second sentence of the text indicates that baleen whale accessory spleens may not be useless, not that they aren't "operative," or functional.

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QUESTION 6

Choice C is the best answer because it most logically completes the text's discussion of the factors that influence peoples' decisions to move to a different state. As used in this context, "overshadowed by" means to be surpassed by or caused to seem less important than other factors affecting a move. The text indicates that, according to a US tax policy expert, when people think about an interstate move, state taxes have little effect on their decisions, while employment opportunities, housing availability, and climate have a very strong effect. This context suggests that people consider these other factors to be more important than state taxes.

Choice A is incorrect because the text indicates that state taxes aren't as important a consideration as other factors when people are thinking of moving to another state. The context doesn't suggest that state taxes are "consistent with," or in agreement with these other factors. *Choice B* is incorrect because it wouldn't make sense in context to say that state taxes are "representative of," or typical of, other factors. Taxes aren't an example of employment opportunities, housing availability, and climate, which are the other factors listed in the text. *Choice D* is incorrect because it wouldn't make sense in context to say that state taxes are "irrelevant to," or unconnected or unimportant to other factors. State taxes are irrelevant to peoples' decisions, not to other factors. In other words, although the text suggests that state taxes may be irrelevant to people considering a move to another state, the other factors mentioned in the text, such as employment opportunities, are unable to have an opinion about state taxes. Furthermore, the text indicates that significant differences in state taxes have almost no effect on peoples' choices to move, but they aren't completely unimportant.

QUESTION 7

Choice B is the best answer because it most logically completes the text's discussion of the author's claim about the relationship between Neanderthals and Homo sapiens. As used in this context, "tenuous" means lacking substance or strength. The text states that the author's claim isn't convincing because it doesn't consider certain pieces of evidence—relevant recent discoveries. The context conveys the idea that the author's claim is weak.

Choice A is incorrect because the text doesn't suggest that the author's claim is "disorienting," or confusing; rather than indicating that the claim is hard to grasp, the text focuses on the idea that it has a weakness that makes it unconvincing. *Choice C* is incorrect because describing the claim as "nuanced," or subtle, wouldn't make sense in context. The text emphasizes that the claim is unconvincing because it didn't consider certain key archaeological finds; it doesn't suggest that what's in the claim seems subtle. *Choice D* is incorrect because the text faults the claim for not considering certain key archaeological finds; it doesn't suggest that the author's claim is "unoriginal," or imitative and lacking originality.

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QUESTION 8

Choice D is the best answer because it accurately states the text's main purpose. The poem begins with the speaker urging a child to "go forth" with her encouragement ("my heart's desire"). The speaker goes on to suggest that new experiences ("Great reaches, yet unknown") lie ahead for the son that "life is calling" him to seek out. Thus, the main purpose is to encourage a child to embrace the experiences available to him in his life.

Choice A is incorrect because the speaker encourages the child to pursue new experiences ("Great reaches") without knowing exactly what those experiences will be ("yet unknown") or suggesting that they should match the speaker's own accomplishments. *Choice B* is incorrect because the speaker focuses on positive possibilities for her son ("Great reaches, yet unknown") and her enthusiastic encouragement to embrace those possibilities ("life is calling you!"), while there is no mention of raising a child or associated struggles. *Choice C* is incorrect because the speaker frames the possibilities for her son in a positive light when she says that "great reaches, yet unknown" are waiting for him, and this positive outlook for the son is consistent throughout the text.

QUESTION 9

Choice D is the best answer because it best describes the function of the underlined sentence in the text's overall portrayal of how the women in Ohiyesa's tribe harvested maple syrup. The text states that the women used an axe to strike the maple trees in order to find out which ones would produce sap. The underlined sentence compares the trees to people, with the sap described as the trees' "life-blood." Some of the trees are ready to give out their sap, while others are unwilling to do so. Using personification, the sentence provides greater detail about the aspect of the maple trees—their potential to give sap—that the women are evaluating.

Choice A is incorrect because the personalities of the women are not discussed in the text. Although the underlined sentence does mention "individual characters," this reference is not to the women in the text but rather to the maple trees, which the sentence compares to people with individual character traits. *Choice B* is incorrect because the underlined sentence focuses on the trees' willingness or refusal to yield sap, not on the beneficial relationship between the women and the trees. Additionally, although the text does suggest that the women and their tribe benefit from the maple trees since the trees allow the women to harvest syrup, there is nothing in the text to suggest that the trees benefit from this relationship in turn. *Choice C* is incorrect because the underlined sentence is comparing maple trees to humans, not addressing the influence of the natural environment on how the actual humans in the text, the women, behave.

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QUESTION 10

Choice A is the best answer because based on Text 2, it represents how Behrenfeld and colleagues would most likely respond to the "conventional wisdom" discussed in Text 1. The conventional wisdom cited holds the opinion that when there is species diversity within a phytoplankton population, "one species should emerge after outcompeting the rest"—that is, after being so successful in competing for resources that the other species vanish from the population. However, Text 2 explains that according to Behrenfeld and colleagues, phytoplankton are so small and spaced so far apart in the water that there is "much less" direct competition for resources within phytoplankton populations than scientists had previously thought.

Choice B is incorrect because Text 2 never discusses whether routine replenishment of ocean nutrients affects competition between phytoplankton species. *Choice C* is incorrect because the interspecies competition discussed in both texts is specifically between phytoplankton species, and neither text considers whether phytoplankton compete for resources with larger nonphytoplankton species. *Choice D* is incorrect because according to Text 2, Behrenfeld and colleagues argue that water density decreases, not increases, competition between phytoplankton species.

QUESTION 11

Choice A is the best answer because it presents an explanation about a helicopter that is directly supported by the text. The text states that Mars's atmosphere is much less dense than Earth's, and as a result, the air on Mars doesn't provide the resistance required to support the blades of a helicopter built for Earth and to keep the helicopter aloft. In other words, a helicopter built for Earth can't fly on Mars because of the differences in the two planets' atmospheres.

Choice B is incorrect because instead of stating that the blades of helicopters built for Earth are too large to work on Mars, the text indicates that the helicopter built to fly on Mars actually has even longer blades than a helicopter built for Earth. *Choice C* is incorrect because the text never addresses the role of gravity on Mars or on Earth; instead, it focuses on atmospheric conditions. *Choice D* is incorrect because the text doesn't indicate that helicopters built for Earth are too small to operate in the conditions on Mars. In fact, the text states that the size of the helicopter built to fly on Mars is the same size as a helicopter built for Earth, even though it has longer blades that rotate faster.

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QUESTION 12

Choice A is the best answer because it best states the main idea of the text. According to the text, jalis' traditional role has been to maintain information about families' histories and significant events. The text goes on to say that although technological changes have altered jalis' role somewhat, jalis are still valued for preserving the histories of their communities.

Choice B is incorrect because the text says nothing about jalis' views of the various tasks they perform. There is no information to support the idea that many jalis prefer teaching to other tasks. *Choice C* is incorrect because the text doesn't describe jalis as being sources of entertainment. Rather, jalis are presented as valued sources of knowledge. Additionally, the text gives no indication of how long jalis have been serving their communities. *Choice D* is incorrect because the main focus of the text is on jalis' role and their continued value despite the effects of technology, not on what technology can now do. Although the text indicates that jalis' role has changed as a result of technological changes, the text doesn't present any specific information about technology performing tasks that jalis once performed.

QUESTION 13

Choice A is the best answer because it most accurately states the main idea of the text. According to the text, Eugene Wigner hypothesized that a crystal could exist that would be composed of electrons and have a honeycomb-like shape. The text goes on to say that the existence of the Wigner crystal remained unconfirmed until Feng Wang and colleagues were able to make an impression of one using two semiconductors and an ultrathin sheet of graphene. Thus, the main idea is that researchers have obtained the most definitive evidence to date of the existence of the Wigner crystal.

Choice B is incorrect because the text focuses on one kind of crystal—the Wigner crystal—and doesn't discuss crystalline structures in general. And although the text conveys that Wang and colleagues figured out a way to capture an image of a Wigner crystal, it doesn't address the idea of applying this approach to other types of crystals. *Choice C* is incorrect because the text describes in general the process Wang and colleagues followed to obtain an impression of the Wigner crystal; it doesn't address the relative importance of each component in that process. *Choice D* is incorrect because the text doesn't state that researchers had a hard time getting an impression of the Wigner crystal because of its honeycomb structure. Nothing in the text indicates why it took so long to prove the existence of this crystal or take an impression of it.

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QUESTION 14

Choice D is the best answer because it describes data from the graph that support the researchers' conclusion that there is a growing interest among CEOs in connecting with more departments in their companies. The graph shows the average number of individuals reporting directly to CEOs during three different time periods: the individuals are divided into managers and department leaders. The average number of department leaders directly reporting to their CEO during the 1991–1995 period was slightly more than three, during the 1996–2001 period it was four, and during the 2001–2008 period it was almost seven. Thus, the average number of department leaders reporting directly to their CEO rose over the three periods studied, which suggests that CEOs were connecting with more departments.

Choice A is incorrect because the average number of managers and department leaders reporting directly to their CEO rose for both categories between the 1991–1995 and 2001–2008 periods; thus, it isn't true that the average numbers didn't fluctuate. *Choice B* is incorrect because the average number of managers reporting directly to their CEO was highest in the 2001–2008 period, not in the 1996–2001 period. *Choice C* is incorrect. Although it correctly describes a feature of the graph, the observation that more department leaders than managers are reporting to CEOs does not by itself address the question of whether CEOs are connecting with more departments over time—to address that question, one needs to know whether the number of department leaders reporting to CEOs is increasing over time.

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QUESTION 15

Choice C is the best answer because it presents a finding that, if true, would weaken Foster's hypothesis that damage to eelgrass roots improves the health of eelgrass meadows by boosting genetic diversity. The text indicates that sea otters damage eelgrass roots but that eelgrass meadows near Vancouver Island, where there's a large otter population, are comparatively healthy. When Foster and her colleagues compared the Vancouver Island eelgrass meadows to those that don't have established otter populations, the researchers found that the Vancouver Island meadows are more genetically diverse than the other meadows are. This finding led Foster to hypothesize that damage to the eelgrass roots encourages eelgrass reproduction, thereby improving genetic diversity and the health of the meadows. If, however, other meadows not included in the study are less healthy the larger the local otter population is and the longer the otters have been in residence, that would suggest that damage to the eelgrass roots, which would be expected to increase with the size and residential duration of the otter population, isn't leading meadows to be healthier. Such a finding would therefore weaken Foster's hypothesis.

Choice A is incorrect because finding that small, recently introduced otter populations are near other eelgrass meadows in the study wouldn't weaken Foster's hypothesis. If otter populations were small and only recently established, they wouldn't be expected to have caused much damage to eelgrass roots, so even if those eelgrass meadows were less healthy than the Vancouver Island meadows, that wouldn't undermine Foster's hypothesis. In fact, it would be consistent with Foster's hypothesis since it would suggest that the greater damage caused by larger, more established otter populations is associated with healthier meadows. *Choice B* is incorrect because the existence of areas with otters but without eelgrass meadows wouldn't reveal anything about whether the damage that otters cause to eelgrass roots ultimately benefits eelgrass meadows. *Choice D* is incorrect because the health of plants other than eelgrass would have no bearing on Foster's hypothesis that damage to eelgrass roots leads to greater genetic diversity and meadow health. It would be possible for otters to have a negative effect on other plants while nevertheless improving the health of eelgrass meadows by damaging eelgrass roots.

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QUESTION 16

Choice B is the best answer because it most logically completes the text's discussion of Zelda Fitzgerald's contributions to literature. The text begins by saying that many scholars view Zelda mainly in terms of her marriage to F. Scott Fitzgerald and "don't recognize Zelda as a writer in her own right." The text then mentions a novel and "numerous short stories" that she wrote and that such scholars tend to ignore. Therefore, those scholars who focus on Zelda only as an inspiration for F. Scott's writings risk misrepresenting the full range of Zelda's contributions to literature.

Choice A is incorrect. Although the text does mention that Zelda Fitzgerald "likely influenced" her husband's literary work, its focus is on Zelda's own writing, not on her husband's writing or factors that might have influenced it. *Choice C* is incorrect because the text does not discuss F. Scott and Zelda Fitzgerald's opinions of each other's works. *Choice D* is incorrect. Although the text does suggest that F. Scott Fitzgerald's works were "likely influenced in part" by his marriage to Zelda, it does not discuss autobiographical interpretations of the works of either F. Scott or Zelda.

QUESTION 17

Choice B is the best answer because it presents the conclusion that most logically completes the text's discussion of the study by Versace and colleagues. The text indicates that newborn animals of some species are attracted to faces and to stimuli that resemble faces. These species, the text says, share two characteristics: they're social and they practice parental care, meaning that parents care for their young. The text goes on to describe Versace and colleagues' experiment, which showed that although Testudo tortoises aren't social and don't practice parental care, tortoise hatchlings were attracted to a stimulus that resembles a face. Since Versace and colleagues have shown that a species that isn't social and doesn't practice parental care nevertheless has the innate characteristic of being attracted to face-like stimuli, it follows that this characteristic shouldn't be assumed to be an adaptation related to social interaction or parental care.

Choice A is incorrect because the text indicates that the tortoise hatchlings were attracted to the face-like stimuli (even though their species is solitary and doesn't practice parental care), not that they perceived the stimuli as threatening. *Choice C* is incorrect because the phenomenon discussed in the text is an attraction to faces and face-like stimuli on the part of newborn animals, which can't show any learned characteristics since they were just born. Additionally, the text tells us that the tortoises Versace and colleagues studied aren't social and don't practice parental care, so any findings about those tortoises wouldn't be relevant to the question of whether an attraction to faces in social species that practice parental care is innate or learned. *Choice D* is incorrect because the text gives no indication that adult tortoises were presented with face-like stimuli and, if adults were in fact tested, no information about how they responded is provided. Since no information about adult tortoises' responses is provided, no conclusion comparing those responses to the responses of newly hatched tortoises can be supported.

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QUESTION 18

Choice A is the best answer because it most logically completes the text. The text explains that the *Cantares Mexicanos* contains poems about the Aztec Empire from before the Spanish invasion. Furthermore, it indicates that notes in the collection attest that some of these poems predate the Spanish invasion, while some customs depicted are likely Spanish in origin. The implication is that some poems were composed before the invasion but the references to Spanish customs could have come about only after the invasion, and thus that the collection includes content that predates the invasion and also content from after the invasion.

Choice B is incorrect because the text clearly indicates that the collection is in Nahuatl, not Spanish, so the compilers' unfamiliarity with Spanish is irrelevant to whether the collection contains material composed after the Spanish invasion. *Choice C* is incorrect because the text mentions only the Aztec Empire and Spain: there is no information about the relationship of Aztec literature to any traditions other than its own or Spain's. *Choice D* is incorrect because the text states that some of the poems make "inarguable references" to common Spanish customs, which conflicts with the idea that these references can reasonably be attributed to mere coincidence.

QUESTION 19

Choice A is the best answer because it presents the conclusion that most logically follows from the text's discussion of the study of capuchin monkeys' cognitive abilities. The text explains that the study failed to distinguish between outcomes for the tasks performed by the capuchin monkeys, such that simpler tasks requiring less dexterity, or skill, were judged by the same criteria as tasks that demanded more dexterity. Because the study didn't account for this discrepancy, the researchers might have assumed that observed differences in performance were due to the abilities of the monkeys rather than the complexity of the tasks. In other words, the results may suggest cognitive differences among the monkeys even though such differences may not really exist.

Choice B is incorrect because the text focuses on the fact that the tasks assigned to the capuchin monkeys in the study varied in difficulty and that the variety wasn't taken into consideration. The text doesn't suggest that the capuchin monkeys couldn't perform certain tasks, just that some tasks were more difficult to do. *Choice C* is incorrect because the text doesn't suggest that the study's results are indicative of the abilities of capuchin monkeys but not of other monkey species; in fact, the text suggests that the results may not even be an accurate reflection of capuchin monkeys' abilities. *Choice D* is incorrect because the text doesn't indicate that the researchers compared results for artificial tasks with those for tasks encountered in the wild, although the tasks described in the text—sliding a panel and putting a straw in a bottle—are presumably artificial.

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QUESTION 20

Choice A is the best answer. The convention being tested is finite and nonfinite verb forms within a sentence. A main clause requires a finite verb to perform the action of the subject (in this case, "embryos"), and this choice supplies the clause with the finite present tense verb "enter" to indicate how the embryos achieve diapause.

Choice B is incorrect because the nonfinite to-infinitive "to enter" doesn't supply the main clause with a finite verb. *Choice C* is incorrect because the nonfinite participle "having entered" doesn't supply the main clause with a finite verb. *Choice D* is incorrect because the nonfinite participle "entering" doesn't supply the main clause with a finite verb.

QUESTION 21

Choice B is the best answer. The convention being tested is the use of verbs to express tense. In this choice, the past perfect verb "had doubled" properly indicates that the doubling of the organization's initial membership occurred during a specific period before the present (between the organization's founding in 1967 and the end of the 1990s).

Choice A is incorrect because the present perfect verb "has doubled" doesn't indicate that the organization's doubling of its initial membership occurred during a specific period in the past. *Choice C* is incorrect because the present tense verb "doubles" doesn't indicate that the organization's doubling of its initial membership occurred during a specific period in the past. *Choice D* is incorrect because the future tense verb "will double" doesn't indicate that the organization's doubling of its initial membership occurred during a specific period in the past.

QUESTION 22

Choice A is the best answer. The convention being tested is punctuation use between sentences. In this choice, the period after "configurations" is used correctly to mark the boundary between one sentence ("The intense… configurations") and another ("TMAO…fish"). The supplementary phrase ("ensuring…configurations") modifies the main clause of the first sentence ("The chemical…effect"), and "TMAO" is the subject of the second sentence.

Choice B is incorrect because it results in a run-on sentence. The sentences ("The intense…configurations" and "TMAO…fish") are fused without punctuation and/or a conjunction. *Choice C* is incorrect because it results in a comma splice. A comma can't be used in this way to mark the boundary between sentences. *Choice D* is incorrect. Without a comma preceding it, the conjunction "and" can't be used in this way to join sentences.

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QUESTION 23

Choice C is the best answer. The convention being tested is the use of verbs to express tense. In this choice, the present tense verb "experiences" is consistent with the other present tense verbs (e.g., "connects" and "prepares") used to describe the events in Truong's novels. Furthermore, it's conventional to use the present tense when discussing a literary work.

Choice A is incorrect because the past tense verb "experienced" isn't consistent with the other present tense verbs used to describe the events in Truong's novels. *Choice B* is incorrect because the past perfect tense verb "had experienced" isn't consistent with the other present tense verbs used to describe the events in Truong's novels. *Choice D* is incorrect because the future progressive tense verb "will be experiencing" isn't consistent with the other present tense verbs used to describe the events in Truong's novels.

QUESTION 24

Choice C is the best answer. The convention being tested is the use of plural and possessive nouns. The singular possessive noun "screw's" and the plural noun "threads" correctly indicate that there is only one screw and it has multiple threads.

Choice A is incorrect because the context requires the plural noun "threads," not the singular possessive noun "thread's." *Choice B* is incorrect because the context requires the singular possessive noun "screw's," not the plural possessive noun "screws'." *Choice D* is incorrect because the context requires the singular possessive noun "screw's" and the plural noun "threads," not the plural noun "screws" or the plural possessive noun "threads.'"

QUESTION 25

Choice C is the best answer. The convention being tested is punctuation between a main clause and a supplementary noun phrase. This choice correctly uses a comma to mark the boundary between the main clause ("scholar… materialism") and the supplementary noun phrase ("an apt assessment") that describes Waid's observation about how *The House of Mirth* depicts the upper classes of New York society.

Choice A is incorrect because a semicolon and the conjunction "and" can't be used in this way to mark the boundary between a main clause and a supplementary noun phrase. *Choice B* is incorrect. Joining the main clause ("scholar…materialism") and the following noun phrase with the conjunction "and" results in a confusing and illogical sentence that suggests that the novel depicts the upper classes of New York society as "an apt assessment," which doesn't make sense in this context. *Choice D* is incorrect because it fails to mark the boundary between the main clause and the supplementary noun phrase with appropriate punctuation.

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QUESTION 26

Choice D is the best answer. The convention being tested is the coordination of main clauses within a sentence. The semicolon is correctly used to join the first main clause ("To humans…prey") and the second main clause ("rather… approach"). Further, the comma after the adverb "rather" is correctly used to separate the adverb from the main clause ("the brightly…approach") it modifies, logically indicating that the information in this clause (how the spider's behavior appears to humans) is contrary to the information in the previous clause (how the spider's behavior does not appear to humans).

Choice A is incorrect because it results in a comma splice. Without a conjunction following it, a comma can't be used in this way to join two main clauses. *Choice B* is incorrect because it results in a run-on sentence. The two main clauses are fused without appropriate punctuation and/or a conjunction. *Choice C* is incorrect. Placing the comma between the first main clause "To humans…prey" and the adverb "rather" illogically indicates that the information in the first main clause is contrary to what came before, which doesn't make sense in this context.

QUESTION 27

Choice C is the best answer. The convention being tested is the use of plural and possessive nouns. The singular possessive noun "playa's" and the plural possessive noun "rocks'" correctly indicate that the sediment is that of one playa (the Racetrack Playa) and that there are multiple rocks that have mysteriously migrated across the sediment.

Choice A is incorrect because the context requires the singular possessive noun "playa's" and the plural possessive noun "rocks'," not the plural noun "playas" and the singular possessive noun "rock's." *Choice B* is incorrect because the context requires the plural possessive noun "rocks'," not the plural noun "rocks." *Choice D* is incorrect because the context requires the singular possessive noun "playa's," not the plural possessive noun "playas'."

QUESTION 28

Choice C is the best answer. The convention being tested is punctuation between a subject and a verb. When, as in this case, a subject ("her 2019 novel *Gingerbread*") is immediately followed by a verb ("offers"), no punctuation is needed.

Choice A is incorrect because no punctuation is needed between the subject and the verb. *Choice B* is incorrect because no punctuation is needed between the subject and the verb. *Choice D* is incorrect because no punctuation is needed between the subject and the verb.

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QUESTION 29

Choice B is the best answer. The sentence explains an advantage of microprobes, noting that because microprobes weigh as little as 50 milligrams, they can explore areas inaccessible to rovers.

Choice A is incorrect. The sentence indicates that rovers can land successfully on Mars despite their weight; it doesn't explain an advantage of microprobes. *Choice C* is incorrect. While the sentence mentions that microprobes have been proposed as an alternative to rovers, it doesn't explain an advantage of microprobes. *Choice D* is incorrect. The sentence emphasizes a similarity between microprobes and rovers; it doesn't explain an advantage of microprobes.

QUESTION 30

Choice C is the best answer. The sentence effectively introduces *Paradise* to an audience unfamiliar with the novel and its author, describing *Paradise* as a historical novel about colonial East Africa and its author as the winner of the 2021 Nobel Prize in Literature.

Choice A is incorrect. While the sentence introduces Abdulrazak Gurnah to an audience unfamiliar with the author, it doesn't effectively introduce *Paradise*. *Choice B* is incorrect. While the sentence provides background information about *Paradise*, it doesn't effectively introduce the novel to an audience unfamiliar with its author. *Choice D* is incorrect. While the sentence provides background information about *Paradise*, it doesn't effectively introduce the novel to an audience unfamiliar with its author.

QUESTION 31

Choice C is the best answer. The sentence emphasizes the relative sizes of the capital cities' populations, noting that even though Hanoi has more people overall, Ulaanbaatar accounts for a larger percentage of the people in its country.

Choice A is incorrect. While the sentence indicates the population size of each capital, it fails to emphasize their sizes relative to each other or to their countries' overall population sizes. *Choice B* is incorrect. While the sentence indicates the population size of each capital, it fails to emphasize their sizes relative to each other or to their countries' overall population sizes. *Choice D* is incorrect. The sentence emphasizes the population sizes of the two countries; it fails to mention the capitals.

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QUESTION 32

Choice D is the best answer. The sentence explains how the House of Wisdom preserved the world's knowledge, noting that the library collected, translated, and printed writings from different countries.

Choice A is incorrect. While the sentence indicates that the House of Wisdom was known for bringing together knowledge from around the world, it doesn't explain how the library preserved this knowledge. *Choice B* is incorrect. The sentence makes a generalization about the scholars who were employed by the House of Wisdom; it doesn't explain how the library preserved the world's knowledge. *Choice C* is incorrect. The sentence identifies two authors whose writings were preserved at the House of Wisdom; it doesn't explain how the library preserved the world's knowledge.

QUESTION 33

Choice D is the best answer. The sentence uses information from the notes to make a generalization about the kind of study Glickman, Brown, and Song conducted. Specifically, the sentence indicates that the study was of a kind that used statistical methods to address questions of authorship within the field of music.

Choice A is incorrect because the sentence summarizes the methodology and findings of a particular analysis of a single song; it doesn't make a generalization about the kind of study conducted. *Choice B* is incorrect because the sentence mentions the data and conclusion of a particular analysis of a single song; it doesn't make a generalization about the kind of study conducted. *Choice C* is incorrect because the sentence focuses on a specific conclusion from a particular analysis of a single song; it doesn't make a generalization about the kind of study conducted.

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Reading and Writing

Module 2

(33 questions)

QUESTION 1

Choice C is the best answer because it logically and precisely completes the text's discussion of *The Mule Bone*, a play that Zora Neale Hurston and Langston Hughes wrote together. In this context, "collaboration" means working together with someone to write a literary work. The text indicates that most writers prefer to work alone and that working together destroyed the friendship between Hurston and Hughes. This establishes that *The Mule Bone* is a relatively rare example of collaboration in literature.

Choice A is incorrect because in this context, "characterization" would mean a literary work's portrayal of characters' psychological experiences and motivations, but the text doesn't discuss characterization in *The Mule Bone* specifically or in collaborative works more generally. *Choice B* is incorrect because in this context, "interpretation" would mean the explanation of a literary work's meaning or significance, but the text doesn't discuss how readers or critics have interpreted *The Mule Bone*; instead, the text discusses how the play was written collaboratively and how the writing process affected the two authors. *Choice D* is incorrect because in this context, "commercialization" would mean writing a literary work in such a way as to ensure its commercial appeal, but the text never discusses commercial appeal as a factor in the writing of *The Mule Bone* specifically or the writing of collaborative works more generally.

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QUESTION 2

Choice B is the best answer because it most logically completes the text's discussion about recycling plastics. In this context, "inadequate" means not satisfactory. The text indicates that the mechanical plastic-recycling process affects the environment and causes "the loss of material quality." The text contrasts that with Chazovachii's chemical plastic-recycling process, which is cleaner and produces a desirable product. The text's emphasis on the negative aspects of mechanical recycling suggests that it is inadequate in terms of environmental impact and the quality of the material the process yields.

Choice A is incorrect because in this context "resilient" would mean able to withstand difficulty and the text does not characterize the plastic-recycling process as having this quality or describe any difficulties that these processes might need to overcome. *Choice C* is incorrect because in this context "dynamic" would mean constantly changing. Although the text suggests that there have been changes in the field of recycling, as is the case with the advent of Chazovachii's chemical recycling process, there is nothing to suggest that the mechanical process itself has changed or is prone to change. *Choice D* is incorrect because in this context "satisfactory" would mean acceptable but not perfect. The text mentions only shortcomings of the mechanical process (environmental effects and lower material quality), so the text more strongly supports a negative view of this process and provides no evidence that it would be considered satisfactory.

QUESTION 3

Choice D is the best answer because it most logically completes the text's discussion of the economist's claim about sales of personal electronic devices. In this context, "invalidate" most nearly means nullify or make invalid. The text indicates that interruptions in the supply of microchips for personal electronics "have challenged" the economist's claim that sales of personal electronics will show strong growth in the coming months. The text goes on to clarify the effect of the delays on the economist's projection, stating that the delays are very likely to extend the time frame over which the projected growth in sales will occur. This context suggests that the delays are unlikely to invalidate the economist's projection entirely—the delays will probably alter the time frame of the projection, not nullify it or make it invalid.

Choice A is incorrect because saying that the delays are unlikely to "dispute," or argue against, the economist's projection wouldn't make sense. Since the delays are an inanimate circumstance, they couldn't argue against a prediction about the sales of personal electronics. *Choice B* is incorrect because saying that the delays are unlikely to "withdraw," or remove from consideration, the economist's projection wouldn't make sense. Although the economist could withdraw her projection because of the delays, the delays themselves couldn't withdraw her projection since they're an inanimate circumstance and thus can't choose to remove something from consideration. *Choice C* is incorrect because there's nothing in the text to suggest that the delays will "underscore," or emphasize, the economist's projection. Instead, the text suggests that the delays are likely to extend the time frame of the economist's projection but not to undermine the projection entirely.

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QUESTION 4

Choice B is the best answer because it most logically completes the text's discussion of the art installation *Anthem*. In this context, "inventive" means characterized by invention and creativity. The text explains that critics' responses to the installation involved praise for Tsang's creative transformation of a space into a dynamic exhibit with huge images and lots of sound. This context conveys that the critics found the piece particularly creative.

Choice A is incorrect because the text indicates that critics praised the installation for being dynamic and including huge images and lots of sound, and it wouldn't make sense to describe such an exhibit as "restrained," or limited and not extravagant or showy. *Choice C* is incorrect because it wouldn't make sense to say that critics found the installation "inexplicable," or incapable of being explained or interpreted, since the critics were able to explain their praise for the installation's transformation of a space with huge images and lots of sound. *Choice D* is incorrect because the text focuses on the idea that critics praised Tsang for creatively transforming a space into a dynamic exhibit, not that they found the installation "mystifying," or bewildering and hard to understand. Nothing in the text suggests that the critics couldn't understand the piece.

QUESTION 5

Choice C is the best answer because it most logically completes the text's discussion of the kinds of mammals alive during the Mesozoic era. As used in this context, "diverse" means to have a significant amount of variety. The text indicates that some scientists have suggested that Mesozoic mammals can't be characterized in a certain way, then contrasts the view put forward by those scientists with Luo's research, which shows that Mesozoic mammals "weren't all ground-dwelling insectivores" and instead were "various." This context suggests that some scientists have viewed Mesozoic mammals as being all alike, or not a very diverse group.

Choice A is incorrect because it wouldn't make sense to say that some scientists have suggested that Mesozoic mammals weren't very "predatory," or that they didn't prey on other animals, since the text establishes a contrast between what some scientists have suggested and Luo's research showing that Mesozoic mammals "weren't all ground-dwelling insectivores." This context suggests that some scientists have regarded Mesozoic mammals as all being insectivores, or animals that prey on insects, not that some scientists have suggested that Mesozoic mammals didn't prey on other animals. *Choice B* is incorrect because it wouldn't make sense to say that some scientists have suggested that Mesozoic mammals weren't very "obscure," or concealed or not well known, since the text establishes a contrast between what some scientists have suggested and Luo's research showing that Mesozoic mammals were a varied group. There's no contrast between saying that the mammals weren't concealed or well known and the mammals being varied. *Choice D* is incorrect because it wouldn't make sense to say that some scientists have suggested that Mesozoic mammals weren't very "localized," or confined to a particular area, since the text establishes a contrast

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between what some scientists have suggested and Luo's research showing that Mesozoic mammals were a varied group. There's no contrast between saying that the mammals weren't localized and the mammals being varied. Although the text mentions mammal fossils found in China, nothing in the discussion of Luo's research addresses the limits of Mesozoic mammal habitats.

QUESTION 6

Choice C is the best answer because it most accurately describes the overall structure of the text. Throughout the text, the speaker characterizes nighttime as if it were a person who wears clothing ("a garment" that is "velvet soft" and "violet blue") and a veil "over her face" and who moves her hands "slowly with their gem-starred light" through her dark hair. Thus, the text is structured as an extended comparison of night to a human being.

Choice A is incorrect because the text never mentions any particular location; instead, it focuses on presenting a single description of night as a person with certain clothing and features. *Choice B* is incorrect because the text doesn't make any reference to the sun or sunrise; instead, it focuses on presenting a single image of night as a person with certain clothing and features. *Choice D* is incorrect. Rather than describing how nighttime changes seasonally (or in any other way), the text presents a single image of night as a person with certain clothing and features.

QUESTION 7

Choice A is the best answer because it best describes how the underlined portion functions in the text as a whole. The text says that the increased production quotas of food processing companies during World War II enabled employees to make better bargains in exchange for their labor. The underlined portion presents an example of this increased bargaining power: employees requested more favorable benefits, and employers complied because they were under pressure to fulfill the demanding terms of their contracts. Thus, the underlined portion of the text elaborates on a claim about labor relations in a particular industry (food processing) made earlier in the text.

Choice B is incorrect because there is no indication in the text that the economic factors that influenced food processing also influenced other parts of the economy; thus, the bargaining described in the underlined portion of the text cannot be called an example of a trend. *Choice C* is incorrect because the underlined portion supports the historical narrative of labor activism in food processing that is sketched in the text, instead of noting an exception to that narrative. *Choice D* is incorrect because while the underlined portion does discuss the demands that workers made in exchange for their labor, it does not discuss the identities of the workers.

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QUESTION 8

Choice D is the best answer because it accurately describes how the underlined sentence functions in the text as a whole. The text establishes that John has a strong imagination and then goes on to describe the St. John River near John's home in the Florida woods. The underlined sentence depicts John sending twigs sailing down the river while he imagines them reaching "Jacksonville, the sea, the wide world," where he wishes he could follow. This suggests that John longs to expand his life experiences beyond the Florida woods.

Choice A is incorrect because the second and third sentences of the text provide an extended description of the riverbank where John likes to go, whereas the underlined sentence describes what John does at that location. *Choice B* is incorrect because the first sentence of the text suggests that John's behavior "was puzzling" to others around him, whereas the underlined sentence concerns the content of John's imaginings. *Choice C* is incorrect because the underlined sentence elaborates on John's imagination but doesn't mention any other children to whom John could be compared.

QUESTION 9

Choice B is the best answer because it presents a statement about Dorian that is directly supported by the text. The narrator of the text says that when Dorian sees his portrait, "his cheeks flushed for a moment with pleasure" and "a look of joy came into his eyes." The narrator goes on to say that Dorian looked at the portrait "in wonder" and presents him as being so entranced by the portrait that he doesn't notice what Hallward is saying to him. These details support the description of Dorian as being delighted by what he sees in the portrait.

Choice A is incorrect because Dorian isn't depicted as interested in Hallward's opinion of the portrait; rather, he is so enraptured by the painting that he's hardly even aware of Hallward. *Choice C* is incorrect because the portrait of Dorian is the only painting mentioned in the text. Although Dorian is depicted as being delighted with this particular portrait, there's no evidence in the text that he likes portraits better than other kinds of paintings. *Choice D* is incorrect because nothing in the text suggests that Dorian is uncertain about Hallward's talent. Instead, the text is focused on Dorian's delight with the portrait.

QUESTION 10

Choice A is the best answer because it uses data from the graph to effectively support Charles and Stephens's claim about how level of information affects voters. The graph shows the probability of voting for both high- and low-information voters in seven categories of political orientation. Charles and Stephens claim that "the more informed voters are about politics…the more likely they are to vote." This statement correctly asserts that the graph shows a higher probability of voting for high-information voters than for low-information voters at each of the seven political orientations. Thus, this statement accurately cites data from the graph that support Charles and Stephens's claim about how level of information affects voters.

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Choice B is incorrect. Although this statement is correct that the only probability in the graph below 50% is for low-information voters categorized as independent (orientation 4), the claim in question is about the relative likelihood that low- and high-information voters will vote, and without some reference to high-information voters, this statement cannot help support such a comparison. *Choice C* is incorrect. Although this statement is correct that the highest probabilities of voting for low-information voters are at the ends of the orientation scale (1 and 7), the claim in question is about the relative likelihood that low- and high-information voters will vote, and without some reference to high-information voters, this statement cannot help support such a comparison. *Choice D* is incorrect because the graph does not give any information about how many people are represented in any of the categories, so this statement is not based on data from the graph. Furthermore, even if we did have this information, the claim is about how level of information affects voters' probability of voting, not whether they're likely to strongly identify with a particular political party.

QUESTION 11

Choice B is the best answer because it describes data from the graph that weaken the student's conclusion about the reduction in the spider population in the enclosure with lizards. The graph shows that the enclosure with lizards and the enclosure without lizards each began with about 85 spiders, and that the number of spiders in each enclosure fell over the 30 days of the study. The student's claim is that the reduction in spiders in the enclosure with lizards is "entirely attributable to the presence of the lizards," meaning that the spider population wouldn't have declined except for the presence of the lizards. This claim is weakened, however, by the fact that the enclosure without lizards also saw a substantial reduction in the number of spiders. Since the number of spiders fell in the enclosure without lizards as well as in the enclosure with lizards, there must be some other factor than just the presence of the lizards that contributed to the reduction in the spider population.

Choice A is incorrect because the fact that the two enclosures started with the same number of spiders is irrelevant to the claim that the reduction in spider population by day 30 in the enclosure with lizards can be entirely attributed to the lizards. *Choice C* is incorrect because the fact that the spider population in the enclosure with lizards fell more between days 1 and 10 than in other periods has nothing to do with the student's claim that the reduction in spiders in that enclosure by day 30 can be entirely attributed to the lizards. *Choice D* is incorrect. Although it's true that on day 30 the spider population was lower in the enclosure with lizards than in the enclosure without lizards, this fact doesn't weaken the student's claim that the reduction in the spider population in the enclosure with lizards can be entirely attributed to the lizards. Indeed, the lower spider population in the enclosure with lizards suggests that the lizards are contributing to the reduction in the spider population, though the fact that the spider population also fell substantially in the other enclosure means that the lizards aren't the only cause of the reduction.

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QUESTION 12

Choice A is the best answer because it presents a finding that, if true, would most strongly support the team's conclusion that cattle were likely raised closer to human settlements than sheep and goats were. The text explains that Vaiglova, Liu, and their colleagues analyzed the chemical composition of sheep, goat, and cattle bones from the Bronze Age in China in order to investigate the animals' domestication, or their adaptation from a wild state to a state in which they existed in close connection with humans. According to the text, the team's analysis showed that sheep and goats of the era fed largely on wild plants, whereas cattle fed on millet—importantly, a crop cultivated by humans. If analysis of the animal bones shows that the cattle's diet also consisted of wheat, another crop cultivated by humans in China during the Bronze Age, the finding would support the team's conclusion by offering additional evidence that cattle during this era fed on human-grown crops—and, by extension, that humans raised cattle relatively close to the settlements where they grew these crops, leaving goats and sheep to roam farther away in areas with wild vegetation, uncultivated by humans.

Choice B is incorrect because if it were true that sheep's and goats' diets consisted of small portions of millet, which the text states was a crop cultivated by humans, the finding would suggest that sheep and goats were raised relatively close to human settlements, weakening the team's conclusion that cattle were likely raised closer to those settlements than sheep and goats were. *Choice C* is incorrect because the finding that cattle generally require more food and nutrients than do sheep and goats wouldn't support the team's conclusion that cattle were likely raised closer to human settlements than sheep and goats were. Nothing in the text suggests that cattle were incapable of obtaining sufficient food and nutrients without access to human-grown crops. Hence, even if cattle's diets are found to have different requirements than the diets of sheep and goats, the cattle could have met those requirements from food located far from human settlements. *Choice D* is incorrect because if it were true that the diets of sheep, goats, and cattle varied based on what the farmers in each Bronze Age settlement could grow, the finding would weaken the team's conclusion that cattle were likely raised closer to human settlements than sheep and goats were, suggesting instead that all three types of animals were raised close enough to human settlements to feed on those settlements' crops.

Shaimaa

QUESTION 13

Choice C is the best answer because it presents the finding that, if true, would best support Suarez, Pérez-Huerta, and Harrell's claim about mosasaurs. The text states that Suarez, Pérez-Huerta, and Harrell's research on mosasaur tooth enamel led them to conclude that mosasaurs were endothermic, which means that they could live in waters at many different temperatures and still maintain a stable body temperature. The researchers claim that endothermy enabled mosasaurs to live in relatively cold waters near the poles. If several mosasaur fossils have been found in areas that were near the poles during the period when mosasaurs were alive and fossils of nonendothermic marine reptiles are rare in such locations, that would support the researchers' claim: it would show that mosasaurs inhabited polar waters but nonendothermic marine mammals tended not to, suggesting that endothermy may have been the characteristic that enabled mosasaurs to include polar waters in their range.

Choice A is incorrect because finding that it's easier to determine mosasaur body temperatures from tooth enamel data than it is to determine nonendothermic reptile body temperatures wouldn't support the researchers' claim. Whether one research process is more difficult than another indicates nothing about the results of those processes and therefore is irrelevant to the issue of where mosasaurs lived and what enabled them to live in those locations. *Choice B* is incorrect because finding roughly equal numbers of mosasaur and nonendothermic marine reptile fossils in areas that were near the poles in the Late Cretaceous would suggest that endothermy didn't give mosasaurs any particular advantage when it came to expanding their range to include relatively cold polar waters, thereby weakening the researchers' claim rather than supporting it. *Choice D* is incorrect because finding that the temperature of seawater in the Late Cretaceous was warmer than seawater today wouldn't weaken the researchers' claim. Seawater in the Late Cretaceous could have been warmer than seawater today but still cold enough for endothermy to be advantageous to mosasaurs, so this finding wouldn't provide enough information to either support or weaken the researchers' claim.

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QUESTION 14

Choice D is the best answer because it presents a finding that, if true, would most directly support the researchers' hypothesis about the connection between the dusky shark population decline and the eastern oyster population decline. The text indicates that although dusky sharks don't usually eat eastern oysters, they do consume cownose rays, which are the main predators of eastern oysters. An increase in the abundance of cownose rays in the region in response to a decline in the abundance of dusky sharks would directly support the researchers' hypothesis: a higher number of cownose rays would consume more eastern oysters, driving down the oyster population.

Choice A is incorrect because a finding that there's an association between a decline in the regional abundance of some of dusky sharks' prey and the regional abundance of dusky sharks wouldn't directly support the researchers' hypothesis that a decline in dusky sharks has led to a decline in eastern oysters in the region. Although such a finding might help explain why shark abundance has declined, it would reveal nothing about whether the shark decline is related to the oyster decline. *Choice B* is incorrect because a finding that eastern oyster abundance tends to be greater when dusky sharks and cownose rays are present than when only dusky sharks are present wouldn't support the researchers' hypothesis that a decline in dusky sharks has led to a decline in eastern oysters in the region. The text indicates that the sharks prey on the rays, which are the main predators of the oysters; if oyster abundance is found to be greater when rays are present than when rays are absent, that would suggest that rays aren't keeping oyster abundance down, and thus that a decline in rays' predators, which would be expected to lead to an increase in the abundance of rays, wouldn't bring about a decline in oyster abundance as the researchers hypothesize. *Choice C* is incorrect because a finding that consumption of eastern oysters by cownose rays increased substantially before dusky sharks declined in regional abundance wouldn't support the researchers' hypothesis that the decline in dusky sharks has led to a decline in eastern oysters in the region. Such a finding would suggest that some factor other than shark abundance led to an increase in rays' consumption of oysters and thus to a decrease in oyster abundance, thereby weakening the researchers' hypothesis.

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QUESTION 15

Choice C is the best answer because it presents a finding that, if true, would weaken the claim made by people who favor the traditional view of voter behavior. According to the text, people who favor that view believe that voting in an election doesn't change a voter's attitude toward the candidates in that election. If Washington and Mullainathan found that two years after an election, attitudes toward the winning candidate were significantly more polarized among subjects who had voted than among subjects who had been too young to vote, that would suggest that the act of voting did have an effect on the voters' attitudes toward the candidates, which would undermine the claim that voting doesn't change voters' attitudes.

Choice A is incorrect because a finding about links between subjects' attitudes and general political orientation, regardless of age and ability to vote, wouldn't address the presence or absence of changes in attitudes among those subjects who did actually vote. Therefore, the finding wouldn't have any bearing on the claim that voting in an election doesn't change a voter's attitude toward the candidates in that election. *Choice B* is incorrect because a finding that positive attitudes toward a winning candidate significantly increased in the two years after the election among subjects who had been too young to vote would involve only people who didn't vote; therefore, the finding wouldn't have any bearing on the claim that when people do vote, the act of voting doesn't change their attitudes toward the candidates. *Choice D* is incorrect because the finding that subjects in both groups were more likely to have negative attitudes than positive attitudes toward the winning candidate two years after an election would reflect all subjects' attitudes at one particular time whether they voted or not, rather than the presence or absence of a change in voters' attitudes after voting. Therefore, the finding would neither weaken nor strengthen the claim that voting in an election doesn't change a voter's attitude toward the candidates.

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QUESTION 16

Choice B is the best answer because it describes data from the graph that support Taylor and colleagues' conclusion that spray coating holds promise for improving the power conversion efficiency of ETLs in perovskite solar cells. The text explains that perovskite solar cells' efficiency at converting light into electricity is diminished by their electron transport layer (ETL), which is applied through spin coating, but that Taylor's team devised a new spray coating method for applying the ETL that improves its power conversion efficiency. The graph displays data on the power conversion efficiency of solar cells in tests conducted by Taylor's team, with bars for both the highest- and lowest-performing ETLs in two data categories: spray coating and spin coating. According to the graph, the lowest-performing ETL applied through spray coating had a power conversion efficiency of between 14% and 16%, while the highest-performing ETL applied through spin coating had a power conversion efficiency of less than 14%. These data confirm that ETLs applied through novel spray coating are more efficient than those applied though traditional spin coating. Thus, the data support Taylor and colleagues' conclusion about spray coating's potential value.

Choice A is incorrect. Although this claim correctly describes the data in the graph by stating that both the lowest-performing ETL applied through spin coating and the lowest-performing ETL applied through spray coating had a power conversion efficiency greater than 10%, this relationship in the data doesn't support or relate to Taylor and colleagues' conclusion that spray coating promises greater efficiency for solar cells than traditional spin coating does. *Choice C* is incorrect. This claim does address the greater power conversion efficiency of the highest-performing ETL applied through spray coating, compared with the highest-performing ETL applied through spin coating. However, it also incorrectly cites the value for the efficiency of the highest-performing ETL applied through spray coating as approximately 13%, instead of a value between 14% and 16%, and the value for the efficiency of the highest-performing ETL applied through spin coating as approximately 11%, instead of a value between 12% and 14%, as shown in the graph. *Choice D* is incorrect because Taylor and colleagues' conclusion is based on the difference in the power conversion efficiency of ETLs applied through spray coating and that of ETLs applied through spin coating, not on the difference between the highest- and lowest-performing ETLs applied through just spray coating.

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QUESTION 17

Choice B is the best answer because it presents a finding that, if true, would most directly support the arts journalist's claim about Enwezor's work as a curator and art historian. In the text, the arts journalist asserts that Enwezor wished not just to focus on modern African artists but also to show "how their work fits into the larger context of global modern art and art history," or how their work relates to artistic developments and work by other artists elsewhere in the world. The description of *Postwar: Art Between the Pacific and the Atlantic, 1945–1965* indicates that Enwezor and Siegel's exhibition brought works by African artists together with works by artists from other countries, thus supporting the arts journalist's claim that Enwezor sought to show works by African artists in a context of global modern art and art history.

Choice A is incorrect because it describes a retrospective that wouldn't support the arts journalist's claim that Enwezor wanted to show how works by modern African artists fit into the larger context of global modern art and art history. The description of *El Anatsui: Triumphant Scale* indicates that the retrospective focused only on the work of a single African artist, El Anatsui. The description doesn't suggest that the exhibition showed how El Anatsui's works fit into a global artistic context. *Choice C* is incorrect because it describes an exhibition that wouldn't support the arts journalist's claim that Enwezor wanted to show how works by modern African artists relate to the larger context of global modern art and art history. The description of *The Short Century: Independence and Liberation Movements in Africa, 1945–1994* indicates that the exhibition showed how African artists were influenced by movements for independence from European colonial powers following the Second World War. Although this suggests that Enwezor intended the exhibition to place works by African artists in a political context, it doesn't indicate that the works were placed in a global artistic context. *Choice D* is incorrect because it describes an exhibition that wouldn't support the arts journalist's claim that Enwezor wanted to show how works by modern African artists relate to the larger context of global modern art and art history. The description of *In/sight: African Photographers, 1940 to the Present* indicates that the exhibition was intended to reveal the broad range of approaches taken by African photographers, not that the exhibition showed how photography by African artists fits into a global artistic context.

QUESTION 18

Choice B is the best answer. The convention being tested is finite and nonfinite verb forms within a sentence. A main clause requires a finite verb to perform the action of the subject (in this case, "people in the Americas"), and this choice supplies the finite past perfect tense verb "have used" to indicate what people in the Americas used the gourd for.

Choice A is incorrect because the nonfinite to-infinitive "to use" doesn't supply the main clause with a finite verb. *Choice C* is incorrect because the nonfinite participle "having used" doesn't supply the main clause with a finite verb. *Choice D* is incorrect because the nonfinite participle "using" doesn't supply the main clause with a finite verb.

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QUESTION 19

Choice C is the best answer. The convention being tested is punctuation between a subordinate clause and a main clause. This choice correctly uses a comma to mark the boundary between the subordinate clause ("While…lifelike") and the main clause ("others look to the past").

Choice A is incorrect because it results in an incomplete sentence with no main clause. *Choice B* is incorrect because it fails to mark the boundary between the subordinate clause ("While…lifelike") and the main clause ("others…past"). *Choice D* is incorrect because it results in an incomplete sentence with no main clause.

QUESTION 20

Choice A is the best answer. The convention being tested is the use of verbs to express tense. In this choice, the simple past tense verb "suggested" properly indicates that Zufall offered her suggestion for the product's name in the past. This verb tense is consistent with the previous sentence's use of a simple past tense verb ("advised") to describe Zufall's advice to McVicker in the 1950s.

Choice B is incorrect because the present tense verb "suggests" doesn't indicate that Zufall offered her suggestion in the past. *Choice C* is incorrect because the past perfect verb "had suggested" isn't consistent with the previous sentence's use of the simple past tense verb "advised" to describe Zufall's advice to McVicker. *Choice D* is incorrect because the past progressive verb "was suggesting" isn't consistent with the previous sentence's use of the simple past tense verb "advised" to describe Zufall's advice to McVicker.

QUESTION 21

Choice D is the best answer. The convention being tested is punctuation use between two supplementary phrases following the coordinate clause ("but she… mycology"). This choice correctly uses a comma to mark the boundary between the supplementary noun phrase ("the study of fungi") that defines the term "mycology" and the supplementary participial phrase ("producing...London") that provides additional information about the extent to which Potter dedicated herself to mycology.

Choice A is incorrect because a semicolon can't be used in this way to join two supplementary phrases following a coordinate clause. *Choice B* is incorrect because it results in a rhetorically unacceptable sentence fragment beginning with "producing." *Choice C* is incorrect. The lack of punctuation results in a sentence that illogically suggests that the study of fungi is producing more than 350 paintings.

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QUESTION 22

Choice A is the best answer. The convention being tested is subject-modifier placement. This choice makes the noun phrase "many critics" the subject of the sentence and places it immediately after the modifying phrase "in assessing… Kurosawa." In doing so, this choice clearly establishes that it is the critics—and not another noun in the sentence—who assess Kurosawa's films.

Choice B is incorrect because it results in a dangling modifier. The placement of the noun phrase "Kurosawa's…sources" immediately after the modifying phrase illogically suggests that his use of Western literary sources is what assesses Kurosawa's films. *Choice C* is incorrect because it results in a dangling modifier. The placement of the function word "there" immediately after the modifying phrase illogically suggests that "there" is what assesses Kurosawa's films. *Choice D* is incorrect because it results in a dangling modifier. The placement of the noun phrase "the focus…critics" immediately after the modifying phrase illogically suggests that the critics' focus is what assesses Kurosawa's films.

QUESTION 23

Choice C is the best answer. The convention being tested is the punctuation of items in a complex series. It's conventional to use a semicolon to separate items in a complex series with internal punctuation, and in this choice, the semicolon after "2009" is conventionally used to separate the first item ("the world's…2009") and the second item ("an online…2010") in the series of things that Hinson helped create. Further, the comma after "Basic" correctly pairs with the comma after "app," and the comma after "network" correctly pairs with the comma after "TV" to set off the supplemental elements ("Chickasaw Basic" and "Chickasaw TV") that provide the names of the app and the TV network, respectively. Altogether, the punctuation in this choice results in a sentence that clearly indicates that Hinson helped make a language app in 2009, an online TV network in 2010, and a language course in 2015.

Choice A is incorrect because it fails to punctuate the complex series in a way that makes clear that Hinson helped make a language app in 2009, an online TV network in 2010, and a language course in 2015. *Choice B* is incorrect because it fails to punctuate the complex series in a way that makes clear that Hinson helped make a language app in 2009, an online TV network in 2010, and a language course in 2015. *Choice D* is incorrect because the comma after "2009" doesn't match the semicolon used to separate the second and third items in the complex series.

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QUESTION 24

Choice C is the best answer. The convention being tested is the punctuation of a supplementary element within a sentence. The dash after "Springs" pairs with the dash after "earth" to separate the supplementary element "in this case, the porous rocks of the hills around Hot Springs" from the rest of the sentence.

Choice A is incorrect because it fails to use appropriate punctuation to separate the supplementary element from the rest of the sentence. *Choice B* is incorrect because a colon can't be paired with a dash in this way to separate the supplementary element from the rest of the sentence. *Choice D* is incorrect because a comma can't be paired with a dash in this way to separate the supplementary element from the rest of the sentence.

QUESTION 25

Choice A is the best answer. The convention being tested is the use of a colon within a sentence. In this choice, the colon is used in a conventional way to introduce the following description of how the number of jams available varied.

Choice B is incorrect because it creates a comma splice. A comma can't be used in this way to join two main clauses ("the number…varied" and "some…six"). *Choice C* is incorrect because it results in an illogical and confusing sentence. Using the conjunction "while" to join the main clause ("the number…varied") with the following clause's description of the number of jams available suggests that the variation in the number of jams is in contrast to some shoppers having twenty-four options. *Choice D* is incorrect because it results in an illogical and confusing sentence. Using "while" in this way suggests that the number of jams available varied during the time in which some shoppers had twenty-four options and others had six. The sentence makes clear, however, that what follows "varied" is a description of the variation, not a separate, simultaneous occurrence.

QUESTION 26

Choice B is the best answer. The convention being tested is the punctuation of items in a complex series (a series including internal punctuation). In this choice, the semicolon after "Lagos" is conventionally used to separate the first item ("*The Joys*…Lagos") and the second item ("*A Kind*…Nigeria") in the series. Further, the comma after "*Marriage*" correctly separates the title "*A Kind of Marriage*" from the supplementary phrase ("a television…Nigeria") that describes it.

Choice A is incorrect because the comma after "Lagos" doesn't match the semicolon used later in the series to separate the second item ("*A Kind*…Nigeria") from the third item ("and…autobiography"). *Choice C* is incorrect because the comma after "Lagos" doesn't match the semicolon used later in the series to separate the second item ("*A Kind*…Nigeria") from the third item ("and… autobiography"). Additionally, a colon can't be used in this way to separate the title "*A Kind of Marriage*" from the supplementary phrase ("a television… Nigeria") that describes it. *Choice D* is incorrect because it fails to use appropriate punctuation to separate the title "*A Kind of Marriage*" from the supplementary phrase ("a television…Nigeria") that describes it.

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QUESTION 27

Choice A is the best answer. "Meanwhile" logically signals that the action described in this sentence (Obinze's move to London to pursue a career) is simultaneous with the action described in the previous sentence (Ifemelu's move to the United States). The first sentence establishes that the actions take place around the same time, referring to the characters' "divergent experiences" following high school.

Choice B is incorrect because "nevertheless" illogically signals that the information in this sentence about Obinze's move to London is true despite the previous information about Ifemelu's move to the United States. Instead, as the first sentence establishes, Obinze's move and Ifemelu's move are related, parallel experiences that occur around the same time. *Choice C* is incorrect because "secondly" illogically signals that the information in this sentence is a second point or reason separate from the previous information about Ifemelu's move to the United States. Instead, as the first sentence establishes, Obinze's move and Ifemelu's move are related, parallel experiences that occur around the same time. *Choice D* is incorrect because "in fact" illogically signals that the information in this sentence emphasizes, modifies, or contradicts the previous information about Ifemelu's move to the United States. Instead, as the first sentence establishes, Obinze's move and Ifemelu's move are related, parallel experiences that occur around the same time.

QUESTION 28

Choice B is the best answer. "For example" logically signals that the information in this sentence—that tadpole shrimp embryos can pause development during extended periods of drought—exemplifies the previous sentence's claim that organisms have evolved surprising adaptations to survive in adverse conditions.

Choice A is incorrect because "in contrast" illogically signals that the information in this sentence contrasts with the claim about organisms in the previous sentence. Instead, it exemplifies this claim. *Choice C* is incorrect because "meanwhile" illogically signals that the information in this sentence is separate from (while occurring simultaneously with) the claim about organisms in the previous sentence. Instead, it exemplifies this claim. *Choice D* is incorrect because "consequently" illogically signals that the information in this sentence is a consequence, or result, of the claim about organisms in the previous sentence. Instead, it exemplifies this claim.

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QUESTION 29

Choice D is the best answer. "In addition" logically signals that the information in this sentence—that the Twentieth Amendment requires newly elected US senators and representatives to be sworn in on January 3 —is separate from and additional to the amendment's mandate concerning presidential inaugurations.

Choice A is incorrect because "instead" illogically signals that the information in the sentence presents an alternative to or substitute for the Twentieth Amendment's mandate concerning presidential inaugurations. Rather, the sentence presents a separate requirement in addition to that one. *Choice B* is incorrect because "for instance" illogically signals that the information in the sentence exemplifies the Twentieth Amendment's mandate concerning presidential inaugurations. Instead, the sentence presents a separate requirement in addition to that one. *Choice C* is incorrect because "specifically" illogically signals that the sentence provides specific, precise details elaborating on the Twentieth Amendment's mandate concerning presidential inaugurations. Instead, the sentence presents a separate requirement in addition to that one.

QUESTION 30

Choice D is the best answer. "Similarly" logically signals that the information in the sentence—that Dove situates Beulah's life in the context of the US Civil Rights Movement—is similar to the previous information about Thomas and the Great Migration. Both sentences support the first sentence's claim that Dove portrays her characters in the context of broader historical narratives.

Choice A is incorrect because "specifically" illogically signals that the information about Beulah in this sentence provides specific details elaborating on the previous information about Thomas. Instead, it's similar to the previous information about Thomas. *Choice B* is incorrect because "thus" illogically signals that the information about Beulah in this sentence is a result or consequence of the previous information about Thomas. Instead, it's similar to the previous information about Thomas. *Choice C* is incorrect because "regardless" illogically signals that the information about Beulah in this sentence is true despite the previous information about Thomas. Instead, it's similar to the previous information about Thomas.

QUESTION 31

Choice A is the best answer. The sentence emphasizes the distance covered by the Philadelphia and Lancaster Turnpike, noting that the turnpike, which connected the two Pennsylvania cities in its name, was sixty-two miles long.

Choice B is incorrect. The sentence emphasizes the significance of the turnpike; it doesn't emphasize the distance that the turnpike covered. *Choice C* is incorrect. While the sentence mentions that the turnpike connected two Pennsylvania cities, it doesn't emphasize the specific distance covered by the turnpike. *Choice D* is incorrect. The sentence emphasizes when the turnpike was built; it doesn't emphasize the distance that the turnpike covered.

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QUESTION 32

Choice C is the best answer. The sentence emphasizes the aim, or goal, of the research study, noting what Terada set out to do: determine whether some of the Moon's oxygen was coming from Earth.

Choice A is incorrect. The sentence focuses on how the Kaguya satellite collected data; it doesn't emphasize the aim of the research study. *Choice B* is incorrect. While the sentence mentions what Terada was curious about before conducting the research study, it doesn't emphasize his study's aim. *Choice D* is incorrect. The sentence presents the research study's conclusion; it doesn't emphasize the study's aim.

QUESTION 33

Choice B is the best answer. The sentence presents both the study and its methodology (that is, the researcher's approach to the problem), explaining that Yuan used computer simulations to study the effect of the mother duck's wake on the ducklings' energy expenditure.

Choice A is incorrect. The sentence describes the findings of Yuan's study; it doesn't present the study and its methodology. *Choice C* is incorrect. While the sentence provides general information about Yuan's study, it doesn't present the study's methodology. *Choice D* is incorrect. The sentence describes the findings of Yuan's study; it doesn't present the study and its methodology.

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Math

Module 1
(27 questions)

QUESTION 1

Choice C is correct. For the given line graph, the percent of cars for sale at a used car lot on a given day is represented on the vertical axis. The percent of cars for sale is the smallest when the height of the line graph is the lowest. The lowest height of the line graph occurs for cars with a model year of 2014.

Choice A is incorrect and may result from conceptual errors. *Choice B* is incorrect and may result from conceptual errors. *Choice D* is incorrect and may result from conceptual errors.

QUESTION 2

Choice C is correct. It's given that 29 out of every 100 beads that the machine produces have a defect. It follows that if the machine produces k beads, then the number of beads that have a defect is $\frac{29}{100}k$, for some constant k. If a bead produced by the machine will be selected at random, the probability of selecting a bead that has a defect is given by the number of beads with a defect, $\frac{29}{100}k$, divided by the number of beads produced by the machine, k. Therefore, the probability of selecting a bead that has a defect is $\frac{\frac{29}{100}k}{k}$, or $\frac{29}{100}$.

Choice A is incorrect and may result from conceptual or computational errors. *Choice B* is incorrect and may result from conceptual or computational errors. *Choice D* is incorrect and may result from conceptual or computational errors.

QUESTION 3

Choice D is correct. It's given that line m is parallel to line n, and line t intersects both lines. It follows that line t is a transversal. When two lines are parallel and intersected by a transversal, exterior angles on the same side of the transversal

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are supplementary. Thus, $x+33=180$. Subtracting 33 from both sides of this equation yields $x=147$. Therefore, the value of x is 147.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice B is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.

QUESTION 4

Choice D is correct. The *y*-intercept of a graph in the *xy*-plane is the point at which the graph crosses the *y*-axis. The graph shown crosses the *y*-axis at the point $(0, 8)$. Therefore, the *y*-intercept of the graph shown is $(0, 8)$.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice B is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.

QUESTION 5

Choice C is correct. It's given that $f(x)$ is the total cost, in dollars, to lease a car from this dealership with a monthly payment of x dollars. Therefore, the total cost, in dollars, to lease the car when the monthly payment is $\$400$ is represented by the value of $f(x)$ when $x=400$. Substituting 400 for x in the equation $f(x)=36x+1{,}000$ yields $f(400)=36(400)+1{,}000$, or $f(400)=15{,}400$. Thus, when the monthly payment is $\$400$, the total cost to lease a car is $\$15{,}400$.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice B is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 6

The correct answer is 180. The perimeter of a polygon is equal to the sum of the lengths of the sides of the polygon. It's given that each side of the square has a length of 45. Since a square is a polygon with 4 sides, the perimeter of this square is $45+45+45+45$, or 180.

QUESTION 7

The correct answer is 5. Multiplying both sides of the given equation by $x+6$ results in $55=x(x+6)$. Applying the distributive property of multiplication to the right-hand side of this equation results in $55=x^2+6x$. Subtracting 55 from both sides of this equation results in $0=x^2+6x-55$. The right-hand side of this equation can be rewritten by factoring. The two values that multiply to -55 and add to 6 are 11 and -5. It follows that the equation $0=x^2+6x-55$ can be rewritten as $0=(x+11)(x-5)$. Setting each factor equal to 0 yields two equations: $x+11=0$ and $x-5=0$. Subtracting 11 from both sides of the equation $x+11=0$ results in $x=-11$. Adding 5 to both sides of the equation $x-5=0$ results in $x=5$. Therefore, the positive solution to the given equation is 5.

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QUESTION 8

Choice A is correct. If the object travels 108 centimeters at a speed of 12 centimeters per second, the time of travel can be determined by dividing the total distance by the speed. This results in $\frac{108 \text{ centimeters}}{12 \text{ centimeters/second}}$, which is 9 seconds.

Choice B is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 9

Choice B is correct. The mean of a data set is the sum of the values in the data set divided by the number of values in the data set. It follows that the mean of data set X is $\frac{5+9+9+13}{4}$, or 9, and the mean of data set Y is $\frac{5+9+9+13+27}{5}$, or 12.6. Since 9 is less than 12.6, the mean of data set X is less than the mean of data set Y.

Alternate approach: Data set Y consists of the 4 values in data set X and one additional value, 27. Since the additional value, 27, is larger than any value in data set X, the mean of data set X is less than the mean of data set Y.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 10

Choice A is correct. It's given that the rocket contained $467{,}000$ kilograms (kg) of propellant before launch and had $362{,}105$ kg remaining exactly 21 seconds after launch. Finding the difference between the amount, in kg, of propellant before launch and the remaining amount, in kg, of propellant after launch gives the amount, in kg, of propellant burned during the 21 seconds: $467{,}000-362{,}105=104{,}895$. Dividing the amount of propellant burned by the number of seconds yields $\frac{104{,}895}{21}=4{,}995$. Thus, an average of $4{,}995$ kg of propellant burned each second after launch.

Choice B is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from finding the amount of propellant burned, rather than the amount of propellant burned each second.

QUESTION 11

Choice B is correct. Multiplying both sides of the given equation by 4 yields $(4)(4x+2)=(4)(12)$, or $16x+8=48$. Therefore, the value of $16x+8$ is 48.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

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QUESTION 12

Choice D is correct. It's given that the equation $h=-4.9t^2+7t+9$ represents this situation, where h is the height, in meters, of the object t seconds after it is kicked. It follows that the height, in meters, from which the object was kicked is the value of h when $t=0$. Substituting 0 for t in the equation $h=-4.9t^2+7t+9$ yields $h=-4.9(0)^2+7(0)+9$, or $h=9$. Therefore, the object was kicked from a height of 9 meters.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice B is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.

QUESTION 13

The correct answer is $\frac{25}{4}$. The given equation can be rewritten in the form $f(x)=a(x-h)^2+k$, where a, h, and k are constants. When $a>0$, h is the value of x for which $f(x)$ reaches its minimum. The given equation can be rewritten as $f(x)=4\left(x^2-\frac{50}{4}x\right)+126$, which is equivalent to $f(x)=4\left(x^2-\frac{50}{4}x+\left(\frac{50}{8}\right)^2-\left(\frac{50}{8}\right)^2\right)+126$. This equation can be rewritten as $f(x)=4\left(\left(x-\frac{50}{8}\right)^2-\left(\frac{50}{8}\right)^2\right)+126$, or $f(x)=4\left(x-\frac{50}{8}\right)^2-4\left(\frac{50}{8}\right)^2+126$, which is equivalent to $f(x)=4\left(x-\frac{25}{4}\right)^2-\frac{121}{4}$. Therefore, $h=\frac{25}{4}$, so the value of x for which $f(x)$ reaches its minimum is $\frac{25}{4}$. Note that 25/4 and 6.25 are examples of ways to enter a correct answer.

QUESTION 14

The correct answer is 182. Let s represent the number of small candles the owner can purchase, and let ℓ represent the number of large candles the owner can purchase. It's given that the owner pays $\$4.90$ per candle to purchase small candles and $\$11.60$ per candle to purchase large candles. Therefore, the owner pays $4.90s$ dollars for s small candles and 11.60ℓ dollars for ℓ large candles, which means the owner pays a total of $4.90s+11.60\ell$ dollars to purchase candles. It's given that the owner budgets $\$2{,}200$ to purchase candles. Therefore, $4.90s+11.60\ell\le 2{,}200$. It's also given that the owner must purchase a minimum of 200 candles. Therefore, $s+\ell\ge 200$. The inequalities $4.90s+11.60\ell\le 2{,}200$ and $s+\ell\ge 200$ can be combined into one compound inequality by rewriting the second inequality so that its left-hand side is equivalent to the left-hand side of the first inequality. Subtracting ℓ from both sides of the inequality $s+\ell\ge 200$ yields $s\ge 200-\ell$. Multiplying both sides of this inequality by 4.90 yields $4.90s\ge 4.90(200-\ell)$, or $4.90s\ge 980-4.90\ell$. Adding 11.60ℓ to both sides of this inequality yields $4.90s+11.60\ell\ge 980-4.90\ell+11.60\ell$, or $4.90s+11.60\ell\ge 980+6.70\ell$. This inequality can be combined with the inequality $4.90s+11.60\ell\le 2{,}200$, which yields the compound inequality

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$980+6.70\ell \le 4.90s+11.60\ell \le 2{,}200$. It follows that $980+6.70\ell \le 2{,}200$. Subtracting 980 from both sides of this inequality yields $6.70\ell \le 2{,}200$. Dividing both sides of this inequality by 6.70 yields approximately $\ell \le 182.09$. Since the number of large candles the owner purchases must be a whole number, the maximum number of large candles the owner can purchase is the largest whole number less than 182.09, which is 182.

QUESTION 15

Choice D is correct. Since f is a linear function, it can be defined by an equation of the form $f(x)=ax+b$, where a and b are constants. It's given that $f(0)=8$. Substituting 0 for x and 8 for $f(x)$ in the equation $f(x)=ax+b$ yields $8=a(0)+b$, or $8=b$. Substituting 8 for b in the equation $f(x)=ax+b$ yields $f(x)=ax+8$. It's given that $f(1)=12$. Substituting 1 for x and 12 for $f(x)$ in the equation $f(x)=ax+8$ yields $12=a(1)+8$, or $12=a+8$. Subtracting 8 from both sides of this equation yields $a=4$. Substituting 4 for a in the equation $f(x)=ax+8$ yields $f(x)=4x+8$. Therefore, an equation that defines f is $f(x)=4x+8$.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice B* is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect and may result from conceptual or calculation errors.

QUESTION 16

Choice A is correct. The function f gives the area of the rectangle, in ft^2, if its width is w ft. Since the value of $f(14)$ is the value of $f(w)$ if $w=14$, it follows that $f(14)=1{,}176$ means that $f(w)$ is $1{,}176$ if $w=14$. In the given context, this means that if the width of the rectangle is 14 ft, then the area of the rectangle is $1{,}176\ \text{ft}^2$.

Choice B is incorrect and may result from conceptual errors. *Choice C* is incorrect and may result from conceptual errors. *Choice D* is incorrect and may result from interpreting $f(w)$ as the width, in ft, of the rectangle if its area is $w\ \text{ft}^2$, rather than as the area, in ft^2, of the rectangle if its width is w ft.

QUESTION 17

Choice B is correct. Since $\overline{PR}$ and $\overline{QS}$ are diameters of the circle shown, $\overline{OS}$, $\overline{OR}$, $\overline{OP}$, and $\overline{OQ}$ are radii of the circle and are therefore congruent. Since $\angle SOP$ and $\angle ROQ$ are vertical angles, they are congruent. Therefore, arc PS and arc QR are formed by congruent radii and have the same angle measure, so they are congruent arcs. Similarly, $\angle SOR$ and $\angle POQ$ are vertical angles, so they are congruent. Therefore, arc SR and arc PQ are formed by congruent radii and have the same angle measure, so they are congruent arcs. Let x represent the length of arc SR. Since arc SR and arc PQ are congruent arcs, the length of arc PQ can also be represented by x. It's given that the length of arc PS is twice the length of arc PQ. Therefore, the length of arc PS can be represented by the expression $2x$. Since arc PS and arc QR are congruent arcs, the length of arc QR can also be represented by $2x$. This gives the expression $x+x+2x+2x$.

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Since it's given that the circumference is 144π, the expression $x+x+2x+2x$ is equal to 144π. Thus $x+x+2x+2x=144\pi$, or $6x=144\pi$. Dividing both sides of this equation by 6 yields $x=24\pi$. Therefore, the length of arc QR is $2(24\pi)$, or 48π.

Choice A is incorrect. This is the length of arc PQ, not arc QR. *Choice C* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 18

Choice C is correct. Let x represent the number of children in a whale-watching tour group. Let y represent the number of adults in this group. Because it's given that 21 people are in a group and the group consists of adults and children, it must be true that $x+y=21$. Since the company's revenue is 60 dollars per child, the total revenue from x children in this group was $60x$ dollars. Since the company's revenue is 80 dollars per adult, the total revenue from y adults in this group was $80y$ dollars. Because it's given that the total revenue for this group was $1{,}440$ dollars, it must be true that $60x+80y=1{,}440$. The equations $x+y=21$ and $60x+80y=1{,}440$ form a linear system of equations that can be solved to find the value of x, which represents the number of children in the group, using the elimination method. Multiplying both sides of the equation $x+y=21$ by 80 yields $80x+80y=1{,}680$. Subtracting $60x+80y=1{,}440$ from $80x+80y=1{,}680$ yields $(80x+80y)-(60x+80y)=1{,}680-1{,}440$, which is equivalent to $80x-60x+80y-80y=240$, or $20x=240$. Dividing both sides of this equation by 20 yields $x=12$. Therefore, 12 people in the group were children.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice B* is incorrect. This is the number of adults in the group, not the number of children in the group. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 19

Choice A is correct. The *x*-intercept of a graph in the *xy*-plane is the point on the graph where $y=0$. It's given that function h is defined by $h(x)=4x+28$. Therefore, the equation representing the graph of $y=h(x)$ is $y=4x+28$. Substituting 0 for y in the equation $y=4x+28$ yields $0=4x+28$. Subtracting 28 from both sides of this equation yields $-28=4x$. Dividing both sides of this equation by 4 yields $-7=x$. Therefore, the *x*-intercept of the graph of $y=h(x)$ in the *xy*-plane is $(-7, 0)$. It's given that the *x*-intercept of the graph of $y=h(x)$ is $(a, 0)$. Therefore, $a=-7$. The *y*-intercept of a graph in the *xy*-plane is the point on the graph where $x=0$. Substituting 0 for x in the equation $y=4x+28$ yields $y=4(0)+28$, or $y=28$. Therefore, the *y*-intercept of the graph of $y=h(x)$ in the *xy*-plane is $(0, 28)$. It's given that the *y*-intercept of the graph of $y=h(x)$ is $(0, b)$. Therefore, $b=28$. If $a=-7$ and $b=28$, then the value of $a+b$ is $-7+28$, or 21.

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Choice B is incorrect. This is the value of b, not $a+b$. *Choice C* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect. This is the value of $-a+b$, not $a+b$.

QUESTION 20

The correct answer is 8. Since each term of the given expression, $2x^3+42x^2+208x$, has a factor of $2x$, the expression can be rewritten as $2x(x^2)+2x(21x)+2x(104)$, or $2x(x^2+21x+104)$. Since the values 8 and 13 have a sum of 21 and a product of 104, the expression $x^2+21x+104$ can be factored as $(x+8)(x+13)$. Therefore, the given expression can be factored as $2x(x+8)(x+13)$. It follows that the factors of the given expression are 2, x, $x+8$, and $x+13$. Of these factors, only $x+8$ and $x+13$ are of the form $x+b$, where b is a positive constant. Therefore, the possible values of b are 8 and 13. Thus, the smallest possible value of b is 8.

QUESTION 21

The correct answer is $\frac{29}{2}$. According to the first equation in the given system, the value of y is -1.5. Substituting -1.5 for y in the second equation in the given system yields $-1.5=x^2+8x+a$. Adding 1.5 to both sides of this equation yields $0=x^2+8x+a+1.5$. If the given system has exactly one distinct real solution, it follows that $0=x^2+8x+a+1.5$ has exactly one distinct real solution. A quadratic equation in the form $0=px^2+qx+r$, where p, q, and r are constants, has exactly one distinct real solution if and only if the discriminant, q^2-4pr, is equal to 0. The equation $0=x^2+8x+a+1.5$ is in this form, where $p=1$, $q=8$, and $r=a+1.5$. Therefore, the discriminant of the equation $0=x^2+8x+a+1.5$ is $(8)^2-4(1)(a+1.5)$, or $58-4a$. Setting the discriminant equal to 0 to solve for a yields $58-4a=0$. Adding $4a$ to both sides of this equation yields $58=4a$. Dividing both sides of this equation by 4 yields $\frac{58}{4}=a$, or $\frac{29}{2}=a$. Therefore, if the given system of equations has exactly one distinct real solution, the value of a is $\frac{29}{2}$. Note that 29/2 and 14.5 are examples of ways to enter a correct answer.

QUESTION 22

Choice B is correct. It's given that $f(x)=(x+6)(x+5)(x-4)$ and $y=f(x)-3$. Substituting $(x+6)(x+5)(x-4)$ for $f(x)$ in the equation $y=f(x)-3$ yields $y=(x+6)(x+5)(x-4)-3$. Substituting -6 for x in this equation yields $y=(-6+6)(-6+5)(-6-4)-3$, or $y=-3$. Substituting -5 for x in the equation $y=(x+6)(x+5)(x-4)-3$ yields $y=(-5+6)(-5+5)(-5-4)-3$, or $y=-3$. Substituting 4 for x in the equation $y=(x+6)(x+5)(x-4)-3$ yields $y=(4+6)(4+5)(4-4)-3$, or $y=-3$. Therefore, when $x=-6$ then $y=-3$, when $x=-5$ then $y=-3$, and when $x=4$ then $y=-3$. Thus, the table of values in choice B represents $y=f(x)-3$.

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Choice A is incorrect. This table represents $y=x-3$ rather than $y=f(x)-3$.
Choice C is incorrect. This table represents $y=x+3$ rather than $y=f(x)-3$.
Choice D is incorrect. This table represents $y=f(x)+3$ rather than $y=f(x)-3$.

QUESTION 23

Choice C is correct. Since the value of $q(x)$ decreases by a fixed percentage, 45%, for every increase in the value of x by 1, the function q is a decreasing exponential function. A decreasing exponential function can be written in the form $q(x)=a\left(1-\frac{p}{100}\right)^x$, where a is the value of $q(0)$ and the value of $q(x)$ decreases by p% for every increase in the value of x by 1. If $q(0)=14$, then $a=14$. Since the value of $q(x)$ decreases by 45% for every increase in the value of x by 1, $p=45$. Substituting 14 for a and 45 for p in the equation $q(x)=a\left(1-\frac{p}{100}\right)^x$ yields $q(x)=14\left(1-\frac{45}{100}\right)^x$, which is equivalent to $q(x)=14(1-0.45)^x$, or $q(x)=14(0.55)^x$.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice B is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect. For this function, the value of $q(x)$ increases, rather than decreases, by 45% for every increase in the value of x by 1.

QUESTION 24

Choice A is correct. An equation for the graph shown can be written in slope-intercept form $y=mx+b$, where m is the slope of the graph and its y-intercept is $(0, b)$. Since the y-intercept of the graph shown is $(0, 2)$, the value of b is 2. Since the graph also passes through the point $(4, 1)$, the slope can be calculated as $\frac{1-2}{4-0}$, or $-\frac{1}{4}$. Therefore, the value of m is $-\frac{1}{4}$. Substituting $-\frac{1}{4}$ for m and 2 for b in the equation $y=mx+b$ yields $y=-\frac{1}{4}x+2$. It's given that an equation for the graph shown is $y=f(x)+14$. Substituting $f(x)+14$ for y in the equation $y=-\frac{1}{4}x+2$ yields $f(x)+14=-\frac{1}{4}x+2$. Subtracting 14 from both sides of this equation yields $f(x)=-\frac{1}{4}x-12$.

Choice B is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 25

Choice B is correct. It's given that right triangle RST is similar to triangle UVW, where S corresponds to V and T corresponds to W. It's given that the side lengths of the right triangle RST are $RS=20$, $ST=48$, and $TR=52$. Corresponding angles in similar triangles are equal. It follows that the measure of angle T is equal to the measure of angle W. The hypotenuse of a right triangle is

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the longest side. It follows that the hypotenuse of triangle RST is side TR. The hypotenuse of a right triangle is the side opposite the right angle. Therefore, angle S is a right angle. The adjacent side of an acute angle in a right triangle is the side closest to the angle that is not the hypotenuse. It follows that the adjacent side of angle T is side ST. The opposite side of an acute angle in a right triangle is the side across from the acute angle. It follows that the opposite side of angle T is side RS. The tangent of an acute angle in a right triangle is the ratio of the length of the opposite side to the length of the adjacent side. Therefore, $\tan T = \frac{RS}{ST}$. Substituting 20 for RS and 48 for ST in this equation yields $\tan T = \frac{20}{48}$, or $\tan T = \frac{5}{12}$. The tangents of two acute angles with equal measures are equal. Since the measure of angle T is equal to the measure of angle W, it follows that $\tan T = \tan W$. Substituting $\frac{5}{12}$ for $\tan T$ in this equation yields $\frac{5}{12} = \tan W$. Therefore, the value of $\tan W$ is $\frac{5}{12}$.

Choice A is incorrect. This is the value of $\sin W$. *Choice C* is incorrect. This is the value of $\cos W$. *Choice D* is incorrect. This is the value of $\frac{1}{\tan W}$.

QUESTION 26

Choice A is correct. It's given that w represents the total wall area, in square feet. Since the walls of the room will be painted twice, the amount of paint, in gallons, needs to cover $2w$ square feet. It's also given that one gallon of paint will cover 220 square feet. Dividing the total area, in square feet, of the surface to be painted by the number of square feet covered by one gallon of paint gives the number of gallons of paint that will be needed. Dividing $2w$ by 220 yields $\frac{2w}{220}$, or $\frac{w}{110}$. Therefore, the equation that represents the total amount of paint P, in gallons, needed to paint the walls of the room twice is $P = \frac{w}{110}$.

Choice B is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect and may result from finding the amount of paint needed to paint the walls once rather than twice. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 27

The correct answer is 9.87. It's given that the number a is 110% greater than the number b. It follows that $a = \left(1 + \frac{110}{100}\right)b$, or $a = 2.1b$. It's also given that the number b is 90% less than 47. It follows that $b = \left(1 - \frac{90}{100}\right)(47)$, or $b = 0.1(47)$, which yields $b = 4.7$. Substituting 4.7 for b in the equation $a = 2.1b$ yields $a = 2.1(4.7)$, which is equivalent to $a = 9.87$. Therefore, the value of a is 9.87.

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Math

Module 2
(27 questions)

QUESTION 1

Choice A is correct. It's given that 20% of the students surveyed responded that they intend to enroll in the study program. Therefore, the proportion of students in Spanish club who intend to enroll in the study program, based on the survey, is 0.20. Since there are 55 total students in Spanish club, the best estimate for the total number of these students who intend to enroll in the study program is $55(0.20)$, or 11.

Choice B is incorrect. This is the best estimate for the percentage, rather than the total number, of students in Spanish club who intend to enroll in the study program. *Choice C* is incorrect. This is the best estimate for the total number of Spanish club students who do not intend to enroll in the study program. *Choice D* is incorrect. This is the total number of students in Spanish club.

QUESTION 2

Choice A is correct. Since Jay walks at a speed of 3 miles per hour for w hours, Jay walks a total of $3w$ miles. Since Jay runs at a speed of 5 miles per hour for r hours, Jay runs a total of $5r$ miles. Therefore, the total number of miles Jay travels can be represented by $3w+5r$. Since the combined total number of miles is 14, the equation $3w+5r=14$ represents this situation.

Choice B is incorrect and may result from conceptual errors. *Choice C* is incorrect and may result from conceptual errors. *Choice D* is incorrect and may result from conceptual errors.

QUESTION 3

Choice A is correct. The line of best fit shown intersects the y-axis at a positive y-value and has a positive slope. The graph of an equation of the form $y=a+bx$, where a and b are constants, intersects the y-axis at a y-value of a and has a

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slope of b. Of the given choices, only choice A represents a line that intersects the y-axis at a positive y-value, 2.8, and has a positive slope, 1.7.

Choice B is incorrect. This equation represents a line that has a negative slope, not a positive slope. *Choice C* is incorrect. This equation represents a line that intersects the y-axis at a negative y-value, not a positive y-value. *Choice D* is incorrect. This equation represents a line that intersects the y-axis at a negative y-value, not a positive y-value, and has a negative slope, not a positive slope.

QUESTION 4

Choice D is correct. Because the graph of $y=f(x)$ is shown, the value of $f(0)$ is the value of y on the graph that corresponds with $x=0$. When $x=0$, the corresponding value of y is 3. Therefore, the value of $f(0)$ is 3.

Choice A is incorrect and may result from conceptual errors. *Choice B* is incorrect and may result from conceptual errors. *Choice C* is incorrect and may result from conceptual errors.

QUESTION 5

Choice B is correct. Applying the commutative property of multiplication, the expression $(m^4q^4z^{-1})(mq^5z^3)$ can be rewritten as $(m^4m)(q^4q^5)(z^{-1}z^3)$. For positive values of x, $(x^a)(x^b)=x^{a+b}$. Therefore, the expression $(m^4m)(q^4q^5)(z^{-1}z^3)$ can be rewritten as $(m^{4+1})(q^{4+5})(z^{-1+3})$, or $m^5q^9z^2$.

Choice A is incorrect and may result from multiplying, not adding, the exponents. *Choice C* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 6

The correct answer is 79. The median of a data set with an odd number of values is the middle value of the set when the values are ordered from least to greatest. Because the given data set consists of nine values that are ordered from least to greatest, the median is the fifth value in the data set. Therefore, the median of the data shown is 79.

QUESTION 7

The correct answer is 55. Subtracting 40 from both sides of the given equation yields $x=55$. Therefore, the value of x is 55.

QUESTION 8

Choice C is correct. Adding the second equation of the given system to the first equation yields $5x+(-4x+y)=15+(-2)$, which is equivalent to $x+y=13$. So the value of $x+y$ is 13.

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Choice A is incorrect and may result from conceptual or calculation errors. *Choice B* is incorrect. This is the value of $-(x+y)$. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 9

Choice A is correct. It's given that the function g models the number of gallons that remain from a full gas tank in a car after driving m miles. In the given function $g(m)=-0.05m+12.1$, the coefficient of m is -0.05. This means that for every increase in the value of m by 1, the value of $g(m)$ decreases by 0.05. It follows that for each mile driven, there is a decrease of 0.05 gallons of gasoline. Therefore, 0.05 gallons of gasoline are used to drive each mile.

Choice B is incorrect and represents the number of gallons of gasoline in a full gas tank. *Choice C* is incorrect and may result from conceptual errors. *Choice D* is incorrect and may result from conceptual errors.

QUESTION 10

Choice C is correct. Multiplying each side of the given equation by y yields the equivalent equation $\frac{y}{7b}=11x$. Dividing each side of this equation by 11 yields $\frac{y}{77b}=x$, or $x=\frac{y}{77b}$.

Choice A is incorrect. This equation is not equivalent to the given equation. *Choice B* is incorrect. This equation is not equivalent to the given equation. *Choice D* is incorrect. This equation is not equivalent to the given equation.

QUESTION 11

Choice B is correct. Since the point (x, y) is an intersection point of the graphs of the given equations in the *xy*-plane, the pair (x, y) should satisfy both equations, and thus is a solution of the given system. According to the first equation, $y=76$. Substituting 76 in place of y in the second equation yields $x^2-5=76$. Adding 5 to both sides of this equation yields $x^2=81$. Taking the square root of both sides of this equation yields two solutions: $x=9$ and $x=-9$. Of these two solutions, only -9 is given as a choice.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect. This is the value of coordinate y, rather than x, of the intersection point (x, y).

QUESTION 12

Choice A is correct. It's given that the point $(x, 53)$ is a solution to the given system of inequalities in the *xy*-plane. This means that the coordinates of the point, when substituted for the variables x and y, make both of the inequalities in the system true. Substituting 53 for y in the inequality $y>14$ yields $53>14$, which is true. Substituting 53 for y in the inequality $4x+y<18$ yields $4x+53<18$. Subtracting 53 from both sides of this inequality yields $4x<-35$.

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Dividing both sides of this inequality by 4 yields $x<-8.75$. Therefore, x must be a value less than -8.75. Of the given choices, only -9 is less than -8.75.

Choice B is incorrect. Substituting -5 for x and 53 for y in the inequality $4x+y<18$ yields $4(-5)+53<18$, or $33<18$, which is not true. *Choice C* is incorrect. Substituting 5 for x and 53 for y in the inequality $4x+y<18$ yields $4(5)+53<18$, or $73<18$, which is not true. *Choice D* is incorrect. Substituting 9 for x and 53 for y in the inequality $4x+y<18$ yields $4(9)+53<18$, or $89<18$, which is not true.

QUESTION 13

The correct answer is 240. It's given that 80% of the 300 seeds sprouted. Therefore, the number of seeds that sprouted can be calculated by multiplying the number of seeds that were planted by $\frac{80}{100}$, which gives $300\left(\frac{80}{100}\right)$, or 240.

QUESTION 14

The correct answer is 2. Substituting 8 for $f(x)$ in the given equation yields $8=4x$. Dividing the left- and right-hand sides of this equation by 4 yields $x=2$. Therefore, the value of x is 2 when $f(x)=8$.

QUESTION 15

Choice B is correct. The given expression has a common factor of 2 in the denominator, so the expression can be rewritten as $\frac{8x(x-7)-3(x-7)}{2(x-7)}$. The three terms in this expression have a common factor of $(x-7)$. Since it's given that $x>7$, x can't be equal to 7, which means $(x-7)$ can't be equal to 0. Therefore, each term in the expression, $\frac{8x(x-7)-3(x-7)}{2(x-7)}$, can be divided by $(x-7)$, which gives $\frac{8x-3}{2}$.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 16

Choice C is correct. It's given that line r is perpendicular to line p in the xy-plane. This means that the slope of line r is the negative reciprocal of the slope of line p. If the equation for line p is rewritten in slope-intercept form $y=mx+b$, where m and b are constants, then m is the slope of the line and $(0, b)$ is its y-intercept. Subtracting $18x$ from both sides of the equation $2y+18x=9$ yields $2y=-18x+9$. Dividing both sides of this equation by 2 yields $y=-9x+\frac{9}{2}$. It follows that the slope of line p is -9. The negative reciprocal of a number is -1 divided by the number. Therefore, the negative reciprocal of -9 is $\frac{-1}{-9}$, or $\frac{1}{9}$. Thus, the slope of line r is $\frac{1}{9}$.

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Choice A is incorrect. This is the slope of line p, not line r. *Choice B* is incorrect. This is the reciprocal, not the negative reciprocal, of the slope of line p. *Choice D* is incorrect. This is the negative, not the negative reciprocal, of the slope of line p.

QUESTION 17

Choice D is correct. The *y*-intercept of a graph in the *ty*-plane is the point where $t=0$. For the given function f, the *y*-intercept of the graph of $y=f(t)$ in the *ty*-plane can be found by substituting 0 for t in the equation $y=8{,}000(0.65)^t$, which gives $y=8{,}000(0.65)^0$. This is equivalent to $y=8{,}000(1)$, or $y=8{,}000$. Therefore, the *y*-intercept of the graph of $y=f(t)$ is $(0, 8{,}000)$. It's given that the function f models the number of coupons a company sent to their customers at the end of each year. Therefore, $f(t)$ represents the estimated number of coupons the company sent to their customers at the end of each year. It's also given that t represents the number of years since the end of 1998. Therefore, $t=0$ represents 0 years since the end of 1998, or the end of 1998. Thus, the best interpretation of the *y*-intercept of the graph of $y=f(t)$ is that the estimated number of coupons the company sent to their customers at the end of 1998 was $8{,}000$.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice B* is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect and may result from conceptual or calculation errors.

QUESTION 18

Choice C is correct. It's given that triangle XYZ is similar to triangle RST, such that X, Y, and Z correspond to R, S, and T, respectively. Since corresponding angles of similar triangles are congruent, it follows that the measure of $\angle Z$ is congruent to the measure of $\angle T$. It's given that the measure of $\angle Z$ is 20°. Therefore, the measure of $\angle T$ is 20°.

Choice A is incorrect and may result from a conceptual error. *Choice B* is incorrect. This is half the measure of $\angle Z$. *Choice D* is incorrect. This is twice the measure of $\angle Z$.

QUESTION 19

Choice B is correct. A system of two linear equations in two variables, x and y, has no solution if the lines represented by the equations in the *xy*-plane are parallel and distinct. Lines represented by equations in standard form, $Ax+By=C$ and $Dx+Ey=F$, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients in the other equation, meaning $\frac{D}{A}=\frac{E}{B}$; and the lines are distinct if the constants are not proportional, meaning $\frac{F}{C}$ is not equal to $\frac{D}{A}$ or $\frac{E}{B}$. The given equation, $y=6x+18$, can be written in standard form by subtracting $6x$ from both sides of the equation to yield $-6x+y=18$. Therefore, the given equation can be written in the form $Ax+By=C$, where $A=-6$, $B=1$, and $C=18$. The equation in choice B,

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$-6x+y=22$, is written in the form $Dx+Ey=F$, where $D=-6$, $E=1$, and $F=22$. Therefore, $\frac{D}{A}=\frac{-6}{-6}$, which can be rewritten as $\frac{D}{A}=1$; $\frac{E}{B}=\frac{1}{1}$, which can be rewritten as $\frac{E}{B}=1$; and $\frac{F}{C}=\frac{22}{18}$, which can be rewritten as $\frac{F}{C}=\frac{11}{9}$. Since $\frac{D}{A}=1$, $\frac{E}{B}=1$, and $\frac{F}{C}$ is not equal to 1, it follows that the given equation and the equation $-6x+y=22$ are parallel and distinct. Therefore, a system of two linear equations consisting of the given equation and the equation $-6x+y=22$ has no solution. Thus, the equation in choice B could be the second equation in the system.

Choice A is incorrect. The equation $-6x+y=18$ and the given equation represent the same line in the *xy*-plane. Therefore, a system of these linear equations would have infinitely many solutions, rather than no solution. *Choice C* is incorrect. The equation $-12x+y=36$ and the given equation represent lines in the *xy*-plane that are distinct and not parallel. Therefore, a system of these linear equations would have exactly one solution, rather than no solution. *Choice D* is incorrect. The equation $-12x+y=18$ and the given equation represent lines in the *xy*-plane that are distinct and not parallel. Therefore, a system of these linear equations would have exactly one solution, rather than no solution.

QUESTION 20

The correct answer is 986. The area, A, of a rectangle is given by $A=\ell w$, where ℓ is the length of the rectangle and w is its width. It's given that the length of the rectangle is 34 centimeters (cm) and the width is 29 cm. Substituting 34 for ℓ and 29 for w in the equation $A=\ell w$ yields $A=(34)(29)$, or $A=986$. Therefore, the area, in square centimeters, of this rectangle is 986.

QUESTION 21

The correct answer is 35. The first equation in the given system of equations defines y as $4x+1$. Substituting $4x+1$ for y in the second equation in the given system of equations yields $4(4x+1)=15x-8$. Applying the distributive property on the left-hand side of this equation yields $16x+4=15x-8$. Subtracting $15x$ from each side of this equation yields $x+4=-8$. Subtracting 4 from each side of this equation yields $x=-12$. Substituting -12 for x in the first equation of the given system of equations yields $y=4(-12)+1$, or $y=-47$. Substituting -12 for x and -47 for y into the expression $x-y$ yields $-12-(-47)$, or 35.

QUESTION 22

Choice D is correct. The number of solutions of a quadratic equation of the form $ax^2+bx+c=0$, where a, b, and c are constants, can be determined by the value of the discriminant, b^2-4ac. If the value of the discriminant is positive, then the quadratic equation has exactly two distinct real solutions. If the value of the discriminant is equal to zero, then the quadratic equation has exactly one real solution. If the value of the discriminant is negative, then the quadratic equation

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has zero real solutions. In the given equation, $5x^2+10x+16=0$, $a=5$, $b=10$, and $c=16$. Substituting these values for a, b, and c in b^2-4ac yields $(10)^2-4(5)(16)$, or -220. Since the value of its discriminant is negative, the given equation has zero real solutions. Therefore, the number of distinct real solutions the given equation has is zero.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice B* is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect and may result from conceptual or calculation errors.

QUESTION 23

Choice B is correct. Since 1 mile is equal to 1,760 yards, 1 square mile is equal to $1{,}760^2$, or 3,097,600, square yards. It's given that the park has an area of 11,863,808 square yards. Therefore, the park has an area of $(11{,}863{,}808 \text{ square yards})\left(\frac{1 \text{ square mile}}{3{,}097{,}600 \text{ square yards}}\right)$, or $\frac{11{,}863{,}808}{3{,}097{,}600}$ square miles. Thus, the area, in square miles, of the park is 3.83.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect. This is the square root of the area of the park in square yards, not the area of the park in square miles. *Choice D* is incorrect and may result from converting 11,863,808 yards to miles, rather than converting 11,863,808 square yards to square miles.

QUESTION 24

Choice C is correct. The graph of the equation $(x-h)^2+(y-k)^2=r^2$ in the *xy*-plane is a circle with center (h,k) and a radius of length r. The radius of a circle is the distance from the center of the circle to any point on the circle. If a circle in the *xy*-plane intersects the *y*-axis at exactly one point, then the perpendicular distance from the center of the circle to this point on the *y*-axis must be equal to the length of the circle's radius. It follows that the *x*-coordinate of the circle's center must be equivalent to the length of the circle's radius. In other words, if the graph of $(x-h)^2+(y-k)^2=r^2$ is a circle that intersects the *y*-axis at exactly one point, then $r=|h|$ must be true. The equation in choice C is $(x-4)^2+(y-9)^2=16$, or $(x-4)^2+(y-9)^2=4^2$. This equation is in the form $(x-h)^2+(y-k)^2=r^2$, where $h=4$, $k=9$, and $r=4$, and represents a circle in the *xy*-plane with center $(4,9)$ and radius of length 4. Substituting 4 for r and 4 for h in the equation $r=|h|$ yields $4=|4|$, or $4=4$, which is true. Therefore, the equation in choice C represents a circle in the *xy*-plane that intersects the *y*-axis at exactly one point.

Choice A is incorrect. This is the equation of a circle that does not intersect the *y*-axis at any point. *Choice B* is incorrect. This is an equation of a circle that intersects the *x*-axis, not the *y*-axis, at exactly one point. *Choice D* is incorrect. This is the equation of a circle with the center located on the *y*-axis and thus intersects the *y*-axis at exactly two points, not exactly one point.

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QUESTION 25

Choice C is correct. Since angles B and E each have the same measure and angles C and F each have the same measure, triangles ABC and DEF are similar, where side BC corresponds to side EF. To determine whether two similar triangles are congruent, it is sufficient to determine whether one pair of corresponding sides are congruent. Therefore, to determine whether triangles ABC and DEF are congruent, it is sufficient to determine whether sides BC and EF have equal length. Thus, knowing the lengths of BC and EF is sufficient to determine whether triangle ABC is congruent to triangle DEF.

Choice A is incorrect and may result from conceptual errors. *Choice B* is incorrect and may result from conceptual errors. *Choice D* is incorrect. The given information is sufficient to determine that triangles ABC and DEF are similar, but not whether they are congruent.

QUESTION 26

Choice B is correct. The histograms shown have the same shape, but data set A contains values between 20 and 60 and data set B contains values between 10 and 50. Thus, the mean of data set A is greater than the mean of data set B. Therefore, the smallest possible difference between the mean of data set A and the mean of data set B is the difference between the smallest possible mean of data set A and the greatest possible mean of data set B. In data set A, since there are 3 integers in the interval greater than or equal to 20 but less than 30, 4 integers greater than or equal to 30 but less than 40, 7 integers greater than or equal to 40 but less than 50, and 9 integers greater than or equal to 50 but less than 60, the smallest possible mean for data set A is $\frac{(3 \cdot 20)+(4 \cdot 30)+(7 \cdot 40)+(9 \cdot 50)}{23}$. In data set B, since there are 3 integers greater than or equal to 10 but less than 20, 4 integers greater than or equal to 20 but less than 30, 7 integers greater than or equal to 30 but less than 40, and 9 integers greater than or equal to 40 but less than 50, the largest possible mean for data set B is $\frac{(3 \cdot 19)+(4 \cdot 29)+(7 \cdot 39)+(9 \cdot 49)}{23}$. Therefore, the smallest possible difference between the mean of data set A and the mean of data set B is $\frac{(3 \cdot 20)+(4 \cdot 30)+(7 \cdot 40)+(9 \cdot 50)}{23} - \frac{(3 \cdot 19)+(4 \cdot 29)+(7 \cdot 39)+(9 \cdot 49)}{23}$, which is equivalent to $\frac{(3 \cdot 20)-(3 \cdot 19)+(4 \cdot 30)-(4 \cdot 29)+(7 \cdot 40)-(7 \cdot 39)+(9 \cdot 50)-(9 \cdot 49)}{23}$. This expression can be rewritten as $\frac{3(20-19)+4(30-29)+7(40-39)+9(50-49)}{23}$, or $\frac{23}{23}$, which is equal to 1. Therefore, the smallest possible difference between the mean of data set A and the mean of data set B is 1.

Choice A is incorrect. This is the smallest possible difference between the ranges, not the means, of the data sets. *Choice C* is incorrect. This is the difference between the greatest possible mean, not the smallest possible mean, of data set A and the greatest possible mean of data set B. *Choice D* is incorrect. This is the smallest possible difference between the sum of the values in data set A and

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the sum of the values in data set B, not the smallest possible difference between the means.

QUESTION 27

The correct answer is 113. It's given that the legs of a right triangle have lengths 24 centimeters and 21 centimeters. In a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the two legs. It follows that if h represents the length, in centimeters, of the hypotenuse of the right triangle, $h^2 = 24^2 + 21^2$. This equation is equivalent to $h^2 = 1{,}017$. Taking the square root of each side of this equation yields $h = \sqrt{1{,}017}$. This equation can be rewritten as $h = \sqrt{9 \cdot 113}$, or $h = \sqrt{9} \cdot \sqrt{113}$. This equation is equivalent to $h = 3\sqrt{113}$. It's given that the length of the triangle's hypotenuse, in centimeters, can be written in the form $3\sqrt{d}$. It follows that the value of d is 113.

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Test begins on the next page.

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Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

In the early 1800s, the Cherokee scholar Sequoyah created the first script, or writing system, for an Indigenous language in the United States. Because it represented the sounds of spoken Cherokee so accurately, his script was easy to learn and thus quickly achieved _____ use: by 1830, over 90 percent of the Cherokee people could read and write it.

Which choice completes the text with the most logical and precise word or phrase?

A) widespread
B) careful
C) unintended
D) infrequent

2

When Mexican-American archaeologist Zelia Maria Magdalena Nuttall published her 1886 research paper on sculptures found at the ancient Indigenous city of Teotihuacan in present-day Mexico, other researchers readily _____ her work as groundbreaking; this recognition stemmed from her convincing demonstration that the sculptures were much older than had previously been thought.

Which choice completes the text with the most logical and precise word or phrase?

A) acknowledged
B) ensured
C) denied
D) underestimated

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3

Like other tribal nations, the Muscogee (Creek) Nation is self-governing; its National Council generates laws regulating aspects of community life such as land use and healthcare, while the principal chief and cabinet officials _____ those laws by devising policies and administering services in accordance with them.

Which choice completes the text with the most logical and precise word or phrase?

A) implement
B) presume
C) improvise
D) mimic

4

In the Indigenous intercropping system known as the Three Sisters, maize, squash, and beans form an _____ web of relations: maize provides the structure on which the bean vines grow; the squash vines cover the soil, discouraging competition from weeds; and the beans aid their two "sisters" by enriching the soil with essential nitrogen.

Which choice completes the text with the most logical and precise word or phrase?

A) indecipherable
B) ornamental
C) obscure
D) intricate

5

The artisans of the Igun Eronmwon guild in Benin City, Nigeria, typically _____ the bronze- and brass-casting techniques that have been passed down through their families since the thirteenth century, but they don't strictly observe every tradition; for example, guild members now use air-conditioning motors instead of handheld bellows to help heat their forges.

Which choice completes the text with the most logical and precise word or phrase?

A) experiment with
B) adhere to
C) improve on
D) grapple with

6

Some economic historians _____ that late nineteenth- and early twentieth-century households in the United States experienced an economy of scale when it came to food purchases—they assumed that large households spent less on food per person than did small households. Economist Trevon Logan showed, however, that a close look at the available data disproves this supposition.

Which choice completes the text with the most logical and precise word or phrase?

A) surmised
B) contrived
C) questioned
D) regretted

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7

The work of Kiowa painter T.C. Cannon derives its power in part from the tension among his _____ influences: classic European portraiture, with its realistic treatment of faces; the American pop art movement, with its vivid colors; and flatstyle, the intertribal painting style that rejects the effect of depth typically achieved through shading and perspective.

Which choice completes the text with the most logical and precise word or phrase?

A) complementary
B) unknown
C) disparate
D) interchangeable

8

New and interesting research conducted by Suleiman A. Al-Sweedan and Moath Alhaj is inspired by their observation that though there have been many studies of the effect of high altitude on blood chemistry, there is a _____ studies of the effect on blood chemistry of living in locations below sea level, such as the California towns of Salton City and Seeley.

Which choice completes the text with the most logical and precise word or phrase?

A) quarrel about
B) paucity of
C) profusion of
D) verisimilitude in

9

The following text is from Sarah Orne Jewett's 1899 short story "Martha's Lady." Martha is employed by Miss Pyne as a maid.

> Miss Pyne sat by the window watching, in her best dress, looking stately and calm; she seldom went out now, and it was almost time for the carriage. Martha was just coming in from the garden with the strawberries, and with more flowers in her apron. It was a bright cool evening in June, the golden robins sang in the elms, and the sun was going down behind the apple-trees at the foot of the garden. The beautiful old house stood wide open to the long-expected guest.

Which choice best states the main purpose of the text?

A) To convey the worries brought about by a new guest
B) To describe how the characters have changed over time
C) To contrast the activity indoors with the stillness outside
D) To depict the setting as the characters await a visitor's arrival

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10

Astronomers are confident that the star Betelgeuse will eventually consume all the helium in its core and explode in a supernova. They are much less confident, however, about when this will happen, since that depends on internal characteristics of Betelgeuse that are largely unknown. Astrophysicist Sarafina El-Badry Nance and colleagues recently investigated whether acoustic waves in the star could be used to determine internal stellar states but concluded that this method could not sufficiently reveal Betelgeuse's internal characteristics to allow its evolutionary state to be firmly fixed.

Which choice best describes the function of the second sentence in the overall structure of the text?

A) It explains how the work of Nance and colleagues was received by others in the field.

B) It presents the central finding reported by Nance and colleagues.

C) It identifies the problem that Nance and colleagues attempted to solve but did not.

D) It describes a serious limitation of the method used by Nance and colleagues.

11

The following text is from Jane Austen's 1811 novel *Sense and Sensibility*. Elinor lives with her younger sisters and her mother, Mrs. Dashwood.

> Elinor, this eldest daughter, whose advice was so effectual, possessed a strength of understanding, and coolness of judgment, which qualified her, though only nineteen, to be the counsellor of her mother, and enabled her frequently to counteract, to the advantage of them all, that eagerness of mind in Mrs. Dashwood which must generally have led to imprudence. She had an excellent heart;—her disposition was affectionate, and her feelings were strong; but she knew how to govern them: it was a knowledge which her mother had yet to learn; and which one of her sisters had resolved never to be taught.

According to the text, what is true about Elinor?

A) Elinor often argues with her mother but fails to change her mind.

B) Elinor can be overly sensitive with regard to family matters.

C) Elinor thinks her mother is a bad role model.

D) Elinor is remarkably mature for her age.

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12

Believing that living in an impractical space can heighten awareness and even improve health, conceptual artists Madeline Gins and Shusaku Arakawa designed an apartment building in Japan to be more fanciful than functional. A kitchen counter is chest-high on one side and knee-high on the other; a ceiling has a door to nowhere. The effect is disorienting but invigorating: after four years there, filmmaker Nobu Yamaoka reported significant health benefits.

Which choice best states the main idea of the text?

A) Although inhabiting a home surrounded by fanciful features such as those designed by Gins and Arakawa can be rejuvenating, it is unsustainable.

B) Designing disorienting spaces like those in the Gins and Arakawa building is the most effective way to create a physically stimulating environment.

C) As a filmmaker, Yamaoka has long supported the designs of conceptual artists such as Gins and Arakawa.

D) Although impractical, the design of the apartment building by Gins and Arakawa may improve the well-being of the building's residents.

13

In a research paper, a student criticizes some historians of modern African politics, claiming that they have evaluated Patrice Lumumba, the first prime minister of what is now the Democratic Republic of the Congo, primarily as a symbol rather than in terms of his actions.

Which quotation from a work by a historian would best illustrate the student's claim?

A) "Lumumba is a difficult figure to evaluate due to the starkly conflicting opinions he inspired during his life and continues to inspire today."

B) "The available information makes it clear that Lumumba's political beliefs and values were largely consistent throughout his career."

C) "Lumumba's practical accomplishments can be passed over quickly; it is mainly as the personification of Congolese independence that he warrants scholarly attention."

D) "Many questions remain about Lumumba's ultimate vision for an independent Congo; without new evidence coming to light, these questions are likely to remain unanswered."

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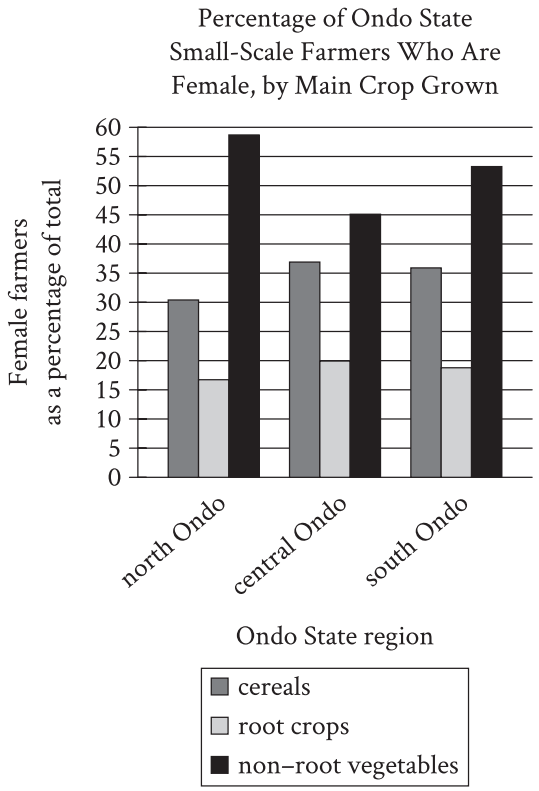
14



Geographer Adebayo Oluwole Eludoyin and his colleagues surveyed small-scale farmers in three locations in Ondo State, Nigeria—which has mountainous terrain in the north, an urbanized center, and coastal terrain in the south—to learn more about their practices, like the types of crops they mainly cultivated. In some regions, female farmers were found to be especially prominent in the cultivation of specific types of crops and even constituted the majority of farmers who cultivated those crops; for instance, _____

Which choice most effectively uses data from the graph to complete the example?

A) most of the farmers who mainly cultivated cereals and most of the farmers who mainly cultivated non–root vegetables in south Ondo were women.

B) more women in central Ondo mainly cultivated root crops than mainly cultivated cereals.

C) most of the farmers who mainly cultivated non–root vegetables in north and south Ondo were women.

D) a relatively equal proportion of women across the three regions of Ondo mainly cultivated cereals.

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15

Given that stars and planets initially form from the same gas and dust in space, some astronomers have posited that host stars (such as the Sun) and their planets (such as those in our solar system) are composed of the same materials, with the planets containing equal or smaller quantities of the materials that make up the host star. This idea is also supported by evidence that rocky planets in our solar system are composed of some of the same materials as the Sun.

Which finding, if true, would most directly weaken the astronomers' claim?

A) Most stars are made of hydrogen and helium, but when cooled they are revealed to contain small amounts of iron and silicate.

B) A nearby host star is observed to contain the same proportion of hydrogen and helium as that of the Sun.

C) Evidence emerges that the amount of iron in some rocky planets is considerably higher than the amount in their host star.

D) The method for determining the composition of rocky planets is discovered to be less effective when used to analyze other kinds of planets.

16

In the twentieth century, ethnographers made a concerted effort to collect Mexican American folklore, but they did not always agree about that folklore's origins. Scholars such as Aurelio Espinosa claimed that Mexican American folklore derived largely from the folklore of Spain, which ruled Mexico and what is now the southwestern United States from the sixteenth to early nineteenth centuries. Scholars such as Américo Paredes, by contrast, argued that while some Spanish influence is undeniable, Mexican American folklore is mainly the product of the ongoing interactions of various cultures in Mexico and the United States.

Which finding, if true, would most directly support Paredes's argument?

A) The folklore that the ethnographers collected included several songs written in the form of a *décima*, a type of poem originating in late sixteenth-century Spain.

B) Much of the folklore that the ethnographers collected had similar elements from region to region.

C) Most of the folklore that the ethnographers collected was previously unknown to scholars.

D) Most of the folklore that the ethnographers collected consisted of *corridos*—ballads about history and social life—of a clearly recent origin.

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17

In the early nineteenth century, some Euro-American farmers in the northeastern United States used agricultural techniques developed by the Haudenosaunee (Iroquois) people centuries earlier, but it seems that few of those farmers had actually seen Haudenosaunee farms firsthand. Barring the possibility of several farmers of the same era independently developing techniques that the Haudenosaunee people had already invented, these facts most strongly suggest that _____

Which choice most logically completes the text?

A) those farmers learned the techniques from other people who were more directly influenced by Haudenosaunee practices.

B) the crops typically cultivated by Euro-American farmers in the northeastern United States were not well suited to Haudenosaunee farming techniques.

C) Haudenosaunee farming techniques were widely used in regions outside the northeastern United States.

D) Euro-American farmers only began to recognize the benefits of Haudenosaunee farming techniques late in the nineteenth century.

18

If some artifacts recovered from excavations of the settlement of Kuulo Kataa, in modern Ghana, date from the thirteenth century CE, that may lend credence to claims that the settlement was founded before or around that time. There is other evidence, however, strongly supporting a fourteenth century CE founding date for Kuulo Kataa. If both the artifact dates and the fourteenth century CE founding date are correct, that would imply that _____

Which choice most logically completes the text?

A) artifacts from the fourteenth century CE are more commonly recovered than are artifacts from the thirteenth century CE.

B) the artifacts originated elsewhere and eventually reached Kuulo Kataa through trade or migration.

C) Kuulo Kataa was founded by people from a different region than had previously been assumed.

D) excavations at Kuulo Kataa may have inadvertently damaged some artifacts dating to the fourteenth century CE.

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19

A team of biologists led by Jae-Hoon Jung, Antonio D. Barbosa, and Stephanie Hutin investigated the mechanism that allows *Arabidopsis thaliana* (thale cress) plants to accelerate flowering at high temperatures. They replaced the protein ELF3 in the plants with a similar protein found in another species (stiff brome) that, unlike *A. thaliana*, displays no acceleration in flowering with increased temperature. A comparison of unmodified *A. thaliana* plants with the altered plants showed no difference in flowering at 22° Celsius, but at 27° Celsius, the unmodified plants exhibited accelerated flowering while the altered ones did not, which suggests that _____

Which choice most logically completes the text?

A) temperature-sensitive accelerated flowering is unique to *A. thaliana*.

B) *A. thaliana* increases ELF3 production as temperatures rise.

C) ELF3 enables *A. thaliana* to respond to increased temperatures.

D) temperatures of at least 22° Celsius are required for *A. thaliana* to flower.

20

A member of the Cherokee Nation, Mary Golda Ross is renowned for her contributions to NASA's Planetary Flight Handbook, which _____ detailed mathematical guidance for missions to Mars and Venus.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) provided

B) having provided

C) to provide

D) providing

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21

Typically, underlines, scribbles, and notes left in the margins by a former owner lower a book's _____ when the former owner is a famous poet like Walt Whitman, such markings, known as marginalia, can be a gold mine to literary scholars.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) value, but
B) value
C) value,
D) value but

22

After the United Kingdom began rolling out taxes equivalent to a few cents on single-use plastic grocery bags in 2011, plastic-bag consumption decreased by up to ninety _____ taxes are subject to what economists call the "rebound effect": as the change became normalized, plastic-bag use started to creep back up.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) percent, such
B) percent and such
C) percent. Such
D) percent such

23

As British scientist Peter Whibberley has observed, "the Earth is not a very good timekeeper." Earth's slightly irregular rotation rate means that measurements of time must be periodically adjusted. Specifically, an extra "leap second" (the 86,401st second of the day) is _____ time based on the planet's rotation lags a full nine-tenths of a second behind time kept by precise atomic clocks.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) added, whenever
B) added; whenever
C) added. Whenever
D) added whenever

24

Bengali author Toru Dutt's *A Sheaf Gleaned in French Fields* (1876), a volume of English translations of French poems, _____ scholars' understanding of the transnational and multilingual contexts in which Dutt lived and worked.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) has enhanced
B) are enhancing
C) have enhanced
D) enhance

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25

Journalists have dubbed Gil Scott-Heron the "godfather of rap," a title that has appeared in hundreds of articles about him since the 1990s. Scott-Heron himself resisted the godfather _____ feeling that it didn't encapsulate his devotion to the broader African American blues music tradition as well as "bluesologist," the moniker he preferred.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) nickname, however
B) nickname, however;
C) nickname, however,
D) nickname; however,

26

From afar, African American fiber artist Bisa Butler's portraits look like paintings, their depictions of human faces, bodies, and clothing so intricate that it seems only a fine brush could have rendered them. When viewed up close, however, the portraits reveal themselves to be _____ stitching barely visible among the thousands of pieces of printed, microcut fabric.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) quilts, and the
B) quilts, the
C) quilts; the
D) quilts. The

27

Most conifers (trees belonging to the phylum Coniferophyta) are evergreen. That is, they keep their green leaves or needles year-round. However, not all conifer species are evergreen. Larch trees, _____ lose their needles every fall.

Which choice completes the text with the most logical transition?

A) for instance,
B) nevertheless,
C) meanwhile,
D) in addition,

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28

While researching a topic, a student has taken the following notes:

- Sam Maloof (1916–2009) was an American woodworker and furniture designer.
- He was the son of Lebanese immigrants.
- He received a "genius grant" from the John D. and Catherine T. MacArthur Foundation in 1985.
- The Museum of Fine Arts in Boston, Massachusetts, owns a rocking chair that Maloof made from walnut wood.
- The armrests and the seat of the chair are sleek and contoured, and the back consists of seven spindle-like slats.

The student wants to describe the rocking chair to an audience unfamiliar with Sam Maloof. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) With its sleek, contoured armrests and seat, the walnut rocking chair in Boston's Museum of Fine Arts is just one piece of furniture created by American woodworker Sam Maloof.

B) Sam Maloof was born in 1916 and died in 2009, and during his life, he made a chair that you can see if you visit the Museum of Fine Arts in Boston.

C) Furniture designer Sam Maloof was a recipient of one of the John D. and Catherine T. MacArthur Foundation's "genius grants."

D) The rocking chair is made from walnut, and it has been shaped such that its armrests and seat are sleek and contoured.

29

While researching a topic, a student has taken the following notes:

- In the late 1890s, over 14,000 unique varieties of apples were grown in the US.
- The rise of industrial agriculture in the mid-1900s narrowed the range of commercially grown crops.
- Thousands of apple varieties considered less suitable for commercial growth were lost.
- Today, only 15 apple varieties dominate the market, making up 90% of apples purchased in the US.
- The Lost Apple Project, based in Washington State, attempts to find and grow lost apple varieties.

The student wants to emphasize the decline in unique apple varieties in the US and specify why this decline occurred. Which choice most effectively uses relevant information from the notes to accomplish these goals?

A) The Lost Apple Project is dedicated to finding some of the apple varieties lost following a shift in agricultural practices in the mid-1900s.

B) While over 14,000 apple varieties were grown in the US in the late 1890s, only 15 unique varieties make up most of the apples sold today.

C) Since the rise of industrial agriculture, US farmers have mainly grown the same few unique apple varieties, resulting in the loss of thousands of varieties less suitable for commercial growth.

D) As industrial agriculture rose to prominence in the mid-1900s, the number of crops selected for cultivation decreased dramatically.

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30

While researching a topic, a student has taken the following notes:

- Cecilia Vicuña is a multidisciplinary artist.
- In 1971, her first solo art exhibition, *Pinturas, poemas y explicaciones,* was shown at the Museo Nacional de Bellas Artes in Santiago, Chile.
- Her poetry collection *Precario/Precarious* was published in 1983 by Tanam Press.
- Her poetry collection *Instan* was published in 2002 by Kelsey St. Press.
- She lives part time in Chile, where she was born, and part time in New York.

The student wants to introduce the artist's 1983 poetry collection. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Before she published the books *Precario/Precarious* (1983) and *Instan* (2002), Cecilia Vicuña exhibited visual art at the Museo Nacional de Bellas Artes in Santiago, Chile.

B) Cecilia Vicuña is a true multidisciplinary artist whose works include numerous poetry collections and visual art exhibitions.

C) Published in 1983 by Tanam Press, *Precario/Precarious* is a collection of poetry by the multidisciplinary artist Cecilia Vicuña.

D) In 1971, Cecilia Vicuña exhibited her first solo art exhibition, *Pinturas, poemas y explicaciones*, in Chile, her country of birth.

31

While researching a topic, a student has taken the following notes:

- When medical students mention their patients on social media, they may violate patient confidentiality.
- Terry Kind led a study to determine how many medical schools have student policies that mention social media use.
- Kind and her team reviewed 132 medical school websites, examining publicly available student policies.
- Only thirteen medical schools had guidelines that explicitly mention social media, and only five defined what constitutes acceptable social media use.

The student wants to emphasize the study's methodology. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The student policies of 132 medical schools can be found online, according to research by Terry Kind.

B) To find out how many medical schools have guidelines about student social media use, Terry Kind and her team examined the student policies of 132 medical schools.

C) Out of 132 medical schools, only thirteen had student policies that mentioned social media, and only five specified what use was acceptable.

D) Terry Kind and her team wanted to know how many medical schools have student social media policies in place about protecting patient confidentiality.

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32

While researching a topic, a student has taken the following notes:

- The Gullah are a group of African Americans who have lived in parts of the southeastern United States since the 18th century.
- Gullah culture is influenced by West African and Central African traditions.
- Louise Miller Cohen is a Gullah historian, storyteller, and preservationist.
- She founded the Gullah Museum of Hilton Head Island, South Carolina, in 2003.
- Vermelle Rodrigues is a Gullah historian, artist, and preservationist.
- She founded the Gullah Museum of Georgetown, South Carolina, in 2003.

The student wants to emphasize the duration and purpose of Cohen's and Rodrigues's work. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) At the Gullah Museums in Hilton Head Island and Georgetown, South Carolina, visitors can learn more about the Gullah people who have lived in the region for centuries.

B) Louise Miller Cohen and Vermelle Rodrigues have worked to preserve the culture of the Gullah people, who have lived in the United States since the 18th century.

C) Since 2003, Louise Miller Cohen and Vermelle Rodrigues have worked to preserve Gullah culture through their museums.

D) Influenced by the traditions of West and Central Africa, Gullah culture developed in parts of the southeastern United States in the 18th century.

33

While researching a topic, a student has taken the following notes:

- In North America, woodlands have expanded into areas that were once grasslands.
- Thomas Rogers and F. Leland Russell of Wichita State University investigated whether woodland expansion is related to changes in climate.
- Rogers and Russell analyzed core samples from oak trees on a site that was not wooded in the past and indexed the age of the trees with historical climate data to see if tree populations and climate were correlated.
- Tree population growth was associated with dry intervals.
- Droughts may have played a role in woodland expansion.

The student wants to emphasize the aim of the research study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Thomas Rogers and F. Leland Russell, researchers at Wichita State University, wanted to know if woodland expansion is related to changes in climate.

B) Thanks to the work done by Thomas Rogers and F. Leland Russell, we now know that droughts may have played a role in woodland expansion.

C) Wichita State University researchers have determined that tree population growth was associated with dry intervals.

D) Thomas Rogers and F. Leland Russell analyzed core samples from oak trees on a site that was not wooded in the past, indexing the age of the trees with historical climate data.

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

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Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

According to botanists, a viburnum plant experiencing insect damage may develop erineum—a discolored, felty growth—on its leaf blades. A _____ viburnum plant, on the other hand, will have leaves with smooth surfaces and uniformly green coloration.

Which choice completes the text with the most logical and precise word or phrase?

A) struggling
B) beneficial
C) simple
D) healthy

2

Nigerian American author Teju Cole's _____ his two passions—photography and the written word—culminates in his 2017 book, *Blind Spot*, which evocatively combines his original photographs from his travels with his poetic prose.

Which choice completes the text with the most logical and precise word or phrase?

A) indifference to
B) enthusiasm for
C) concern about
D) surprise at

3

Novelist N. K. Jemisin declines to _____ the conventions of the science fiction genre in which she writes, and she has suggested that her readers appreciate her work precisely because of this willingness to thwart expectations and avoid formulaic plots and themes.

Which choice completes the text with the most logical and precise word or phrase?

A) question
B) react to
C) perceive
D) conform to

4

In *Nature Poem* (2017), Kumeyaay poet Tommy Pico portrays his _____ the natural world by honoring the centrality of nature within his tribe's traditional beliefs while simultaneously expressing his distaste for being in wilderness settings himself.

Which choice completes the text with the most logical and precise word or phrase?

A) responsiveness to
B) ambivalence toward
C) renunciation of
D) mastery over

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CONTINUE

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5

The following text is from the 1924 poem "Cycle" by D'Arcy McNickle, who was a citizen of the Confederated Salish and Kootenai Tribes.

There shall be new roads wending,
A new beating of the drum—

Men's eyes shall have fresh seeing,
Grey lives reprise their span—
But under the new sun's being,
Completing what night began,

There'll be the same backs bending,
The same sad feet shall drum—
When this night finds its ending
And day shall have come.....

Which choice best states the main purpose of the text?

A) To consider how the repetitiveness inherent in human life can be both rewarding and challenging

B) To question whether activities completed at one time of day are more memorable than those completed at another time of day

C) To refute the idea that joy is a more commonly experienced emotion than sadness is

D) To demonstrate how the experiences of individuals relate to the experiences of their communities

6

The following text is adapted from Jane Austen's 1814 novel *Mansfield Park*. The speaker, Tom, is considering staging a play at home with a group of his friends and family.

> We mean nothing but a little amusement among ourselves, just to vary the scene, and exercise our powers in something new. We want no audience, no publicity. We may be trusted, I think, in choosing some play most perfectly unexceptionable; and I can conceive no greater harm or danger to any of us in conversing in the elegant written language of some respectable author than in chattering in words of our own.

Which choice best states the main purpose of the text?

A) To offer Tom's assurance that the play will be inoffensive and involve only a small number of people

B) To clarify that the play will not be performed in the manner Tom had originally intended

C) To elaborate on the idea that the people around Tom lack the skills to successfully stage a play

D) To assert that Tom believes the group performing the play will be able to successfully promote it

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CONTINUE

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7

Musician Joni Mitchell, who is also a painter, uses images she creates for her album covers to emphasize ideas expressed in her music. For the cover of her album *Turbulent Indigo* (1994), Mitchell painted a striking self-portrait that closely resembles Vincent van Gogh's *Self-Portrait with Bandaged Ear* (1889). The image calls attention to the album's title song, in which Mitchell sings about the legacy of the postimpressionist painter. In that song, Mitchell also hints that she feels a strong artistic connection to Van Gogh—an idea that is reinforced by her imagery on the cover.

Which choice best describes the overall structure of the text?

A) It presents a claim about Mitchell, then gives an example supporting that claim.

B) It discusses Van Gogh's influence on Mitchell, then considers Mitchell's influence on other artists.

C) It describes a similarity between two artists, then notes a difference between them.

D) It describes the songs on *Turbulent Indigo*, then explains how they relate to the album's cover.

8

Text 1

Astronomer Mark Holland and colleagues examined four white dwarfs—small, dense remnants of past stars—in order to determine the composition of exoplanets that used to orbit those stars. Studying wavelengths of light in the white dwarf atmospheres, the team reported that traces of elements such as lithium and sodium support the presence of exoplanets with continental crusts similar to Earth's.

Text 2

Past studies of white dwarf atmospheres have concluded that certain exoplanets had continental crusts. Geologist Keith Putirka and astronomer Siyi Xu argue that those studies unduly emphasize atmospheric traces of lithium and other individual elements as signifiers of the types of rock found on Earth. The studies don't adequately account for different minerals made up of various ratios of those elements, and the possibility of rock types not found on Earth that contain those minerals.

Based on the texts, how would Putirka and Xu (Text 2) most likely characterize the conclusion presented in Text 1?

A) As unexpected, because it was widely believed at the time that white dwarf exoplanets lack continental crusts

B) As premature, because researchers have only just begun trying to determine what kinds of crusts white dwarf exoplanets had

C) As questionable, because it rests on an incomplete consideration of potential sources of the elements detected in white dwarf atmospheres

D) As puzzling, because it's unusual to successfully detect lithium and sodium when analyzing wavelengths of light in white dwarf atmospheres

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9

Utah is home to Pando, a colony of about 47,000 quaking aspen trees that all share a single root system. Pando is one of the largest single organisms by mass on Earth, but ecologists are worried that its growth is declining in part because of grazing by animals. The ecologists say that strong fences could prevent deer from eating young trees and help Pando start thriving again.

According to the text, why are ecologists worried about Pando?

A) It isn't growing at the same rate it used to.

B) It isn't producing young trees anymore.

C) It can't grow into new areas because it is blocked by fences.

D) Its root system can't support many more new trees.

10

For many years, the only existing fossil evidence of mixopterid eurypterids—an extinct family of large aquatic arthropods known as sea scorpions and related to modern arachnids and horseshoe crabs—came from four species living on the paleocontinent of Laurussia. In a discovery that expands our understanding of the geographical distribution of mixopterids, paleontologist Bo Wang and others have identified fossilized remains of a new mixopterid species, *Terropterus xiushanensis*, that lived over 400 million years ago on the paleocontinent of Gondwana.

According to the text, why was Wang and his team's discovery of the *Terropterus xiushanensis* fossil significant?

A) The fossil constitutes the first evidence found by scientists that mixopterids lived more than 400 million years ago.

B) The fossil helps establish that mixopterids are more closely related to modern arachnids and horseshoe crabs than previously thought.

C) The fossil helps establish a more accurate timeline of the evolution of mixopterids on the paleocontinents of Laurussia and Gondwana.

D) The fossil constitutes the first evidence found by scientists that mixopterids existed outside the paleocontinent of Laurussia.

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CONTINUE

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11

The novelist Toni Morrison was the first Black woman to work as an editor at the publishing company Random House, from 1967 to 1983. A scholar asserts that one of Morrison's likely aims during her time as an editor was to strengthen the presence of Black writers on the list of Random House's published authors.

Which finding, if true, would most strongly support the scholar's claim?

A) The percentage of authors published by Random House who were Black rose in the early 1970s and stabilized throughout the decade.

B) Black authors who were interviewed in the 1980s and 1990s were highly likely to cite Toni Morrison's novels as a principal influence on their work.

C) The novels written by Toni Morrison that were published after 1983 sold significantly more copies and received wider critical acclaim than the novels she wrote that were published before 1983.

D) Works that were edited by Toni Morrison during her time at Random House displayed stylistic characteristics that distinguished them from works that were not edited by Morrison.

12

"The Poet Walt Whitman" is an 1887 essay by José Martí, a Cuban author and political activist, originally written in Spanish. In the essay, Martí explores the value of literature, arguing that a society's spiritual well-being depends on the character of its literary culture: _____

Which quotation from a translation of "The Poet Walt Whitman" most effectively illustrates the claim?

A) "Poetry, which brings together or separates, which fortifies or brings anguish, which shores up or demolishes souls, which gives or robs men of faith and vigor, is more necessary to a people than industry itself, for industry provides them with a means of subsistence, while literature gives them the desire and strength for life."

B) "Every society brings to literature its own form of expression, and the history of the nations can be told with greater truth by the stages of literature than by chronicles and decades."

C) "Where will a race of men go when they have lost the habit of thinking with faith about the scope and meaning of their actions? The best among them, those who consecrate Nature with their sacred desire for the future, will lose, in a sordid and painful annihilation, all stimulus to alleviate the ugliness of humanity."

D) "Listen to the song of this hardworking and satisfied nation; listen to Walt Whitman. The exercise of himself exalts him to majesty, tolerance exalts him to justice, and order to joy."

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13

Estimates of Tyrannosaurid Bite Force

Study	Year	Estimation method	Approximate bite force (newtons)
Cost et al.	2019	muscular and skeletal modeling	35,000–63,000
Gignac and Erickson	2017	tooth-bone interaction analysis	8,000–34,000
Meers	2002	body-mass scaling	183,000–235,000
Bates and Falkingham	2012	muscular and skeletal modeling	35,000–57,000

The largest tyrannosaurids—the family of carnivorous dinosaurs that includes *Tarbosaurus*, *Albertosaurus*, and, most famously, *Tyrannosaurus rex*—are thought to have had the strongest bites of any land animals in Earth's history. Determining the bite force of extinct animals can be difficult, however, and paleontologists Paul Barrett and Emily Rayfield have suggested that an estimate of dinosaur bite force may be significantly influenced by the methodology used in generating that estimate.

Which choice best describes data from the table that support Barrett and Rayfield's suggestion?

A) The study by Meers used body-mass scaling and produced the lowest estimated maximum bite force, while the study by Cost et al. used muscular and skeletal modeling and produced the highest estimated maximum.

B) In their study, Gignac and Erickson used tooth-bone interaction analysis to produce an estimated bite force range with a minimum of 8,000 newtons and a maximum of 34,000 newtons.

C) The bite force estimates produced by Bates and Falkingham and by Cost et al. were similar to each other, while the estimates produced by Meers and by Gignac and Erickson each differed substantially from any other estimate.

D) The estimated maximum bite force produced by Cost et al. exceeded the estimated maximum produced by Bates and Falkingham, even though both groups of researchers used the same method to generate their estimates.

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14

Number and Origin of Clamshell Tools Found at Different Levels Below the Surface in Neanderthal Cave

Depth of tools found below surface in cave (meters)	Clamshells that Neanderthals collected from the beach	Clamshells that Neanderthals harvested from the seafloor
3–4	99	33
6–7	1	0
4–5	2	0
2–3	7	0
5–6	18	7

Studying tools unearthed at a cave site on the western coast of Italy, archaeologist Paola Villa and colleagues have determined that prehistoric Neanderthal groups fashioned them from shells of clams that they harvested from the seafloor while wading or diving or that washed up on the beach. Clamshells become thin and eroded as they wash up on the beach, while those on the seafloor are smooth and sturdy, so the research team suspects that Neanderthals prized the tools made with seafloor shells. However, the team also concluded that those tools were likely more challenging to obtain, noting that _____

Which choice most effectively uses data from the table to support the research team's conclusion?

A) at each depth below the surface in the cave, the difference in the numbers of tools of each type suggests that shells were easier to collect from the beach than to harvest from the seafloor.

B) the highest number of tools were at a depth of 3–4 meters below the surface, which suggests that the Neanderthal population at the site was highest during the related period of time.

C) at each depth below the surface in the cave, the difference in the numbers of tools of each type suggests that Neanderthals preferred to use clamshells from the beach because of their durability.

D) the higher number of tools at depths of 5–6 meters below the surface in the cave than at depths of 4–5 meters below the surface suggests that the size of clam populations changed over time.

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15

Average Number and Duration of Torpor Bouts and Arousal Episodes for Alaska Marmots and Arctic Ground Squirrels, 2008–2011

Feature	Alaska marmots	Arctic ground squirrels
torpor bouts	12	10.5
duration per bout	13.81 days	16.77 days
arousal episodes	11	9.5
duration per episode	21.2 hours	14.2 hours

When hibernating, Alaska marmots and Arctic ground squirrels enter a state called torpor, which minimizes the energy their bodies need to function. Often a hibernating animal will temporarily come out of torpor (called an arousal episode) and its metabolic rate will rise, burning more of the precious energy the animal needs to survive the winter. Alaska marmots hibernate in groups and therefore burn less energy keeping warm during these episodes than they would if they were alone. A researcher hypothesized that because Arctic ground squirrels hibernate alone, they would likely exhibit longer bouts of torpor and shorter arousal episodes than Alaska marmots.

Which choice best describes data from the table that support the researcher's hypothesis?

A) The Alaska marmots' arousal episodes lasted for days, while the Arctic ground squirrels' arousal episodes lasted less than a day.

B) The Alaska marmots and the Arctic ground squirrels both maintained torpor for several consecutive days per bout, on average.

C) The Alaska marmots had shorter torpor bouts and longer arousal episodes than the Arctic ground squirrels did.

D) The Alaska marmots had more torpor bouts than arousal episodes, but their arousal episodes were much shorter than their torpor bouts.

16

Ratified by more than 90 countries, the Nagoya Protocol is an international agreement ensuring that Indigenous communities are compensated when their agricultural resources and knowledge of wild plants and animals are utilized by agricultural corporations. However, the protocol has shortcomings. For example, it allows corporations to insist that their agreements with communities to conduct research on the commercial uses of the communities' resources and knowledge remain confidential. Therefore, some Indigenous advocates express concern that the protocol may have the unintended effect of _____

Which choice most logically completes the text?

A) diminishing the monetary reward that corporations might derive from their agreements with Indigenous communities.

B) limiting the research that corporations conduct on the resources of the Indigenous communities with which they have signed agreements.

C) preventing independent observers from determining whether the agreements guarantee equitable compensation for Indigenous communities.

D) discouraging Indigenous communities from learning new methods for harvesting plants and animals from their corporate partners.

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17

The domestic sweet potato (*Ipomoea batatas*) descends from a wild plant native to South America. It also populates the Polynesian Islands, where evidence confirms that Native Hawaiians and other Indigenous peoples were cultivating the plant centuries before seafaring first occurred over the thousands of miles of ocean separating them from South America. To explain how the sweet potato was first introduced in Polynesia, botanist Pablo Muñoz-Rodríguez and colleagues analyzed the DNA of numerous varieties of the plant, concluding that Polynesian varieties diverged from South American ones over 100,000 years ago. Given that Polynesia was peopled only in the last three thousand years, the team concluded that _____

Which choice most logically completes the text?

A) the cultivation of the sweet potato in Polynesia likely predates its cultivation in South America.

B) Polynesian peoples likely acquired the sweet potato from South American peoples only within the last three thousand years.

C) human activity likely played no role in the introduction of the sweet potato in Polynesia.

D) Polynesian sweet potato varieties likely descend from a single South American variety that was domesticated, not wild.

18

Atoms in a synchrotron, a type of circular particle accelerator, travel faster and faster until they _____ a desired energy level, at which point they are diverted to collide with a target, smashing the atoms.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) will reach

B) reach

C) had reached

D) are reaching

19

Even though bats prefer very sweet nectar, the plants that attract them have evolved to produce nectar that is only moderately sweet. A recent study _____ why: making sugar is energy-intensive, and it is more advantageous for plants to make a large amount of low-sugar nectar than a small amount of high-sugar nectar.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) explains

B) explaining

C) having explained

D) to explain

20

Former First Lady of the United States Eleanor Roosevelt and Indian activist and educator Hansa Mehta were instrumental in drafting the United Nations' Universal Declaration of Human Rights, a document that _____ the basic freedoms to which all people are entitled.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) have outlined

B) were outlining

C) outlines

D) outline

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21

The life spans of rockfish vary greatly by species. For instance, the colorful calico rockfish (*Sebastes dalli*) can survive for a little over a _____ the rougheye rockfish (*Sebastes aleutianus*) boasts a maximum life span of about two centuries.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) decade: while
B) decade. While
C) decade; while
D) decade, while

22

The Lion Light system, developed by Kenyan inventor Richard Turere, consists of LED lights installed around the perimeter of livestock pastures. Powered with _____ the blinking LEDs keep lions away at night, thus protecting the livestock without risking harm to the endangered lions.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) energy collected, by solar panels, during the day
B) energy collected by solar panels during the day
C) energy collected by solar panels during the day,
D) energy, collected by solar panels during the day,

23

Materials scientist Marie-Agathe Charpagne and her colleagues believed they could improve on the multicomponent alloy NiCoCr, an equal-proportions mixture of nickel (Ni), cobalt (Co), and chromium (Cr), by replacing chromium with ruthenium _____ the alloy that resulted, NiCoRu, turned out to be an unsuitable replacement for NiCoCr.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) (Ru)
B) (Ru) but
C) (Ru),
D) (Ru), but

24

The Progressive Era in the United States witnessed the rise of numerous Black women's clubs, local organizations that advocated for racial and gender equality. Among the clubs' leaders _____ Josephine St. Pierre Ruffin, founder of the Women's Era Club of Boston.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) was
B) were
C) are
D) have been

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25

Based on genetic evidence, archaeologists have generally agreed that reindeer domestication began in the eleventh century CE. However, since uncovering fragments of a 2,000-year-old reindeer training harness in northern Siberia, _____ may have begun much earlier.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) researcher Robert Losey has argued that domestication
B) researcher Robert Losey's argument is that domestication
C) domestication, researcher Robert Losey has argued,
D) the argument researcher Robert Losey has made is that domestication

26

Hegra is an archaeological site in present-day Saudi Arabia and was the second largest city of the Nabataean Kingdom (fourth century BCE to first century CE). Archaeologist Laila Nehmé recently traveled to Hegra to study its ancient _____ into the rocky outcrops of a vast desert, these burial chambers seem to blend seamlessly with nature.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) tombs. Built
B) tombs, built
C) tombs and built
D) tombs built

27

When external forces are applied to common glass made from silicates, energy builds up around minuscule defects in the material, resulting in fractures. Recently, engineer Erkka Frankberg of Tampere University in Finland used the chemical _____ to make a glassy solid that can withstand higher strain than silicate glass can before fracturing.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) compound, aluminum oxide
B) compound aluminum oxide,
C) compound, aluminum oxide,
D) compound aluminum oxide

28

Etched into Peru's Nazca Desert are line drawings so large that they can only be fully seen from high above. Archaeologists have known of the lines since the 1920s, when a researcher spotted some from a nearby foothill, and they have been studying the markings ever since. _____ archaeologists' efforts are aided by drones that capture high-resolution aerial photographs of the lines.

Which choice completes the text with the most logical transition?

A) Currently,
B) In comparison,
C) Still,
D) However,

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29

Archaeologist Sue Brunning explains why the seventh-century ship burial site at Sutton Hoo in England was likely the tomb of a king. First, the gold artifacts inside the ship suggest that the person buried with them was a wealthy and respected leader. _____ the massive effort required to bury the ship would likely only have been undertaken for a king.

Which choice completes the text with the most logical transition?

A) Instead,

B) Still,

C) Specifically,

D) Second,

30

The more diverse and wide ranging an animal's behaviors, the larger and more energy demanding the animal's brain tends to be. _____ from an evolutionary perspective, animals that perform only basic actions should allocate fewer resources to growing and maintaining brain tissue. The specialized subtypes of ants within colonies provide an opportunity to explore this hypothesis.

Which choice completes the text with the most logical transition?

A) Subsequently,

B) Besides,

C) Nevertheless,

D) Thus,

31

When designing costumes for film, American artist Suttirat Larlarb typically custom fits the garments to each actor. _____ for the film *Sunshine*, in which astronauts must reignite a dying Sun, she designed a golden spacesuit and had a factory reproduce it in a few standard sizes; lacking a tailor-made quality, the final creations reflected the ungainliness of actual spacesuits.

Which choice completes the text with the most logical transition?

A) Nevertheless,

B) Thus,

C) Likewise,

D) Moreover,

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32

While researching a topic, a student has taken the following notes:

- Shaun Tan is an Australian author.
- In 2008, he published *Tales from Outer Suburbia*, a book of fifteen short stories.
- The stories describe surreal events occurring in otherwise ordinary suburban neighborhoods.
- In 2018, he published *Tales from the Inner City*, a book of twenty-five short stories.
- The stories describe surreal events occurring in otherwise ordinary urban settings.

The student wants to emphasize a similarity between the two books by Shaun Tan. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Shaun Tan's book *Tales from Outer Suburbia*, which describes surreal events occurring in otherwise ordinary places, contains fewer short stories than *Tales from the Inner City* does.

B) *Tales from Outer Suburbia* was published in 2008, and *Tales from the Inner City* was published in 2018.

C) Unlike *Tales from the Inner City*, Shaun Tan's book *Tales from Outer Suburbia* is set in suburban neighborhoods.

D) Shaun Tan's books *Tales from Outer Suburbia* and *Tales from the Inner City* both describe surreal events occurring in otherwise ordinary places.

33

While researching a topic, a student has taken the following notes:

- The factors that affect clutch size (the number of eggs laid at one time) have been well studied in birds but not in lizards.
- A team led by Shai Meiri of Tel Aviv University investigated which factors influence lizard clutch size.
- Meiri's team obtained clutch-size and habitat data for over 3,900 lizard species and analyzed the data with statistical models.
- Larger clutch size was associated with environments in higher latitudes that have more seasonal change.
- Lizards in higher-latitude environments may lay larger clutches to take advantage of shorter windows of favorable conditions.

The student wants to emphasize the aim of the research study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Researchers wanted to know which factors influence lizard egg clutch size because such factors have been well studied in birds but not in lizards.

B) After they obtained data for over 3,900 lizard species, researchers determined that larger clutch size was associated with environments in higher latitudes that have more seasonal change.

C) We now know that lizards in higher-latitude environments may lay larger clutches to take advantage of shorter windows of favorable conditions.

D) Researchers obtained clutch-size and habitat data for over 3,900 lizard species and analyzed the data with statistical models.

STOP

If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.

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Math

27 QUESTIONS

DIRECTIONS

The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

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- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

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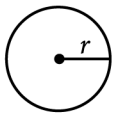
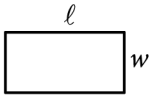
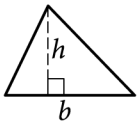


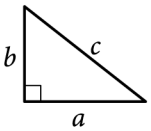
$A = \pi r^2$
$C = 2\pi r$



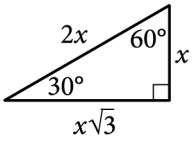
$A = \ell w$

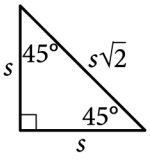


$A = \frac{1}{2}bh$



$c^2 = a^2 + b^2$





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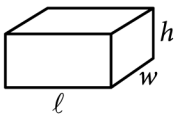
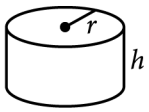
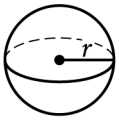


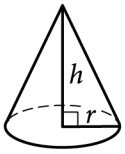
$V = \ell wh$



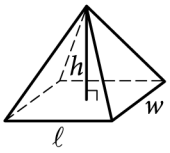
$V = \pi r^2 h$



$V = \frac{4}{3}\pi r^3$



$V = \frac{1}{3}\pi r^2 h$



$V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π.

The sum of the measures in degrees of the angles of a triangle is 180.

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For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find **more than one correct answer**, write and circle only one answer.
- Your answer can be up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer, but no more.
- If your answer is a **fraction** that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a **decimal** that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

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1

$k + 12 = 336$

What is the solution to the given equation?

A) 28

B) 324

C) 348

D) 4,032

2

The function f is defined by $f(x) = x^3 + 15$. What is the value of $f(2)$?

A) 20

B) 21

C) 23

D) 24

3

Sean rents a tent at a cost of \$11 per day plus a onetime insurance fee of \$10. Which equation represents the total cost c, in dollars, to rent the tent with insurance for d days?

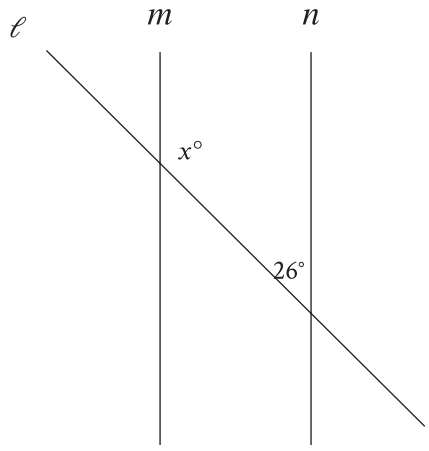
A) $c = 11(d + 10)$

B) $c = 10(d + 11)$

C) $c = 11d + 10$

D) $c = 10d + 11$

4



Note: Figure not drawn to scale.

In the figure shown, line m is parallel to line n. What is the value of x ?

A) 13

B) 26

C) 52

D) 154

5

John paid a total of \$165 for a microscope by making a down payment of \$37 plus p monthly payments of \$16 each. Which of the following equations represents this situation?

A) $16p - 37 = 165$

B) $37p - 16 = 165$

C) $16p + 37 = 165$

D) $37p + 16 = 165$

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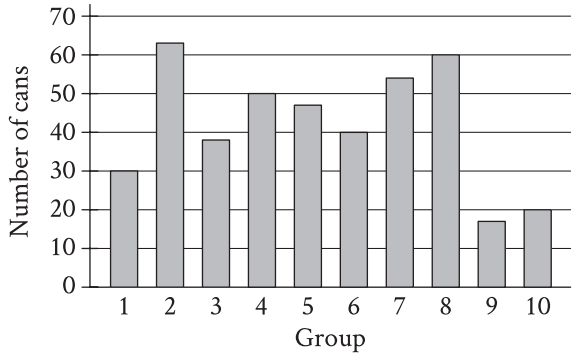
6

If $y = 5x + 10$, what is the value of y when $x = 8$?

7



The bar graph shows the distribution of 419 cans collected by 10 different groups for a food drive. How many cans were collected by group 6?

8

The table gives the distribution of votes for a new school mascot and grade level for 80 students.

Mascot	Grade level			
	Sixth	Seventh	Eighth	Total
Badger	4	9	9	22
Lion	9	2	9	20
Longhorn	4	6	4	14
Tiger	6	9	9	24
Total	23	26	31	80

If one of these students is selected at random, what is the probability of selecting a student whose vote for new mascot was for a lion?

A) $\frac{1}{9}$

B) $\frac{1}{5}$

C) $\frac{1}{4}$

D) $\frac{2}{3}$

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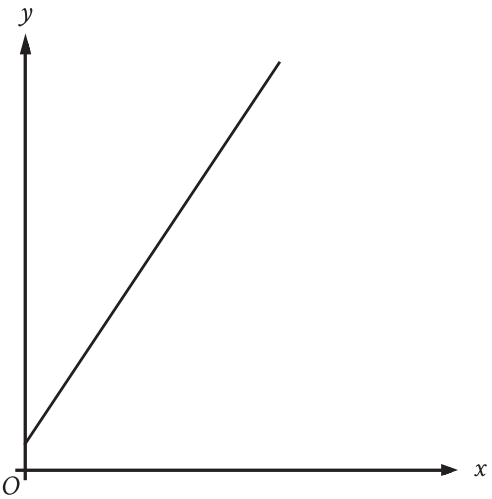
9



The graph represents the total charge, in dollars, by an electrician for x hours of work. The electrician charges a onetime fee plus an hourly rate. What is the best interpretation of the slope of the graph?

A) The electrician's hourly rate

B) The electrician's onetime fee

C) The maximum amount that the electrician charges

D) The total amount that the electrician charges

10

Square X has a side length of 12 centimeters. The perimeter of square Y is 2 times the perimeter of square X. What is the length, in centimeters, of one side of square Y?

A) 6

B) 10

C) 14

D) 24

11

What is the equation of the line that passes through the point (0, 5) and is parallel to the graph of $y = 7x + 4$ in the xy-plane?

A) $y = 5x$

B) $y = 7x + 5$

C) $y = 7x$

D) $y = 5x + 7$

12

In the linear function h, $h(0) = 41$ and $h(1) = 40$. Which equation defines h ?

A) $h(x) = -x + 41$

B) $h(x) = -x$

C) $h(x) = -41x$

D) $h(x) = -41$

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13

The function $f(t) = 60{,}000(2)^{\frac{t}{410}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

14

The function f is defined by $f(x) = (x - 6)(x - 2)(x + 6)$. In the xy-plane, the graph of $y = g(x)$ is the result of translating the graph of $y = f(x)$ up 4 units. What is the value of $g(0)$?

15

A candle is made of 17 ounces of wax. When the candle is burning, the amount of wax in the candle decreases by 1 ounce every 4 hours. If 6 ounces of wax remain in this candle, for how many hours has it been burning?

A) 3
B) 6
C) 24
D) 44

16

$$14j + 5k = m$$

The given equation relates the numbers j, k, and m. Which equation correctly expresses k in terms of j and m ?

A) $k = \frac{m - 14j}{5}$

B) $k = \frac{1}{5}m - 14j$

C) $k = \frac{14j - m}{5}$

D) $k = 5m - 14j$

17

Triangle FGH is similar to triangle JKL, where angle F corresponds to angle J and angles G and K are right angles. If $\sin(F) = \frac{308}{317}$, what is the value of $\sin(J)$?

A) $\frac{75}{317}$
B) $\frac{308}{317}$
C) $\frac{317}{308}$
D) $\frac{317}{75}$

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18

The product of two positive integers is 546. If the first integer is 11 greater than twice the second integer, what is the smaller of the two integers?

A) 7
B) 14
C) 39
D) 78

19

$$y \leq x + 7$$
$$y \geq -2x - 1$$

Which point (x, y) is a solution to the given system of inequalities in the xy-plane?

A) $(-14, 0)$
B) $(0, -14)$
C) $(0, 14)$
D) $(14, 0)$

20

$$\sqrt{(x-2)^2} = \sqrt{3x + 34}$$

What is the smallest solution to the given equation?

21

The regular price of a shirt at a store is \$11.70. The sale price of the shirt is 80% less than the regular price, and the sale price is 30% greater than the store's cost for the shirt. What was the store's cost, in dollars, for the shirt?

22

A sample of oak has a density of 807 kilograms per cubic meter. The sample is in the shape of a cube, where each edge has a length of 0.90 meters. To the nearest whole number, what is the mass, in kilograms, of this sample?

A) 588
B) 726
C) 897
D) 1,107

23

For $x > 0$, the function f is defined as follows:

$f(x)$ equals 201% of x

Which of the following could describe this function?

A) Decreasing exponential
B) Decreasing linear
C) Increasing exponential
D) Increasing linear

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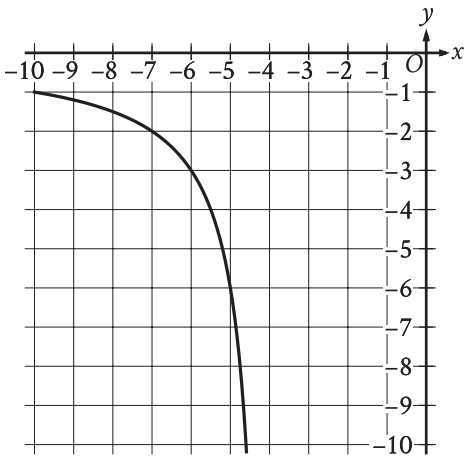
24



The rational function f is defined by an equation in the form $f(x) = \frac{a}{x + b}$, where a and b are constants. The partial graph of $y = f(x)$ is shown. If $g(x) = f(x + 4)$, which equation could define function g ?

A) $g(x) = \frac{6}{x}$

B) $g(x) = \frac{6}{x + 4}$

C) $g(x) = \frac{6}{x + 8}$

D) $g(x) = \frac{6(x + 4)}{x + 4}$

25

Which expression is equivalent to $\frac{y + 12}{x - 8} + \frac{y(x - 8)}{x^2y - 8xy}$?

A) $\frac{xy + y + 4}{x^3y - 16x^2y + 64xy}$

B) $\frac{xy + 9y + 12}{x^2y - 8xy + x - 8}$

C) $\frac{xy^2 + 13xy - 8y}{x^2y - 8xy}$

D) $\frac{xy^2 + 13xy - 8y}{x^3y - 16x^2y + 64xy}$

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26

Poll Results

Angel Cruz	483
Terry Smith	320

The table shows the results of a poll. A total of 803 voters selected at random were asked which candidate they would vote for in the upcoming election. According to the poll, if 6,424 people vote in the election, by how many votes would Angel Cruz be expected to win?

A) 163

B) 1,304

C) 3,864

D) 5,621

27

The graph of $x^2 + x + y^2 + y = \frac{199}{2}$ in the xy-plane is a circle. What is the length of the circle's radius?

STOP

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Math

27 QUESTIONS

DIRECTIONS

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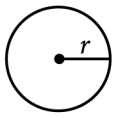

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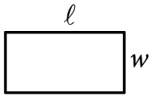
$C = 2\pi r$



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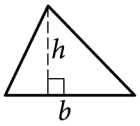
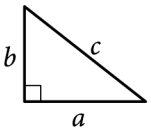


$A = \frac{1}{2}bh$



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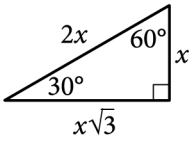
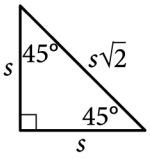
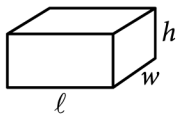
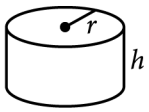




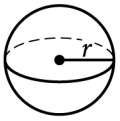
Special Right Triangles



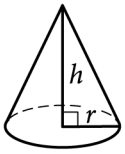
$V = \ell wh$



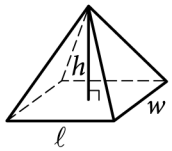
$V = \pi r^2 h$



$V = \frac{4}{3}\pi r^3$



$V = \frac{1}{3}\pi r^2 h$



$V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π.

The sum of the measures in degrees of the angles of a triangle is 180.

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For multiple-choice questions, solve each problem, choose the correct answer from the choices provided, and then circle your answer in this book. Circle only one answer for each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

For student-produced response questions, solve each problem and write your answer next to or under the question in the test book as described below.

- Once you've written your answer, circle it clearly. You will not receive credit for anything written outside the circle, or for any questions with more than one circled answer.
- If you find **more than one correct answer**, write and circle only one answer.
- Your answer can be up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer, but no more.
- If your answer is a **fraction** that is too long (over 5 characters for positive, 6 characters for negative), write the decimal equivalent.
- If your answer is a **decimal** that is too long (over 5 characters for positive, 6 characters for negative), truncate it or round at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), write it as an improper fraction (7/2) or its decimal equivalent (3.5).
- Don't include **symbols** such as a percent sign, comma, or dollar sign in your circled answer.

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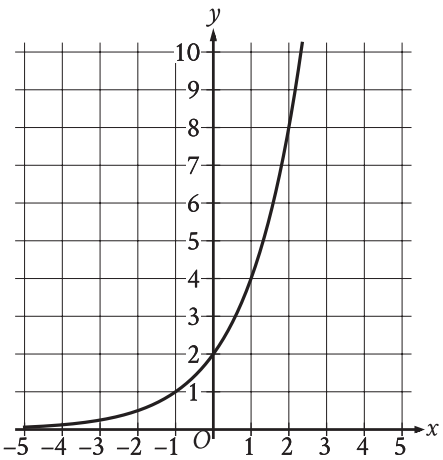
1

Isabel grows potatoes in her garden. This year, she harvested 760 potatoes and saved 10% of them to plant next year. How many of the harvested potatoes did Isabel save to plant next year?

A) 66
B) 76
C) 84
D) 86

2



What is the y-intercept of the graph shown?

A) (0, 0)
B) (0, 2)
C) (2, 0)
D) (2, 2)

3

What length, in centimeters, is equivalent to a length of 51 meters? (1 meter = 100 centimeters)

A) 0.051
B) 0.51
C) 5,100
D) 51,000

4

A bus is traveling at a constant speed along a straight portion of road. The equation $d = 30t$ gives the distance d, in feet from a road marker, that the bus will be t seconds after passing the marker. How many feet from the marker will the bus be 2 seconds after passing the marker?

A) 30
B) 32
C) 60
D) 90

5

Which expression is equivalent to $20w - (4w + 3w)$?

A) $10w$
B) $13w$
C) $19w$
D) $21w$

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6

If $6 + x = 9$, what is the value of $18 + 3x$?

7

$$y = x^2 - 14x + 22$$

The given equation relates the variables x and y. For what value of x does the value of y reach its minimum?

8

Which expression is equivalent to $9x^2 + 5x$?

A) $x(9x + 5)$

B) $5x(9x + 1)$

C) $9x(x + 5)$

D) $x^2(9x + 5)$

9

In triangle ABC, the measure of angle B is 52° and the measure of angle C is 17°. What is the measure of angle A ?

A) 21°

B) 35°

C) 69°

D) 111°

10

$$x = 8$$
$$y = x^2 + 8$$

The graphs of the equations in the given system of equations intersect at the point (x, y) in the xy-plane. What is the value of y ?

A) 8

B) 24

C) 64

D) 72

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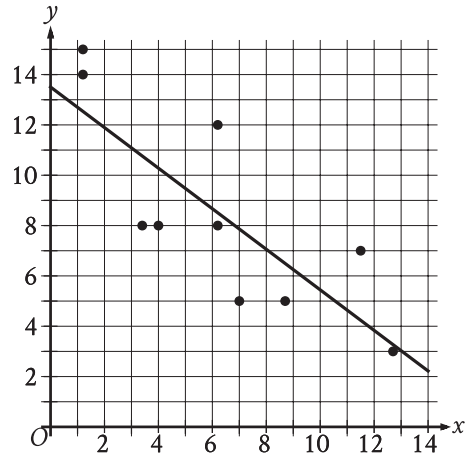
11

The scatterplot shows the relationship between two variables, x and y. A line of best fit is also shown.



Which of the following equations best represents the line of best fit shown?

A) $y = 13.5 + 0.8x$

B) $y = 13.5 - 0.8x$

C) $y = -13.5 + 0.8x$

D) $y = -13.5 - 0.8x$

12

The function f is defined by $f(x) = 8\sqrt{x}$. For what value of x does $f(x) = 48$?

A) 6

B) 8

C) 36

D) 64

13

A circle has center O, and points R and S lie on the circle. In triangle ORS, the measure of $\angle ROS$ is 88°. What is the measure of $\angle RSO$, in degrees?

14

$$x(x + 1) - 56 = 4x(x - 7)$$

What is the sum of the solutions to the given equation?

15

$$y = 3x$$
$$2x + y = 12$$

The solution to the given system of equations is (x, y). What is the value of $5x$?

A) 24

B) 15

C) 12

D) 5

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16

A cube has an edge length of 41 inches. What is the volume, in cubic inches, of the cube?

A) 164

B) 1,681

C) 10,086

D) 68,921

17

$$p(t) = 90{,}000(1.06)^t$$

The given function p models the population of Lowell t years after a census. Which of the following functions best models the population of Lowell m months after the census?

A) $r(m) = \frac{90{,}000}{12}(1.06)^m$

B) $r(m) = 90{,}000\left(\frac{1.06}{12}\right)^m$

C) $r(m) = 90{,}000\left(\frac{1.06}{12}\right)^{\frac{m}{12}}$

D) $r(m) = 90{,}000(1.06)^{\frac{m}{12}}$

18

$$6x + 7y = 28$$
$$2x + 2y = 10$$

The solution to the given system of equations is (x, y). What is the value of y ?

A) -2

B) 7

C) 14

D) 18

19

The minimum value of x is 12 less than 6 times another number n. Which inequality shows the possible values of x ?

A) $x \leq 6n - 12$

B) $x \geq 6n - 12$

C) $x \leq 12 - 6n$

D) $x \geq 12 - 6n$

20

Data set A consists of the heights of 75 buildings and has a mean of 32 meters. Data set B consists of the heights of 50 buildings and has a mean of 62 meters. Data set C consists of the heights of the 125 buildings from data sets A and B. What is the mean, in meters, of data set C?

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21

The graph of $9x - 10y = 19$ is translated down 4 units in the xy-plane. What is the x-coordinate of the x-intercept of the resulting graph?

22

Two variables, x and y, are related such that for each increase of 1 in the value of x, the value of y increases by a factor of 4. When $x = 0$, $y = 200$. Which equation represents this relationship?

A) $y = 4(x)^{200}$

B) $y = 4(200)^{x}$

C) $y = 200(x)^{4}$

D) $y = 200(4)^{x}$

23

$$x^2 - 2x - 9 = 0$$

One solution to the given equation can be written as $1 + \sqrt{k}$, where k is a constant. What is the value of k ?

A) 8

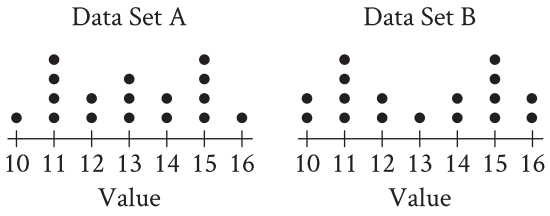
B) 10

C) 20

D) 40

24

The dot plots represent the distributions of values in data sets A and B.



Which of the following statements must be true?

I. The median of data set A is equal to the median of data set B.

II. The standard deviation of data set A is equal to the standard deviation of data set B.

A) I only

B) II only

C) I and II

D) Neither I nor II

25

An isosceles right triangle has a perimeter of $94 + 94\sqrt{2}$ inches. What is the length, in inches, of one leg of this triangle?

A) 47

B) $47\sqrt{2}$

C) 94

D) $94\sqrt{2}$

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26

$$-9x^2 + 30x + c = 0$$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c ?

A) 3

B) 0

C) −25

D) −53

27

$$\frac{3}{2}y - \frac{1}{4}x = \frac{2}{3} - \frac{3}{2}y$$

$$\frac{1}{2}x + \frac{3}{2} = py + \frac{9}{2}$$

In the given system of equations, p is a constant. If the system has no solution, what is the value of p ?

STOP

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Reading and Writing

Module 1
(33 questions)

QUESTION 1

Choice A is the best answer because it most logically completes the text's discussion of the writing system created by Sequoyah. In this context, "widespread" means widely accepted or practiced. The text indicates that because Sequoyah's script accurately represented the spoken sounds of the Cherokee language and was easy to learn, nearly all Cherokee people were able to read and write it soon after it was created. This context demonstrates that the script was widely used by the Cherokee people.

Choice B is incorrect. In this context, "careful" would mean exercised with care and attentive concern. Although the work of creating a writing system likely involved great care, the text indicates that the system was "easy to learn," which conflicts with the idea that using this system requires a noteworthy amount of care. *Choice C* is incorrect because in this context "unintended" means not deliberate. The idea that using Sequoyah's script was unintentional conflicts directly with the claim that it was easy to learn and used by "over 90% of the Cherokee people" by 1830. In fact, because one had to learn this system, it's not clear how one could use it unintentionally. *Choice D* is incorrect because in this context "infrequent" means rare or not occurring often, which conflicts directly with the claim that "over 90% of the Cherokee people" were using Sequoyah's script by 1830.

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QUESTION 2

Choice A is the best answer because it most logically completes the text's discussion of Nuttall's 1886 research paper. In this context, "acknowledged" means recognized as having a certain status. The text indicates that other researchers recognized Nuttall's work as groundbreaking because of its "convincing demonstration" related to the age of the ancient sculptures. In other words, the researchers recognized the groundbreaking status of Nuttall's work.

Choice B is incorrect because in this context, "ensured" would mean to have guaranteed or made sure something was the case. The text states that other researchers gave Nuttall's work recognition after it was published, but there's no indication that they contributed to the work or had any involvement that would have allowed them to make sure the work would be groundbreaking. *Choice C* is incorrect because the text doesn't suggest that other researchers "denied," or refused to admit or accept, that Nuttall's work was groundbreaking; on the contrary, it indicates that researchers praised the work, recognizing it as groundbreaking due to its "convincing demonstration" related to the age of the ancient sculptures. *Choice D* is incorrect because the text doesn't suggest that other researchers "underestimated," or undervalued, Nuttall's work; on the contrary, it indicates that researchers praised the work, recognizing it as groundbreaking due to its "convincing demonstration" related to the age of the ancient sculptures.

QUESTION 3

Choice A is the best answer because it most logically completes the text's discussion of self-government among the Muscogee (Creek) Nation. In this context, "implement" means to carry out or put into effect. The text states that the National Council generates laws, while the principal chief and cabinet officials are responsible for "devising policies and administering services in accordance with" those laws. This context suggests that the principal chief and cabinet officials implement the laws: they put the laws into effect by creating policies and administering services that accord with those laws.

Choice B is incorrect because "presume" in this context would mean to assume based on incomplete information, and the text does not suggest that the principal chief and cabinet officials either made assumptions about the content of the laws or had incomplete information about them. *Choice C* is incorrect because in this context "improvise" would mean to create something without preparation, and the text does not suggest that the principal chief and cabinet officials create policies and administer services without advance preparation. *Choice D* is incorrect because nothing in the text suggests that the principal chief and cabinet officials "mimic," or imitate, the laws generated by the National Council. To mimic laws would mean to generate new laws that are imitations of existing laws, but the text indicates that the National Council, not the principal chief and cabinet officials, is responsible for generating laws. Instead of generating laws, the principal chief and cabinet officials put laws into effect by "devising policies and administering services in accordance with" the laws.

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QUESTION 4

Choice D is the best answer because it most logically completes the text's discussion of the Three Sisters intercropping system. As used in this context, "intricate" would mean made up of complexly related elements. The text indicates that in the Three Sisters system, maize, squash, and beans form a "web of relations" in which the crops interact in various ways. The text's description of these interactions—the bean vines growing on the maize stalks, the squash vines keeping weeds away, and the beans adding nutrients that the maize and squash use—provides context suggesting that this "web of relations" is intricate.

Choice A is incorrect because describing the relationship among the crops in the Three Sisters system as "indecipherable," or impossible to comprehend, would not make sense in context. Although the text presents the relationship as complex, the text's description of the role that each crop plays makes it clear that the relationship is well understood, not indecipherable. *Choice B* is incorrect because the text discusses the practical benefits that each plant in the Three Sisters system provides to other members of the system, showing that the relationship among the crops that make up the system is not "ornamental," or mainly serving a decorative purpose. *Choice C* is incorrect because describing the relationship among the crops in the Three Sisters system as "obscure," or unknown or poorly understood, would not make sense in context. Although the text presents the relationship as complex, the text's description of the role that each crop plays makes it clear that the relationship is well understood, not obscure.

QUESTION 5

Choice B is the best answer because it most logically completes the text's discussion of bronze- and brass-casting techniques used by the Igun Eronmwon guild. In this context "adhere to" would mean to act in accordance with. The text states that although members of the Igun Eronmwon guild typically do something with techniques that have been passed down since the thirteenth century, they "don't strictly observe every tradition." By establishing a contrast with not always following traditions, the context suggests that guild members do typically adhere to traditional techniques.

Choice A is incorrect because in this context "experiment with" would mean to do something new with. Although using motors rather than manual bellows is presented as a new approach, the text establishes a contrast between what the guild members typically do with techniques that have been passed down over centuries and the idea that the members "don't strictly observe every tradition." The phrase "experiment with" wouldn't support the contrast because regularly trying new things with the techniques would be an example of not strictly following all traditions. *Choice C* is incorrect because in this context "improve on" would mean to make better. Although using motors rather than manual bellows might be an improved approach, the text establishes a contrast between what the guild members typically do with techniques that have been passed down over centuries and the idea that the members "don't strictly observe every tradition." The phrase "improve on" wouldn't support the contrast because regularly making changes to the techniques would be an example of not strictly following all traditions. *Choice D* is incorrect because in this context "grapple with" would mean to try hard to solve a

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difficult problem. Although bronze- and brass-casting are likely challenging tasks, nothing in the text suggests that the guild members have any particular difficulties with the techniques passed down since the thirteenth century.

QUESTION 6

Choice A is the best answer because it most logically completes the text's discussion of late nineteenth- and early twentieth-century household food purchases. In this context, "surmised" means formed an idea or assumption with little evidence. The text explains that certain economic historians "assumed" that large and small households spent different amounts on food per person, but that another economist found this supposition to be false based on evidence from available data. This context suggests that the economic historians made an incorrect assumption without enough consideration of evidence.

Choice B is incorrect. In this context, "contrived" would mean brought about or created through trickery. Nothing in the text suggests that the economic historians were deliberately trying to trick people with a claim about food purchasing behaviors in late nineteenth- and early twentieth-century households; the text simply suggests that they made an assumption about those behaviors that another historian believes isn't supported by the available data. *Choice C* is incorrect because the text indicates that it's Logan and not the economic historians who "questioned," or doubted, the assumption that large and small households in the late nineteenth and early twentieth centuries spent different amounts on food per person; the economic historians are the ones who made that assumption to begin with. *Choice D* is incorrect because nothing in the text suggests that some economic historians "regretted," or felt sad or remorseful about, the food purchasing behaviors of late nineteenth- and early twentieth-century households. The text focuses on the idea that the economic historians made an assumption about those behaviors that may not be supported by available data, not on the historians' emotional response to what households did in the past.

QUESTION 7

Choice C is the best answer because it most logically completes the text's discussion of the artistic styles that have influenced Cannon's work. As used in this context, "disparate" means distinct or dissimilar. The text indicates that a tension exists among the styles that have influenced Cannon's work and goes on to describe how those styles differ: classic European portraiture favors realism, American pop art uses vivid colors, and intertribal flatstyle rejects the use of shading and perspective to achieve depth. This context suggests that the styles that have influenced Cannon's work are disparate.

Choice A is incorrect because the text indicates that there is a tension among the influences on Cannon's artwork, so it wouldn't make sense to say that the influences are "complementary," or that they complete one another or make up for one another's deficiencies. *Choice B* is incorrect because it wouldn't make sense to characterize Cannon's influences as "unknown," or not familiar; it's clear that the influences are known because the text goes on to list them. *Choice D* is incorrect because the text indicates that there is a tension among the influences on Cannon's work, not that they are "interchangeable," or capable of being used in one another's place.

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QUESTION 8

Choice B is the best answer because it most logically and precisely completes the text's discussion of studies of altitude's effect on blood chemistry. In this context, "paucity of" means lack of. In describing the inspiration behind Al-Sweedan and Alhaj's research, the text uses the word "though" to suggest a contrasting relationship between two types of studies: those examining the effect on blood chemistry of living at a high altitude and those examining the effect on blood chemistry of living in locations below sea level. This contrasting relationship and the text's use of the word "many" provide context suggesting that there are few, if any, examples of the second type of study, whereas there are numerous examples of the first type.

Choice A is incorrect because it wouldn't make sense in context for there to be a "quarrel about," or open disagreement about, studies of the effect on blood chemistry of living in locations below sea level. The text's use of the words "though" and "many" suggests a contrasting relationship in terms of amount between two types of studies: those examining the effect on blood chemistry of living at a high altitude and those examining the effect on blood chemistry of living in locations below sea level. There's nothing in the text to suggest that the contrast between the two types of studies involves the extent to which researchers broadly agree or disagree about the contents of either type. *Choice C* is incorrect because it wouldn't make sense in context for there to be a "profusion of," or great abundance of, studies of the effect on blood chemistry of living in locations below sea level. The text's use of the words "though" and "many" suggests a contrasting relationship in terms of amount between two types of studies: those examining the effect on blood chemistry of living at a high altitude and those examining the effect on blood chemistry of living in locations below sea level. Rather than logically completing this contrast, "profusion of" would indicate that the two types of studies are similar in terms of amount, with many examples existing of both types. *Choice D* is incorrect because it wouldn't make sense in context for there to be a "verisimilitude in," or appearance of truth in, studies of the effect on blood chemistry of living in locations below sea level. The text's use of the words "though" and "many" suggests a contrasting relationship in terms of amount between two types of studies: those examining the effect on blood chemistry of living at a high altitude and those examining the effect on blood chemistry of living in locations below sea level. There's nothing in the text to suggest that the contrast between the two types of studies involves the extent to which either type of study presents an appearance of truth.

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QUESTION 9

Choice D is the best answer because it most accurately reflects the main purpose of the text. The text portrays Miss Pyne as awaiting the arrival of a carriage while Martha brings strawberries and flowers from the garden into the house. The text also describes the surroundings of the scene, stating that Miss Pyne looks "stately and calm," the evening is bright and cool, and birds are singing in the garden as the sun sets. Then the last sentence states that the house was "wide open to the long-expected guest," which strongly suggests that Miss Pyne's anticipation and Martha's activities were in preparation for the guest who is expected to arrive in the carriage. Thus, the text depicts the setting and conveys what these characters are doing as they await the arrival of their visitor.

Choice A is incorrect because there is nothing in the text to indicate that the characters feel any worry about the guest's arrival. The text indicates that the guest was "long-expected," but characterizing Miss Pyne as "stately and calm" conflicts with the idea that the characters are worried about the guest. *Choice B* is incorrect because the text describes a moment in time when two characters are awaiting the arrival of a visitor rather than an extended period over which characters could be seen changing. *Choice C* is incorrect. Although the text describes the activity indoors (Miss Pyne sitting calmly), it describes a higher level of activity, not stillness, outside (Martha bringing fruit and flowers and birds singing).

QUESTION 10

Choice C is the best answer because it best describes how the second sentence functions in the text as a whole. The first sentence establishes something astronomers believe with some certainty: that Betelgeuse will "explode in a supernova." The second sentence then introduces a problem: astronomers aren't certain *when* Betelgeuse will explode because they don't have enough information about the star's internal characteristics. Finally, the third sentence indicates that researcher Sarafina El-Badry Nance and colleagues investigated a possible method of obtaining the necessary information about Betelgeuse's internal characteristics, though they found that the method wouldn't be sufficient. Thus, the function of the second sentence is to identify the problem that Nance and colleagues attempted to solve but didn't.

Choice A is incorrect because the second sentence doesn't indicate how other astronomers or astrophysicists responded to the work done by Nance and colleagues; the text doesn't address this information at all. *Choice B* is incorrect because the second sentence introduces the general problem Nance and colleagues hoped to solve, not the central finding they ultimately reported. It's the third sentence that presents Nance and colleagues' conclusion that a potential method for determining internal stellar states would be insufficient. *Choice D* is incorrect because the second sentence introduces the general problem Nance and colleagues hoped to solve, not a serious limitation of how they tried to solve it. It's the third sentence that introduces Nance and colleagues, but no serious limitation of their approach to studying a method of determining internal stellar states is described.

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QUESTION 11

Choice D is the best answer because it provides a detail about Elinor that is established in the text. The text indicates that although Elinor is "only nineteen," she gives good advice and exhibits such a high level of understanding and judgment that she serves as "the counsellor of her mother." Thus, Elinor is mature beyond her years.

Choice A is incorrect because it isn't supported by the text: although the text says that Elinor advises her mother and often counteracts her mother's impulses, there's no mention of Elinor arguing with her mother or failing to change her mother's mind. *Choice B* is incorrect because it isn't supported by the text: although the text mentions that Elinor has strong feelings, it doesn't indicate that she's excessively sensitive when it comes to family issues. *Choice C* is incorrect because it isn't supported by the text: there's no mention of what Elinor thinks about her mother and no suggestion that she thinks her mother is a bad role model. Because she's described as having "an excellent heart," Elinor likely doesn't think ill of her mother.

QUESTION 12

Choice D is the best answer because it most accurately states the main idea of the text. According to the text, conceptual artists Gins and Arakawa have designed an apartment building that is disorienting because of several unconventional elements, such as uneven kitchen counters and "a door to nowhere." The text goes on to suggest that there may be benefits to this kind of design because filmmaker Yamaoka lived in the apartment building for four years and reported health benefits. Thus, although the design is impractical, it may improve the well-being of the apartment building's residents.

Choice A is incorrect. Although the text mentions that Yamaoka lived in the apartment for four years, it doesn't address how long someone can beneficially live in a home surrounded by fanciful features or whether doing so can be sustained. *Choice B* is incorrect. Although the text mentions the potential benefits of living in a home with disorienting design features, it doesn't suggest that this is the most effective method to create a physically stimulating environment. *Choice C* is incorrect because the text refers to Yamaoka to support the claim that Gins and Arakawa's apartment building design may be beneficial, but the text doesn't indicate that Yamaoka supports the designs of other conceptual artists.

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QUESTION 13

Choice C is the best answer because it illustrates the student's claim about some historians viewing Lumumba primarily as a symbol. This quotation argues that Lumumba "warrants" (or deserves) "scholarly attention" as a symbol and not for his "practical accomplishments"—that is, his actions as prime minister—which "can be passed over quickly," or dismissed as being of comparatively little importance. Thus, the quotation expresses the view that the student criticizes some historians for holding.

Choice A is incorrect. Although this quotation touches on the difficulty of evaluating Lumumba's legacy, it doesn't address how historians of modern African politics view him as a symbol. *Choice B* is incorrect. While this quotation mentions Lumumba's political beliefs, it doesn't discuss historians viewing him as a symbol. *Choice D* is incorrect. This quotation touches on Lumumba's vision for his country, but it doesn't discuss historians viewing him as a symbol.

QUESTION 14

Choice C is the best answer because it uses data from the graph to effectively complete the example of Eludoyin and his colleagues' findings concerning female farmers in some regions of Ondo State, Nigeria. The graph presents values for the percentage of Ondo State small-scale farmers who are female, by type of crop and region. The graph shows that of the farmers mainly cultivating non-root vegetables, approximately 57% in north Ondo and approximately 54% in south Ondo are female; in other words, most of those farmers are female, which exemplifies the idea that female farmers make up the majority (more than half) of the farmers cultivating specific types of crops in some regions.

Choice A is incorrect because it inaccurately cites data from the graph: the graph shows that in south Ondo, most of the farmers mainly cultivating non-root vegetables are women (approximately 54%), but that only about 35% (less than half) of the farmers mainly cultivating cereals are women. *Choice B* is incorrect because it inaccurately cites data from the graph: the graph shows that more women in central Ondo mainly cultivate cereals than mainly cultivate root crops (approximately 36% and 20%, respectively). Additionally, it doesn't effectively complete the example because the graph shows that female farmers don't make up the majority (more than half) of the farmers for any type of crop in central Ondo. *Choice D* is incorrect because it doesn't effectively complete the example; it simply states that a relatively equal proportion of women across the three regions mainly cultivate cereals, which doesn't address the value for that proportion and thus doesn't show that a majority (more than half) of the farmers cultivating certain crops are female.

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QUESTION 15

Choice C is the best answer because it presents a finding that, if true, would weaken the astronomers' claim about the makeup of host stars and their planets. The text explains that because stars and planets begin forming from the same gas and dust, astronomers believe planets should be composed of the same materials as their host stars, but in equal or smaller quantities. The finding that the amount of iron in some rocky planets is much higher than the amount in their host star would weaken the astronomers' claim because it would show that some planets contain the same material as their host star, but in higher quantities.

Choice A is incorrect because a finding only about the makeup of stars, whether they've cooled or not, would provide no information about the makeup of planets. Thus, it wouldn't have any bearing on the claim that planets and their host stars are composed of the same materials in differing quantities. *Choice B* is incorrect because a finding about two host stars having similar proportions of certain materials wouldn't provide any information about the makeup of planets. Thus, it wouldn't be relevant to the claim that planets and their host stars are composed of the same materials in differing quantities. *Choice D* is incorrect because the text indicates that the astronomers' claim is based on a fact—that stars and planets begin forming from the same gas and dust in space—which would remain true regardless of the effectiveness of a method for analysis of compositions. The text does cite analysis of rocky planets in our solar system and the Sun, but only as a single piece of evidence that is consistent with the claim and not as the source of the claim; the finding that the method used for that analysis is less effective in other scenarios wouldn't weaken a claim that's based on knowledge of how stars and planets initially form.

QUESTION 16

Choice D is the best answer because it presents a finding that, if true, would support Paredes's argument about the origin of Mexican American folklore. The text describes a disagreement among scholars about whether Mexican American folklore mostly derived from the folklore of Spain (the view held by Espinosa and others) or originated in Mexico and the United States through ongoing cultural interactions there (the view held by Paredes and others). If Mexican American folklore collected in the twentieth century mostly consists of ballads about history and social life that originated recently, then that would support Paredes's argument by suggesting that the folklore mostly arose after Spanish rule ended in the early nineteenth century and that the folklore reflects cultural interactions in Mexico and the United States rather than traditions from Spain.

Choice A is incorrect because the inclusion of songs influenced by sixteenth-century Spanish poetry among Mexican American folklore collected in the twentieth century would not support Paredes's view that the folklore was the result of cultural interactions in Mexico and the United States rather than an offshoot of Spanish folklore. If anything, the presence of such songs among the folklore collected in the twentieth century would weaken Paredes's argument, since it would reflect the influence of Spanish culture on the folklore. *Choice B* is incorrect because the mere presence of similarities in Mexican American

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folklore across regions would not be sufficient to draw a conclusion about where the folklore originated, let alone to support Paredes's argument that the folklore reflects various cultural interactions in Mexico and the United States. In fact, Paredes would likely expect there to be regional variations in folklore as different cultures have interacted in different places. *Choice C* is incorrect because scholars' previous ignorance of the folklore would have no bearing on Paredes's argument that Mexican American folklore mostly reflects cultural interactions in Mexico and the United States; the folklore's actual origins exist regardless of the scholars' awareness.

QUESTION 17

Choice A is the best answer because it most logically completes the text's discussion of Euro-American farmers' use of Haudenosaunee agricultural techniques. According to the text, some Euro-American farmers were using these techniques in the early nineteenth century despite few of the farmers having seen Haudenosaunee farms. One explanation for these facts might be that the farmers developed techniques on their own that already had been invented centuries earlier by the Haudenosaunee people, but the text explicitly bars, or rules out, this explanation. If Euro-American farmers didn't learn these techniques from direct observation of Haudenosaunee practices and didn't invent the techniques independently, then the most logical explanation is that they learned the techniques from other people who were more directly influenced by Haudenosaunee practices than the farmers themselves were. Once they learned about Haudenosaunee agricultural practices, Euro-American farmers could then apply those practices to their own farming.

Choice B is incorrect because the fact that some Euro-American farmers in the northeastern United States were using Haudenosaunee techniques suggests that the techniques were likely useful for the crops the farmers raised, not that the crops typically cultivated by the farmers were not well suited to Haudenosaunee farming techniques. If the farmers' crops were ill suited to the techniques, it's unlikely that the farmers would have used those techniques. *Choice C* is incorrect because the text indicates only that Haudenosaunee agricultural techniques were used by Euro-American farmers in the northeastern United States, not that these techniques were widely used outside this region. *Choice D* is incorrect because the text states that some Euro-American farmers were using Haudenosaunee farming techniques early in the nineteenth century. This suggests that some Euro-American farmers were beginning to recognize the benefits of these techniques near the start of the century, not that such farmers only began to recognize the benefits of the techniques much later.

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QUESTION 18

Choice B is the best answer because it most logically completes the text's discussion of artifacts and Kuulo Kataa's founding date. If it were true both that Kuulo Kataa was founded in the fourteenth century CE and that artifacts found in excavations of the settlement are from the thirteenth century CE, it would be reasonable to conclude that the artifacts weren't created in the Kuulo Kataa settlement. That would suggest, then, that the artifacts originated somewhere else and eventually reached the settlement through trading or as people migrated.

Choice A is incorrect because the existence of thirteenth-century CE artifacts recovered during excavations of a settlement founded in the fourteenth century CE isn't logically connected to artifacts from one century being more commonly recovered than artifacts from another century. Rather than suggesting anything about how frequently artifacts from different times are found, the existence of artifacts confirmed as predating the settlement's founding suggests that those items arrived in Kuulo Kataa during or after its establishment. *Choice C* is incorrect because the text focuses on time periods and says nothing about which region the founders of Kuulo Kataa have been thought to come from; similarly, the text doesn't suggest anything about where the thirteenth-century CE artifacts originated other than not from Kuulo Kataa. Therefore, it isn't logical to conclude that the mere existence of artifacts confirmed as predating the Kuulo Kataa settlement suggests that the founders of the settlement came from a particular region other than one previously assumed. *Choice D* is incorrect because the existence of artifacts from the thirteenth century CE at a site dated to the fourteenth century CE doesn't imply that fourteenth-century objects were damaged during excavations. There's nothing in the text to suggest that any objects were damaged; rather, the existence of artifacts confirmed as predating the settlement's founding suggests that those items were brought to Kuulo Kataa during or after its establishment.

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QUESTION 19

Choice C is the best answer because it most logically completes the text's discussion of accelerated flowering in *A. thaliana* plants. The text indicates that *A. thaliana* plants show accelerated flowering at high temperatures. To investigate the mechanism for this accelerated flowering, biologists replaced the ELF3 protein in one group of *A. thaliana* plants with a similar protein found in another plant species that doesn't show accelerated flowering. The team then compared these modified plants to *A. thaliana* plants that retained their original ELF3 protein. The text states that the two samples of plants showed no difference in flowering at 22° Celsius, but at 27° Celsius the unaltered plants with ELF3 showed accelerated flowering while the plants without ELF3 didn't. If accelerated flowering at the higher temperature occurred in the *A. thaliana* plants with ELF3 but not in the plants without the protein, then ELF3 likely enables *A. thaliana* to respond to increased temperatures.

Choice A is incorrect because the text doesn't mention whether any plants other than *A. thaliana* and stiff brome show temperature-sensitive flowering, so there is no support for the idea that this type of flowering is unique to *A. thaliana. Choice B* is incorrect because the text discusses the effects of ELF3 and not the production of it. There's nothing in the text to suggest that the amount of ELF3 in *A. thaliana* varies with temperature. *Choice D* is incorrect. While the text states that there was no difference in the flowering of modified and unmodified *A. thaliana* plants at 22° Celsius, there's no suggestion that *A. thaliana* only begins to flower at 22° Celsius; the text doesn't mention a specific temperature threshold required for *A. thaliana* flowering.

QUESTION 20

Choice A is the best answer. The convention being tested is the use of finite and nonfinite verb forms within a sentence. Relative clauses, such as the one beginning with "which," require a finite verb, a verb that can function as the main verb of a clause. This choice correctly supplies the clause with the finite past tense verb "provided."

Choice B is incorrect because the nonfinite participle "having provided" doesn't supply the clause with a finite verb. *Choice C* is incorrect because the nonfinite to-infinitive "to provide" doesn't supply the clause with a finite verb. *Choice D* is incorrect because the nonfinite participle "providing" doesn't supply the clause with a finite verb.

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QUESTION 21

Choice A is the best answer. The convention being tested is the coordination of clauses within a sentence. This choice correctly uses a comma and the coordinating conjunction "but" to join a main clause ("Typically…value") and a subordinate clause ("when…Whitman") that precedes a main clause ("such… scholars").

Choice B is incorrect because it results in a run-on sentence. A main clause is fused without punctuation and/or a conjunction to a subordinate clause that precedes a main clause. *Choice C* is incorrect because it results in a comma splice. A comma can't be used in this way to mark the boundary between a main clause and a subordinate clause that precedes a main clause. *Choice D* is incorrect. Without a comma preceding it, the conjunction "but" can't be used in this way to join a main clause and a subordinate clause that precedes a main clause.

QUESTION 22

Choice C is the best answer. The convention being tested is punctuation use between sentences. In this choice, the period after "percent" is used correctly to mark the boundary between one sentence ("After… percent") and another ("Such…up").

Choice A is incorrect because it results in a comma splice. A comma can't be used in this way to mark the boundary between sentences. *Choice B* is incorrect. Without a comma preceding it, the conjunction "and" can't be used in this way to join sentences. *Choice D* is incorrect because it results in a run-on sentence. The sentences ("After…percent" and "Such…up") are fused without punctuation and/or a conjunction.

QUESTION 23

Choice D is the best answer. The convention being tested is punctuation between a verb and a preposition. When, as in this case, a verb ("is added") is immediately followed by a preposition ("whenever"), no punctuation is needed.

Choice A is incorrect because no punctuation is needed between the verb and the preposition. *Choice B* is incorrect because no punctuation is needed between the verb and the preposition. *Choice C* is incorrect because no punctuation is needed between the verb and the preposition.

QUESTION 24

Choice A is the best answer. The convention being tested is subject-verb agreement. The singular verb "has enhanced" agrees in number with the singular subject "*A Sheaf Gleaned in French Fields*," which is the title of a book of poems.

Choice B is incorrect because the plural verb "are enhancing" doesn't agree in number with the singular subject "*A Sheaf Gleaned in French Fields*." *Choice C* is incorrect because the plural verb "have enhanced" doesn't agree in number with the singular subject "*A Sheaf Gleaned in French Fields*." *Choice D* is incorrect because the plural verb "enhance" doesn't agree in number with the singular subject "*A Sheaf Gleaned in French Fields*."

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QUESTION 25

Choice C is the best answer. The convention being tested is punctuation use between a main clause and two supplementary elements. In this choice, the commas after "nickname" and "however" are correctly used to separate the supplementary adverb "however" from the main clause ("Scott-Heron… nickname") on one side and the supplementary participial phrase ("feeling… bluesologist") on the other.

Choice A is incorrect because it fails to mark the boundary between the supplementary adverb "however" and the supplementary phrase ("feeling… bluesologist"). *Choice B* is incorrect because a semicolon can't be used in this way to join the supplementary adverb "however" and the supplementary phrase ("feeling…bluesologist"). *Choice D* is incorrect because a semicolon can't be used in this way to join the main clause ("Scott-Heron…nickname") and the supplementary word and phrase ("however" and "feeling…bluesologist"). Moreover, placing the semicolon after "nickname" illogically signals that the following information (Scott-Heron's feeling that the nickname didn't encapsulate his devotion to the blues tradition) is contrary to the information in the previous clause (Scott-Heron's resistance to the nickname).

QUESTION 26

Choice B is the best answer. The convention being tested is punctuation use between a main clause and a supplementary phrase. This choice correctly uses a comma to mark the boundary between the main clause ("the portraits...quilts") and the supplementary noun phrase ("the stitching...fabric") that provides a further description of how the portraits can be identified as quilts.

Choice A is incorrect. A comma and the conjunction "and" can't be used in this way to join a main clause and a supplementary noun phrase. *Choice C* is incorrect because a semicolon can't be used in this way to join a main clause and a supplementary noun phrase. *Choice D* is incorrect because it results in a rhetorically unacceptable sentence fragment beginning with "the stitching."

QUESTION 27

Choice A is the best answer. "For instance" logically signals that the information in this sentence—that larch trees lose their needles every fall—is an example supporting the claim in the previous sentence (that not all conifer species keep their leaves or needles year-round).

Choice B is incorrect because "nevertheless" illogically signals that the information in this sentence is true in spite of the claim about conifer species in the previous sentence. Instead, it's an example supporting that claim. *Choice C* is incorrect because "meanwhile" illogically signals that the information in this sentence is separate from (while occurring simultaneously with) the claim about conifer species in the previous sentence. Instead, it's an example supporting that claim. *Choice D* is incorrect because "in addition" illogically signals that the information in this sentence is merely an additional fact related to the claim about conifer species in the previous sentence. Instead, it's an example supporting that claim.

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QUESTION 28

Choice A is the best answer. The sentence effectively describes the rocking chair to an audience unfamiliar with Sam Maloof, noting its sleek, contoured armrests and seat and explaining that Sam Maloof (the walnut chair's creator) was an American woodworker.

Choice B is incorrect. While the sentence explains who Sam Maloof was and mentions a chair, it doesn't describe the chair. *Choice C* is incorrect. While the sentence explains who Sam Maloof was, it doesn't describe the rocking chair. *Choice D* is incorrect. While the sentence describes the rocking chair, it doesn't explain who Sam Maloof was.

QUESTION 29

Choice C is the best answer. The sentence emphasizes the decline in unique apple varieties in the US and specifies why this decline occurred, noting that thousands of apple varieties were lost because US farmers started mainly growing the same few unique varieties.

Choice A is incorrect. The sentence introduces the Lost Apple Project; it doesn't emphasize the decline in unique apple varieties in the US and specify why this decline occurred. *Choice B* is incorrect. While the sentence emphasizes the decline in unique apple varieties in the US, it doesn't explain why this decline occurred. *Choice D* is incorrect. The sentence emphasizes the general decline of crop varieties in the mid-1900s; it doesn't emphasize the specific decline in unique apple varieties in the US.

QUESTION 30

Choice C is the best answer. The sentence effectively introduces the poetry collection *Precario/Precarious*, noting that it is a collection by Vicuña that was published in 1983 by Tanam Press.

Choice A is incorrect. While the sentence mentions the 1983 poetry collection *Precario/Precarious*, it focuses mainly on Vicuña's visual art. *Choice B* is incorrect. The sentence doesn't introduce the 1983 poetry collection *Precario/Precarious*; instead, it introduces Vicuña. *Choice D* is incorrect. The sentence emphasizes the location of Vicuña's 1971 exhibition *Pinturas, poemas y explicaciones*; it doesn't introduce the 1983 poetry collection *Precario/Precarious.*

QUESTION 31

Choice B is the best answer. The sentence effectively emphasizes Kind's methodology: examining the student policies of 132 medical schools for guidelines about student social media use.

Choice A is incorrect. The sentence specifies how many medical schools' student policies are available online; it doesn't emphasize the study's methodology. *Choice C* is incorrect. The sentence emphasizes the study's results, not the study's methodology. *Choice D* is incorrect. The sentence emphasizes the aim of the study, not the study's methodology.

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QUESTION 32

Choice C is the best answer. The sentence emphasizes both the duration (the length of time) and the purpose of Cohen's and Rodrigues's work by noting that the women have been working since 2003 to preserve Gullah culture.

Choice A is incorrect. While the sentence emphasizes what visitors to Cohen's and Rodrigues's museums can learn, it doesn't mention the duration or purpose of the women's work. *Choice B* is incorrect. While the sentence emphasizes the purpose of Cohen's and Rodrigues's work, it doesn't mention the duration of that work (the length of time the women have been working to preserve Gullah culture). *Choice D* is incorrect. While the sentence emphasizes where and when Gullah culture developed, it doesn't mention the duration or purpose of Cohen's and Rodrigues's work.

QUESTION 33

Choice A is the best answer. The sentence effectively emphasizes the aim, or goal, of the research study (in other words, what the researchers hoped to learn from the study): Rogers and Russell wanted to know if woodland expansion is related to changes in climate.

Choice B is incorrect. The sentence emphasizes the researchers' findings; it doesn't emphasize the aim of the study. *Choice C* is incorrect. The sentence emphasizes the results of the study; it doesn't emphasize the aim. *Choice D* is incorrect. The sentence emphasizes the methodology of the study; it doesn't emphasize the aim.

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Reading and Writing

Module 2

(33 questions)

QUESTION 1

Choice D is the best answer because it most logically completes the text's discussion of damage to viburnum plants. In this context, "healthy" would mean not distressed or diseased. The text states that insect damage may cause viburnum plants to be discolored and have abnormal growths. In the next sentence, the phrase "on the other hand" indicates a contrast with the description of plants suffering from damage. Thus, the context contrasts the appearance of healthy, undamaged plants with the appearance of damaged plants.

Choice A is incorrect because in this context, "struggling" would mean working against difficulties. The text first describes viburnum plants experiencing damage by insects, and the phrase "on the other hand" then establishes a contrast with that description. It wouldn't make sense to contrast struggling viburnum plants with those being damaged by insects, because in both cases the plants would be experiencing difficulties. *Choice B* is incorrect because in this context, "beneficial" would mean producing good or helpful effects. The text doesn't discuss how viburnum plants affect other things or suggest that the plants are helpful in some way; rather, it focuses on how viburnum plants are affected by certain conditions. *Choice C* is incorrect because in this context "simple" would mean plain or uncomplicated. The text doesn't discuss whether certain viburnum plants are complicated or uncomplicated; rather, it focuses on how viburnum plants are affected by certain conditions.

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QUESTION 2

Choice B is the best answer because it most logically completes the text's discussion of Cole's book *Blind Spot*. In this context, "enthusiasm for" means excitement about. The text explains that *Blind Spot* consists of original photographs as well as poetic prose—two elements that correspond to Cole's passions, identified in the text, for photography and the written word. This context suggests that Cole's excitement about photography and writing led him to create a book that successfully combines the two mediums.

Choice A is incorrect because describing Cole as feeling "indifference to" his two passions wouldn't make sense in context. If Cole is indifferent to his passions, that would mean he doesn't care about photography or writing—in which case they wouldn't be his passions at all. *Choice C* is incorrect because there's nothing in the text to suggest that Cole feels "concern about," or uneasiness about, his passions. The text's use of the word "culminates" indicates that *Blind Spot* represents a triumphant climax of Cole's passions, not a work that results from his sense of discomfort with photography and writing. *Choice D* is incorrect because there's nothing in the text to suggest that Cole feels "surprise at," or astonished by, his passions. The text indicates that Cole's feeling about his passions "culminates" in a book that "evocatively" combines photographs and writing, suggesting that Cole has a long-standing and skillful relationship to his passions, not that he is startled by them.

QUESTION 3

Choice D is the best answer because it most logically completes the text's discussion of Jemisin's writing. In this context, "conform to" means to act in accordance with something. The text suggests that in her science fiction writing, Jemisin's willingness to go against expectations and not use plots and themes that seem to follow a formula reflects how she treats the standard practices of the genre. This context conveys that Jemisin chooses not to act in accordance with those conventions.

Choice A is incorrect. In this context, "question" would mean doubt or object to. The text indicates that Jemisin is willing to go against expectations and not use formulaic plots and themes in her science fiction writing, suggesting that she may actually object to those conventions of the genre, not that she chooses not to question them. *Choice B* is incorrect because the text indicates that in her science fiction writing, Jemisin is willing to go against expectations and not use formulaic plots and themes. Rather than suggesting that Jemisin chooses not to "react to," or act in response to, the standard practices of the genre, this context suggests that she is acting in response to such conventions by deliberately avoiding them. *Choice C* is incorrect. In this context, "perceive" would mean become aware of or understand. The text indicates that in her science fiction writing, Jemisin is willing to go against expectations and not use formulaic plots and themes. This context conveys that Jemisin is aware of and deliberately avoids those conventions of the genre, not that she chooses not to be aware of them.

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QUESTION 4

Choice B is the best answer because it most logically completes the text's description of how Pico feels about the natural world. In this context, to say that Pico portrays his "ambivalence toward" nature would mean that he portrays his mixed feelings about nature. The text explains that Pico "honors the centrality of nature" and also makes it clear that he doesn't enjoy being in nature. This context suggests that Pico feels ambivalence toward nature.

Choice A is incorrect because saying that Pico portrays his "responsiveness to" nature would mean that he portrays himself as quick to react to nature, which isn't supported by the text. Instead, the text focuses on Pico's mixed feelings toward nature, describing him as both honoring nature's role in his tribe's beliefs and expressing his personal dislike for being in nature. *Choice C* is incorrect because saying that Pico portrays his "renunciation of" nature would mean that he portrays himself as rejecting nature, which isn't supported by the text. The text conveys that Pico demonstrates both positive and negative responses toward nature, not that he's giving it up completely. *Choice D* is incorrect because saying that Pico portrays his "mastery over" nature would mean that he portrays himself as having control over nature, which isn't supported by the text. The text focuses on Pico's mixed feelings about nature; nothing in the text suggests that Pico feels mastery over nature.

QUESTION 5

Choice A is the best answer because it accurately states the main purpose of the text. The text begins by discussing the promise of the future, with positive references to renewal such as "new roads," "new beating of the drum," and "fresh seeing." But with the "new sun," the text continues, there will still be "the same backs bending" and "the same sad feet" drumming, indicating that these difficulties will follow people into this new day. The poem thus considers both the rewards and challenges associated with the repetitiveness of human life.

Choice B is incorrect because the text doesn't say anything about how memorable activities are, let alone compare the memorability of activities completed at different times of the day. *Choice C* is incorrect. Although the text contrasts hope with difficulty, it does not compare the relative frequency of joyful feelings with that of sad feelings. *Choice D* is incorrect because the text makes no distinction between the experiences of individuals and the experiences of their communities.

QUESTION 6

Choice A is the best answer because it most accurately portrays the main purpose of the text. At the beginning of the text, Tom asserts that he and the other people staging the play are doing so only for "a little amusement among ourselves" and aren't interested in attracting an audience or any attention with the production. Then, Tom promises that the play they chose is modest and appropriate, and he further reasons that using the well-written prose of "some respectable author" is better than using their own words. Overall, the main purpose of the text is to convey Tom's promise that the play will be inoffensive and involve only a few people.

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Choice B is incorrect because the text doesn't indicate that Tom had earlier intentions for the play's performance or that anything has changed since the group first decided to stage a play. Instead, the text focuses on how harmless the entire endeavor will be. *Choice C* is incorrect. Although Tom mentions that using the words of a "respectable author" will be better than using their own words, he never addresses the idea that the people around him generally aren't skilled enough to stage a play. *Choice D* is incorrect because in the text Tom specifically says that they "want no audience, no publicity," which indicates that they don't plan on promoting the play at all.

QUESTION 7

Choice A is the best answer because it accurately describes the organization of the elements within the text. The text begins with the claim that Joni Mitchell's album covers use images she creates in order to emphasize ideas embedded in her albums. It then goes on to provide an example of how Mitchell's self-portrait on the cover of *Turbulent Indigo* resembles a painting by Van Gogh, which the text indicates helps emphasize the strong connection Mitchell feels toward Van Gogh, a connection that is also expressed in the album's title song.

Choice B is incorrect because there are no references in the text to artists other than Joni Mitchell and Van Gogh. *Choice C* is incorrect because there is nothing in the text that calls attention to any similarities or differences between Joni Mitchell and Van Gogh. Instead, it mentions that Mitchell feels a strong "artistic connection" to Van Gogh. *Choice D* is incorrect because the text discusses the cover before referring to any songs, and it only references one song from the album not all the songs.

QUESTION 8

Choice C is the best answer because it reflects how Putirka and Xu (Text 2) would likely characterize the conclusion presented in Text 1. Text 1 discusses a study by Mark Holland and colleagues in which they detected traces of lithium and sodium in the atmospheres of four white dwarf stars. The team claims that this supports the idea that exoplanets with continental crusts like Earth's once orbited these stars. Text 2 introduces Putirka and Xu, who indicate that sodium and lithium are present in several different minerals and that some of those minerals might exist in types of rock that are not found on Earth. Therefore, Putirka and Xu would likely describe the conclusion in Text 1 as questionable because it does not consider that lithium and sodium are also found in rocks that are not like Earth's continental crust.

Choice A is incorrect because the texts do not indicate how widely held any of the viewpoints described are. *Choice B* is incorrect because neither text discusses how new this area of study is. *Choice D* is incorrect because neither text discusses how likely lithium and sodium are to be detected by analyzing wavelengths of light.

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QUESTION 9

Choice A is the best answer because it presents an explanation that is directly stated in the text for why ecologists are worried about Pando. The text states that Pando is a colony of about 47,000 quaking aspen trees that represents one of the largest organisms on Earth. According to the text, ecologists are worried that Pando's growth is declining, partly because animals are feeding on the trees. In other words, the ecologists are worried that Pando isn't growing at the same rate it used to.

Choice B is incorrect. Rather than indicating that Pando isn't producing young trees anymore, the text reveals that Pando is indeed producing young trees, stating that those trees can be protected from grazing deer by strong fences. *Choice C* is incorrect because the text states that fences can be used to prevent deer from eating Pando's young trees, not that Pando itself can't grow in new areas because it's blocked by fences. *Choice D* is incorrect because the text offers no evidence that Pando's root system is incapable of supporting new trees or is otherwise a cause of worry for ecologists.

QUESTION 10

Choice D is the best answer because it states why Wang and his team's discovery of the *Terropterus xiushanensis* fossil was significant. The text explains that up until Wang and his team's discovery, the only fossil evidence of mixopterids came from the paleocontinent of Laurussia. Wang and his team, however, identified fossil remains of a mixopterid species from the paleocontinent Gondwana. Therefore, the team's discovery was significant because the fossil remains of a mixopterid species were outside of the paleocontinent Laurussia.

Choice A is incorrect. Although the text states that Wang and his team identified fossilized remains of a mixopterid species that lived more than 400 million years ago, it doesn't indicate that mixopterid fossils previously found by scientists dated to a more recent period than that. *Choice B* is incorrect. Although the text states that mixopterids are related to modern arachnids and horseshoe crabs, it doesn't suggest that the fossil discovered by Wang and his team confirmed that this relationship is closer than scientists had previously thought. *Choice C* is incorrect because the team's fossil established the presence of mixopterids on Gondwana, not on Laurussia. Moreover, the text only discusses the fossil in relation to the geographical distribution of mixopterids, not in relation to their evolution.

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QUESTION 11

Choice A is the best answer because it presents a finding that, if true, would support the scholar's claim about Toni Morrison's likely goal of strengthening the presence of Black writers on Random House's list of published authors. The text explains that Morrison was the first Black woman to be an editor for Random House and that she was an editor there from 1967 to 1983. If it were true that Random House published a higher percentage of works by Black authors throughout the 1970s—during most of Morrison's time working there—than it had previously published, that would suggest that Morrison may have made a deliberate effort to strengthen the presence of Black authors on the list of Random House's published authors, thus supporting the scholar's claim.

Choice B is incorrect because the scholar's claim is about Morrison's work as an editor at a publishing company and her likely effort to strengthen the presence of Black writers on that company's list of published authors. It might be true that Black authors interviewed in the 1980s and 1990s often cited Morrison's novels as an influence on their work, but that finding would simply suggest something about how those authors approached their work; it wouldn't show that Morrison intended to increase the number of Black writers among the published authors specifically at Random House. *Choice C* is incorrect because the scholar's claim is about Morrison's work as an editor at a publishing company, not about her work as a novelist. Therefore, a finding that Morrison's novels published after 1983 sold more copies and were more widely acclaimed than her earlier novels would have no bearing on the claim that as an editor Morrison made an effort to ensure that more Black writers were present on Random House's list of published authors. *Choice D* is incorrect. Although the text discusses Morrison's work as an editor at Random House, the scholar's claim focuses on Morrison's likely effort in that role to increase the number of Black writers present on Random House's list of published authors, not on the influence she may have had on the content of the works she edited. Without knowing whether Morrison's stylistic influence led to more publications or if Morrison applied her influence specifically to works by Black writers, the finding that works edited by Morrison could be identified by stylistic characteristics would have no bearing on the claim that Morrison intended to strengthen the presence of Black writers among the published authors at Random House.

QUESTION 12

Choice A is the best answer because it most effectively illustrates the claim that Martí argues that a society's spiritual well-being depends on the character of its literary culture. In the quotation, Martí asserts that poetry is "more necessary to a people than industry itself" and that it has the power to provide people with "faith and vigor." He also adds that literature gives people "the desire and strength for life." Therefore, this quotation shows that Martí believes that literature is a societal necessity because it uplifts people and nourishes their spiritual well-being.

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Choice B is incorrect. Although this quotation emphasizes the importance of literature, it focuses on how the nature of a society is reflected in that society's literature rather than on literature's value for people's spiritual well-being. *Choice C* is incorrect. Although this quotation involves an element of spirituality, it doesn't discuss literature. The quotation instead focuses on humanity's actions. *Choice D* is incorrect because this quotation mainly focuses on the importance of Walt Whitman rather than on the value of literature in general.

QUESTION 13

Choice C is the best answer because it accurately describes data from the table that support Barrett and Rayfield's suggestion about bite force estimates. According to the text, Barrett and Rayfield believe that estimates of dinosaur bite force may be strongly influenced by the methods used to produce them—that is, that different methods may produce significantly different results. The table shows that the studies by Bates and Falkingham and by Cost et al. used the same estimation method (muscular and skeletal modeling) and produced similar bite force estimates (approximately 35,000–57,000 newtons and 35,000–63,000 newtons, respectively). The study by Meers, however, used body-mass scaling and produced a much higher bite force estimate (183,000–235,000 newtons), while the study by Gignac and Erickson used tooth-bone interaction analysis and produced a much lower bite force estimate (8,000–34,000 newtons). The fact that one method produced similar estimates in two different studies and that two different methods used in other studies produced substantially different estimates supports the idea that dinosaur bite force estimates are significantly influenced by the methodology used to produce them.

Choice A is incorrect because it inaccurately describes data from the table. The table does show that the studies by Meers and by Cost et al. used different estimation methods and produced very different ranges of estimated dinosaur bite force, which would support Barrett and Rayfield's suggestion that different methodologies may produce significantly different estimates. However, the table doesn't show that the study by Meers produced the lowest estimated maximum bite force while the study by Cost et al. produced the highest. In fact, the study by Meers estimated a maximum bite force of approximately 235,000 newtons, which is the highest of all the estimated maximums. *Choice B* is incorrect. Although the data from Gignac and Ericson's study are accurately described, a single set of findings from one study using only one methodology can't show that different methodologies may produce significantly different dinosaur bite force estimates, as Barrett and Rayfield suggest. *Choice D* is incorrect. Although the table shows that the maximum bite force estimated by Cost et al. was higher than that estimated by Bates and Falkingham, the difference is relatively small; in fact, both teams estimated a minimum bite force of approximately 35,000 newtons and a maximum bite force close to approximately 60,000 newtons. Because these findings demonstrate that a single methodology (muscular and skeletal modeling) produced similar overall results in two studies, the findings don't support Barrett and Rayfield's suggestion that different methodologies may produce significantly different dinosaur bite force estimates.

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QUESTION 14

Choice A is the best answer because it most effectively uses data from the table to support the researchers' conclusion about the harvesting of clamshells by Neanderthals for use as tools. The text explains that Neanderthals used clamshells to make tools and that the sturdiest, and therefore the most desirable, shells for this purpose are found on the seafloor, not on the beach. However, the researchers also concluded that the clamshell tools made from shells from the seafloor are rarer than those made from shells from the beach. Meanwhile the table shows that at each depth, the number of tools made from shells from the beach exceeds the number made from the more desirable shells from the seafloor. The fact that the more desirable shells are less common suggests that it was significantly more difficult to harvest shells from the seafloor than from the beach.

Choice B is incorrect because knowing which depth represents the period of time with the highest Neanderthal population does not help answer the question of why the Neanderthals consistently made more tools from the less desirable shells from the beach than they made from the more desirable shells from the seafloor. *Choice C* is incorrect because it claims that the beach shells are more durable than the seafloor shells, which contradicts the text's description of shells from the seafloor as smoother and sturdier than shells from the beach. *Choice D* is incorrect because knowing which depth has the most artifacts or whether the clam population fluctuated does not help explain why tools made from the less desirable shells from the beach outnumber tools made from the more desirable shells from the seafloor.

QUESTION 15

Choice C is the best answer because it describes data from the table that support the researcher's hypothesis. According to the text, the researcher hypothesized that Arctic ground squirrels would exhibit longer torpor bouts and shorter arousal episodes than Alaska marmots do—or, put the other way, that the marmots would show shorter torpor bouts and longer arousal episodes than the ground squirrels do. The table shows data about torpor bouts and arousal episodes for the two species from 2008 to 2011. According to the table, the average duration of torpor bouts was 13.81 days for Alaska marmots, shorter than the average of 16.77 days for Arctic ground squirrels, and the average duration of arousal episodes was 21.2 hours for Alaska marmots, longer than the average of 14.2 hours for Arctic ground squirrels. Thus, the table supports the researcher's hypothesis by showing that Alaska marmots had shorter bouts of torpor and longer arousal episodes than Arctic ground squirrels did.

Choice A is incorrect because it inaccurately describes data from the table and doesn't support the researcher's hypothesis. The table shows that the average duration of arousal episodes was less than a day for both Alaska marmots (21.2 hours) and Arctic ground squirrels (14.2 hours). Additionally, information about arousal episodes for Alaska marmots and Arctic ground squirrels isn't sufficient to support a hypothesis involving comparisons of both arousal episodes and torpor bouts for those animals. *Choice B* is incorrect because it doesn't support the researcher's hypothesis, which involves comparisons of arousal episodes

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as well as torpor bouts for Alaska marmots and Arctic ground squirrels. Noting that both animals had torpor bouts lasting several days, on average, doesn't address arousal episodes at all, nor does it reveal how the animals' torpor bouts compared. *Choice D* is incorrect because it doesn't support the researcher's hypothesis. Although the table does show that Alaska marmots had more torpor bouts (12) than arousal episodes (11) and that their arousal episodes were much shorter than their torpor bouts (21.2 hours and 13.81 days, respectively), comparing data across only Alaska marmot behaviors isn't sufficient to support a hypothesis about torpor and arousal behaviors of both Alaska marmots and Arctic ground squirrels.

QUESTION 16

Choice C is the best answer because it most logically completes the argument about an unintended effect of the Nagoya Protocol. The text explains that the Nagoya Protocol is an agreement ensuring that Indigenous communities are compensated when their agricultural resources and knowledge are used by corporations. The text then states that the protocol allows corporations to keep their agreements with Indigenous communities confidential, about which some Indigenous advocates express concern. Choice C, when inserted into the blank, gives a good justification for the advocates' concern: such secrecy could mean that the public is unable to determine whether participating Indigenous communities were properly compensated under these agreements.

Choice A is incorrect. The text suggests that because corporations can keep their agreements with Indigenous communities confidential, Indigenous communities, not corporations, might not be compensated fairly. *Choice B* is incorrect because the text doesn't suggest that the ability of corporations to keep their agreements with Indigenous communities confidential would place limits on how much research corporations can undertake. *Choice D* is incorrect because the text doesn't indicate that Indigenous communities aim to learn new harvesting methods from their corporate partners. Rather, the text suggests that corporations use the knowledge of Indigenous communities for their research.

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QUESTION 17

Choice C is the best answer because it most logically completes the text's discussion of the sweet potato in Polynesia. The text indicates that the sweet potato is found in Polynesia but originated in South America, and that the sweet potato was being cultivated by Native Hawaiians and other Indigenous peoples in Polynesia long before sea voyages between South America and Polynesia began. The text goes on to note that research by Muñoz-Rodríguez and colleagues has established that the Polynesian varieties of sweet potato split from South American varieties more than 100,000 years ago, which is thousands of years before humans settled in Polynesia. If Polynesian peoples were cultivating the sweet potato before sea voyages between Polynesia and South America began, and if Polynesian varieties of sweet potato diverged from South American varieties well before people were in Polynesia, it can reasonably be concluded that humans didn't play a role in bringing the sweet potato to Polynesia.

Choice A is incorrect. The text doesn't provide any information about when the sweet potato began to be cultivated in South America, so there's no support for the conclusion that cultivation began in Polynesia before it began in South America. *Choice B* is incorrect because the text indicates that the sweet potato was being cultivated in Polynesia long before sea journeys between Polynesia and South America began. Therefore, it wouldn't be reasonable to conclude that Polynesian peoples acquired the sweet potato from South American peoples. Additionally, the text indicates that the Polynesian varieties of sweet potato diverged from the South American varieties thousands of years before people settled in Polynesia, which suggests that the sweet potato was already present in Polynesia when people arrived. *Choice D* is incorrect because the text states that the domestic sweet potato, which is found in Polynesia, descends from a wild South American plant, not from a domesticated South American plant. The only people that the text describes as cultivating the sweet potato are Native Hawaiians and other Indigenous peoples of Polynesia.

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QUESTION 18

Choice B is the best answer. The convention being tested is the use of verbs to express tense in a sentence. In this choice, the present tense verb "reach" is consistent with the present tense verbs "travel" and "are diverted" used to describe how atoms move through the synchrotron.

Choice A is incorrect because the future tense verb "will reach" is inconsistent with the present tense verbs used to describe how atoms move through the synchrotron. Though the atoms' movement is a recurring action and "will reach" can also be used to indicate a habitual or recurring action, it creates a logical inconsistency in this sentence when paired with the present tense verbs "travel" and "are diverted." *Choice C* is incorrect because the past perfect tense verb "had reached" is inconsistent with the present tense verbs used to describe how atoms move through the synchrotron. *Choice D* is incorrect because the present progressive tense verb "are reaching" is inconsistent with the present tense verbs used to describe how atoms move through the synchrotron. While both verbs occur in the present, the present progressive tense suggests that the action is currently in progress. This creates a logical inconsistency when paired with the present tense verbs "travel" and "are diverted," which offer a general description of the tendencies of the atoms' movement, rather than a description of an action that is currently in progress.

QUESTION 19

Choice A is the best answer. The convention being tested is the use of finite and nonfinite verb forms within a sentence. A main clause requires a finite verb to perform the action of the subject (in this case, "a recent study"), and this choice supplies the finite present tense verb "explains" to indicate that the study explains why plants that attract bats have evolved to produce moderately sweet nectar.

Choice B is incorrect because the nonfinite participle "explaining" doesn't supply the main clause with a finite verb. *Choice C* is incorrect because the nonfinite participle "having explained" doesn't supply the main clause with a finite verb. *Choice D* is incorrect because the nonfinite to-infinitive "to explain" doesn't supply the main clause with a finite verb.

QUESTION 20

Choice C is the best answer. The convention being tested is subject-verb agreement. The singular verb "outlines" agrees in number with the singular subject "document."

Choice A is incorrect because the plural verb "have outlined" doesn't agree in number with the singular subject "document." *Choice B* is incorrect because the plural verb "were outlining" doesn't agree in number with the singular subject "document." *Choice D* is incorrect because the plural verb "outline" doesn't agree in number with the singular subject "document."

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QUESTION 21

Choice D is the best answer. The convention being tested is punctuation between a main clause and a subordinate clause. This choice correctly uses a comma to mark the boundary between the main clause ("the colorful…decade") and the subordinate clause ("while…centuries") that provides contrasting information about the life span of rougheye rockfish.

Choice A is incorrect because a colon can't be used in this way to join a main clause and a subordinate clause. *Choice B* is incorrect because it results in a rhetorically unacceptable sentence fragment beginning with "while." *Choice C* is incorrect because a semicolon can't be used in this way to join a main clause and a subordinate clause.

QUESTION 22

Choice C is the best answer. The convention being tested is punctuation between a supplementary phrase and a main clause. This choice correctly uses a comma to mark the boundary between the supplementary phrase ("powered…day"), which describes how the LEDs are powered, and the main clause ("the blinking… night").

Choice A is incorrect because it fails to mark the boundary between the supplementary phrase and the main clause with appropriate punctuation. Furthermore, placing commas around the phrase "by solar panels" suggests that it could be removed without affecting the coherence of the sentence, which isn't the case. *Choice B* is incorrect because it fails to mark the boundary between the supplementary phrase and the main clause with appropriate punctuation. *Choice D* is incorrect. Placing commas around the phrase "collected by solar panels during the day" suggests that it could be removed without affecting the coherence of the sentence, which isn't the case.

QUESTION 23

Choice D is the best answer. The convention being tested is the coordination of main clauses. This choice correctly uses a comma and the coordinating conjunction "but" to join the first main clause ("Materials… Ru") and the second main clause ("the alloy…NiCoCr").

Choice A is incorrect because it results in a run-on sentence. The two main clauses are fused without punctuation and/or a conjunction. *Choice B* is incorrect because when coordinating two longer main clauses such as these, it's conventional to use a comma before the coordinating conjunction. *Choice C* is incorrect because it results in a comma splice. Without a conjunction following it, a comma can't be used in this way to join two main clauses.

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QUESTION 24

Choice A is the best answer. The convention being tested here is subject-verb agreement. The singular verb "was" agrees in number with the singular subject "Josephine St. Pierre Ruffin."

Choice B is incorrect because the plural verb "were" doesn't agree in number with the singular subject "Josephine St. Pierre Ruffin." *Choice C* is incorrect because the plural verb "are" doesn't agree in number with the singular subject "Josephine St. Pierre Ruffin." *Choice D* is incorrect because the plural verb "have been" doesn't agree in number with the singular subject "Josephine St. Pierre Ruffin."

QUESTION 25

Choice A is the best answer. The convention being tested is subject-modifier placement. This choice makes the noun phrase "researcher Robert Losey" the subject of the sentence and places it immediately after the modifying phrase "since…Siberia." In doing so, this choice clearly establishes that researcher Robert Losey— and not another noun in the sentence—is who uncovered fragments of a 2,000-year-old reindeer training harness in northern Siberia.

Choice B is incorrect because it results in a dangling modifier. The placement of the noun phrase "researcher Robert Losey's argument" immediately after the modifying phrase illogically suggests that the "argument" is what uncovered fragments of a 2,000-year-old reindeer training harness in northern Siberia. *Choice C* is incorrect because it results in a dangling modifier. The placement of the noun "domestication" immediately after the modifying phrase illogically suggests that "domestication" is what uncovered fragments of a 2,000-year-old reindeer training harness in northern Siberia. *Choice D* is incorrect because it results in a dangling modifier. The placement of the noun phrase "the argument" immediately after the modifying phrase illogically suggests that the "argument" is what uncovered fragments of a 2,000-year-old reindeer training harness in northern Siberia.

QUESTION 26

Choice A is the best answer. The convention being tested is punctuation use between sentences. In this choice, the period after "tombs" is used correctly to mark the boundary between one sentence ("Archaeologist…tombs") and another ("Built…nature").

Choice B is incorrect because it results in a comma splice. A comma can't be used in this way to mark the boundary between sentences. *Choice C* is incorrect. Without a comma preceding it, the conjunction "and" can't be used in this way to join the two sentences. *Choice D* is incorrect because it results in a run-on sentence. The sentences ("Archaeologist…tombs" and "Built…nature") are fused without punctuation and/or a conjunction.

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QUESTION 27

Choice D is the best answer. The convention being tested is the use of punctuation around noun phrases. No punctuation is needed because the noun phrase "aluminum oxide" is a restrictive appositive, meaning that it provides essential identifying information about the noun phrase before it, "the chemical compound."

Choice A is incorrect because no punctuation is needed. *Choice B* is incorrect because no punctuation is needed. *Choice C* is incorrect because the noun phrase "aluminum oxide" is a restrictive appositive. Setting the phrase off with commas suggests that it could be removed without affecting the coherence of the sentence, which isn't the case.

QUESTION 28

Choice A is the best answer. "Currently" logically signals that the archaeologists' use of drones (a current technology) to photograph the lines is the present-day continuation of the ongoing archaeological research described in the previous sentence.

Choice B is incorrect because "in comparison" illogically signals that the action described in this sentence offers a comparison to the ongoing archaeological research described in the previous sentence. Instead, the use of drones is the present-day continuation of that research. *Choice C* is incorrect because "still" illogically signals that the action described in this sentence occurs despite the ongoing archaeological research described in the previous sentence. Instead, the use of drones is the present-day continuation of that research. *Choice D* is incorrect because "however" illogically signals that the action described in this sentence occurs either despite or in contrast to the ongoing archaeological research described in the previous sentence. Instead, the use of drones is the present-day continuation of that research.

QUESTION 29

Choice D is the best answer. "Second" logically signals that the information in this sentence—that the effort to bury the ship would likely only have been made for a king—joins the information in the previous sentence ("first…") in supporting Brunning's claim that the burial site was likely the tomb of a king.

Choice A is incorrect because "instead" illogically signals that the information in this sentence presents an alternative or substitute to the previous information about the gold artifacts inside the ship. Rather, this sentence presents a second piece of information that supports Brunning's claim. *Choice B* is incorrect because "still" illogically signals that the information in this sentence exists in contrast to or despite the previous information about the gold artifacts inside the ship. Instead, this sentence presents a second piece of information that supports Brunning's claim. *Choice C* is incorrect because "specifically" illogically signals that the information in this sentence specifies or elaborates on the previous information about the gold artifacts inside the ship. Instead, this sentence presents a second piece of information that supports Brunning's claim.

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QUESTION 30

Choice D is the best answer. "Thus" logically signals that the claim in this sentence—that animals performing only basic actions should allocate relatively few resources to their brain tissue—is a consequence of the previous sentence's claim about the energy demands of animal brains (namely, that the more diverse an animal's behaviors, the more energy its brain needs).

Choice A is incorrect because "subsequently" illogically signals that the claim in this sentence occurs later in a chronological sequence of events than the previous sentence's claim about the energy demands of animal brains. Instead, the second claim is a consequence of the first. *Choice B* is incorrect because "besides" illogically signals that the claim in this sentence provides a separate point in addition to, or apart from, the previous sentence's claim about the energy demands of animal brains. Instead, the second claim is a consequence of the first. *Choice C* is incorrect because "nevertheless" illogically signals that the claim in this sentence is true in spite of the previous sentence's claim about the energy demands of animal brains. Instead, the second claim is a consequence of the first.

QUESTION 31

Choice A is the best answer. "Nevertheless" logically signals that the information in this sentence—that the spacesuits Suttirat Larlarb designed for the film *Sunshine* were made in standard sizes in a factory—presents a notable exception to Larlarb's typical approach of custom-fitting garments to actors, which is described in the previous sentence.

Choice B is incorrect because "thus" illogically signals that the information in this sentence is a result or consequence of the previous information about Larlarb's typical approach of custom-fitting garments to actors. Instead, it presents a notable exception to Larlarb's typical approach. *Choice C* is incorrect because "likewise" illogically signals that the information in this sentence is similar to the previous information about Larlarb's typical approach of custom-fitting garments to actors. Instead, it presents a notable exception to Larlarb's typical approach. *Choice D* is incorrect because "moreover" illogically signals that the information in this sentence merely adds to the previous information about Larlarb's typical approach of custom-fitting garments to actors. Instead, it presents a notable exception to Larlarb's typical approach.

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QUESTION 32

Choice D is the best answer. The sentence uses "both" to emphasize a thematic similarity between Tan's two books, noting that both *Tales from Outer Suburbia* and *Tales from the Inner City* describe surreal events occurring in otherwise ordinary places.

Choice A is incorrect. The sentence emphasizes a difference (one contains fewer stories than the other), not a similarity, between the two books. *Choice B* is incorrect. The sentence indicates that Tan's books were published ten years apart; it doesn't emphasize a similarity between the two books. *Choice C* is incorrect. The sentence uses "unlike" to emphasize a difference between *Tales from Outer Suburbia* and *Tales from the Inner City*; it doesn't emphasize a similarity between the two books.

QUESTION 33

Choice A is the best answer. The sentence emphasizes the aim of the research study by highlighting what the researchers conducting the study wanted to know—specifically, which factors influence clutch size among lizards.

Choice B is incorrect because the sentence emphasizes what researchers determined at the end of the study, not what the study's aim was. *Choice C* is incorrect because the sentence emphasizes a finding from the research study, not the aim of the study. *Choice D* is incorrect because the sentence emphasizes the research study's methodology, not its aim.

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Math

Module 1

(27 questions)

QUESTION 1

Choice B is correct. Subtracting 12 from both sides of the given equation yields $k = 324$. Therefore, the solution to the given equation is 324.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 2

Choice C is correct. The value of $f(2)$ is the value of $f(x)$ when $x = 2$. Substituting 2 for x in the given function yields $f(2) = (2)^3 + 15$, or $f(2) = 8 + 15$, which is equivalent to $f(2) = 23$. Therefore, the value of $f(2)$ is 23.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice B* is incorrect. This is the value of $f(2)$ when $f(x) = x(3) + 15$, rather than $f(x) = x^3 + 15$. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 3

Choice C is correct. It's given that the cost of renting a tent is $\$11$ per day for d days. Multiplying the rental cost by the number of days yields $\$11d$, which represents the cost of renting the tent for d days before the insurance is added. Adding the onetime insurance fee of $\$10$ to the rental cost of $\$11d$ gives the total cost c, in dollars, which can be represented by the equation $c = 11d + 10$.

Choice A is incorrect. This equation represents the total cost to rent the tent if the insurance fee was charged every day. *Choice B* is incorrect. This equation represents the total cost to rent the tent if the daily fee was $\$(d + 11)$ for 10 days. *Choice D* is incorrect. This equation represents the total cost to rent the tent if the daily fee was $\$10$ and the onetime fee was $\$11$.

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QUESTION 4

Choice D is correct. The sum of consecutive interior angles between two parallel lines and on the same side of the transversal is 180 degrees. Since it's given that line m is parallel to line n, it follows that $x+26=180$. Subtracting 26 from both sides of this equation yields 154. Therefore, the value of x is 154.

Choice A is incorrect. This is half of the given angle measure. *Choice B* is incorrect. This is the value of the given angle measure. *Choice C* is incorrect. This is twice the value of the given angle measure.

QUESTION 5

Choice C is correct. It's given that John made a $\$16$ payment each month for p months. The total amount of these payments can be represented by the expression $16p$. The down payment can be added to that amount to find the total amount John paid, yielding the expression $16p+37$. It's given that John paid a total of $\$165$. Therefore, the expression for the total amount John paid can be set equal to that amount, yielding the equation $16p+37=165$.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice B* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 6

The correct answer is 50. Substituting 8 for x in the given equation yields $y=5(8)+10$, or $y=50$. Therefore, the value of y is 50 when $x=8$.

QUESTION 7

The correct answer is 40. The height of each bar in the bar graph shown represents the number of cans collected by the group specified at the bottom of the bar. The bar for group 6 reaches a height of 40. Therefore, group 6 collected 40 cans.

QUESTION 8

Choice C is correct. If one of these students is selected at random, the probability of selecting a student whose vote for the new mascot was for a lion is given by the number of votes for a lion divided by the total number of votes. The given table indicates that the number of votes for a lion is 20 votes, and the total number of votes is 80 votes. The table gives the distribution of votes for 80 students, and the table shows a total of 80 votes were counted. It follows that each of the 80 students voted exactly once. Thus, the probability of selecting a student whose vote for the new mascot was for a lion is $\frac{20}{80}$, or $\frac{1}{4}$.

Choice A is incorrect and may result from conceptual or computational errors. *Choice B* is incorrect and may result from conceptual or computational errors. *Choice D* is incorrect and may result from conceptual or computational errors.

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QUESTION 9

Choice A is correct. It's given that the electrician charges a onetime fee plus an hourly rate. It's also given that the graph represents the total charge, in dollars, for x hours of work. This graph shows a linear relationship in the *xy*-plane. Thus, the total charge y, in dollars, for x hours of work can be represented as $y = mx + b$, where m is the slope and $(0, b)$ is the *y*-intercept of the graph of the equation in the *xy*-plane. Since the given graph represents the total charge, in dollars, by an electrician for x hours of work, it follows that its slope is m, or the electrician's hourly rate.

Choice B is incorrect. The electrician's onetime fee is represented by the *y*-coordinate of the *y*-intercept, not the slope, of the graph. *Choice C* is incorrect and may result from conceptual errors. *Choice D* is incorrect and may result from conceptual errors.

QUESTION 10

Choice D is correct. The perimeter, P, of a square can be found using the formula $P = 4s$, where s is the length of each side of the square. It's given that square X has a side length of 12 centimeters. Substituting 12 for s in the formula for the perimeter of a square yields $P = 4(12)$, or $P = 48$. Therefore, the perimeter of square X is 48 centimeters. It's also given that the perimeter of square Y is 2 times the perimeter of square X. Therefore, the perimeter of square Y is $2(48)$, or 96, centimeters. Substituting 96 for P in the formula $P = 4s$ gives $96 = 4s$. Dividing both sides of this equation by 4 gives $24 = s$. Therefore, the length of one side of square Y is 24 centimeters.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice B* is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect and may result from conceptual or calculation errors.

QUESTION 11

Choice B is correct. The equation of a line in the *xy*-plane can be written in slope-intercept form $y = mx + b$, where m is the slope of the line and $(0, b)$ is its *y*-intercept. It's given that the line passes through the point $(0, 5)$. Therefore, $b = 5$. It's also given that the line is parallel to the graph of $y = 7x + 4$, which means the line has the same slope as the graph of $y = 7x + 4$. The slope of the graph of $y = 7x + 4$ is 7. Therefore, $m = 7$. Substituting 7 for m and 5 for b in the equation $y = mx + b$ yields $y = 7x + 5$.

Choice A is incorrect. The graph of this equation passes through the point $(0, 0)$, not $(0, 5)$, and has a slope of 5, not 7. *Choice C* is incorrect. The graph of this equation passes through the point $(0, 0)$, not $(0, 5)$. *Choice D* is incorrect. The graph of this equation passes through the point $(0, 7)$, not $(0, 5)$, and has a slope of 5, not 7.

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QUESTION 12

Choice A is correct. An equation defining a linear function can be written in the form $h(x)=ax+b$, where a and b are constants. It's given that $h(0)=41$. Substituting 0 for x and 41 for $h(x)$ in the equation $h(x)=ax+b$ yields $41=a(0)+b$, or $b=41$. Substituting 41 for b in the equation $h(x)=ax+b$ yields $h(x)=ax+41$. It's also given that $h(1)=40$. Substituting 1 for x and 40 for $h(x)$ in the equation $h(x)=ax+41$ yields $40=a(1)+41$, or $40=a+41$. Subtracting 41 from the left- and right-hand sides of this equation yields $-1=a$. Substituting -1 for a in the equation $h(x)=ax+41$ yields $h(x)=-1x+41$, or $h(x)=-x+41$.

Choice B is incorrect. Substituting 0 for x and 41 for $h(x)$ in this equation yields $41=-0$, which isn't a true statement. *Choice C* is incorrect. Substituting 0 for x and 41 for $h(x)$ in this equation yields $41=-41(0)$, or $41=0$, which isn't a true statement. *Choice D* is incorrect. Substituting 41 for $h(x)$ in this equation yields $41=-41$, which isn't a true statement.

QUESTION 13

The correct answer is 410. It's given that t minutes after an initial observation, the number of bacteria in a population is $60{,}000(2)^{\frac{t}{410}}$. This expression consists of the initial number of bacteria, $60{,}000$, multiplied by the expression $2^{\frac{t}{410}}$. The time it takes for the number of bacteria to double is the increase in the value of t that causes the expression $2^{\frac{t}{410}}$ to double. Since the base of the expression $2^{\frac{t}{410}}$ is 2, the expression $2^{\frac{t}{410}}$ will double when the exponent increases by 1. Since the exponent of the expression $2^{\frac{t}{410}}$ is $\frac{t}{410}$, the exponent will increase by 1 when t increases by 410. Therefore the time, in minutes, it takes for the number of bacteria in the population to double is 410.

QUESTION 14

The correct answer is 76. It's given that the graph of $y=g(x)$ is the result of translating the graph of $y=f(x)$ up 4 units in the xy-plane. It follows that the graph of $y=g(x)$ is the same as the graph of $y=f(x)+4$. Substituting $g(x)$ for y in the equation $y=f(x)+4$ yields $g(x)=f(x)+4$. It's given that $f(x)=(x-6)(x-2)(x+6)$. Substituting $(x-6)(x-2)(x+6)$ for $f(x)$ in the equation $g(x)=f(x)+4$ yields $g(x)=(x-6)(x-2)(x+6)+4$. Substituting 0 for x in this equation yields $g(0)=(0-6)(0-2)(0+6)+4$, or $g(0)=76$. Thus, the value of $g(0)$ is 76.

QUESTION 15

Choice D is correct. It's given that the candle starts with 17 ounces of wax and has 6 ounces of wax remaining after a period of time has passed. The amount of wax the candle has lost during the time period can be found by subtracting the

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remaining amount of wax from the amount of wax the candle was made of, which yields $17-6$ ounces, or 11 ounces. This means the candle loses 11 ounces of wax during that period of time. It's given that the amount of wax decreases by 1 ounce every 4 hours. If h represents the number of hours the candle has been burning, it follows that $\frac{1}{4}=\frac{11}{h}$. Multiplying both sides of this equation by $4h$ yields $h=44$. Therefore, the candle has been burning for 44 hours.

Choice A is incorrect and may result from using the equation $\frac{1}{4}=\frac{h}{11}$ rather than $\frac{1}{4}=\frac{11}{h}$ to represent the situation, and then rounding to the nearest whole number. *Choice B* is incorrect. This is the amount of wax, in ounces, remaining in the candle, not the number of hours it has been burning. *Choice C* is incorrect and may result from using the equation $\frac{1}{4}=\frac{6}{h}$ rather than $\frac{1}{4}=\frac{11}{h}$ to represent the situation.

QUESTION 16

Choice A is correct. Subtracting $14j$ from each side of the given equation results in $5k=m-14j$. Dividing each side of this equation by 5 results in $k=\frac{m-14j}{5}$.

Choice B is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 17

Choice B is correct. If two triangles are similar, then their corresponding angles are congruent. It's given that right triangle FGH is similar to right triangle JKL and angle F corresponds to angle J. It follows that angle F is congruent to angle J and, therefore, the measure of angle F is equal to the measure of angle J. The sine ratios of angles of equal measure are equal. Since the measure of angle F is equal to the measure of angle J, $\sin(F)=\sin(J)$. It's given that $\sin(F)=\frac{308}{317}$. Therefore, $\sin(J)$ is $\frac{308}{317}$.

Choice A is incorrect. This is the value of $\cos(J)$, not the value of $\sin(J)$. *Choice C* is incorrect. This is the reciprocal of the value of $\sin(J)$, not the value of $\sin(J)$. *Choice D* is incorrect. This is the reciprocal of the value of $\cos(J)$, not the value of $\sin(J)$.

QUESTION 18

Choice B is correct. Let x be the first integer and let y be the second integer. If the first integer is 11 greater than twice the second integer, then $x=2y+11$. If the product of the two integers is 546, then $xy=546$. Substituting $2y+11$ for x in this equation results in $(2y+11)y=546$. Distributing the y to both terms in the parentheses results in $2y^2+11y=546$. Subtracting 546 from both sides of this equation results in $2y^2+11y-546=0$. The left-hand side of this equation can be factored by finding two values whose product is $2(-546)$, or $-1{,}092$, and whose sum is 11. The two values whose product is $-1{,}092$ and whose sum is 11 are 39 and -28. Thus, the equation $2y^2+11y-546=0$ can be rewritten as

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$2y^2+28y-39y-546=0$, which is equivalent to $2y(y-14)+39(y-14)=0$, or $(2y+39)(y-14)=0$. By the zero product property, it follows that $2y+39=0$ and $y-14=0$. Subtracting 39 from both sides of the equation $2y+39=0$ yields $2y=-39$. Dividing both sides of this equation by 2 yields $y=-\frac{39}{2}$. Since y is a positive integer, the value of y is not $-\frac{39}{2}$. Adding 14 to both sides of the equation $y-14=0$ yields $y=14$. Substituting 14 for y in the equation $xy=546$ yields $x(14)=546$. Dividing both sides of this equation by 14 results in $x=39$. Therefore, the two integers are 14 and 39, so the smaller of the two integers is 14.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect. This is the larger of the two integers. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 19

Choice D is correct. A point (x, y) is a solution to a system of inequalities in the *xy*-plane if substituting the *x*-coordinate and the *y*-coordinate of the point for x and y, respectively, in each inequality makes both of the inequalities true. Substituting the *x*-coordinate and the *y*-coordinate of choice D, 14 and 0, for x and y, respectively, in the first inequality in the given system, $y\le x+7$, yields $0\le 14+7$, or $0\le 21$, which is true. Substituting 14 for x and 0 for y in the second inequality in the given system, $y\ge -2x-1$, yields $0\ge -2(14)-1$, or $0\ge -29$, which is true. Therefore, the point $(14, 0)$ is a solution to the given system of inequalities in the *xy*-plane.

Choice A is incorrect. Substituting -14 for x and 0 for y in the inequality $y\le x+7$ yields $0\le -14+7$, or $0\le -7$, which is not true. *Choice B* is incorrect. Substituting 0 for x and -14 for y in the inequality $y\ge -2x-1$ yields $-14\ge -2(0)-1$, or $-14\ge -1$, which is not true. *Choice C* is incorrect. Substituting 0 for x and 14 for y in the inequality $y\le x+7$ yields $14\le 0+7$, or $14\le 7$, which is not true.

QUESTION 20

The correct answer is -3. Squaring both sides of the given equation yields $(x-2)^2=3x+34$, which can be rewritten as $x^2-4x+4=3x+34$. Subtracting $3x$ and 34 from both sides of this equation yields $x^2-7x-30=0$. This quadratic equation can be rewritten as $(x-10)(x+3)=0$. According to the zero product property, $(x-10)(x+3)$ equals zero when either $x-10=0$ or $x+3=0$. Solving each of these equations for x yields $x=10$ or $x=-3$. Therefore, the given equation has two solutions, 10 and -3. Of these two solutions, -3 is the smallest solution to the given equation.

QUESTION 21

The correct answer is 1.8. It's given that the regular price of a shirt at a store is $\$11.70$, and the sale price of the shirt is 80% less than the regular price. It

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follows that the sale price of the shirt is $\$11.70\left(1-\frac{80}{100}\right)$, or $\$11.70(1-0.8)$, which is equivalent to $\$2.34$. It's also given that the sale price of the shirt is 30% greater than the store's cost for the shirt. Let x represent the store's cost for the shirt. It follows that $2.34=\left(1+\frac{30}{100}\right)x$, or $2.34=1.3x$. Dividing both sides of this equation by 1.3 yields $x=1.80$. Therefore, the store's cost, in dollars, for the shirt is 1.80. Note that 1.8 and 9/5 are examples of ways to enter a correct answer.

QUESTION 22

Choice A is correct. It's given that the sample is in the shape of a cube with edge lengths of 0.9 meters. Therefore, the volume of the sample is 0.90^3, or 0.729, cubic meters. It's also given that the sample has a density of 807 kilograms per 1 cubic meter. Therefore, the mass of this sample is $0.729 \text{ cubic meters}\left(\frac{807 \text{ kilograms}}{1 \text{ cubic meter}}\right)$, or 588.303 kilograms. Rounding this mass to the nearest whole number gives 588 kilograms. Therefore, to the nearest whole number, the mass, in kilograms, of this sample is 588.

Choice B is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 23

Choice D is correct. It's given that for $x>0$, $f(x)$ is equal to 201% of x. This is equivalent to $f(x)=\frac{201}{100}x$, or $f(x)=2.01x$, for $x>0$. This function indicates that as x increases, $f(x)$ also increases, which means f is an increasing function. Furthermore, $f(x)$ increases at a constant rate of 2.01 for each increase of x by 1. A function with a constant rate of change is linear. Thus, the function f can be described as an increasing linear function.

Choice A is incorrect and may result from conceptual errors. *Choice B* is incorrect and may result from conceptual errors. *Choice C* is incorrect. This could describe the function $f(x)=(2.01)^x$, where $f(x)$ is equal to 201% of $f(x-1)$, not x, for $x>0$.

QUESTION 24

Choice C is correct. It's given that $f(x)=\frac{a}{x+b}$ and that the graph shown is a partial graph of $y=f(x)$. Substituting y for $f(x)$ in the equation $f(x)=\frac{a}{x+b}$ yields $y=\frac{a}{x+b}$. The graph passes through the point $(-7, -2)$. Substituting -7 for x and -2 for y in the equation $y=\frac{a}{x+b}$ yields $-2=\frac{a}{-7+b}$. Multiplying each side of this equation by $-7+b$ yields $-2(-7+b)=a$, or $14-2b=a$. The graph also passes through the point $(-5, -6)$. Substituting -5 for x and -6 for y in the equation $y=\frac{a}{x+b}$ yields $-6=\frac{a}{-5+b}$. Multiplying each side of this equation by

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$-5+b$ yields $-6(-5+b)=a$, or $30-6b=a$. Substituting $14-2b$ for a in this equation yields $30-6b=14-2b$. Adding $6b$ to each side of this equation yields $30=14+4b$. Subtracting 14 from each side of this equation yields $16=4b$. Dividing each side of this equation by 4 yields $4=b$. Substituting 4 for b in the equation $14-2b=a$ yields $14-2(4)=a$, or $6=a$. Substituting 6 for a and 4 for b in the equation $f(x)=\frac{a}{x+b}$ yields $f(x)=\frac{6}{x+4}$. It's given that $g(x)=f(x+4)$. Substituting $x+4$ for x in the equation $f(x)=\frac{6}{x+4}$ yields $f(x+4)=\frac{6}{x+4+4}$, which is equivalent to $f(x+4)=\frac{6}{x+8}$. It follows that $g(x)=\frac{6}{x+8}$.

Choice A is incorrect. This could define function g if $g(x)=f(x-4)$. *Choice B* is incorrect. This could define function g if $g(x)=f(x)$. *Choice D* is incorrect. This could define function g if $g(x)=f(x)\cdot(x+4)$.

QUESTION 25

Choice C is correct. Factoring the denominator in the second term of the given expression gives $\frac{y+12}{x-8}+\frac{y(x-8)}{xy(x-8)}$. This expression can be rewritten with common denominators by multiplying the first term by $\frac{xy}{xy}$, giving $\frac{xy(y+12)}{xy(x-8)}+\frac{y(x-8)}{xy(x-8)}$. Adding these two terms yields $\frac{xy(y+12)+y(x-8)}{xy(x-8)}$. Using the distributive property to rewrite this expression gives $\frac{xy^2+12xy+xy-8y}{x^2y-8xy}$. Combining the like terms in the numerator of this expression gives $\frac{xy^2+13xy-8y}{x^2y-8xy}$.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice B is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 26

Choice B is correct. It's given that 483 out of 803 voters responded that they would vote for Angel Cruz. Therefore, the proportion of voters from the poll who responded they would vote for Angel Cruz is $\frac{483}{803}$. It's also given that there are a total of $6,424$ voters in the election. Therefore, the total number of people who would be expected to vote for Angel Cruz is $6,424\left(\frac{483}{803}\right)$, or $3,864$. Since $3,864$ of the $6,424$ total voters would be expected to vote for Angel Cruz, it follows that $6,424-3,864$, or $2,560$ voters would be expected not to vote for Angel Cruz. The difference in the number of votes for and against Angel Cruz is $3,864-2,560$, or $1,304$ votes. Therefore, if $6,424$ people vote in the election, Angel Cruz would be expected to win by $1,304$ votes.

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Choice A is incorrect. This is the difference in the number of voters from the poll who responded that they would vote for and against Angel Cruz. *Choice C* is incorrect. This is the total number of people who would be expected to vote for Angel Cruz. *Choice D* is incorrect. This is the difference between the total number of people who vote in the election and the number of voters from the poll.

QUESTION 27

The correct answer is 10. It's given that the graph of $x^2+x+y^2+y=\frac{199}{2}$ in the *xy*-plane is a circle. The equation of a circle in the *xy*-plane can be written in the form $(x-h)^2+(y-k)^2=r^2$, where the coordinates of the center of the circle are (h,k) and the length of the radius of the circle is r. The term $(x-h)^2$ in this equation can be obtained by adding the square of half the coefficient of x to both sides of the given equation to complete the square. The coefficient of x is 1. Half the coefficient of x is $\frac{1}{2}$. The square of half the coefficient of x is $\frac{1}{4}$. Adding $\frac{1}{4}$ to each side of $(x^2+x)+(y^2+y)=\frac{199}{2}$ yields $\left(x^2+x+\frac{1}{4}\right)+(y^2+y)=\frac{199}{2}+\frac{1}{4}$, or $\left(x+\frac{1}{2}\right)^2+(y^2+y)=\frac{199}{2}+\frac{1}{4}$. Similarly, the term $(y-k)^2$ can be obtained by adding the square of half the coefficient of y to both sides of this equation, which yields $\left(x+\frac{1}{2}\right)^2+\left(y^2+y+\frac{1}{4}\right)=\frac{199}{2}+\frac{1}{4}+\frac{1}{4}$, or $\left(x+\frac{1}{2}\right)^2+\left(y+\frac{1}{2}\right)^2=\frac{199}{2}+\frac{1}{4}+\frac{1}{4}$. This equation is equivalent to $\left(x+\frac{1}{2}\right)^2+\left(y+\frac{1}{2}\right)^2=100$, or $\left(x+\frac{1}{2}\right)^2+\left(y+\frac{1}{2}\right)^2=10^2$. Therefore, the length of the circle's radius is 10.

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Math

Module 2
(27 questions)

QUESTION 1

Choice B is correct. The number of harvested potatoes Isabel saved to plant next year can be calculated by multiplying the total number of potatoes Isabel harvested, 760, by the proportion of potatoes she saved. Since she saved 10% of the potatoes she harvested, the proportion of potatoes she saved is $\frac{10}{100}$, or 0.1. Multiplying 760 by this proportion gives $760(0.1)$, or 76, potatoes that she saved to plant next year.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 2

Choice B is correct. The *y*-intercept of a graph in the *xy*-plane is the point at which the graph crosses the *y*-axis. The graph shown crosses the *y*-axis at the point $(0, 2)$. Therefore, the *y*-intercept of the graph shown is $(0, 2)$.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 3

Choice C is correct. Since 1 meter is equal to 100 centimeters, 51 meters is equal to $51 \text{ meters}\left(\frac{100 \text{ centimeters}}{1 \text{ meter}}\right)$, or $5{,}100$ centimeters.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice B is incorrect and may result from dividing, rather than multiplying, 51 by 100. *Choice D* is incorrect. This is the length, in millimeters rather than centimeters, that is equivalent to a length of 51 meters.

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QUESTION 4

Choice C is correct. It's given that t represents the number of seconds after the bus passes the marker. Substituting 2 for t in the given equation $d=30t$ yields $d=30(2)$, or $d=60$. Therefore, the bus will be 60 feet from the marker 2 seconds after passing it.

Choice A is incorrect. This is the distance, in feet, the bus will be from the marker 1 second, not 2 seconds, after passing it. *Choice B* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect. This is the distance, in feet, the bus will be from the marker 3 seconds, not 2 seconds, after passing it.

QUESTION 5

Choice B is correct. Combining like terms inside the parentheses of the given expression, $20w-(4w+3w)$, yields $20w-(7w)$. Combining like terms in this resulting expression yields $13w$.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 6

The correct answer is 27. Multiplying both sides of the given equation by 3 yields $3(6+x)=3(9)$, or $18+3x=27$. Therefore, the value of $18+3x$ is 27.

QUESTION 7

The correct answer is 7. When an equation is of the form $y=ax^2+bx+c$, where a, b, and c are constants, the value of y reaches its minimum when $x=-\frac{b}{2a}$. Since the given equation is of the form $y=ax^2+bx+c$, it follows that $a=1$, $b=-14$, and $c=22$. Therefore, the value of y reaches its minimum when $x=-\frac{(-14)}{2(1)}$, or $x=7$.

QUESTION 8

Choice A is correct. Since x is a factor of each term in the given expression, the expression is equivalent to $x(9x)+x(5)$, or $x(9x+5)$.

Choice B is incorrect. This expression is equivalent to $45x^2+5x$, not $9x^2+5x$.
Choice C is incorrect. This expression is equivalent to $9x^2+45x$, not $9x^2+5x$.
Choice D is incorrect. This expression is equivalent to $9x^3+5x^2$, not $9x^2+5x$.

QUESTION 9

Choice D is correct. The sum of the angle measures of a triangle is $180°$. Adding the measures of angles B and C gives $52+17=69°$. Therefore, the measure of angle A is $180-69=111°$.

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Choice A is incorrect and may result from subtracting the sum of the measures of angles B and C from 90°, instead of from 180°. *Choice B* is incorrect and may result from subtracting the measure of angle C from the measure of angle B. *Choice C* is incorrect and may result from adding the measures of angles B and C but not subtracting the result from 180°.

QUESTION 10

Choice D is correct. Since the graphs of the equations in the given system intersect at the point (x, y), the point (x, y) represents a solution to the given system of equations. The first equation of the given system of equations states that $x = 8$. Substituting 8 for x in the second equation of the given system of equations yields $y = 8^2 + 8$, or $y = 72$. Therefore, the value of y is 72.

Choice A is incorrect. This is the value of x, not y. *Choice B* is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect and may result from conceptual or calculation errors.

QUESTION 11

Choice B is correct. The line of best fit shown intersects the *y*-axis at a positive *y*-value and has a negative slope. The graph of an equation of the form $y = a + bx$, where a and b are constants, intersects the *y*-axis at a *y*-value of a and has a slope of b. Of the given choices, only choice B represents a line that intersects the *y*-axis at a positive *y*-value, 13.5, and has a negative slope, -0.8.

Choice A is incorrect. This equation represents a line that has a positive slope, not a negative slope. *Choice C* is incorrect. This equation represents a line that intersects the *y*-axis at a negative *y*-value, not a positive *y*-value, and has a positive slope, not a negative slope. *Choice D* is incorrect. This equation represents a line that intersects the *y*-axis at a negative *y*-value, not a positive *y*-value.

QUESTION 12

Choice C is correct. It's given that $f(x) = 8\sqrt{x}$. Substituting 48 for $f(x)$ in this equation yields $48 = 8\sqrt{x}$. Dividing both sides of this equation by 8 yields $6 = \sqrt{x}$. This can be rewritten as $\sqrt{x} = 6$. Squaring both sides of this equation yields $x = 36$. Therefore, the value of x for which $f(x) = 48$ is 36.

Choice A is incorrect. If $x = 6$, $f(x) = 8\sqrt{6}$, not 48. *Choice B* is incorrect. If $x = 8$, $f(x) = 8\sqrt{8}$, not 48. *Choice D* is incorrect. If $x = 64$, $f(x) = 8\sqrt{64}$, which is equivalent to 64, not 48.

QUESTION 13

The correct answer is 46. It's given that O is the center of a circle and that points R and S lie on the circle. Therefore, $\overline{OR}$ and $\overline{OS}$ are radii of the circle. It follows that $OR = OS$. If two sides of a triangle are congruent, then the angles opposite them are congruent. It follows that the angles $\angle RSO$ and $\angle ORS$, which are across from the sides of equal length, are congruent. Let x° represent the

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measure of $\angle RSO$. It follows that the measure of $\angle ORS$ is also $x°$. It's given that the measure of $\angle ROS$ is $88°$. Because the sum of the measures of the interior angles of a triangle is $180°$, the equation $x° + x° + 88° = 180°$, or $2x + 88 = 180$, can be used to find the measure of $\angle RSO$. Subtracting 88 from both sides of this equation yields $2x = 92$. Dividing both sides of this equation by 2 yields $x = 46$. Therefore, the measure of $\angle RSO$, in degrees, is 46.

QUESTION 14

The correct answer is $\frac{29}{3}$. Applying the distributive property to the left-hand side of the given equation, $x(x+1)-56$, yields $x^2 + x - 56$. Applying the distributive property to the right-hand side of the given equation, $4x(x-7)$, yields $4x^2 - 28x$. Thus, the equation becomes $x^2 + x - 56 = 4x^2 - 28x$. Combining like terms on the left- and right-hand sides of this equation yields $0 = (4x^2 - x^2) + (-28x - x) + 56$, or $3x^2 - 29x + 56 = 0$. For a quadratic equation in the form $ax^2 + bx + c = 0$, where a, b, and c are constants, the quadratic formula gives the solutions to the equation in the form $x = \frac{\left(-b \pm \sqrt{b^2 - 4ac}\right)}{2a}$. Substituting 3 for a, -29 for b, and 56 for c from the equation $3x^2 - 29x + 56 = 0$ into the quadratic formula yields $x = \frac{\left(29 \pm \sqrt{(-29)^2 - 4(3)(56)}\right)}{2(3)}$, or $x = \frac{29}{6} \pm \frac{13}{6}$. It follows that the solutions to the given equation are $\frac{29}{6} + \frac{13}{6}$ and $\frac{29}{6} - \frac{13}{6}$. Adding these two solutions gives the sum of the solutions: $\frac{29}{6} + \frac{13}{6} + \frac{29}{6} - \frac{13}{6}$, which is equivalent to $\frac{29}{6} + \frac{29}{6}$, or $\frac{29}{3}$. Note that 29/3, 9.666, and 9.667 are examples of ways to enter a correct answer.

QUESTION 15

Choice C is correct. It's given by the first equation in the system that $y = 3x$. Substituting $3x$ for y in the equation $2x + y = 12$ yields $2x + 3x = 12$, or $5x = 12$.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice B is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 16

Choice D is correct. The volume, V, of a cube can be found using the formula $V = s^3$, where s is the edge length of the cube. It's given that a cube has an edge length of 41 inches. Substituting 41 inches for s in this equation yields $V = 41^3$ cubic inches, or $V = 68{,}921$ cubic inches. Therefore, the volume of the cube is $68{,}921$ cubic inches.

Choice A is incorrect. This is the perimeter, in inches, of the cube. *Choice B* is incorrect. This is the area, in square inches, of a face of the cube. *Choice C* is incorrect. This is the surface area, in square inches, of the cube.

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QUESTION 17

Choice D is correct. It's given that the function p models the population of Lowell t years after a census. Since there are 12 months in a year, m months after the census is equivalent to $\frac{m}{12}$ years after the census. Substituting $\frac{m}{12}$ for t in the equation $p(t)=90{,}000(1.06)^t$ yields $p\left(\frac{m}{12}\right)=90{,}000(1.06)^{\frac{m}{12}}$. Therefore, the function r that best models the population of Lowell m months after the census is $r(m)=90{,}000(1.06)^{\frac{m}{12}}$.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice B* is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect and may result from conceptual or calculation errors.

QUESTION 18

Choice A is correct. The given system of linear equations can be solved by the elimination method. Multiplying each side of the second equation in the given system by 3 yields $(2x+2y)(3)=(10)(3)$, or $6x+6y=30$. Subtracting this equation from the first equation in the given system yields $(6x+7y)-(6x+6y)=(28)-(30)$, which is equivalent to $(6x-6x)+(7y-6y)=28-30$, or $y=-2$.

Choice B is incorrect. This is the value of x, not the value of y. *Choice C* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 19

Choice B is correct. It's given that the minimum value of x is 12 less than 6 times another number n. Therefore, the possible values of x are all greater than or equal to the value of 12 less than 6 times n. The value of 6 times n is given by the expression $6n$. The value of 12 less than $6n$ is given by the expression $6n-12$. Therefore, the possible values of x are all greater than or equal to $6n-12$. This can be shown by the inequality $x\geq 6n-12$.

Choice A is incorrect. This inequality shows the possible values of x if the maximum, not the minimum, value of x is 12 less than 6 times n. *Choice C* is incorrect. This inequality shows the possible values of x if the maximum, not the minimum, value of x is 6 times n less than 12, not 12 less than 6 times n. *Choice D* is incorrect. This inequality shows the possible values of x if the minimum value of x is 6 times n less than 12, not 12 less than 6 times n.

QUESTION 20

The correct answer is 44. The mean of a data set is computed by dividing the sum of the values in the data set by the number of values in the data set. It's given that data set A consists of the heights of 75 buildings and has a mean of 32 meters. This can be represented by the equation $\frac{x}{75}=32$, where x represents the sum of the heights of the buildings, in meters, in data set A. Multiplying both sides of this equation by 75 yields $x=75(32)$, or $x=2{,}400$ meters. Therefore, the sum

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of the heights of the buildings in data set A is $2,400$ meters. It's also given that data set B consists of the heights of 50 buildings and has a mean of 62 meters. This can be represented by the equation $\frac{y}{50}=62$, where y represents the sum of the heights of the buildings, in meters, in data set B. Multiplying both sides of this equation by 50 yields $y=50(62)$, or $y=3,100$ meters. Therefore, the sum of the heights of the buildings in data set B is $3,100$ meters. Since it's given that data set C consists of the heights of the 125 buildings from data sets A and B, it follows that the mean of data set C is the sum of the heights of the buildings, in meters, in data sets A and B divided by the number of buildings represented in data sets A and B, or $\frac{2,400+3,100}{125}$, which is equivalent to 44 meters. Therefore, the mean, in meters, of data set C is 44.

QUESTION 21

The correct answer is $\frac{59}{9}$. When the graph of an equation in the form $Ax+By=C$, where A, B, and C are constants, is translated down k units in the xy-plane, the resulting graph can be represented by the equation $Ax+B(y+k)=C$. It's given that the graph of $9x-10y=19$ is translated down 4 units in the xy-plane. Therefore, the resulting graph can be represented by the equation $9x-10(y+4)=19$, or $9x-10y-40=19$. Adding 40 to both sides of this equation yields $9x-10y=59$. The x-coordinate of the x-intercept of the graph of an equation in the xy-plane is the value of x in the equation when $y=0$. Substituting 0 for y in the equation $9x-10y=59$ yields $9x-10(0)=59$, or $9x=59$. Dividing both sides of this equation by 9 yields $x=\frac{59}{9}$. Therefore, the x-coordinate of the x-intercept of the resulting graph is $\frac{59}{9}$. Note that 59/9, 6.555, and 6.556 are examples of ways to enter a correct answer.

QUESTION 22

Choice D is correct. Since the value of y increases by a constant factor, 4, for each increase of 1 in the value of x, the relationship between x and y is exponential. An exponential relationship between x and y can be represented by an equation of the form $y=a(b)^x$, where a is the value of x when $y=0$ and y increases by a factor of b for each increase of 1 in the value of x. Since $y=200$ when $x=0$, $a=200$. Since y increases by a factor of 4 for each increase of 1 in the value of x, $b=4$. Substituting 200 for a and 4 for b in the equation $y=a(b)^x$ yields $y=200(4)^x$. Thus, the equation $y=200(4)^x$ represents the relationship between x and y.

Choice A is incorrect and may result from conceptual errors. *Choice B* is incorrect. This equation represents a relationship where for each increase of 1 in the value of x, the value of y increases by a factor of 200, not 4, and when $x=0$, y is equal to 4, not 200. *Choice C* is incorrect and may result from conceptual errors.

QUESTION 23

Choice B is correct. Adding 9 to each side of the given equation yields $x^2-2x=9$. To complete the square, adding 1 to each side of this equation

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yields $x^2-2x+1=9+1$, or $(x-1)^2=10$. Taking the square root of each side of this equation yields $x-1=\pm\sqrt{10}$. Adding 1 to each side of this equation yields $x=1\pm\sqrt{10}$. Since it's given that one of the solutions to the equation can be written as $1+\sqrt{k}$, the value of k must be 10.

Alternate approach: It's given that $1+\sqrt{k}$ is a solution to the given equation. It follows that $x=1+\sqrt{k}$. Substituting $1+\sqrt{k}$ for x in the given equation yields $(1+\sqrt{k})^2-2(1+\sqrt{k})-9=0$, or $(1+\sqrt{k})(1+\sqrt{k})-2(1+\sqrt{k})-9=0$. Expanding the products on the left-hand side of this equation yields $1+2\sqrt{k}+k-2-2\sqrt{k}-9=0$, or $k-10=0$. Adding 10 to each side of this equation yields $k=10$.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 24

Choice A is correct. The median of a data set with an odd number of values that are in ascending or descending order is the middle value of the data set. Since the distribution of the values of both data set A and data set B form symmetric dot plots, and each data set has an odd number of values, it follows that the median is given by the middle value in each of the dot plots. Thus, the median of data set A is 13, and the median of data set B is 13. Therefore, statement I is true. Data set A and data set B have the same frequency for each of the values 11, 12, 14, and 15. Data set A has a frequency of 1 for values 10 and 16, whereas data set B has a frequency of 2 for values 10 and 16. Standard deviation is a measure of the spread of a data set; it is larger when there are more values further from the mean, and smaller when there are more values closer to the mean. Since both distributions are symmetric with an odd number of values, the mean of each data set is equal to its median. Thus, each data set has a mean of 13. Since more of the values in data set A are closer to 13 than data set B, it follows that data set A has a smaller standard deviation than data set B. Thus, statement II is false. Therefore, only statement I must be true.

Choice B is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 25

Choice B is correct. It's given that the right triangle is isosceles. In an isosceles right triangle, the two legs have equal lengths, and the length of the hypotenuse is $\sqrt{2}$ times the length of one of the legs. Let ℓ represent the length, in inches, of each leg of the isosceles right triangle. It follows that the length of the hypotenuse is $\ell\sqrt{2}$ inches. The perimeter of a figure is the sum of the lengths of the sides of the figure. Therefore, the perimeter of the isosceles right triangle is $\ell+\ell+\ell\sqrt{2}$ inches. It's given that the perimeter of the triangle is $94+94\sqrt{2}$ inches. It follows that $\ell+\ell+\ell\sqrt{2}=94+94\sqrt{2}$. Factoring the left-hand side of this equation yields $(1+1+\sqrt{2})\ell=94+94\sqrt{2}$, or $(2+\sqrt{2})\ell=94+94\sqrt{2}$. Dividing both sides of this

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equation by $2+\sqrt{2}$ yields $\ell=\frac{94+94\sqrt{2}}{2+\sqrt{2}}$. Rationalizing the denominator of the right-hand side of this equation by multiplying the right-hand side of the equation by $\frac{2-\sqrt{2}}{2-\sqrt{2}}$ yields $\ell=\frac{(94+94\sqrt{2})(2-\sqrt{2})}{(2+\sqrt{2})(2-\sqrt{2})}$. Applying the distributive property to the numerator and to the denominator of the right-hand side of this equation yields $\ell=\frac{188-94\sqrt{2}+188\sqrt{2}-94\sqrt{4}}{4-2\sqrt{2}+2\sqrt{2}-\sqrt{4}}$. This is equivalent to $\ell=\frac{94\sqrt{2}}{2}$, or $\ell=47\sqrt{2}$. Therefore, the length, in inches, of one leg of the isosceles right triangle is $47\sqrt{2}$.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect. This is the length, in inches, of the hypotenuse. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 26

Choice C is correct. It's given that the equation $-9x^2+30x+c=0$ has exactly one solution. A quadratic equation of the form $ax^2+bx+c=0$ has exactly one solution if and only if its discriminant, $-4ac+b^2$, is equal to zero. It follows that for the given equation, $a=-9$ and $b=30$. Substituting -9 for a and 30 for b into b^2-4ac yields $30^2-4(-9)(c)$, or $900+36c$. Since the discriminant must equal zero, $900+36c=0$. Subtracting $36c$ from both sides of this equation yields $900=-36c$. Dividing each side of this equation by -36 yields $-25=c$. Therefore, the value of c is -25.

Choice A is incorrect. If the value of c is 3, this would yield a discriminant that is greater than zero. Therefore, the given equation would have two solutions, rather than exactly one solution. *Choice B* is incorrect. If the value of c is 0, this would yield a discriminant that is greater than zero. Therefore, the given equation would have two solutions, rather than exactly one solution. *Choice D* is incorrect. If the value of c is -53, this would yield a discriminant that is less than zero. Therefore, the given equation would have no real solutions, rather than exactly one solution.

QUESTION 27

The correct answer is 6. A system of two linear equations in two variables, x and y, has no solution if the lines represented by the equations in the xy-plane are parallel and distinct. Lines represented by equations in standard form, $Ax+By=C$ and $Dx+Ey=F$, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients in the other equation, meaning $\frac{D}{A}=\frac{E}{B}$; and the lines are distinct if the constants are not proportional, meaning $\frac{F}{C}$ is not equal to $\frac{D}{A}$ or $\frac{E}{B}$. The first equation in the given system is $\frac{3}{2}y-\frac{1}{4}x=\frac{2}{3}-\frac{3}{2}y$. Multiplying each side of this equation by 12 yields $18y-3x=8-18y$. Adding $18y$ to each side of this equation yields $36y-3x=8$, or $-3x+36y=8$. The second equation in the given system is $\frac{1}{2}x+\frac{3}{2}=py+\frac{9}{2}$.

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Multiplying each side of this equation by 2 yields $x+3=2py+9$. Subtracting $2py$ from each side of this equation yields $x+3-2py=9$. Subtracting 3 from each side of this equation yields $x-2py=6$. Therefore, the two equations in the given system, written in standard form, are $-3x+36y=8$ and $x-2py=6$. As previously stated, if this system has no solution, the lines represented by the equations in the *xy*-plane are parallel and distinct, meaning the proportion $\frac{1}{-3}=\frac{-2p}{36}$, or $-\frac{1}{3}=-\frac{p}{18}$, is true and the proportion $\frac{6}{8}=\frac{1}{-3}$ is not true. The proportion $\frac{6}{8}=\frac{1}{-3}$ is not true. Multiplying each side of the true proportion, $-\frac{1}{3}=-\frac{p}{18}$, by -18 yields $6=p$. Therefore, if the system has no solution, then the value of p is 6.

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Test begins on the next page.

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Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

The spacecraft OSIRIS-REx briefly made contact with the asteroid 101955 Bennu in 2020. NASA scientist Daniella DellaGiustina reports that despite facing the unexpected obstacle of a surface mostly covered in boulders, OSIRIS-REx successfully _____ a sample of the surface, gathering pieces of it to bring back to Earth.

Which choice completes the text with the most logical and precise word or phrase?

A) attached
B) collected
C) followed
D) replaced

2

Research conducted by planetary scientist Katarina Miljkovic suggests that the Moon's surface may not accurately _____ early impact events. When the Moon was still forming, its surface was softer, and asteroid or meteoroid impacts would have left less of an impression; thus, evidence of early impacts may no longer be present.

Which choice completes the text with the most logical and precise word or phrase?

A) reflect
B) receive
C) evaluate
D) mimic

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3

Handedness, a preferential use of either the right or left hand, typically is easy to observe in humans. Because this trait is present but less _____ in many other animals, animal-behavior researchers often employ tasks specially designed to reveal individual animals' preferences for a certain hand or paw.

Which choice completes the text with the most logical and precise word or phrase?

A) recognizable
B) intriguing
C) significant
D) useful

4

It is by no means _____ to recognize the influence of Dutch painter Hieronymus Bosch on Ali Banisadr's paintings; indeed, Banisadr himself cites Bosch as an inspiration. However, some scholars have suggested that the ancient Mesopotamian poem *Epic of Gilgamesh* may have had a far greater impact on Banisadr's work.

Which choice completes the text with the most logical and precise word or phrase?

A) substantial
B) satisfying
C) unimportant
D) appropriate

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5

The following text is adapted from Susan Glaspell's 1912 short story "'Out There.'" An elderly shop owner is looking at a picture that he recently acquired and hopes to sell.

> It did seem that the picture failed to fit in with the rest of the shop. A persuasive young fellow who claimed he was closing out his stock let the old man have it for what he called a song. It was only a little out-of-the-way store which subsisted chiefly on the framing of pictures. The old man looked around at his views of the city, his pictures of cats and dogs, his flaming bits of landscape. "Don't belong in here," he fumed.
>
> And yet the old man was secretly proud of his acquisition. There was a hidden dignity in his scowling as he shuffled about pondering the least ridiculous place for the picture.

Which choice best states the main purpose of the text?

A) To reveal the shop owner's conflicted feelings about the new picture

B) To convey the shop owner's resentment of the person he got the new picture from

C) To describe the items that the shop owner most highly prizes

D) To explain differences between the new picture and other pictures in the shop

6

The following text is from the 1923 poem "Black Finger" by Angelina Weld Grimké, a Black American writer. A cypress is a type of evergreen tree.

> I have just seen a most beautiful thing,
> Slim and still,
> Against a gold, gold sky,
> A straight black cypress,
> Sensitive,
> Exquisite,
> A black finger
> Pointing upwards.
> Why, beautiful still finger, are you black?
> And why are you pointing upwards?

Which choice best describes the overall structure of the text?

A) The speaker assesses a natural phenomenon, then questions the accuracy of her assessment.

B) The speaker describes a distinctive sight in nature, then ponders what meaning to attribute to that sight.

C) The speaker presents an outdoor scene, then considers a human behavior occurring within that scene.

D) The speaker examines her surroundings, then speculates about their influence on her emotional state.

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7

The following text is from Walt Whitman's 1860 poem "Calamus 24."

> I HEAR it is charged against me that I seek to destroy institutions;
> But really I am neither for nor against institutions
> (What indeed have I in common with them?—Or what with the destruction of them?),
> Only I will establish in the Mannahatta [Manhattan] and in every city of These States, inland and seaboard,
> And in the fields and woods, and above every keel [ship] little or large, that dents the water,
> Without edifices, or rules, or trustees, or any argument,
> The institution of the dear love of comrades.

Which choice best describes the overall structure of the text?

A) The speaker questions an increasingly prevalent attitude, then summarizes his worldview.

B) The speaker regrets his isolation from others, then predicts a profound change in society.

C) The speaker concedes his personal shortcomings, then boasts of his many achievements.

D) The speaker addresses a criticism leveled against him, then announces a grand ambition of his.

8

The mimosa tree evolved in East Asia, where the beetle *Bruchidius terrenus* preys on its seeds. In 1785, mimosa trees were introduced to North America, far from any *B. terrenus*. But evolutionary links between predators and their prey can persist across centuries and continents. Around 2001, *B. terrenus* was introduced in southeastern North America near where botanist Shu-Mei Chang and colleagues had been monitoring mimosa trees. Within a year, 93 percent of the trees had been attacked by the beetles.

Which choice best describes the function of the third sentence in the overall structure of the text?

A) It states the hypothesis that Chang and colleagues had set out to investigate using mimosa trees and *B. terrenus*.

B) It presents a generalization that is exemplified by the discussion of the mimosa trees and *B. terrenus*.

C) It offers an alternative explanation for the findings of Chang and colleagues.

D) It provides context that clarifies why the species mentioned spread to new locations.

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9

Text 1
Conventional wisdom long held that human social systems evolved in stages, beginning with hunter-gatherers forming small bands of members with roughly equal status. The shift to agriculture about 12,000 years ago sparked population growth that led to the emergence of groups with hierarchical structures: associations of clans first, then chiefdoms, and finally, bureaucratic states.

Text 2
In a 2021 book, anthropologist David Graeber and archaeologist David Wengrow maintain that humans have always been socially flexible, alternately forming systems based on hierarchy and collective ones with decentralized leadership. The authors point to evidence that as far back as 50,000 years ago some hunter-gatherers adjusted their social structures seasonally, at times dispersing in small groups but also assembling into communities that included esteemed individuals.

Based on the texts, how would Graeber and Wengrow (Text 2) most likely respond to the "conventional wisdom" presented in Text 1?

A) By conceding the importance of hierarchical systems but asserting the greater significance of decentralized collective societies

B) By disputing the idea that developments in social structures have followed a linear progression through distinct stages

C) By acknowledging that hierarchical roles likely weren't a part of social systems before the rise of agriculture

D) By challenging the assumption that groupings of hunter-gatherers were among the earliest forms of social structure

10

The following text is adapted from Frances Hodgson Burnett's 1911 novel *The Secret Garden*. Mary, a young girl, recently found an overgrown hidden garden.

> Mary was an odd, determined little person, and now she had something interesting to be determined about, she was very much absorbed, indeed. She worked and dug and pulled up weeds steadily, only becoming more pleased with her work every hour instead of tiring of it. It seemed to her like a fascinating sort of play.

Which choice best states the main idea of the text?

A) Mary hides in the garden to avoid doing her chores.

B) Mary is getting bored with pulling up so many weeds in the garden.

C) Mary is clearing out the garden to create a space to play.

D) Mary feels very satisfied when she's taking care of the garden.

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11

The following text is from Ezra Pound's 1909 poem "Hymn III," based on the work of Marcantonio Flaminio.

As a fragile and lovely flower unfolds its gleaming foliage on the breast of the fostering earth, if the dew and the rain draw it forth;
So doth my tender mind flourish, if it be fed with the sweet dew of the fostering spirit,
Lacking this, it beginneth straightway to languish, even as a floweret born upon dry earth, if the dew and the rain tend it not.

Based on the text, in what way is the human mind like a flower?

A) It becomes increasingly vigorous with the passage of time.
B) It draws strength from changes in the weather.
C) It requires proper nourishment in order to thrive.
D) It perseveres despite challenging circumstances.

12

The following text is adapted from Jack London's 1903 novel *The Call of the Wild*. Buck is a sled dog living with John Thornton in Yukon, Canada.

> Thornton alone held [Buck]. The rest of mankind was as nothing. Chance travellers might praise or pet him; but he was cold under it all, and from a too demonstrative man he would get up and walk away. When Thornton's partners, Hans and Pete, arrived on the long-expected raft, Buck refused to notice them till he learned they were close to Thornton; after that he tolerated them in a passive sort of way, accepting favors from them as though he favored them by accepting.

Which choice best states the main idea of the text?

A) Buck has become less social since he began living with Thornton.
B) Buck mistrusts humans and does his best to avoid them.
C) Buck has been especially well liked by most of Thornton's friends.
D) Buck holds Thornton in higher regard than any other person.

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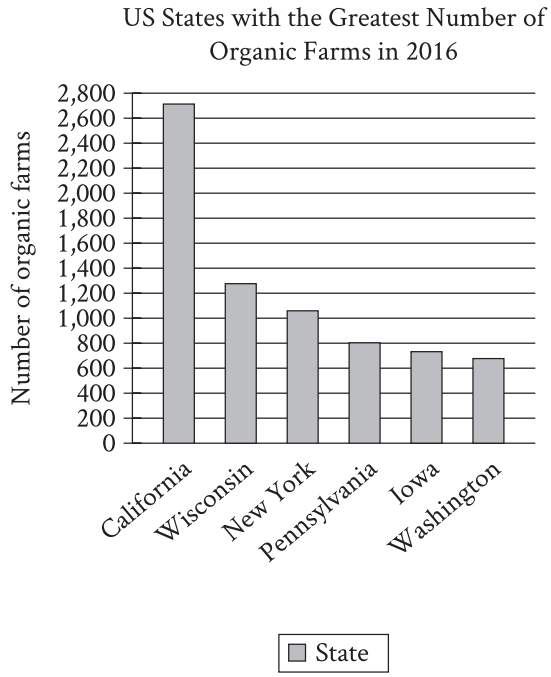
13



Organic farming is a method of growing food that tries to reduce environmental harm by using natural forms of pest control and avoiding fertilizers made with synthetic materials. Organic farms are still a small fraction of the total farms in the United States, but they have been becoming more popular. According to the US Department of Agriculture, in 2016 California had between 2,600 and 2,800 organic farms and _____

Which choice most effectively uses data from the graph to complete the text?

A) Washington had between 600 and 800 organic farms.

B) New York had fewer than 800 organic farms.

C) Wisconsin and Iowa each had between 1,200 and 1,400 organic farms.

D) Pennsylvania had more than 1,200 organic farms.

14

Biologist Valentina Gómez-Bahamón and her team have investigated two subspecies of the fork-tailed flycatcher bird that live in the same region in Colombia, but one subspecies migrates south for part of the year, and the other doesn't. The researchers found that, due to slight differences in feather shape, the feathers of migratory forked-tailed flycatcher males make a sound during flight that is higher pitched than that made by the feathers of nonmigratory males. The researchers hypothesize that fork-tailed flycatcher females are attracted to the specific sound made by the males of their own subspecies, and that over time the females' preference will drive further genetic and anatomical divergence between the subspecies.

Which finding, if true, would most directly support Gómez-Bahamón and her team's hypothesis?

A) The feathers located on the wings of the migratory fork-tailed flycatchers have a narrower shape than those of the nonmigratory birds, which allows them to fly long distances.

B) Over several generations, the sound made by the feathers of migratory male fork-tailed flycatchers grows progressively higher pitched relative to that made by the feathers of nonmigratory males.

C) Fork-tailed flycatchers communicate different messages to each other depending on whether their feathers create high-pitched or low-pitched sounds.

D) The breeding habits of the migratory and nonmigratory fork-tailed flycatchers remained generally the same over several generations.

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15

Ablation Rates for Three Elements in Cosmic Dust, by Dust Source

Element	SPC	AST	HTC	OCC
iron	20%	28%	90%	98%
potassium	44%	74%	97%	100%
sodium	45%	75%	99%	100%

Earth's atmosphere is bombarded by cosmic dust originating from several sources: short-period comets (SPCs), particles from the asteroid belt (ASTs), Halley-type comets (HTCs), and Oort cloud comets (OCCs). Some of the dust's material vaporizes in the atmosphere in a process called ablation, and the faster the particles move, the higher the rate of ablation. Astrophysicist Juan Diego Carrillo-Sánchez led a team that calculated average ablation rates for elements in the dust (such as iron and potassium) and showed that material in slower-moving SPC or AST dust has a lower rate than the same material in faster-moving HTC or OCC dust. For example, whereas the average ablation rate for iron from AST dust is 28%, the average rate for _____

Which choice most effectively uses data from the table to complete the example?

A) iron from SPC dust is 20%.

B) sodium from OCC dust is 100%.

C) iron from HTC dust is 90%.

D) sodium from AST dust is 75%.

16

Art collectives, like the United States- and Vietnam-based collective The Propeller Group or Cuba's Los Carpinteros, are groups of artists who agree to work together: perhaps for stylistic reasons, or to advance certain shared political ideals, or to help mitigate the costs of supplies and studio space. Regardless of the reasons, art collectives usually involve some collaboration among the artists. Based on a recent series of interviews with various art collectives, an arts journalist claims that this can be difficult for artists who are often used to having sole control over their work.

Which quotation from the interviews best illustrates the journalist's claim?

A) "The first collective I joined included many amazingly talented artists, and we enjoyed each other's company, but because we had a hard time sharing credit and responsibility for our work, the collective didn't last."

B) "We work together, but that doesn't mean that individual projects are equally the work of all of us. Many of our projects are primarily the responsibility of whoever originally proposed the work to the group."

C) "Having worked as a member of a collective for several years, it's sometimes hard to recall what it was like to work alone without the collective's support. But that support encourages my individual expression rather than limits it."

D) "Sometimes an artist from outside the collective will choose to collaborate with us on a project, but all of those projects fit within the larger themes of the work the collective does on its own."

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17

Effects of Mycorrhizal Fungi on 3 Plant Species

Plant species	Mycorrhizal host	Average mass of plants grown in soil containing mycorrhizal fungi (in grams)	Average mass of plants grown in soil treated to kill fungi (in grams)
Corn	yes	15.1	3.8
Marigold	yes	10.2	2.4
Broccoli	no	7.5	7

Mycorrhizal fungi in soil benefits many plants, substantially increasing the mass of some. A student conducted an experiment to illustrate this effect. The student chose three plant species for the experiment, including two that are mycorrhizal hosts (species known to benefit from mycorrhizal fungi) and one nonmycorrhizal species (a species that doesn't benefit from and may even be harmed by mycorrhizal fungi). The student then grew several plants from each species both in soil containing mycorrhizal fungi and in soil that had been treated to kill mycorrhizal and other fungi. After several weeks, the student measured the plants' average mass and was surprised to discover that _____

Which choice most effectively uses data from the table to complete the statement?

A) broccoli grown in soil containing mycorrhizal fungi had a slightly higher average mass than broccoli grown in soil that had been treated to kill fungi.

B) corn grown in soil containing mycorrhizal fungi had a higher average mass than broccoli grown in soil containing mycorrhizal fungi.

C) marigolds grown in soil containing mycorrhizal fungi had a much higher average mass than marigolds grown in soil that had been treated to kill fungi.

D) corn had the highest average mass of all three species grown in soil that had been treated to kill fungi, while marigolds had the lowest.

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18

Several artworks found among the ruins of the ancient Roman city of Pompeii depict a female figure fishing with a cupid nearby. Some scholars have asserted that the figure is the goddess Venus, since she is known to have been linked with cupids in Roman culture, but University of Leicester archaeologist Carla Brain suggests that cupids may have also been associated with fishing generally. The fact that a cupid is shown near the female figure, therefore, _____

Which choice most logically completes the text?

A) is not conclusive evidence that the figure is Venus.

B) suggests that Venus was often depicted fishing.

C) eliminates the possibility that the figure is Venus.

D) would be difficult to account for if the figure is not Venus.

19

Literary agents estimate that more than half of all nonfiction books credited to a celebrity or other public figure are in fact written by ghostwriters, professional authors who are paid to write other _____ but whose names never appear on book covers.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) people's stories

B) peoples story's

C) peoples stories

D) people's story's

20

Like other amphibians, the wood frog (*Rana sylvatica*) is unable to generate its own heat, so during periods of subfreezing temperatures, it _____ by producing large amounts of glucose, a sugar that helps prevent damaging ice from forming inside its cells.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) had survived

B) survived

C) would survive

D) survives

21

After a spate of illnesses as a child, Wilma Rudolph was told she might never walk again. Defying all odds, Rudolph didn't just walk, she _____ the 1960 Summer Olympics in Rome, she won both the 100- and 200-meter dashes and clinched first place for her team in the 4×100-meter relay, becoming the first US woman to win three gold medals in a single Olympics.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) ran—fast—during

B) ran—fast during

C) ran—fast, during

D) ran—fast. During

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22

In many of her landscape paintings from the 1970s and 1980s, Lebanese American artist Etel Adnan worked to capture the essence of California's fog-shrouded Mount Tamalpais region through abstraction, using splotches of color to represent the area's features. Interestingly, the triangle representing the mountain itself _____ among the few defined figures in her paintings.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) are
B) have been
C) were
D) is

23

Seneca sculptor Marie Watt's blanket art comes in a range of shapes and sizes. In 2004, Watt sewed strips of blankets together to craft a 10-by-13-inch _____ in 2014, she arranged folded blankets into two large stacks and then cast them in bronze, creating two curving 18-foot-tall blue-bronze pillars.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) sampler later,
B) sampler;
C) sampler,
D) sampler, later,

24

African American Percy Julian was a scientist and entrepreneur whose work helped people around the world to see. Named in 1999 as one of the greatest achievements by a US chemist in the past hundred years, _____ led to the first mass-produced treatment for glaucoma.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) Julian synthesized the alkaloid physostigmine in 1935; it
B) in 1935 Julian synthesized the alkaloid physostigmine, which
C) Julian's 1935 synthesis of the alkaloid physostigmine
D) the alkaloid physostigmine was synthesized by Julian in 1935 and

25

The Arctic-Alpine Botanic Garden in Norway and the Jardim Botânico of Rio de Janeiro in Brazil are two of many botanical gardens around the world dedicated to growing diverse plant _____ fostering scientific research; and educating the public about plant conservation.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) species, both native and nonnative,
B) species, both native and nonnative;
C) species; both native and nonnative,
D) species both native and nonnative,

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26

Sociologist Alton Okinaka sits on the review board tasked with adding new sites to the Hawai'i Register of Historic Places, which includes Pi'ilanihale Heiau and the 'Ōpaeka'a Road Bridge. Okinaka doesn't make such decisions _____ all historical designations must be approved by a group of nine other experts from the fields of architecture, archaeology, history, and Hawaiian culture.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) single-handedly, however;
B) single-handedly; however,
C) single-handedly, however,
D) single-handedly however

27

In 1968, US Congressman John Conyers introduced a bill to establish a national holiday in honor of Dr. Martin Luther King Jr. The bill didn't make it to a vote, but Conyers was determined. He teamed up with Shirley Chisholm, the first Black woman to be elected to Congress, and they resubmitted the bill every session for the next fifteen years. _____ in 1983, the bill passed.

Which choice completes the text with the most logical transition?

A) Instead,
B) Likewise,
C) Finally,
D) Additionally,

28

Geoscientists have long considered Hawaii's Mauna Loa volcano to be Earth's largest shield volcano by volume, measuring approximately 74,000 cubic kilometers. _____ according to a 2020 study by local geoscientist Michael Garcia, Hawaii's Pūhāhonu shield volcano is significantly larger, boasting a volume of about 148,000 cubic kilometers.

Which choice completes the text with the most logical transition?

A) Secondly,
B) Consequently,
C) Moreover,
D) However,

29

Samuel Coleridge-Taylor was a prominent classical music composer from England who toured the US three times in the early 1900s. The child of a West African father and an English mother, Coleridge-Taylor emphasized his mixed-race ancestry. For example, he referred to himself as Anglo-African. _____ he incorporated the sounds of traditional African music into his classical music compositions.

Which choice completes the text with the most logical transition?

A) In addition,
B) Actually,
C) However,
D) Regardless,

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30

In 2019, researcher Patricia Jurado Gonzalez and food historian Nawal Nasrallah prepared a stew from a 4,000-year-old recipe found on a Mesopotamian clay tablet. When they tasted the dish, known as *pašrūtum* ("unwinding"), they found that it had a mild taste and inspired a sense of calm. _____ the researchers, knowing that dishes were sometimes named after their intended effects, theorized that the dish's name, "unwinding," referred to its function: to help ancient diners relax.

Which choice completes the text with the most logical transition?

A) Therefore,

B) Alternately,

C) Nevertheless,

D) Likewise,

31

While researching a topic, a student has taken the following notes:

- Chemical leavening agents cause carbon dioxide to be released within a liquid batter, making the batter rise as it bakes.
- Baking soda and baking powder are chemical leavening agents.
- Baking soda is pure sodium bicarbonate.
- To produce carbon dioxide, baking soda needs to be mixed with liquid and an acidic ingredient such as honey.
- Baking powder is a mixture of sodium bicarbonate and an acid.
- To produce carbon dioxide, baking powder needs to be mixed with liquid but not with an acidic ingredient.

The student wants to emphasize a difference between baking soda and baking powder. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) To make batters rise, bakers use chemical leavening agents such as baking soda and baking powder.

B) Baking soda and baking powder are chemical leavening agents that, when mixed with other ingredients, cause carbon dioxide to be released within a batter.

C) Baking soda is pure sodium bicarbonate, and honey is a type of acidic ingredient.

D) To produce carbon dioxide within a liquid batter, baking soda needs to be mixed with an acidic ingredient, whereas baking powder does not.

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32

While researching a topic, a student has taken the following notes:

- Soo Sunny Park is a Korean American artist who uses light as her primary medium of expression.
- She created her work *Unwoven Light* in 2013.
- *Unwoven Light* featured a chain-link fence fitted with iridescent plexiglass tiles.
- When light passed through the fence, colorful prisms formed.

The student wants to describe *Unwoven Light* to an audience unfamiliar with Soo Sunny Park. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Park's 2013 installation *Unwoven Light*, which included a chain-link fence and iridescent tiles made from plexiglass, featured light as its primary medium of expression.

B) Korean American light artist Soo Sunny Park created *Unwoven Light* in 2013.

C) The chain-link fence in Soo Sunny Park's *Unwoven Light* was fitted with tiles made from iridescent plexiglass.

D) In *Unwoven Light*, a 2013 work by Korean American artist Soo Sunny Park, light formed colorful prisms as it passed through a fence Park had fitted with iridescent tiles.

33

While researching a topic, a student has taken the following notes:

- Cambodia's Angkor Wat was built in the 1100s to honor the Hindu god Vishnu.
- It has been a Buddhist temple since the sixteenth century.
- Decorrelation stretch analysis is a novel digital imaging technique that enhances the contrast between colors in a photograph.
- Archaeologist Noel Hidalgo Tan applied decorrelation stretch analysis to photographs he had taken of Angkor Wat's plaster walls.
- Tan's analysis revealed hundreds of images unknown to researchers.

The student wants to present Tan's research to an audience unfamiliar with Angkor Wat. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Tan photographed Angkor Wat's plaster walls and then applied decorrelation stretch analysis to the photographs.

B) Decorrelation stretch analysis is a novel digital imaging technique that Tan used to enhance the contrast between colors in a photograph.

C) Using a novel digital imaging technique, Tan revealed hundreds of images hidden on the walls of Angkor Wat, a Cambodian temple.

D) Built to honor a Hindu god before becoming a Buddhist temple, Cambodia's Angkor Wat concealed hundreds of images on its plaster walls.

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

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Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

The fashion resale market, in which consumers purchase secondhand clothing from stores and online sellers, generated nearly $30 billion globally in 2019. Expecting to see continued growth, some analysts _____ that revenues will more than double by 2028.

Which choice completes the text with the most logical and precise word or phrase?

A) produced
B) denied
C) worried
D) predicted

2

Artificially delivering biomolecules to plant cells is an important component of protecting plants from pathogens, but it is difficult to transmit biomolecules through the layers of the plant cell wall. Markita del Carpio Landry and her colleagues have shown that it may be possible to _____ this problem by transmitting molecules through carbon nanotubes, which can cross cell walls.

Which choice completes the text with the most logical and precise word or phrase?

A) conceptualize
B) neglect
C) illustrate
D) overcome

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3

Particle physicists like Ayana Holloway Arce and Aida El-Khadra spend much of their time _____ what is invisible to the naked eye: using sophisticated technology, they closely examine the behavior of subatomic particles, the smallest detectable parts of matter.

Which choice completes the text with the most logical and precise word or phrase?

A) selecting
B) inspecting
C) creating
D) deciding

4

Anthropologist Kristian J. Carlson and colleagues examined the fossilized clavicle and shoulder bones of a 3.6-million-year-old early hominin known as "Little Foot." They found that these bones were _____ the clavicle and shoulder bones of modern apes that are frequent climbers, such as gorillas and chimpanzees, suggesting that Little Foot had adapted to life in the trees.

Which choice completes the text with the most logical and precise word or phrase?

A) surpassed by
B) comparable to
C) independent of
D) obtained from

5

Rydra Wong, the protagonist of Samuel R. Delany's 1966 novel *Babel-17*, is a poet, an occupation which, in Delany's work, is not _____: nearly a dozen of the characters that populate his novels are poets or writers.

Which choice completes the text with the most logical and precise word or phrase?

A) infallible
B) atypical
C) lucrative
D) tedious

6

For a 2020 exhibition, photographer and neurobiologist Okunola Jeyifous _____ a series of new images based on a series of alphabet posters from the 1970s known as the "Black ABCs," which featured Black children from Chicago. Jeyifous photographed the now-adult models and layered the photos over magnified images of the models' cells, resulting in what he called "micro and macro portraiture."

Which choice completes the text with the most logical and precise word or phrase?

A) validated
B) created
C) challenged
D) restored

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CONTINUE

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7

In addition to being an accomplished psychologist himself, Francis Cecil Sumner was a _____ increasing the opportunity for Black students to study psychology, helping to found the psychology department at Howard University, a historically Black university, in 1930.

Which choice completes the text with the most logical and precise word or phrase?

A) proponent of
B) supplement to
C) beneficiary of
D) distraction for

8

Whether the reign of a French monarch such as Hugh Capet or Henry I was historically consequential or relatively uneventful, its trajectory was shaped by questions of legitimacy and therefore cannot be understood without a corollary understanding of the factors that allowed the monarch to _____ his right to hold the throne.

Which choice completes the text with the most logical and precise word or phrase?

A) reciprocate
B) annotate
C) buttress
D) disengage

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CONTINUE

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9

Some bird species don't raise their own chicks. Instead, adult females lay their eggs in other nests, next to another bird species' own eggs. Female cuckoos have been seen quickly laying eggs in the nests of other bird species when those birds are out looking for food. After the eggs hatch, the noncuckoo parents will typically raise the cuckoo chicks as if they were their own offspring, even if the cuckoos look very different from the other chicks.

Which choice best describes the function of the underlined sentence in the text as a whole?

A) It introduces a physical feature of female cuckoos that is described later in the text.

B) It describes the appearance of the cuckoo nests mentioned earlier in the text.

C) It offers a detail about how female cuckoos carry out the behavior discussed in the text.

D) It explains how other birds react to the female cuckoo behavior discussed in the text.

10

Cats can judge unseen people's positions in space by the sound of their voices and thus react with surprise when the same person calls to them from two different locations in a short span of time. Saho Takagi and colleagues reached this conclusion by measuring cats' levels of surprise based on their ear and head movements while the cats heard recordings of their owners' voices from two speakers spaced far apart. Cats exhibited a low level of surprise when owners' voices were played twice from the same speaker, but they showed a high level of surprise when the voice was played once each from the two different speakers.

According to the text, how did the researchers determine the level of surprise displayed by the cats in the study?

A) They watched how each cat moved its ears and head.

B) They examined how each cat reacted to the voice of a stranger.

C) They studied how each cat physically interacted with its owner.

D) They tracked how each cat moved around the room.

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11

A student performs an experiment testing her hypothesis that a slightly acidic soil environment is more beneficial for the growth of the plant *Brassica rapa parachinensis* (a vegetable commonly known as choy sum) than a neutral soil environment. She plants sixteen seeds of choy sum in a mixture of equal amounts of coffee grounds (which are highly acidic) and potting soil and another sixteen seeds in potting soil without coffee grounds as the control for the experiment. The two groups of seeds were exposed to the same growing conditions and monitored for three weeks.

Which finding, if true, would most directly weaken the student's hypothesis?

A) The choy sum planted in the soil without coffee grounds were significantly taller at the end of the experiment than the choy sum planted in the mixture of soil and coffee grounds.

B) The choy sum grown in the soil without coffee grounds weighed significantly less at the end of the experiment than the choy sum grown in the mixture of soil and coffee grounds.

C) The choy sum seeds planted in the soil without coffee grounds sprouted significantly later in the experiment than did the seeds planted in the mixture of soil and coffee grounds.

D) Significantly fewer of the choy sum seeds planted in the soil without coffee grounds sprouted plants than did the seeds planted in the mixture of soil and coffee grounds.

12

"The Young Girl" is a 1920 short story by Katherine Mansfield. In the story, the narrator takes an unnamed seventeen-year-old girl and her younger brother out for a meal. In describing the teenager, Mansfield frequently contrasts the character's pleasant appearance with her unpleasant attitude, as when Mansfield writes of the teenager, _____

Which quotation from "The Young Girl" most effectively illustrates the claim?

A) "I heard her murmur, 'I can't bear flowers on a table.' They had evidently been giving her intense pain, for she positively closed her eyes as I moved them away."

B) "While we waited she took out a little, gold powder-box with a mirror in the lid, shook the poor little puff as though she loathed it, and dabbed her lovely nose."

C) "I saw, after that, she couldn't stand this place a moment longer, and, indeed, she jumped up and turned away while I went through the vulgar act of paying for the tea."

D) "She didn't even take her gloves off. She lowered her eyes and drummed on the table. When a faint violin sounded she winced and bit her lip again. Silence."

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CONTINUE

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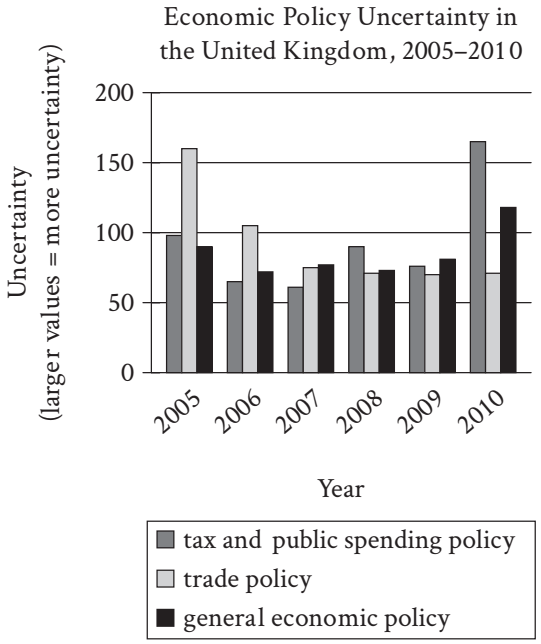
13



High levels of public uncertainty about which economic policies a country will adopt can make planning difficult for businesses, but measures of such uncertainty have not tended to be very detailed. Recently, however, economist Sandile Hlatshwayo analyzed trends in news reports to derive measures not only for general economic policy uncertainty but also for uncertainty related to specific areas of economic policy, like tax or trade policy. One revelation of her work is that a general measure may not fully reflect uncertainty about specific areas of policy, as in the case of the United Kingdom, where general economic policy uncertainty _____

Which choice most effectively uses data from the graph to illustrate the claim?

A) aligned closely with uncertainty about tax and public spending policy in 2005 but differed from uncertainty about tax and public spending policy by a large amount in 2009.

B) was substantially lower than uncertainty about tax and public spending policy each year from 2005 to 2010.

C) reached its highest level between 2005 and 2010 in the same year that uncertainty about trade policy and tax and public spending policy reached their lowest levels.

D) was substantially lower than uncertainty about trade policy in 2005 and substantially higher than uncertainty about trade policy in 2010.

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CONTINUE

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14

Linguist Deborah Tannen has cautioned against framing contentious issues in terms of two highly competitive perspectives, such as pro versus con. According to Tannen, this debate-driven approach can strip issues of their complexity and, when used in front of an audience, can be less informative than the presentation of multiple perspectives in a noncompetitive format. To test Tannen's hypothesis, students conducted a study in which they showed participants one of three different versions of local news commentary about the same issue. Each version featured a debate between two commentators with opposing views, a panel of three commentators with various views, or a single commentator.

Which finding from the students' study, if true, would most strongly support Tannen's hypothesis?

A) On average, participants perceived commentators in the debate as more knowledgeable about the issue than commentators in the panel.

B) On average, participants perceived commentators in the panel as more knowledgeable about the issue than the single commentator.

C) On average, participants who watched the panel correctly answered more questions about the issue than those who watched the debate or the single commentator did.

D) On average, participants who watched the single commentator correctly answered more questions about the issue than those who watched the debate did.

15

King Lear is a circa 1606 play by William Shakespeare. In the play, the character of King Lear attempts to test his three daughters' devotion to him. He later expresses regret for his actions, as is evident when he _____

Which choice most effectively uses a quotation from *King Lear* to illustrate the claim?

A) says of himself, "I am a man / more sinned against than sinning."

B) says during a growing storm, "This tempest will not give me leave to ponder / On things would hurt me more."

C) says to himself while striking his head, "Beat at this gate that let thy folly in / And thy dear judgement out!"

D) says of himself, "I will do such things— / What they are yet, I know not; but they shall be / The terrors of the earth!"

16

Many of William Shakespeare's tragedies address broad themes that still appeal to today's audiences. For instance, *Romeo and Juliet*, which is set in the Italy of Shakespeare's time, tackles the themes of parents versus children and love versus hate, and the play continues to be read and produced widely around the world. But understanding Shakespeare's so-called history plays can require a knowledge of several centuries of English history. Consequently, _____

Which choice most logically completes the text?

A) many theatergoers and readers today are likely to find Shakespeare's history plays less engaging than the tragedies.

B) some of Shakespeare's tragedies are more relevant to today's audiences than twentieth-century plays.

C) *Romeo and Juliet* is the most thematically accessible of all Shakespeare's tragedies.

D) experts in English history tend to prefer Shakespeare's history plays to his other works.

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CONTINUE

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17

Ancestral Puebloans, the civilization from which present-day Pueblo tribes descended, emerged as early as 1500 B.C.E. in an area of what is now the southwestern United States and dispersed suddenly in the late 1200s C.E., abandoning established villages with systems for farming crops and turkeys. Recent analysis comparing turkey remains at Mesa Verde, one such village in southern Colorado, to samples from modern turkey populations in the Rio Grande Valley of north central New Mexico determined that the latter birds descended in part from turkeys cultivated at Mesa Verde, with shared genetic markers appearing only after 1280. Thus, researchers concluded that _____

Which choice most logically completes the text?

A) conditions of the terrains in the Rio Grande Valley and Mesa Verde had greater similarities in the past than they do today.

B) some Ancestral Puebloans migrated to the Rio Grande Valley in the late 1200s and carried farming practices with them.

C) Indigenous peoples living in the Rio Grande Valley primarily planted crops and did not cultivate turkeys before 1280.

D) the Ancestral Puebloans of Mesa Verde likely adopted the farming practices of Indigenous peoples living in other regions.

18

One challenge when researching whether holding elected office changes a person's behavior is the problem of ensuring that the experiment has an appropriate control group. To reveal the effect of holding office, researchers must compare people who hold elected office with people who do not hold office but who are otherwise similar to the office-holders. Since researchers are unable to control which politicians win elections, they therefore _____

Which choice most logically completes the text?

A) struggle to find valid data about the behavior of politicians who do not currently hold office.

B) can only conduct valid studies with people who have previously held office rather than people who presently hold office.

C) should select a control group of people who differ from office holders in several significant ways.

D) will find it difficult to identify a group of people who can function as an appropriate control group for their studies.

19

In his groundbreaking book *Bengali Harlem and the Lost Histories of South Asian America*, Vivek Bald uses newspaper articles, census records, ships' logs, and memoirs to tell the _____ who made New York City their home in the early twentieth century.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) story's of the South Asian immigrants

B) story's of the South Asian immigrants'

C) stories of the South Asian immigrants

D) stories' of the South Asian immigrant's

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CONTINUE

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20

In her two major series "Memory Test" and "Autobiography," painter Howardena Pindell explored themes _____ healing, self-discovery, and memory by cutting and sewing back together pieces of canvas and inserting personal artifacts, such as postcards, into some of the paintings.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) of
B) of,
C) of—
D) of:

21

Both Sona Charaipotra, an Indian American, and Dhonielle Clayton, an African American, grew up frustrated by the lack of diverse characters in books for young people. In 2011, these two writers joined forces to found CAKE Literary, a book packaging _____ specializes in the creation and promotion of stories told from diverse perspectives for children and young adults.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) company,
B) company that
C) company
D) company, that

22

A study led by scientist Rebecca Kirby at the University of Wisconsin–Madison found that black bears that eat human food before hibernation have increased levels of a rare carbon isotope, _____ due to the higher ^{13}C levels in corn and cane sugar. Bears with these elevated levels were also found to have much shorter hibernation periods on average.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) carbon-13, (^{13}C)
B) carbon-13 (^{13}C)
C) carbon-13, (^{13}C),
D) carbon-13 (^{13}C),

23

In 2010, archaeologist Noel Hidalgo Tan was visiting the twelfth-century temple of Angkor Wat in Cambodia when he noticed markings of red paint on the temple _____ the help of digital imaging techniques, he discovered the markings to be part of an elaborate mural containing over 200 paintings.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) walls, with
B) walls with
C) walls so with
D) walls. With

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CONTINUE
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24

Working from an earlier discovery of Charpentier's, chemists Emmanuelle Charpentier and Jennifer Doudna—winners of the 2020 Nobel Prize in Chemistry—re-created and then reprogrammed the so-called "genetic scissors" of a species of DNA-cleaving bacteria _____ a tool that is revolutionizing the field of gene technology.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) to forge
B) forging
C) forged
D) and forging

25

In 2016, engineer Vanessa Galvez oversaw the installation of 164 bioswales, vegetated channels designed to absorb and divert stormwater, along the streets of Queens, New York. By reducing the runoff flowing into city sewers, _____

Which choice completes the text so that it conforms to the conventions of Standard English?

A) the mitigation of both street flooding and the resulting pollution of nearby waterways has been achieved by bioswales.
B) the bioswales have mitigated both street flooding and the resulting pollution of nearby waterways.
C) the bioswales' mitigation of both street flooding and the resulting pollution of nearby waterways has been achieved.
D) both street flooding and the resulting pollution of nearby waterways have been mitigated by bioswales.

26

A study published by Rice University geoscientist Ming Tang in 2019 offers a new explanation for the origin of Earth's _____ structures called arcs, towering ridges that form when a dense oceanic plate subducts under a less dense continental plate, melts in the mantle below, and then rises and bursts through the continental crust above.

Which choice completes the text so that it conforms to the conventions of Standard English?

A) continents geological
B) continents: geological
C) continents; geological
D) continents. Geological

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27

During a 2021 launch, Rocket Labs' Electron rocket experienced an unexpected failure: its second-stage booster shut down suddenly after ignition. _____ instead of downplaying the incident, Rocket Labs' CEO publicly acknowledged what happened and apologized for the loss of the rocket's payload, which had consisted of two satellites.

Which choice completes the text with the most logical transition?

A) Afterward,

B) Additionally,

C) Indeed,

D) Similarly,

28

When soil becomes contaminated by toxic metals, it can be removed from the ground and disposed of in a landfill. _____ contaminated soil can be detoxified via phytoremediation: plants that can withstand high concentrations of metals absorb the pollutants and store them in their shoots, which are then cut off and safely disposed of, preserving the health of the plants.

Which choice completes the text with the most logical transition?

A) Alternatively,

B) Specifically,

C) For example,

D) As a result,

29

While researching a topic, a student has taken the following notes:

- The calendar used by most of the world (the Gregorian calendar) has 365 days.
- Because 365 days can't be divided evenly by 7 (the number of days in a week), calendar dates fall on a different day of the week each year.
- The Hanke-Henry permanent calendar, developed as an alternative to the Gregorian calendar, has 364 days.
- Because 364 can be divided evenly by 7, calendar dates fall on the same day of the week each year, which supports more predictable scheduling.

The student wants to explain an advantage of the Hanke-Henry calendar. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) The Gregorian calendar has 365 days, which is one day longer than the Hanke-Henry permanent calendar.

B) Adopting the Hanke-Henry permanent calendar would help solve a problem with the Gregorian calendar.

C) Designed so calendar dates would occur on the same day of the week each year, the Hanke-Henry calendar supports more predictable scheduling than does the Gregorian calendar.

D) The Hanke-Henry permanent calendar was developed as an alternative to the Gregorian calendar, which is currently the most-used calendar in the world.

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30

While researching a topic, a student has taken the following notes:

- The Haudenosaunee Confederacy is a nearly 1,000-year-old alliance of six Native nations in the northeastern US.
- The members are bound by a centuries-old agreement known as the Great Law of Peace.
- Historian Bruce Johansen is one of several scholars who believe that the principles of the Great Law of Peace influenced the US Constitution.
- This theory is called the influence theory.
- Johansen cites the fact that Benjamin Franklin and Thomas Jefferson both studied the Haudenosaunee Confederacy.

The student wants to present the influence theory to an audience unfamiliar with the Haudenosaunee Confederacy. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Historian Bruce Johansen believes that the Great Law of Peace was very influential.

B) The influence theory is supported by the fact that Benjamin Franklin and Thomas Jefferson both studied the Haudenosaunee Confederacy.

C) The influence theory holds that the principles of the Great Law of Peace, a centuries-old agreement binding six Native nations in the northeastern US, influenced the US Constitution.

D) Native people, including the members of the Haudenosaunee Confederacy, influenced the founding of the US in many different ways.

31

While researching a topic, a student has taken the following notes:

- In 1999, astronomer Todd Henry studied the differences in surface temperature between the Sun and nearby stars.
- His team mapped all stars within 10 parsecs (approximately 200 trillion miles) of the Sun.
- The surface temperature of the Sun is around 9,800°F, which classifies it as a G star.
- 327 of the 357 stars in the study were classified as K or M stars, with surface temperatures under 8,900°F (cooler than the Sun).
- 11 of the 357 stars in the study were classified as A or F stars, with surface temperatures greater than 10,300°F (hotter than the Sun).

The student wants to emphasize how hot the Sun is relative to nearby stars. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) At around 9,800°F, which classifies it as a G star, the Sun is hotter than most but not all of the stars within 10 parsecs of it.

B) Astronomer Todd Henry determined that the Sun, at around 9,800°F, is a G star, and several other stars within a 10-parsec range are A or F stars.

C) Of the 357 stars within ten parsecs of the Sun, 327 are classified as K or M stars, with surface temperatures under 8,900°F.

D) While most of the stars within 10 parsecs of the Sun are classified as K, M, A, or F stars, the Sun is classified as a G star due to its surface temperature of 9,800°F.

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CONTINUE

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32

While researching a topic, a student has taken the following notes:

- The *Atlantic Monthly* magazine was first published in 1857.
- The magazine focused on politics, art, and literature.
- In 2019, historian Cathryn Halverson published the book *Faraway Women and the "Atlantic Monthly."*
- Its subject is female authors whose autobiographies appeared in the magazine in the early 1900s.
- One of the authors discussed is Juanita Harrison.

The student wants to introduce Cathryn Halverson's book to an audience already familiar with the *Atlantic Monthly*. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) Cathryn Halverson's *Faraway Women and the "Atlantic Monthly"* discusses female authors whose autobiographies appeared in the magazine in the early 1900s.

B) A magazine called the *Atlantic Monthly*, referred to in Cathryn Halverson's book title, was first published in 1857.

C) *Faraway Women and the "Atlantic Monthly"* features contributors to the *Atlantic Monthly*, first published in 1857 as a magazine focusing on politics, art, and literature.

D) An author discussed by Cathryn Halverson is Juanita Harrison, whose autobiography appeared in the *Atlantic Monthly* in the early 1900s.

33

While researching a topic, a student has taken the following notes:

- The magnificent frigatebird (*fregata magnificens*) is a species of seabird that feeds mainly on fish, tuna, squid, and other small sea animals.
- It is unusual among seabirds in that it doesn't dive into the water for prey.
- One way it acquires food is by using its hook-tipped bill to snatch prey from the surface of the water.
- Another way it acquires food is by taking it from weaker birds by force.
- This behavior is known as kleptoparasitism.

The student wants to emphasize a similarity between the two ways a magnificent frigatebird acquires food. Which choice most effectively uses relevant information from the notes to accomplish this goal?

A) A magnificent frigatebird never dives into the water, instead using its hook-tipped bill to snatch prey from the surface.

B) Neither of a magnificent frigatebird's two ways of acquiring food requires the bird to dive into the water.

C) Of the magnificent frigatebird's two ways of acquiring food, only one is known as kleptoparasitism.

D) In addition to snatching prey from the water with its hook-tipped bill, a magnificent frigatebird takes food from other birds by force.

STOP

If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.

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Math

27 QUESTIONS

DIRECTIONS

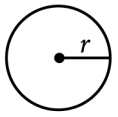
The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

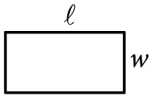
Unless otherwise indicated:

- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

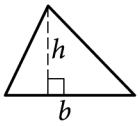
REFERENCE



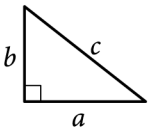
$A = \pi r^2$
$C = 2\pi r$



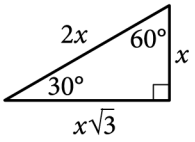
$A = \ell w$

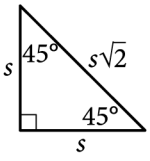


$A = \frac{1}{2}bh$

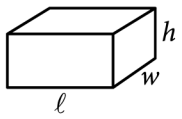


$c^2 = a^2 + b^2$

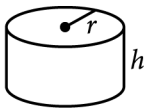




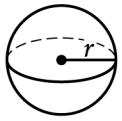
Special Right Triangles



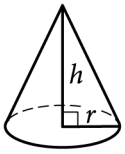
$V = \ell wh$



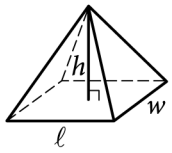
$V = \pi r^2 h$



$V = \frac{4}{3}\pi r^3$



$V = \frac{1}{3}\pi r^2 h$



$V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π.

The sum of the measures in degrees of the angles of a triangle is 180.

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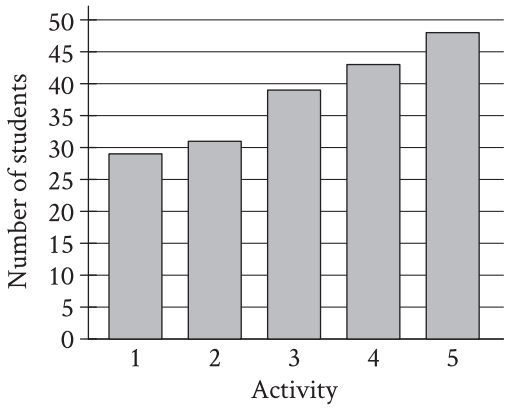
1



A group of students voted on five after-school activities. The bar graph shows the number of students who voted for each of the five activities. How many students chose activity 3?

A) 25

B) 39

C) 48

D) 50

2

What percentage of 300 is 75?

A) 25%

B) 50%

C) 75%

D) 225%

3

$$\frac{x^2}{25} = 36$$

What is a solution to the given equation?

A) 6

B) 30

C) 450

D) 900

4

3 more than 8 times a number x is equal to 83. Which equation represents this situation?

A) $(3)(8)x = 83$

B) $8x = 83 + 3$

C) $3x + 8 = 83$

D) $8x + 3 = 83$

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5

Hana deposited a fixed amount into her bank account each month. The function $f(t) = 100 + 25t$ gives the amount, in dollars, in Hana's bank account after t monthly deposits. What is the best interpretation of 25 in this context?

A) With each monthly deposit, the amount in Hana's bank account increased by $25.

B) Before Hana made any monthly deposits, the amount in her bank account was $25.

C) After 1 monthly deposit, the amount in Hana's bank account was $25.

D) Hana made a total of 25 monthly deposits.

6

A customer spent $27 to purchase oranges at $3 per pound. How many pounds of oranges did the customer purchase?

7

Nasir bought 9 storage bins that were each the same price. He used a coupon for $63 off the entire purchase. The cost for the entire purchase after using the coupon was $27. What was the original price, in dollars, for 1 storage bin?

8

x	$f(x)$
0	29
1	32
2	35

For the linear function f, the table shows three values of x and their corresponding values of $f(x)$. Which equation defines $f(x)$?

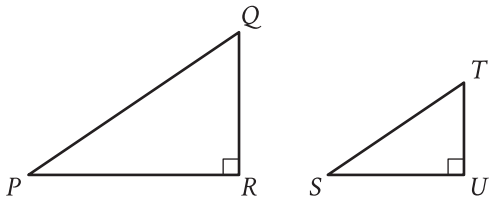
A) $f(x) = 3x + 29$

B) $f(x) = 29x + 32$

C) $f(x) = 35x + 29$

D) $f(x) = 32x + 35$

9



Note: Figures not drawn to scale.

Right triangles PQR and STU are similar, where P corresponds to S. If the measure of angle Q is 18°, what is the measure of angle S ?

A) 18°

B) 72°

C) 82°

D) 162°

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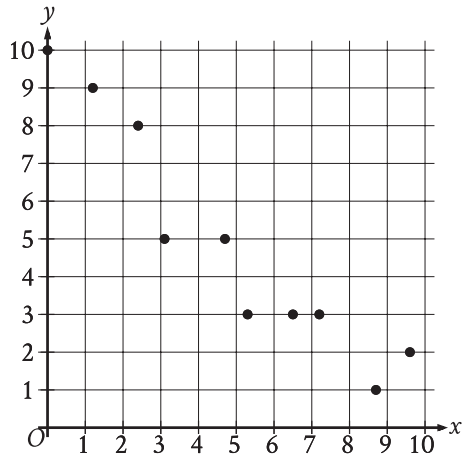
10

The scatterplot shows the relationship between two variables, x and y.



Which of the following equations is the most appropriate linear model for the data shown?

A) $y = 0.9 + 9.4x$

B) $y = 0.9 - 9.4x$

C) $y = 9.4 + 0.9x$

D) $y = 9.4 - 0.9x$

11

$$2.5b + 5r = 80$$

The given equation describes the relationship between the number of birds, b, and the number of reptiles, r, that can be cared for at a pet care business on a given day. If the business cares for 16 reptiles on a given day, how many birds can it care for on this day?

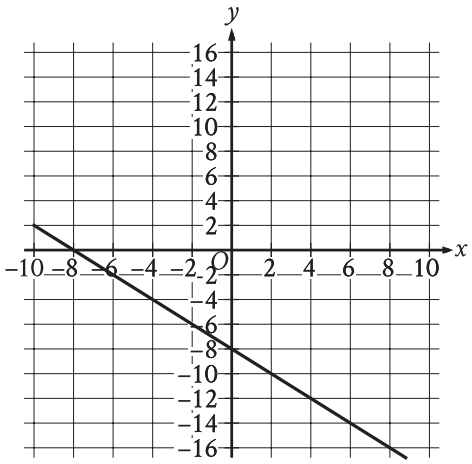
A) 0

B) 5

C) 40

D) 80

12



What is an equation of the graph shown?

A) $y = -2x - 8$

B) $y = x - 8$

C) $y = -x - 8$

D) $y = 2x - 8$

13

If $\frac{x}{8} = 5$, what is the value of $\frac{8}{x}$?

14

$$24x + y = 48$$
$$6x + y = 72$$

The solution to the given system of equations is (x, y). What is the value of y ?

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15

Line t in the xy-plane has a slope of $-\frac{1}{3}$ and passes through the point $(9, 10)$. Which equation defines line t?

A) $y = 13x - \frac{1}{3}$

B) $y = 9x + 10$

C) $y = -\frac{x}{3} + 10$

D) $y = -\frac{x}{3} + 13$

16

The function $f(x) = 206(1.034)^x$ models the value, in dollars, of a certain bank account by the end of each year from 1957 through 1972, where x is the number of years after 1957. Which of the following is the best interpretation of "$f(5)$ is approximately equal to 243" in this context?

A) The value of the bank account is estimated to be approximately 5 dollars greater in 1962 than in 1957.

B) The value of the bank account is estimated to be approximately 243 dollars in 1962.

C) The value, in dollars, of the bank account is estimated to be approximately 5 times greater in 1962 than in 1957.

D) The value of the bank account is estimated to increase by approximately 243 dollars every 5 years between 1957 and 1972.

17

For a certain rectangular region, the ratio of its length to its width is 35 to 10. If the width of the rectangular region increases by 7 units, how must the length change to maintain this ratio?

A) It must decrease by 24.5 units.

B) It must increase by 24.5 units.

C) It must decrease by 7 units.

D) It must increase by 7 units.

18

Square P has a side length of x inches. Square Q has a perimeter that is 176 inches greater than the perimeter of square P. The function f gives the area of square Q, in square inches. Which of the following defines f?

A) $f(x) = (x + 44)^2$

B) $f(x) = (x + 176)^2$

C) $f(x) = (176x + 44)^2$

D) $f(x) = (176x + 176)^2$

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19

$$\frac{14x}{7y} = 2\sqrt{w + 19}$$

The given equation relates the distinct positive real numbers w, x, and y. Which equation correctly expresses w in terms of x and y ?

A) $w = \sqrt{\frac{x}{y}} - 19$

B) $w = \sqrt{\frac{28x}{14y}} - 19$

C) $w = \left(\frac{x}{y}\right)^2 - 19$

D) $w = \left(\frac{28x}{14y}\right)^2 - 19$

20

Point O is the center of a circle. The measure of arc RS on this circle is 100°. What is the measure, in degrees, of its associated angle ROS ?

21

The expression $6\sqrt[5]{3^5x^{45}} \cdot \sqrt[8]{2^8x}$ is equivalent to ax^b, where a and b are positive constants and $x > 1$.

What is the value of $a + b$?

22

A right triangle has sides of length $2\sqrt{2}$, $6\sqrt{2}$, and $\sqrt{80}$ units. What is the area of the triangle, in square units?

A) $8\sqrt{2} + \sqrt{80}$

B) 12

C) $24\sqrt{80}$

D) 24

23

The expression $4x^2 + bx - 45$, where b is a constant, can be rewritten as $(hx + k)(x + j)$, where h, k, and j are integer constants. Which of the following must be an integer?

A) $\frac{b}{h}$

B) $\frac{b}{k}$

C) $\frac{45}{h}$

D) $\frac{45}{k}$

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24

$$y = 2x^2 - 21x + 64$$
$$y = 3x + a$$

In the given system of equations, a is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y), in the xy-plane. What is the value of x ?

A) -8

B) -6

C) 6

D) 8

25

An isosceles right triangle has a hypotenuse of length 58 inches. What is the perimeter, in inches, of this triangle?

A) $29\sqrt{2}$

B) $58\sqrt{2}$

C) $58 + 58\sqrt{2}$

D) $58 + 116\sqrt{2}$

26

In the xy-plane, a parabola has vertex $(9, -14)$ and intersects the x-axis at two points. If the equation of the parabola is written in the form $y = ax^2 + bx + c$, where a, b, and c are constants, which of the following could be the value of $a + b + c$?

A) -23

B) -19

C) -14

D) -12

27

Function f is defined by $f(x) = -a^x + b$, where a and b are constants. In the xy-plane, the graph of $y = f(x) - 15$ has a y-intercept at $\left(0, -\frac{99}{7}\right)$. The product of a and b is $\frac{65}{7}$. What is the value of a?

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

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Math

27 QUESTIONS

DIRECTIONS

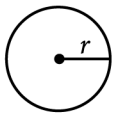
The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

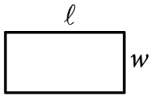
Unless otherwise indicated:

- All variables and expressions represent real numbers.
- Figures provided are drawn to scale.
- All figures lie in a plane.
- The domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE



$A = \pi r^2$
$C = 2\pi r$



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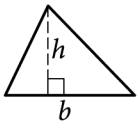

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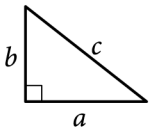
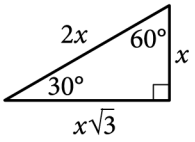
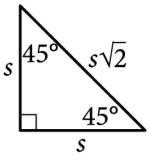
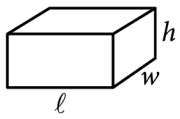


$c^2 = a^2 + b^2$

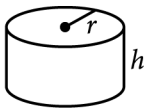




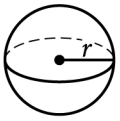
Special Right Triangles



$V = \ell wh$



$V = \pi r^2 h$



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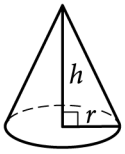
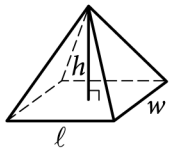


$V = \frac{1}{3}\pi r^2 h$



$V = \frac{1}{3}\ell wh$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π.

The sum of the measures in degrees of the angles of a triangle is 180.

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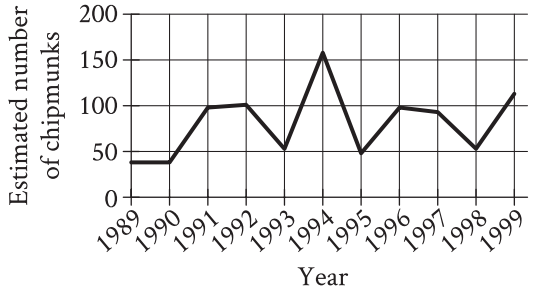
1

The line graph shows the estimated number of chipmunks in a state park on April 1 of each year from 1989 to 1999.



Based on the line graph, in which year was the estimated number of chipmunks in the state park the greatest?

A) 1989

B) 1994

C) 1995

D) 1998

2

A fish swam a distance of 5,104 yards. How far did the fish swim, in miles? (1 mile = 1,760 yards)

A) 0.3

B) 2.9

C) 3,344

D) 6,864

3

Which expression is equivalent to $12x^3 - 5x^3$?

A) $7x^6$

B) $17x^3$

C) $7x^3$

D) $17x^6$

4

$$x + y = 18$$
$$5y = x$$

What is the solution (x, y) to the given system of equations?

A) (15, 3)

B) (16, 2)

C) (17, 1)

D) (18, 0)

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5

The point $(8, 2)$ in the xy-plane is a solution to which of the following systems of inequalities?

A) $x > 0$
$y > 0$

B) $x > 0$
$y < 0$

C) $x < 0$
$y > 0$

D) $x < 0$
$y < 0$

6

$$|x - 5| = 10$$

What is one possible solution to the given equation?

7

$$f(x) = 7x + 1$$

The function gives the total number of people on a company retreat with x managers. What is the total number of people on a company retreat with 7 managers?

8

$$h(x) = x^2 - 3$$

Which table gives three values of x and their corresponding values of $h(x)$ for the given function h?

A)

x	1	2	3
$h(x)$	4	5	6

B)

x	1	2	3
$h(x)$	−2	1	6

C)

x	1	2	3
$h(x)$	−1	1	3

D)

x	1	2	3
$h(x)$	−2	1	3

9

The function f is defined by $f(x) = 270(0.1)^x$. What is the value of $f(0)$?

A) 0
B) 1
C) 27
D) 270

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10

To estimate the proportion of a population that has a certain characteristic, a random sample was selected from the population. Based on the sample, it is estimated that the proportion of the population that has the characteristic is 0.49, with an associated margin of error of 0.04. Based on this estimate and margin of error, which of the following is the most appropriate conclusion about the proportion of the population that has the characteristic?

A) It is plausible that the proportion is between 0.45 and 0.53.

B) It is plausible that the proportion is less than 0.45.

C) The proportion is exactly 0.49.

D) It is plausible that the proportion is greater than 0.53.

11

A moving truck can tow a trailer if the combined weight of the trailer and the boxes it contains is no more than 4,600 pounds. What is the maximum number of boxes this truck can tow in a trailer with a weight of 500 pounds if each box weighs 120 pounds?

A) 34

B) 35

C) 38

D) 39

12

$$-4x^2 - 7x = -36$$

What is the positive solution to the given equation?

A) $\frac{7}{4}$

B) $\frac{9}{4}$

C) 4

D) 7

13

The table summarizes the distribution of color and shape for 100 tiles of equal area.

	Red	Blue	Yellow	Total
Square	10	20	25	55
Pentagon	20	10	15	45
Total	30	30	40	100

If one of these tiles is selected at random, what is the probability of selecting a red tile? (Express your answer as a decimal or fraction, not as a percent.)

14

$$f(x) = 2x + 3$$

For the given function f, the graph of $y = f(x)$ in the xy-plane is parallel to line j. What is the slope of line j?

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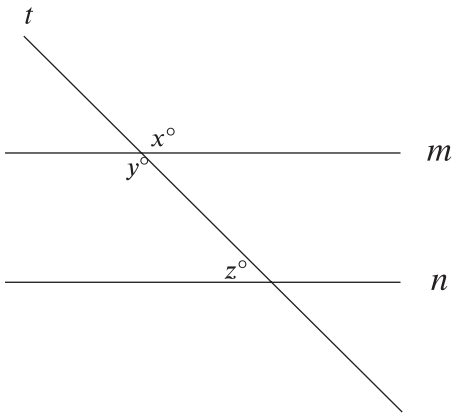
15

A proposal for a new library was included on an election ballot. A radio show stated that 3 times as many people voted in favor of the proposal as people who voted against it. A social media post reported that 15,000 more people voted in favor of the proposal than voted against it. Based on these data, how many people voted against the proposal?

A) 7,500
B) 15,000
C) 22,500
D) 45,000

16



Note: Figure not drawn to scale.

In the figure, lines m and n are parallel. If $x = 6k + 13$ and $y = 8k - 29$, what is the value of z ?

A) 3
B) 21
C) 41
D) 139

17

$$-3x + 21px = 84$$

In the given equation, p is a constant. The equation has no solution. What is the value of p ?

A) 0
B) $\frac{1}{7}$
C) $\frac{4}{3}$
D) 4

18

$$f(x) = (x - 10)(x + 13)$$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

A) −130
B) −13
C) $-\frac{23}{2}$
D) $-\frac{3}{2}$

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19

The function $f(x) = \frac{1}{9}(x-7)^2 + 3$ gives a metal ball's height above the ground $f(x)$, in inches, x seconds after it started moving on a track, where $0 \le x \le 10$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy-plane?

A) The metal ball's minimum height was 3 inches above the ground.
B) The metal ball's minimum height was 7 inches above the ground.
C) The metal ball's height was 3 inches above the ground when it started moving.
D) The metal ball's height was 7 inches above the ground when it started moving.

20

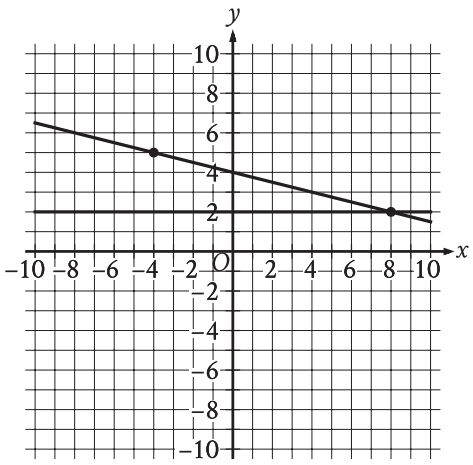
In triangle JKL, $\cos(K) = \frac{24}{51}$ and angle J is a right angle. What is the value of $\cos(L)$?

21

$$-x^2 + bx - 676 = 0$$

In the given equation, b is a positive integer. The equation has no real solution. What is the greatest possible value of b ?

22



If a new graph of three linear equations is created using the system of equations shown and the equation $x + 4y = -16$, how many solutions (x, y) will the resulting system of three equations have?

A) Zero
B) Exactly one
C) Exactly two
D) Infinitely many

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23

$$f(x) = 5{,}470(0.64)^{\frac{x}{12}}$$

The function f gives the value, in dollars, of a certain piece of equipment after x months of use. If the value of the equipment decreases each year by p% of its value the preceding year, what is the value of p ?

A) 4

B) 5

C) 36

D) 64

24

The dot plot represents the 15 values in data set A. Data set B is created by adding 56 to each of the values in data set A. Which of the following correctly compares the medians and the ranges of data sets A and B?

A) The median of data set B is equal to the median of data set A, and the range of data set B is equal to the range of data set A.

B) The median of data set B is equal to the median of data set A, and the range of data set B is greater than the range of data set A.

C) The median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.

D) The median of data set B is greater than the median of data set A, and the range of data set B is greater than the range of data set A.

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25

The equation $x^2 + (y - 1)^2 = 49$ represents circle A. Circle B is obtained by shifting circle A down 2 units in the xy-plane. Which of the following equations represents circle B?

A) $(x - 2)^2 + (y - 1)^2 = 49$

B) $x^2 + (y - 3)^2 = 49$

C) $(x + 2)^2 + (y - 1)^2 = 49$

D) $x^2 + (y + 1)^2 = 49$

26

Two identical rectangular prisms each have a height of 90 centimeters (cm). The base of each prism is a square, and the surface area of each prism is K cm^2. If the prisms are glued together along a square base, the resulting prism has a surface area of $\frac{92}{47}K$ cm^2. What is the side length, in cm, of each square base?

A) 4

B) 8

C) 9

D) 16

27

210 is $p\%$ greater than 30. What is the value of p ?

STOP

If you finish before time is called, you may check your work on this module only. Do not turn to any other module in the test.

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- You may work on only one module at a time.
- If you finish a module before time is called, check your work on that module only. You may NOT turn to any other module.

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- Circle only one answer to each question. If you change your mind, completely erase the circle. You will not get credit for questions with more than one answer circled, or for questions with no answers circled.

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Reading and Writing

Module 1

(33 questions)

QUESTION 1

Choice B is the best answer because it most logically completes the text's discussion of the OSIRIS-REx spacecraft's contact with the asteroid 101955 Bennu. In this context, "collected" means acquired and took away. The text indicates that although the boulders on the asteroid's surface caused some unforeseen problems, OSIRIS-REx was able to gather a sample to return to Earth. This context suggests that OSIRIS-REx successfully collected a sample of 101955 Bennu.

Choice A is incorrect because in this context "attached" means connected or affixed. The text indicates that OSIRIS-REx gathered pieces of 101955 Bennu to bring to Earth; it doesn't suggest that the spacecraft attached anything to the asteroid. *Choice C* is incorrect because in this context "followed" means tracked or traveled behind and the text discusses OSIRIS-REx's brief encounter with 101955 Bennu during which the spacecraft gathered a sample to bring to Earth. The text doesn't suggest that the spacecraft tracked the sample, and it's not clear what it would mean for the spacecraft to travel behind the sample it collected. *Choice D* is incorrect because in this context "replaced" means put back or returned. The text indicates that OSIRIS-REx gathered pieces of 101955 Bennu to bring to Earth but doesn't suggest that anything was returned to the asteroid.

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QUESTION 2

Choice A is the best answer because it most logically completes the text's discussion of the Moon's surface. In this context, "reflect" means show or make apparent. The text states that because the surface of the Moon was softer when the Moon was still forming than it is now, early asteroid and meteoroid impacts "would have left less of an impression" and, as a result, evidence of them may no longer exist. This context supports the idea that the surface of the Moon may not accurately show signs of early impact events.

Choice B is incorrect because it wouldn't make sense to say that the surface of the Moon may not accurately "receive," or acquire or experience, early impacts from asteroids or meteoroids. The text indicates that the impacts have already occurred, and it isn't clear how the Moon's surface could be accurate or inaccurate in experiencing them. *Choice C* is incorrect because it wouldn't make sense to say that the surface of the Moon may not accurately "evaluate," or determine the significance or condition of, early impacts from asteroids or meteoroids, since that would suggest that it's possible for the Moon's surface to make a decision of any kind. *Choice D* is incorrect. In this context, "mimic" would mean to deliberately simulate or closely imitate something. It wouldn't make sense to say that the surface of the Moon may not accurately mimic early asteroid and meteoroid impacts, since that would suggest that it's possible for the Moon to deliberately imitate something.

QUESTION 3

Choice A is the best answer because it most logically completes the text's discussion about handedness in animals. As used in this context, "recognizable" means apparent or identifiable. The text indicates that handedness is "easy to observe in humans," but that animal-behavior researchers use special tasks to determine handedness in other animals. This context and the use of "less" before the blank indicate that compared with handedness in humans, handedness in other animals is less recognizable.

Choice B is incorrect because there's nothing in the text to suggest that handedness is less "intriguing," or fascinating, in nonhuman animals than it is in humans. The text focuses on how easy it is to observe handedness in humans as compared with other animals; the text doesn't suggest that handedness is more fascinating in humans. *Choice C* is incorrect because there's nothing in the text to suggest that handedness is less "significant," or important or meaningful, in nonhuman animals than it is in humans. The text focuses on how easy it is to observe handedness in humans as compared with other animals; the text doesn't suggest that handedness is more significant in humans. *Choice D* is incorrect because "useful," or functional or helpful, wouldn't make sense in context. The text focuses on the ease with which researchers can determine whether an animal or person is right- or left-handed, not on how useful handedness in nonhuman animals is compared with handedness in humans.

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QUESTION 4

Choice C is the best answer because it most logically completes the text's discussion of the influences on Banisadr's work. As used in this context, "unimportant" means trivial or lacking value. "It is by no means" establishes that the word that goes in the blank is contradicted by other information; the material that follows "indeed" later in that sentence provides the contradicting information—namely, that Banisadr himself cites Bosch as an inspiration. In other words, the sentence indicates that Bosch's influence on Banisadr is significant, and thus recognizing that influence is by no means unimportant.

Choice A is incorrect because it wouldn't make sense to say that recognizing Bosch's influence on Banisadr isn't "substantial," or meaningful. The text states that Banisadr himself cites Bosch as an influence. *Choice B* is incorrect because it wouldn't make sense to say that it isn't "satisfying," or pleasing, to recognize Bosch's influence on Banisadr. The text states that Banisadr himself cites Bosch as an influence. *Choice D* is incorrect because it wouldn't make sense to say that recognizing Bosch's influence on Banisadr isn't "appropriate," or suitable. The text indicates that Banisadr himself notes that Bosch's work has had an effect on him.

QUESTION 5

Choice A is the best answer because it most accurately describes the main purpose of the text. The text begins by stating that the new picture "failed to fit in" with the other items that the shop owner has. The text goes on to illustrate that point by describing the other pictures the shop owner has, indicating that the shop owner is fuming because he doesn't think the new picture belongs in the store. In the second paragraph, however, the text indicates that the shop owner is "secretly proud of his acquisition." The main purpose of the text is thus to reveal the shop owner's conflicted feelings about the new picture.

Choice B is incorrect because the text doesn't suggest that the shop owner resents the young man who sold him the new picture; in fact, the text gives no indication of the owner's feelings about the young man at all. *Choice C* is incorrect. Although the text indicates that the new picture is different from the other items in the shop, there's no suggestion that the shop owner prizes either the new picture or the pictures of the city, pets, and landscapes more than he prizes any other items. *Choice D* is incorrect because the text doesn't describe what the new picture looks like; rather, the text identifies some of the other kinds of images that the shop owner has and states that they're different from the new picture without explaining how they're different.

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QUESTION 6

Choice B is the best answer because it most accurately describes the overall structure of the text. First, the speaker describes observing a "most beautiful" sight: a tree ("black cypress") standing out from the golden sky behind it, looking like a person's finger "pointing upwards" and appearing "sensitive" and "exquisite." Then the speaker wonders about the image's meaning, asking why the finger is black and why it's pointing upward. Thus, the text moves from the speaker's description of a distinctive sight in nature to her pondering about what meaning to attribute to that sight.

Choice A is incorrect because the speaker assesses a natural sight—a "black cypress" tree standing "against a gold, gold sky" like a pointed finger—but doesn't question the accuracy of her own assessment. Although she wonders why the finger, which is really a tree, is black and why it's pointing, the speaker doesn't suggest that her belief that the tree resembles a finger is wrong. *Choice C* is incorrect. Although the speaker describes seeing a "black cypress" tree standing "against a gold, gold sky" like a pointed finger, she wonders about that natural image (asking why the finger, which is really a tree, is black and why it's pointing) and doesn't give any indication that any people are present in the scene. *Choice D* is incorrect. Although the speaker examines and wonders about one thing in her surroundings—a "black cypress" tree standing "against a gold, gold sky" like a pointed finger—she doesn't address her own emotional state or consider how it's affected by her surroundings.

QUESTION 7

Choice D is the best answer because it best describes the overall structure of the text. The speaker begins by stating that he has heard that others are accusing him of seeking to destroy institutions. The speaker then addresses this criticism by stating that he is "neither for nor against institutions." Instead, the speaker states that his ultimate goal is to instill "the institution of the dear love of comrades" everywhere in the country. Therefore, the overall structure of the text is best described as an address of criticism followed by an announcement of a grand ambition.

Choice A is incorrect. While the speaker does address an opinion of him that he believes to be untrue, he doesn't indicate that this attitude has become increasingly prevalent. The speaker also concludes by explaining his goal for the future rather than his current worldview. *Choice B* is incorrect because the text doesn't portray the speaker as isolated or regretful, and the speaker gestures toward a hope for societal change but doesn't offer an explicit prediction that it will happen. *Choice C* is incorrect because the speaker addresses a criticism of him that he believes to be false; he doesn't admit any personal shortcomings. Moreover, the speaker concludes by stating a goal he has rather than showcasing his achievements.

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QUESTION 8

Choice B is the best answer because it most accurately describes the function of the third sentence within the overall structure of the text. The third sentence makes a generalization, asserting that evolutionary links between predators and prey can persist across great expanses of time and distance. This generalization is exemplified by the text's discussion of the relationship between mimosa trees and *B. terrenus* beetles. When mimosa trees were introduced to North America in 1785, no *B. terrenus* beetles were present, so the relationship between the trees and the beetles that exists in their native East Asia was disrupted. When the beetles were introduced to North America more than 200 years later, however, they quickly attacked mimosa trees, illustrating the generalization that links between predators and prey "can persist across centuries and continents."

Choice A is incorrect because the third sentence doesn't indicate that Chang and colleagues were investigating any particular hypothesis. According to the text, Chang and colleagues were simply monitoring mimosa trees when the beetles happened to be introduced to the area. *Choice C* is incorrect because the third sentence offers a generalization about the relationship between predators and prey, not an explanation for the findings of Chang and colleagues that differs from an explanation presented elsewhere in the text. *Choice D* is incorrect because the third sentence doesn't discuss any particular species (either the species mentioned elsewhere in the text or any other) and doesn't help explain why species spread to new locations.

QUESTION 9

Choice B is the best answer because it describes the most likely way that Graeber and Wengrow (Text 2) would respond to the "conventional wisdom" presented in Text 1. According to Text 1, the conventional wisdom about human social systems is that they developed through stages, beginning with hunter-gatherer bands, then moving to clan associations, then chiefdoms, and finally arriving at states with bureaucratic structures. Text 2 indicates that Graeber and Wengrow believe that human social systems have been flexible, shifting between different types of structures, including both hierarchical and collective systems, and that these shifts may have even occurred seasonally. This suggests that Graeber and Wengrow would dispute the idea that developments in social structures have followed a linear progression through distinct stages.

Choice A is incorrect because nothing in Text 2 suggests that Graeber and Wengrow believe that decentralized collective societies are more significant than hierarchical systems. Text 2 is focused on Graeber and Wengrow's view that humans have flexibly shifted among various social structures, not on the importance of particular structures relative to others. *Choice C* is incorrect because Text 2 doesn't include any information suggesting that Graeber and Wengrow believe that hierarchies didn't emerge until after the rise of agriculture. In fact, Text 2 indicates that Graeber and Wengrow cite evidence suggesting that some hunter-gatherer groups formed social structures with hierarchical elements ("communities that included esteemed individuals") 50,000 years ago, long before the rise of agriculture, which Text 1 says occurred around 12,000 years ago.

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Choice D is incorrect because there's no information in Text 2 suggesting that Graeber and Wengrow would challenge the assumption that groupings of hunter-gatherers were among the earliest forms of social structure. Although Text 1 does indicate that hunter-gatherer groups are assumed to be the earliest human social system, Text 2 says only that Graeber and Wengrow believe that some hunter-gatherer groups made use of different social structures at different times. Text 2 doesn't imply that Graeber and Wengrow doubt that hunter-gatherer groups preceded most other social structures.

QUESTION 10

Choice D is the best answer because it most accurately states the main idea of the text. The text describes Mary's activities in an overgrown hidden garden, saying that she was "very much absorbed" and was "only becoming more pleased with her work every hour" rather than getting tired of it. She also thinks of garden activities as a "fascinating sort of play." Thus, the main idea of the text is that Mary feels very satisfied when taking care of the garden.

Choice A is incorrect because the text never makes any mention of Mary's chores. *Choice B* is incorrect because the text indicates that Mary finds pulling up weeds to be fascinating, not boring. *Choice C* is incorrect because Mary thinks of garden activities in and of themselves as play, not as something necessary to do to create a space to play.

QUESTION 11

Choice C is the best answer because it presents a description of how the human mind is like a flower that is directly supported by the text. The text compares the needs of a "fragile and lovely flower" to those of the speaker's "tender mind": both need to be fed if they're going to survive. Without such feeding, they'll "beginneth straightway to languish," or weaken. Thus, the text suggests that the human mind is like a flower in that they both need proper nourishment in order to thrive.

Choice A is incorrect because the text doesn't address the passage of time or describe either the human mind or a flower as becoming increasingly vigorous. *Choice B* is incorrect because the text doesn't suggest that human minds or flowers draw strength from changes in weather. The references to rain in the text pertain to a flower's need for water rather than the general effects of changing weather. *Choice D* is incorrect because the text doesn't suggest that the human mind or a flower will persist regardless of challenging circumstances. In fact, the text indicates that they'll both languish right away if not given what they need.

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QUESTION 12

Choice D is the best answer because it most accurately states the main idea of the text. After establishing that Buck views most people "as nothing," the text explains that Buck won't acknowledge people other than Thornton unless they appear friendly toward Thornton, and even then he's only reluctantly accepting. Thus, the text focuses on the idea that Thornton has a special status in Buck's mind, with Buck holding him in higher regard than other people.

Choice A is incorrect because the text conveys that Buck isn't social with people other than Thornton but doesn't address Buck's life or temperament before he lived with Thornton. *Choice B* is incorrect because the text conveys that Buck doesn't really care about people other than Thornton and is aloof toward them. However, there's no indication that Buck mistrusts and avoids people generally; indeed, he accepts Thornton, who is a human. *Choice C* is incorrect because the text refers to random travelers praising and petting Buck and Thornton's partners giving Buck favors, but there's no indication that any of these people are Thornton's friends or that they have a particular fondness for Buck.

QUESTION 13

Choice A is the best answer because it uses data from the graph to accurately complete the text. The graph shows the number of organic farms located in each of six US states in 2016: between 2,600 and 2,800 in California; between 1,200 and 1,400 in Wisconsin; between 1,000 and 1,200 in New York; approximately 800 in Pennsylvania; and between 600 and 800 in both Iowa and Washington. The last sentence of the text provides information about the number of organic farms in 2016, first describing the number in California. The best completion of the sentence is the choice that accurately describes the number of organic farms in 2016 in another state, which the assertion that Washington had between 600 and 800 organic farms provides.

Choice B is incorrect because it doesn't accurately reflect the data from the graph. The graph indicates that there were between 1,000 and 1,200 organic farms in New York, not fewer than 800 organic farms. *Choice C* is incorrect because it doesn't accurately reflect the data from the graph. While the graph indicates that there were between 1,200 and 1,400 organic farms in Wisconsin in 2016, there were only between 600 and 800 in Iowa. *Choice D* is incorrect because it doesn't accurately reflect the data from the graph. The graph indicates that in 2016 there were approximately 800 organic farms in Pennsylvania, not more than 1,200.

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QUESTION 14

Choice B is the best answer because it presents a finding that, if true, would most directly support Gómez-Bahamón and her team's hypothesis about fork-tailed flycatchers. The text indicates that although two subspecies of the birds live in the same region, the tail feathers of the migrating males make a higher-pitched sound than the tail feathers of the nonmigrating males do. Gómez-Bahamón and her team hypothesize that female fork-tailed flycatchers are attracted to the particular sound made by the tail feathers of males of their own subspecies, which will bring about additional "genetic and anatomical divergence" between the two subspecies. If it were found that the pitch generated by the tail feathers of migrating males is getting higher over successive generations, it would indicate that the shape of the migrating subspecies' tail feathers is diverging further from that of the nonmigrating subspecies. And if females continue to prefer the sounds of the males of their own subspecies, the females of the migrating subspecies will become acclimated to increasingly higher pitches over subsequent generations, causing further divergence between the subspecies. Thus, if it were found that migrating males' tail feathers were producing higher pitches over time, that would support the researchers' hypothesis.

Choice A is incorrect because the researchers' hypothesis is that female flycatchers prefer the sounds produced by the tail feathers of males of their own subspecies, which will lead to further divergence between the two subspecies. This finding is about the shape of wing feathers and how that affects long-distance flight, whereas the hypothesis is about the shape of tail feathers and how that relates to female mate preference. *Choice C* is incorrect because the researchers' hypothesis is that female flycatchers prefer the sounds produced by the tail feathers of males of their own subspecies, which will lead to further divergence between the two subspecies. This finding focuses on how the tail feather sounds communicate different messages, which doesn't address differences between the subspecies or female preferences. *Choice D* is incorrect because the researchers' hypothesis is that female flycatchers prefer the sounds produced by the tail feathers of males of their own subspecies, which will lead to further divergence between the two subspecies. The finding that breeding habits haven't changed for either subspecies does not, by itself, suggest anything about female preferences or divergence between the two subspecies.

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QUESTION 15

Choice C is the best answer because it most effectively completes the example regarding the ablation rate of iron. The table shows the ablation rates for three elements—iron, potassium, and sodium—found in cosmic dust that comes from one of four sources. The text says that the ablation rate for a given element in slower-moving SPC or AST dust was lower than the ablation rate for that same element in faster-moving HTC or OCC dust. The text then presents the first part of an example of this pattern, describing an ablation rate of 28% for iron in AST dust. The information that iron from HTC dust had an ablation rate of 90% is therefore the most effective way to complete this example—the comparison of a relatively low ablation rate for iron in slower-moving AST dust with a relatively high ablation rate for iron in faster-moving HTC dust illustrates the tendency of ablation rates for a given element to be lower in slower-moving dust than in faster-moving dust.

Choice A is incorrect because the text indicates that SPC dust, like AST dust, moves relatively slowly; a comparison of the ablation rates of iron from two slower-moving dust sources could not be an example of the difference between ablation rates in slower-moving dust and faster-moving dust, which is the pattern that the example is supposed to illustrate. *Choice B* is incorrect because the example in the text is supposed to illustrate the difference in the ablation rates of the same element from slower-moving dust and faster-moving dust, and the first part of the example provides data about the ablation rate of iron, which means the second part of the example must also be about the ablation rate of iron, not the ablation rate of sodium. *Choice D* is incorrect because the example in the text is supposed to illustrate the difference in the ablation rates of the same element from slower-moving dust and faster-moving dust, and the first part of the example provides data about the ablation rate of iron, which means the second part of the example must also be about the ablation rate of iron, not the ablation rate of sodium. Additionally, any ablation rate from AST dust would be ineffective in this example since AST dust is referenced in the first part of the example and thus additional data focused on AST dust would not illustrate a variation across dust types.

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QUESTION 16

Choice A is the best answer because it presents the quotation that best illustrates the journalist's claim. By indicating that a collective didn't continue because it was hard to share credit and responsibilities within the group even though the company was enjoyable, the quotation shows that working collaboratively can be difficult for artists who are used to having complete control over their work.

Choice B is incorrect because the quotation indicates that members of a collective are able to collaborate together and have agreed on a fair way to manage their responsibilities; this doesn't demonstrate the challenge of sharing control among members of a collective. *Choice C* is incorrect because the quotation highlights the support and encouragement of individual expression an artist experiences due to working in a collective; these positive aspects don't demonstrate the challenge of sharing control among members of a collective. *Choice D* is incorrect because the quotation doesn't address any challenges of sharing control among members of a collective; it simply indicates that artists sometimes choose to work with collectives without having to be a member. Therefore, the quotation doesn't illustrate the journalist's claim.

QUESTION 17

Choice A is the best answer because it most effectively uses data from the table to complete the statement. The text explains that mycorrhizal hosts are plants that benefit from the presence of mycorrhizal fungi in the soil and that some such plants produce more mass when grown in the presence of these fungi, while for nonmycorrhizal species the fungi either have no effect or may be harmful. The experiment included two mycorrhizal hosts (corn and marigold) and one nonmycorrhizal species (broccoli). Given the claim in the text that nonmycorrhizal species will see either no difference or a decrease in mass when exposed to mycorrhizal fungi, the student would likely have been surprised by the higher average mass for broccoli grown in the presence of the fungi than the broccoli grown in the soil treated to kill fungi.

Choice B is incorrect. Although this choice accurately describes the corn data from the table, the fact that the mycorrhizal host corn is more massive in the presence of the fungi likely fits with what the student expected and would therefore not be surprising. *Choice C* is incorrect. Although this choice accurately describes the marigold data from the table, the fact that the mycorrhizal host marigold is more massive in the presence of the fungi is likely what the student expected and thus would not be surprising. *Choice D* is incorrect because it does not accurately represent the data in the table—when grown in soil treated to kill fungi, corn had an average mass of 3.8 g while broccoli had an average mass of 7g—and because making comparisons among the plants in the no-fungi condition, by itself, does not provide a basis to compare the average mass of mycorrhizal hosts and nonmycorrhizal species grown in the presence of the fungi with those grown in the soil treated to kill fungi.

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QUESTION 18

Choice A is the best answer because it presents the conclusion that most logically completes the text's discussion about the significance of the cupid found at Pompeii. The text indicates that the cupid is near a statue of a female figure who is fishing, and it goes on to indicate that because Venus is associated with cupids, some scholars believe the female figure to be the goddess Venus. But the text then says that, according to archaeologist Carla Brain, cupids may have also been associated with the activity of fishing, which, if true, would suggest that the mere appearance of a cupid near a female figure engaged in fishing does not indicate with certainty that the figure is Venus (that is, the cupid might be associated with fishing, and the figure might be anyone at all).

Choice B is incorrect because the text says nothing about how often Venus was depicted fishing in Roman art: it only implies that in certain instances a female figure may or may not be Venus. *Choice C* is incorrect because Carla Brain's proposed explanation for the presence of the cupids makes no reference to the female figure, and so the possibility that the figure in the artworks is in fact Venus cannot be definitively eliminated. *Choice D* is incorrect because there is nothing in the text to suggest that the only reasonable way to interpret the figure is as Venus.

QUESTION 19

Choice A is the best answer. The convention being tested is the use of plural and possessive nouns. The plural possessive noun "people's" and the plural noun "stories" correctly indicate that there are multiple stories from multiple people.

Choice B is incorrect because the context requires the plural possessive noun "people's" and the plural noun "stories," not the plural noun "peoples" and the singular possessive noun "story's." *Choice C* is incorrect because the context requires the plural possessive noun "people's," not the plural noun "peoples." *Choice D* is incorrect because the context requires the plural noun "stories," not the singular possessive noun "story's."

QUESTION 20

Choice D is the best answer. The convention being tested is the use of verbs to express tense. In this choice, the present tense verb "survives" correctly indicates that the wood frog regularly survives subfreezing temperatures by producing large amounts of glucose.

Choice A is incorrect because the past perfect verb "had survived" doesn't indicate that the wood frog regularly survives subfreezing temperatures by producing large amounts of glucose. *Choice B* is incorrect because the past tense verb "survived" doesn't indicate that the wood frog regularly survives subfreezing temperatures by producing large amounts of glucose. *Choice C* is incorrect because the conditional verb "would survive" doesn't indicate that the wood frog regularly survives subfreezing temperatures by producing large amounts of glucose.

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QUESTION 21

Choice D is the best answer. The convention being tested is punctuation use between sentences. In this choice, the period is used correctly to mark the boundary between one sentence ("Defying…fast") and another sentence that begins with a supplementary phrase ("During…Olympics").

Choice A is incorrect. When a dash is present in a sentence ("ran—fast"), it's not conventional to use another dash ("fast—during") to mark the boundary between sentences because it creates a potentially confusing sentence. In this context, a period, semicolon, or colon would be clear and more conventional. *Choice B* is incorrect because it results in a run-on sentence. The sentences ("Defying…fast") and ("during…Olympics") are fused without punctuation and/or a conjunction. *Choice C* is incorrect because it results in a comma splice. A comma can't be used in this way to mark the boundary between sentences.

QUESTION 22

Choice D is the best answer. The convention being tested is subject-verb agreement. The singular verb "is" agrees in number with the singular subject "the triangle."

Choice A is incorrect because the plural verb "are" doesn't agree in number with the singular subject "the triangle." *Choice B* is incorrect because the plural verb "have been" doesn't agree in number with the singular subject "the triangle." *Choice C* is incorrect because the plural verb "were" doesn't agree in number with the singular subject "the triangle."

QUESTION 23

Choice B is the best answer. The convention being tested is the coordination of main clauses within a sentence. This choice uses a semicolon in a conventional way to join the first main clause ("In 2004…sampler") and the second main clause ("in 2014…pillars").

Choice A is incorrect because it results in a comma splice. Without a conjunction following it, a comma can't be used in this way to join two main clauses. The word "later" is an adverb and cannot be used to join two main clauses unless it is preceded by a conjunction. *Choice C* is incorrect because it results in a comma splice. Without a conjunction following it, a comma can't be used in this way to join two main clauses. *Choice D* is incorrect because it results in a comma splice. Without a conjunction following it, a comma can't be used in this way to join two main clauses. The word "later" is an adverb and cannot be used to join two main clauses unless it is preceded by a conjunction.

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QUESTION 24

Choice C is the best answer. The convention being tested is subject-modifier placement. This choice makes the noun phrase "Julian's 1935 synthesis" the subject of the sentence and places it immediately after the modifying phrase "named…years." In doing so, this choice clearly establishes that Julian's 1935 synthesis of the alkaloid physostigmine—and not another noun in the sentence—was named in 1999 as one of the greatest achievements by a US chemist in the past hundred years.

Choice A is incorrect because it results in a dangling modifier. The placement of the noun "Julian" immediately after the modifying phrase illogically suggests that Julian himself was named as one of the greatest achievements by a US chemist in the past hundred years. *Choice B* is incorrect because it results in a dangling modifier. The placement of the prepositional phrase "in 1935" immediately after the modifying phrase illogically and confusingly suggests that "in 1935" was named as one of the greatest achievements by a US chemist in the past hundred years. *Choice D* is incorrect because it results in a dangling modifier. The placement of the noun phrase "the alkaloid physostigmine" immediately after the modifying phrase illogically and confusingly suggests that the alkaloid physostigmine itself (not the synthesis of it) was named as one of the greatest achievements by a US chemist in the past hundred years.

QUESTION 25

Choice B is the best answer. The convention being tested is the punctuation of items in a complex series (a series including internal punctuation). The semicolon after "nonnative" is correctly used to separate the first item ("growing diverse plant species, both native and nonnative") and the second item ("fostering scientific research") in the series of things that botanical gardens are dedicated to. Further, the comma after "species" is correctly used to separate the noun phrase "diverse plant species" and the supplementary phrase "both native and nonnative" that modifies it.

Choice A is incorrect because a comma (specifically, the comma after "nonnative") can't be used in this way to separate items in a complex series. *Choice C* is incorrect because a semicolon can't be used in this way to separate the noun phrase "diverse plant species" and the supplementary phrase "both native and nonnative" that modifies it. Further, a comma can't be used in this way to separate items in a complex series. *Choice D* is incorrect because it fails to use appropriate punctuation to separate the noun phrase "diverse plant species" and the supplementary phrase "both native and nonnative" that modifies it. Further, a comma can't be used in this way to separate items in a complex series.

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QUESTION 26

Choice A is the best answer. The convention being tested is the punctuation of a supplementary word or phrase between two main clauses. This choice correctly uses a comma to separate the supplementary adverb "however" from the preceding main clause ("Okinaka doesn't…single-handedly") and a semicolon to join the next main clause ("all…culture") to the rest of the sentence. Further, placing the semicolon after "however" correctly indicates that the information in the preceding main clause (Okinaka doesn't make such decisions single-handedly) is contrary to what might be assumed from the information in the previous sentence (Okinaka sits on the review board that adds new sites to the Hawaii Register of Historic Places).

Choice B is incorrect because placing the semicolon after "single-handedly" and the comma after "however" illogically indicates that the information in the next main clause (all historical designations must be approved by a group of experts) is contrary to the information in the previous clause (Okinaka doesn't make such decisions single-handedly). *Choice C* is incorrect because it results in a comma splice. Commas can't be used in this way to punctuate a supplementary word or phrase between two main clauses. *Choice D* is incorrect because it results in a run-on sentence. The two main clauses are fused without punctuation and/or a conjunction.

QUESTION 27

Choice C is the best answer. "Finally" logically signals that the bill passing—following many attempts between 1968 and 1983—is the final, concluding event in the sequence described in the previous sentences.

Choice A is incorrect because "instead" illogically signals that the bill passing is an alternative to one of the events described in the previous sentences. It is the final event in the sequence. *Choice B* is incorrect because "likewise" illogically signals that the bill passing is similar to one of the events described in the previous sentences. Instead, it is the final event in the sequence. *Choice D* is incorrect because "additionally" illogically signals that the bill passing is merely another event described along with the events of the previous sentences. Instead, it is the final, concluding event in the sequence.

QUESTION 28

Choice D is the best answer. "However" logically signals that this sentence, which indicates that the Pūhāhonu volcano may be larger than the Mauna Loa volcano, offers a contrast to or refutation of the previous assumption that Mauna Loa is the largest shield volcano.

Choice A is incorrect because "secondly" illogically signals that this sentence merely offers an additional or secondary point concerning the previous assumption that Mauna Loa is the largest shield volcano. Instead, the sentence offers a contrast to or refutation of that assumption. *Choice B* is incorrect because "consequently" illogically signals that this sentence offers a result or consequence of the previous assumption that Mauna Loa is the largest shield volcano. Instead, the sentence offers a contrast to or refutation of that assumption. *Choice C* is incorrect because "moreover" illogically signals that this sentence merely adds to the previous

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assumption that Mauna Loa is the largest shield volcano. Instead, the sentence offers a contrast to or refutation of that assumption.

QUESTION 29

Choice A is the best answer. "In addition" logically signals that the detail in this sentence—that Coleridge-Taylor included traditional African music in his classical compositions—adds to the information in the previous sentence. Specifically, the previous sentence indicates one way in which Coleridge-Taylor emphasized his mixed-race ancestry, and the claim that follows indicates a second, additional way.

Choice B is incorrect because "actually" illogically signals that the detail in this sentence is surprising in light of the information in the previous sentence. Instead, the detail adds to the information, indicating a second, additional way in which Coleridge-Taylor emphasized his mixed-race ancestry. *Choice C* is incorrect because "however" illogically signals that the detail in this sentence contrasts with the information in the previous sentence. Instead, the detail adds to the information, indicating a second, additional way in which Coleridge-Taylor emphasized his mixed-race ancestry. *Choice D* is incorrect because "regardless" illogically signals that the detail in this sentence is true despite the information in the previous sentence. Instead, the detail adds to the information, indicating a second, additional way in which Coleridge-Taylor emphasized his mixed-race ancestry.

QUESTION 30

Choice A is the best answer. "Therefore" logically signals that the action described in this sentence—the researchers theorizing that the dish was named for its effect on diners—is a result or consequence of the previous observation that the dish had a calming effect.

Choice B is incorrect because "alternately" illogically signals that the action described in this sentence offers an alternative or contrast to the previous observation that the dish had a calming effect. Instead, the action is a result or consequence of that observation. *Choice C* is incorrect because "nevertheless" illogically signals that the action described in this sentence occurs despite the previous observation that the dish had a calming effect. Instead, the action is a result or consequence of that observation. *Choice D* is incorrect because "likewise" illogically signals that this sentence merely adds a second, similar detail to the previous observation that the dish had a calming effect. Instead, this sentence describes an action that is a result or consequence of that observation.

QUESTION 31

Choice D is the best answer. The sentence emphasizes a difference between baking soda and baking powder, noting that baking soda needs to be mixed with an acidic ingredient to produce carbon dioxide but baking powder doesn't.

Choice A is incorrect. The sentence focuses on what bakers use to make batters rise; it doesn't emphasize a difference between baking soda and baking powder. *Choice B* is incorrect. The sentence provides a general description of baking soda

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and baking powder; it doesn't emphasize a difference between them. *Choice C* is incorrect. The sentence explains what baking soda and honey are; it doesn't emphasize a difference between baking soda and baking powder.

QUESTION 32

Choice D is the best answer. The sentence effectively describes *Unwoven Light* to an audience unfamiliar with Park, noting that Soo Sunny Park is a Korean American artist and that the 2013 work consists of colorful prisms formed by light passing through iridescent tiles.

Choice A is incorrect. The sentence describes aspects of *Unwoven Light* but doesn't mention who Park is; it thus doesn't effectively describe the work to an audience unfamiliar with Park. *Choice B* is incorrect. Although the sentence indicates when the work was created and who Park is, it lacks descriptive details and thus doesn't effectively describe *Unwoven Light*. *Choice C* is incorrect. The sentence mentions Park and describes an aspect of *Unwoven Light*—the chain-link fence—but doesn't effectively describe the overall work to an audience unfamiliar with the artist.

QUESTION 33

Choice C is the best answer. The sentence effectively presents Tan's research to an audience unfamiliar with Angkor Wat, explaining the results of the research and identifying Angkor Wat as a temple in Cambodia.

Choice A is incorrect. While the sentence presents Tan's research, it fails to explain what Angkor Wat is for an audience unfamiliar with the temple. *Choice B* is incorrect. The sentence emphasizes the role that decorrelation stretch analysis played in Tan's research; it doesn't present the research, which would require specifying where it was conducted. *Choice D* is incorrect. While the sentence explains what Angkor Wat is, it fails to present Tan's research.

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Reading and Writing

Module 2

(33 questions)

QUESTION 1

Choice D is the best answer because it most logically completes the text's discussion of the fashion resale market's continued growth. As used in this context, "predicted" means forecast, or indicated that something would happen in the future. The text indicates that the fashion resale market made a lot of money in 2019 and that some analysts expected the market to continue to grow. This context suggests that the analysts believed that the fashion resale market was going to make more money than it had already made, with the analysts indicating that revenues would more than double by 2028.

Choice A is incorrect because it wouldn't make sense in context to say that some analysts "produced," or manufactured or brought about, the increase in future revenues of the fashion resale market. The analysts themselves couldn't have brought about the future revenue growth, since, as the text suggests, they were merely in the position of drawing conclusions about future fashion resale market revenue based on 2019 revenue. *Choice B* is incorrect because the text indicates that some analysts expected the fashion resale market to continue to grow in the future, not that they "denied," or rejected, this notion. Nothing in the text supports the idea that these analysts thought the revenues wouldn't grow. *Choice C* is incorrect because the text indicates that some analysts expected the fashion resale market to continue to grow in the future, not that they "worried," or felt concerned, that revenue would significantly increase by 2028. Nothing in the text suggests that the analysts felt concerned about the increase; rather, the text suggests that the increase would represent a favorable outcome, since it would mean that the fashion resale market grew to generate even more revenue.

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QUESTION 2

Choice D is the best answer because it most logically completes the text's discussion of delivering biomolecules to plant cells. In this context, "overcome" means to succeed in dealing with an obstacle. The text suggests that although it's difficult to move biomolecules through plant cell walls, Landry and her colleagues have shown that carbon nanotubes may be useful, since they can cross cell walls. This context conveys that Landry and her colleagues think it's possible, using carbon nanotubes, to succeed in dealing with the obstacle of transmitting biomolecules to plant cells.

Choice A is incorrect because it wouldn't make sense in context to say that Landry and her colleagues have shown that it may be possible to "conceptualize," or form an idea of, the difficulty of transmitting biomolecules through the walls of plant cells. The text presents this difficulty as a known problem that Landry and her colleagues think they may have solved, not as a mysterious occurrence that they have yet to form ideas about. *Choice B* is incorrect because the text suggests that Landry and her colleagues think it may be possible to successfully deal with the problem of transmitting biomolecules through the walls of plant cells, not that Landry and her colleagues think it may be possible to "neglect," or simply to disregard and ignore the problem. *Choice C* is incorrect because it wouldn't make sense in context to say that Landry and her colleagues have shown that it may be possible to "illustrate," or demonstrate, the difficulty of transmitting biomolecules through the walls of plant cells by using carbon nanotubes. According to the text, carbon nanotubes allow molecules to be transmitted to plant cells—something that is otherwise difficult to do. The text therefore presents carbon nanotubes as a way of possibly solving a problem, not as a means of demonstrating the problem.

QUESTION 3

Choice B is the best answer because it most logically completes the text's discussion of the work of particle physicists. In this context, "inspecting" means viewing closely in order to examine. The text indicates that as particle physicists, Arce and El-Khadra's work involves using advanced technology to "closely examine" subatomic particles. In other words, they use technology to inspect small parts of matter that can't be seen by the naked eye.

Choice A is incorrect because nothing in the text suggests that Arce and El-Khadra spend time "selecting," or choosing, subatomic particles for some purpose; the text simply states that the particle physicists use advanced technology to see and study the behavior of those tiny parts of matter. *Choice C* is incorrect because nothing in the text suggests that Arce and El-Khadra spend time "creating" subatomic particles, or bringing them into existence; the text simply states that the particle physicists use advanced technology to see and study the behavior of those tiny parts of matter. *Choice D* is incorrect. In this context, "deciding" would mean making a final choice or judgment about something. It wouldn't make sense to say that particle physicists get to choose what is and isn't visible to the naked eye, especially when the text presents it as fact that subatomic particles are "the smallest detectable parts of matter" and would therefore be invisible. The text focuses on Arce and El-Khadra's close observation of those particles, not on any decisions they might make.

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QUESTION 4

Choice B is the best answer because it most logically completes the text's discussion of the fossilized bones of the hominin known as Little Foot. As used in this context, "comparable to" would mean similar to. The text indicates that the relationship between the fossilized clavicle and shoulder bones of Little Foot and the clavicle and shoulder bones of "frequent climbers," such as chimpanzees and gorillas, suggests that Little Foot had adapted to moving around in trees. This context suggests that the relationship between the fossilized bones of Little Foot and the bones of chimpanzees and gorillas is one of similarity—the Little Foot fossils are likely comparable to the modern ape bones.

Choice A is incorrect because if the fossilized bones of Little Foot were "surpassed by," or exceeded by or made inferior to, the bones of modern apes that are frequent climbers, it wouldn't suggest, as the text says, that Little Foot was adapted to moving around in trees. If anything, learning that Little Foot's clavicle and shoulder bones were surpassed by those of chimpanzees and gorillas would suggest that Little Foot was poorly adapted to climbing. *Choice C* is incorrect because if Little Foot's fossilized clavicle and shoulder bones were "independent of," or not influenced by or affiliated with, the bones of modern apes that climb often, it wouldn't suggest, as the text says, that Little Foot was adapted to moving around in trees. *Choice D* is incorrect because the text indicates that Little Foot's fossilized bones date to 3.6 million years ago, so they couldn't have been "obtained from," or acquired from, the bones of modern apes.

QUESTION 5

Choice B is the best answer because it most logically completes the text's discussion of Samuel R. Delany's character Rydra Wong. As used in this context, "atypical" would mean unrepresentative or not common. The text indicates that Wong is one of "nearly a dozen" characters in Delany's novels who are poets or writers. This context conveys that being a poet isn't an atypical occupation for a character in one of Delany's works.

Choice A is incorrect because "infallible" means to be accurate or without fault, which wouldn't make sense in context. The text focuses on the fact that Delany has written many characters who are poets and writers. This context suggests that the occupation isn't atypical for Delany, not that the occupation isn't infallible, or problematic. *Choice C* is incorrect because "lucrative" means to be profitable, which wouldn't make sense in context. If writing poet characters weren't profitable, it wouldn't be logical to explain this by citing that Delany gave many of his characters the same occupation. *Choice D* is incorrect because "tedious" means to be boring, which wouldn't make sense in context. The text focuses on the fact that Delany has written many characters who are poets and writers. This context suggests that the occupation isn't atypical for Delany, not that the occupation isn't tedious.

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QUESTION 6

Choice B is the best answer because it most logically and precisely completes the text's discussion of Jeyifous's series of images for the 2020 exhibition. In this context, "created" means produced. The text explains that Jeyifous, a photographer and neurobiologist, photographed adults who had appeared as children in posters from the 1970s, then combined those photographs with magnified images of the adults' cells—a process that resulted in what he called "micro and macro portraiture." This context suggests that Jeyifous drew on his dual interests in photography and neurobiology to produce the images for display in the exhibition.

Choice A is incorrect because there's nothing in the text to suggest that Jeyifous "validated," or corroborated, the series of images. The text describes Jeyifous's process for composing the images but doesn't describe Jeyifous making an effort to evaluate the images for their artistic or scientific legitimacy. *Choice C* is incorrect because there's nothing in the text to suggest that Jeyifous "challenged," or disputed, an aspect of the images; rather, the focus of the text is on the inspiration behind the images and the method Jeyifous used to achieve them. *Choice D* is incorrect because the text indicates that Jeyifous made the images himself using a combination of photography and magnified pictures of cells, not that he "restored," or reconditioned, the images from a deteriorated state.

QUESTION 7

Choice A is the best answer because it most logically completes the text's discussion of Francis Cecil Sumner. As used in this context, "proponent of" means supporter of. The text says that Sumner helped to found the psychology department at historically Black Howard University in 1930. This is evidence that Sumner supported increasing the opportunity for Black students to study psychology.

Choice B is incorrect because the phrase "supplement to," or addition to, wouldn't make sense in context. The text discusses Sumner's efforts to increase the number of Black psychology students, but it doesn't make sense to describe him as an addition to his efforts. *Choice C* is incorrect because Sumner was already an accomplished psychologist himself when he helped to found the Howard University psychology department. While Black students were the beneficiaries of his efforts—that is, they received help because of his efforts—it wouldn't make sense in this context to describe Sumner as a "beneficiary of" opportunities, because he was the one doing the helping. *Choice D* is incorrect because founding a psychology department at Howard University wouldn't be a "distraction for" Sumner's aim to increase the opportunity for Black students to study psychology—that is, it wouldn't be something that draws Sumner's attention away from that goal, but rather the opposite.

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QUESTION 8

Choice C is the best answer because it most logically completes the text's discussion of the legitimacy of the reigns of French monarchs such as Hugh Capet and Henry I. As used in this context, "buttress" means to strengthen or defend. The text indicates that regardless of whether a French monarch's reign was significant or uneventful, each monarch faced questions about his right to the throne. The text goes on to say that in order to understand the path of a French monarch's reign, it's important to understand what contributed to the monarch's ability to "hold the throne." This context suggests that French monarchs such as Hugh Capet and Henry I had to buttress, or defend, their right to be monarch.

Choice A is incorrect. Saying that a monarch who is faced with questions about the legitimacy of his reign was able to "reciprocate" his right to the French throne would mean that he either returned his right to the throne or that he responded in kind to the challenge. Neither of these meanings would make sense in context because the text focuses on people who did reign as French monarchs and defended their right to do so. *Choice B* is incorrect because it wouldn't make sense in context to discuss factors that enabled a monarch to "annotate," or add notes to or explain, his right to the French throne. Nothing in the text suggests that the monarchs were writing notes about their right to the throne; instead, faced with questions about the legitimacy of their reign, the monarchs defended their right. *Choice D* is incorrect because it wouldn't make sense in context to discuss factors that enabled a monarch to "disengage," or withdraw his right to the French throne. The text focuses on an examination of people who reigned as French monarchs, not on people who didn't choose to rule.

QUESTION 9

Choice C is the best answer because it best describes how the underlined sentence functions in the text as a whole. The first two sentences establish that birds of some species don't raise their own young; instead, they lay their eggs in the nests of birds of other species. The underlined sentence then states that female cuckoo birds engage in this behavior, having been observed specifically laying their eggs in other nests while the other birds are out finding food. According to the text, the cuckoo chicks are then raised by the other birds. Thus, the underlined sentence provides a particular detail about how female cuckoos carry out the behavior of laying eggs for other birds to raise.

Choice A is incorrect. Rather than mentioning a physical feature of female cuckoos, the underlined sentence introduces a specific behavior of female cuckoos: laying eggs in the nests of birds of other species when the other birds are away. The only reference to physical features is the last sentence's general mention of cuckoo chicks looking different from chicks of other species. *Choice B* is incorrect because the underlined sentence refers to the nests of birds other than cuckoos and doesn't describe how any nests look, cuckoo or otherwise. Instead, the sentence addresses how female cuckoos use other birds' nests. *Choice D* is incorrect because the underlined sentence describes only female cuckoo behavior (laying eggs in the nests of birds of other species when the other birds are away); it's the last sentence of the text that addresses the other birds' reaction, indicating that those birds usually raise the cuckoo chicks once they've hatched.

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QUESTION 10

Choice A is the best answer because it explains how the researchers determined the level of surprise displayed by the cats in the study. The text states that Saho Takagi and colleagues played recordings of the voice of each cat's owner and measured how surprised the cat was by the recording based on how it moved its ears and head.

Choice B is incorrect because, as the text explains, the recordings played for each cat in the study were of the voice of the cat's owner, not a stranger's voice. *Choice C* is incorrect because the text explains that during the study, the cats didn't interact directly with their owners; instead, the cats listened to recordings of their owners' voices. *Choice D* is incorrect because the text doesn't indicate that the researchers monitored the cats' movement around the room in which the study was conducted.

QUESTION 11

Choice A is the best answer because it describes an experimental outcome that would most directly weaken the student's hypothesis. According to the text, the student hypothesizes that *Brassica rapa parachinensis* (choy sum) will benefit more from acidic soil than it will from neutral soil. The text then explains that the student planted 16 choy sum seeds in potting soil with coffee grounds added to increase acidity and another 16 seeds in soil without coffee grounds as a control (a group identical to the experimental group except for the experimental modification being tested). If the hypothesis were correct, the plants in the more acidic soil-and-coffee-grounds mixture would grow faster than those in the control group. However, choice A proposes a scenario in which the plants in soil without coffee grounds were "significantly taller" than those in the more acidic mixture—an outcome that weakens the hypothesis that higher acidity is beneficial to the plants' growth.

Choice B is incorrect. If the choy sum planted in the neutral soil produced less plant matter and therefore weighed less than the choy sum planted in the acidic soil-and-coffee-grounds mixture, this finding would strengthen the student's hypothesis, not weaken it. *Choice C* is incorrect. If seeds planted in neutral soil (without coffee grounds) sprouted significantly later than seeds planted in the acidic soil-and-coffee-grounds mixture, this finding would strengthen, not weaken, the student's hypothesis that acidic soil benefits choy sum. *Choice D* is incorrect. If seeds planted in the neutral soil (without coffee grounds) sprouted significantly fewer plants than seeds planted in the acidic soil-and-coffee-grounds mixture did, this finding would strengthen, not weaken, the student's hypothesis that choy sum benefits from acidic soil.

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QUESTION 12

Choice B is the best answer because it most effectively illustrates the claim in the text that in describing the teenaged girl, Mansfield contrasts the character's pleasant appearance with her unpleasant attitude. In the quotation, Mansfield describes the teenager as having a "lovely nose" (a compliment about her appearance) but also as treating her makeup puff "as though she loathed it" (a judgment suggesting her unpleasant attitude).

Choice A is incorrect because the teenager's reaction to the flowers doesn't make it clear that she has an unpleasant attitude, and nothing in the quotation indicates that any part of her appearance is pleasant. *Choice C* is incorrect because the quotation suggests that the teenager has an unpleasant attitude (being upset with the location and leaving the table before the narrator has paid for the meal) but doesn't give any indication that she has a pleasant appearance. *Choice D* is incorrect because the quotation suggests that the teenager may have an unpleasant attitude (lowering her eyes, wincing, and sitting in silence) but doesn't give any indication that any part of her appearance is pleasant.

QUESTION 13

Choice D is the best answer because it uses data from the graph to effectively illustrate the text's claim about general economic policy uncertainty in the United Kingdom. The graph presents values for economic policy uncertainty in tax and public spending policy, trade policy, and general economic policy in the UK from 2005 to 2010. The graph shows that in 2005, the value for general economic policy uncertainty (approximately 90) was substantially lower than the value for uncertainty about trade policy specifically (approximately 160). It also shows that in 2010, the value for general economic policy uncertainty (approximately 120) was substantially higher than the value for uncertainty about trade policy (approximately 70). The substantial differences between these values in 2005 and 2010 support the claim that a general measure may not fully reflect uncertainty about specific areas of policy.

Choice A is incorrect because the graph shows that the level of general economic policy uncertainty was similar to the level of uncertainty about tax and public spending policy in both 2005 (with values of approximately 90 and 100, respectively) and 2009 (with values of approximately 80 and 75, respectively). *Choice B* is incorrect because the graph shows that general economic policy uncertainty was higher than uncertainty about tax and public spending policy in 2006, 2007, and 2009, not that it was lower each year from 2005 to 2010. *Choice C* is incorrect because the graph shows that general economic policy uncertainty reached its highest level in 2010, which was when uncertainty about tax and public spending policy also reached its highest level, not its lowest level.

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QUESTION 14

Choice C is the best answer because it presents the finding that, if true, would most strongly support Tannen's hypothesis. According to the text, Tannen's hypothesis is that multiple perspectives presented in a noncompetitive format is more informative than a debate between opposing viewpoints is. If participants who saw a panel of three commentators with various views about an issue answered more questions about the issue correctly than did participants who saw a debate, that would support Tannen's hypothesis since it would show that participants who heard multiple varied perspectives were better informed than were participants who heard a debate between opposing viewpoints.

Choice A is incorrect because finding that participants perceived commentators in the debate as more knowledgeable than commentators in the panel is irrelevant to Tannen's hypothesis, which is that presenting multiple perspectives on an issue is more informative to the audience than presenting opposing views of the issue is. Participants' perception of how knowledgeable panelists are has no bearing on how much participants learn from the panelists. *Choice B* is incorrect because finding that participants perceived commentators in the panel as more knowledgeable than a single commentator is irrelevant to Tannen's hypothesis, which is that presenting multiple perspectives on an issue is more informative to the audience than presenting opposing views of the issue is. Participants' perception of how knowledgeable panelists are has no bearing on how much participants learn from the panelists, and Tannen's hypothesis says nothing about how informative single commentators are. *Choice D* is incorrect because finding that participants who watched a single commentator answered more questions correctly than participants who watched the debate did wouldn't be relevant to Tannen's hypothesis, which is that hearing multiple varying perspectives is more informative than hearing a debate. Tannen's hypothesis says nothing about how informative single commentators are.

QUESTION 15

Choice C is the best answer because it most effectively uses a quotation from *King Lear* to illustrate the claim that King Lear expresses regret for his actions. In the quotation, Lear describes striking himself on the head—the same act he's engaged in as he speaks, and one that suggests he's deeply upset with himself. Referring to himself in the second person (with "thy"), the character exclaims "Beat at this gate that let thy folly in / And thy dear judgement out!" Lear refers metaphorically to his own mind as a gate that has allowed folly, or poor judgement, to enter and good judgement to escape. This suggests that Lear regrets his attempts to test his three daughters' devotion to him, regarding those attempts as examples of the folly that has entered the gate of his mind.

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Choice A is incorrect because this quotation doesn't express King Lear's sense of regret over his own actions; instead, it expresses his belief that the harm that others have done to him (or the extent to which they have "sinned against" him) outweighs whatever harm he himself has caused by "sinning." *Choice B* is incorrect because this quotation doesn't express King Lear's sense of regret over his own actions; instead, it expresses his thoughts about an approaching storm ("this tempest"), which he believes "will not give [him] leave to ponder," or time to consider, the harm that he will continue to experience ("things" that "would hurt [him] more"). *Choice D* is incorrect because this quotation expresses King Lear's vow to commit terrible actions (or "things" that "shall be / The terrors of the earth") in the future, not his regret over actions that he's already taken.

QUESTION 16

Choice A is the best answer because it most logically completes the text's discussion of the relative appeal of different kinds of plays by Shakespeare to today's audiences. According to the text, Shakespeare's tragedies address broad themes that continue to appeal to today's audiences. Indeed, the text suggests that these themes are timeless, as illustrated by the example of *Romeo and Juliet*, which the text states is still read and widely performed despite being set in the Italy of Shakespeare's time. In contrast, the text indicates that audiences and readers may need to be familiar with several centuries of English history in order to understand Shakespeare's history plays. Because many theatergoers and readers are unlikely to possess such extensive historical knowledge, it follows that they are likely to find Shakespeare's history plays less engaging than his more accessible tragedies.

Choice B is incorrect because the text never introduces a comparison between Shakespeare's tragedies and twentieth-century plays, only between Shakespeare's tragedies and his history plays. Since twentieth-century plays aren't mentioned, there's no basis in the text for the idea that some of Shakespeare's tragedies are more relevant than twentieth-century plays to today's audiences. *Choice C* is incorrect. Although the text indicates that *Romeo and Juliet* is thematically accessible to today's audiences, it doesn't suggest that *Romeo and Juliet* is more accessible than Shakespeare's other tragedies. Rather, the text presents *Romeo and Juliet* as an example to support the idea that Shakespeare's tragedies hold continued appeal for today's readers and theatergoers. *Choice D* is incorrect. Although experts in English history would likely possess the knowledge needed to understand Shakespeare's history plays, the text never mentions such experts or suggests that they would enjoy the history plays more than Shakespeare's other works.

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QUESTION 17

Choice B is the best answer because it presents the conclusion that most logically follows from the text's discussion of Ancestral Puebloans' migration to the Rio Grande Valley. The text states that in the late 1200s C.E., the Ancestral Puebloan civilization abandoned villages in its original homeland, which included the Mesa Verde site. The text goes on to say that recent genetic analysis has demonstrated that the modern turkey population in the Rio Grande Valley descends partly from the ancient turkeys raised at Mesa Verde, and that the genetic markers shared by the two turkey populations first appeared at Mesa Verde only after 1280 C.E. Therefore, it can reasonably be concluded that some Ancestral Puebloans migrated to the Rio Grande Valley in the late 1200s and carried their agricultural practices—including the farming of turkeys— to their new home.

Choice A is incorrect because the text never compares the condition of the Rio Grande Valley's terrain to that of Mesa Verde's terrain, either in the present or in the past. *Choice C* is incorrect. Although genetic analysis has demonstrated that the modern turkey population in the Rio Grande valley descended in part from the turkey population raised by the Ancestral Puebloans of Mesa Verde before their migration to the valley in 1280, this finding doesn't eliminate the possibility that Indigenous peoples living in the valley before 1280 might also have farmed turkeys. *Choice D* is incorrect. The text doesn't consider the possibility that before their migration to the Rio Grande Valley after 1280, the Ancestral Puebloans of Mesa Verde might have adopted turkey farming from an outside Indigenous civilization in another region; instead, the text provides evidence suggesting that the Ancestral Puebloans brought turkey farming to another region— the Rio Grande Valley—after 1280.

QUESTION 18

Choice D is the best answer because it presents the conclusion that most logically follows from the text's discussion of the challenge researchers face when studying the effects of holding elected office on a person's behavior. The text explains that it's hard for researchers to test for the effects that elected office has on people because finding people to serve as a control group is difficult. The text indicates that a control group needs to be made up of people who share characteristics of the group being tested but don't have the variable being tested (in this case, holding elected office). Because researchers aren't able to influence who wins elections, they're also unable to determine who would serve as an appropriately similar member of a control group. Thus, it logically follows that researchers will find it difficult to identify a group of people who can function as an appropriate control group for their studies.

Choice A is incorrect because the text focuses on the struggle to put together a control group for experiments; it doesn't suggest that finding information about politicians' behavior is difficult. *Choice B* is incorrect because the experiments mentioned in the text are testing the effects of holding elected office on a person's behavior. Studying people who have already held elected office wouldn't provide an opportunity to note any behavioral changes that the position might cause. *Choice C* is incorrect because the text defines people in a control group as those "who are otherwise similar to the office-holders"; selecting people who differ from the office-holders wouldn't fit the criteria for an appropriate control group.

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QUESTION 19

Choice C is the best answer. The convention being tested is the use of plural and possessive nouns. The plural nouns "stories" and "immigrants" correctly indicate that the memoir tells multiple stories of multiple immigrants.

Choice A is incorrect because the context requires the plural noun "stories," not the singular possessive noun "story's." *Choice B* is incorrect because the context requires the plural nouns "stories" and "immigrants," not the singular possessive noun "story's" and the plural possessive noun "immigrants'." *Choice D* is incorrect because the context requires the plural nouns "stories" and "immigrants," not the plural possessive noun "stories'" and the singular possessive noun "immigrant's."

QUESTION 20

Choice A is the best answer. The convention being tested is punctuation between a preposition and its complement. No punctuation is needed between the preposition "of" and its complement, the noun phrase "healing, self-discovery, and memory."

Choice B is incorrect because no punctuation is needed between a preposition and its complement. *Choice C* is incorrect because no punctuation is needed between a preposition and its complement. *Choice D* is incorrect because no punctuation is needed between a preposition and its complement.

QUESTION 21

Choice B is the best answer. The convention being tested is the use and punctuation of an integrated relative clause. This choice correctly uses the relative pronoun "that" and no punctuation to create an integrated relative clause that provides essential information about the noun phrase ("a book packaging company") that it modifies.

Choice A is incorrect because it doesn't use a relative pronoun to link the verb phrase beginning with "specializes" to the noun phrase that it modifies ("a book packaging company"). *Choice C* is incorrect because it doesn't use a relative pronoun to link the verb phrase beginning with "specializes" to the noun phrase that it modifies ("a book packaging company"). *Choice D* is incorrect because no punctuation is needed between the integrated relative clause beginning with "that specializes" and the noun phrase that it modifies ("a book packaging company").

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QUESTION 22

Choice D is the best answer. The convention being tested is the punctuation of a supplementary element within a sentence. The comma after "(^{13}C)" pairs with the comma after "isotope" to separate the supplementary element "carbon-13 (^{13}C)" from the rest of the sentence. This supplementary element defines the "rare carbon isotope," and the pair of commas indicates that this element could be removed without affecting the grammatical coherence of the sentence.

Choice A is incorrect because it fails to use appropriate punctuation to separate the supplementary element "carbon-13 (^{13}C)" from the rest of the sentence. *Choice B* is incorrect because it fails to use appropriate punctuation to separate the supplementary element "carbon-13 (^{13}C)" from the rest of the sentence. *Choice C* is incorrect because it fails to use appropriate punctuation to separate the supplementary element "carbon-13 (^{13}C)" from the rest of the sentence. The comma after "carbon-13" isn't necessary because the parentheses around "^{13}C" already separate this element from the rest of the sentence.

QUESTION 23

Choice D is the best answer. The convention being tested is punctuation use between sentences. In this choice, the period after "walls" is used correctly to mark the boundary between the first sentence ("In...walls") and the second sentence ("With…techniques"), which starts with a supplementary phrase.

Choice A is incorrect because it results in a comma splice. A comma can't be used in this way to mark the boundary between sentences. *Choice B* is incorrect because it results in a run-on sentence. The sentences ("In...walls" and "with... paintings") are fused without punctuation and/or a conjunction. *Choice C* is incorrect. Without a comma preceding it, the conjunction "so" can't be used in this way to join sentences.

QUESTION 24

Choice A is the best answer. The convention being tested is the use of finite and nonfinite verb forms within a sentence. The nonfinite to-infinitive "to forge" is correctly used to form a nonfinite (infinitive) clause that explains why the chemists re-created and reprogrammed the DNA-cleaving bacteria.

Choice B is incorrect. Without a comma separating the main clause ("chemists... bacteria") from the participle "forging," this choice illogically suggests that the bacteria are forging a tool, which doesn't make sense. *Choice C* is incorrect. Without a coordinating conjunction such as "and" placed before it, the finite past tense verb "forged" can't be used in this way to describe the chemists' actions. *Choice D* is incorrect. If read as a finite verb, the present progressive verb "forging" isn't consistent with the past tense verbs used in this sentence to describe the actions of the chemists. If read as a nonfinite verb, the participle "forging" can't be used in this way because there is no following main clause for it to modify.

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QUESTION 25

Choice B is the best answer. The convention being tested is subject-modifier placement. This choice makes the noun phrase "the bioswales" the subject of the sentence and places it immediately after the modifying phrase "By reducing… sewers." In doing so, this choice clearly establishes that the bioswales—and not another noun in the sentence—are reducing runoff flowing into city sewers.

Choice A is incorrect because it results in a dangling modifier. The placement of the noun phrase "the mitigation…waterways" immediately after the modifying phrase results in unclear modification. The resulting sentence makes it hard to determine what is responsible for "reducing the runoff": the bioswales or some other noun in the sentence. *Choice C* is incorrect because it results in a dangling modifier. The placement of the noun phrase "the bioswales' mitigation…waterways" immediately after the modifying phrase results in unclear modification. The resulting sentence makes it hard to determine what is responsible for "reducing the runoff": the bioswales or some other noun in the sentence. *Choice D* is incorrect because it results in a dangling modifier. The placement of the noun phrase "street flooding and the resulting pollution" immediately after the modifying phrase illogically suggests that the "flooding and pollution" are reducing runoff flowing into city sewers.

QUESTION 26

Choice B is the best answer. The convention being tested is punctuation use between a main clause and a supplementary phrase. In this choice, a colon is correctly used to mark the boundary between the main clause ("A study… continents") and the supplementary phrase ("geological…above") and to introduce the following explanation of the origin of Earth's continents.

Choice A is incorrect because it fails to mark the boundary between the main clause ("A study…continents") and the supplementary phrase ("geological… above") with appropriate punctuation. *Choice C* is incorrect because a semicolon can't be used in this way to join the main clause ("A study…continents") and the supplementary phrase ("geological…above"). A semicolon is conventionally used to join two main clauses, whereas a colon is conventionally used to introduce an element that explains or amplifies the information in the preceding clause, making it the better choice in this context. *Choice D* is incorrect because it results in a rhetorically unacceptable sentence fragment beginning with "geological."

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QUESTION 27

Choice A is the best answer. "Afterward" logically signals that the events described in this sentence—the CEO's public acknowledgment and apology—occurred after the rocket booster's failure and are part of a chronological sequence of events.

Choice B is incorrect because "additionally" illogically signals that the events described in this sentence merely occurred in addition to the rocket booster's failure. Instead, they occurred after the rocket booster's failure and are part of a chronological sequence of events. *Choice C* is incorrect because "indeed" illogically signals that the events described in this sentence emphasize or strengthen a statement made in the previous sentence. Instead, they occurred after the rocket booster's failure and are part of a chronological sequence of events. *Choice D* is incorrect because "similarly" illogically signals that the events described in this sentence are similar to the rocket booster's failure. Instead, they occurred after the rocket booster's failure and are part of a chronological sequence of events.

QUESTION 28

Choice A is the best answer. "Alternatively" logically signals that the soil decontamination method described in this sentence—removing toxic metals from the soil via phytoremediation—offers an alternative to the previously described method (removing the contaminated soil from the ground).

Choice B is incorrect because "specifically" illogically signals that the soil decontamination method described in this sentence specifies or elaborates on an aspect of the previously described method (removing the contaminated soil from the ground). Instead, phytoremediation is an alternative to that method. *Choice C* is incorrect because "for example" illogically signals that the soil decontamination method described in this sentence is an example of the previously described method (removing the contaminated soil from the ground). Instead, phytoremediation is an alternative to that method. *Choice D* is incorrect because "as a result" illogically signals that the soil decontamination method described in this sentence is a result or consequence of the previously described method (removing the contaminated soil from the ground). Instead, phytoremediation is an alternative to that method.

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QUESTION 29

Choice C is the best answer. The sentence explains an advantage of the Hanke-Henry calendar, noting that it supports more predictable scheduling than does the Gregorian calendar and describing how it does so (by having calendar dates occur on the same day each year).

Choice A is incorrect. The sentence compares the number of days in the Gregorian and Hanke-Henry calendars; it doesn't explain an advantage of the Hanke-Henry calendar. *Choice B* is incorrect. While the sentence refers to a possible reason to adopt the Hanke-Henry calendar—that doing so would help solve a problem with the Gregorian calendar—it doesn't identify the problem or the solution and thus doesn't explain the advantage of the Hanke-Henry calendar. *Choice D* is incorrect. The sentence describes the origins of the Hanke-Henry calendar; it doesn't explain an advantage of it.

QUESTION 30

Choice C is the best answer. The sentence effectively presents the influence theory to an audience unfamiliar with the Haudenosaunee Confederacy, explaining the theory's position that the Great Law of Peace influenced the US Constitution while avoiding mention of the Haudenosaunee Confederacy itself.

Choice A is incorrect. The sentence broadly emphasizes Johansen's ideas about the Great Law of Peace; it doesn't identify the influence theory or effectively present it. *Choice B* is incorrect. The sentence emphasizes one fact that supports the influence theory; it doesn't effectively present the theory to an audience unfamiliar with the Haudenosaunee Confederacy. *Choice D* is incorrect. The sentence makes a broad generalization about Native people's influence on the founding of the US; it doesn't effectively present the influence theory.

QUESTION 31

Choice A is the best answer. Noting that the Sun (9,800°F) is hotter than most stars within 10 parsecs of it, the sentence emphasizes how hot the Sun is relative to nearby stars.

Choice B is incorrect. The sentence explains that astronomer Todd Henry determined the classifications for the Sun and several other stars nearby; it doesn't emphasize how hot the Sun is relative to nearby stars. *Choice C* is incorrect. The sentence explains that the majority of stars near the Sun are classified as K or M stars; it doesn't indicate the Sun's temperature or emphasize how hot it is relative to nearby stars. *Choice D* is incorrect. While the sentence indicates that the Sun is classified differently than most nearby stars due to its surface temperature, it doesn't emphasize how hot the Sun is relative to nearby stars.

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QUESTION 32

Choice A is the best answer. The sentence effectively introduces Cathryn Halverson's book to an audience already familiar with the *Atlantic Monthly*, noting the title of Halverson's book and describing its content without providing background information about the *Atlantic Monthly*.

Choice B is incorrect. The sentence introduces the *Atlantic Monthly* and mentions that it's referred to in Cathryn Halverson's book title; it doesn't effectively introduce Halverson's book. *Choice C* is incorrect. The sentence assumes that the audience is unfamiliar with the *Atlantic Monthly*, providing background information about the magazine; it doesn't effectively introduce Halverson's book to an audience already familiar with the *Atlantic Monthly*. *Choice D* is incorrect. While the sentence assumes that the audience is familiar with the *Atlantic Monthly*, it doesn't effectively introduce Cathryn Halverson's book.

QUESTION 33

Choice B is the best answer. The sentence emphasizes a similarity between the two ways a magnificent frigatebird acquires food, noting that neither way requires the seabird to dive into the water.

Choice A is incorrect. The sentence describes how a magnificent frigatebird captures prey without diving into water; it doesn't emphasize a similarity between the two ways the seabird acquires food. *Choice C* is incorrect. The sentence notes the term used to describe one of the two ways that magnificent frigatebirds acquire food; it doesn't emphasize a similarity between the two ways. *Choice D* is incorrect. The sentence describes the two ways that a magnificent frigatebird acquires food; it doesn't emphasize a similarity between the two ways.

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Math

Module 1
(27 questions)

QUESTION 1

Choice B is correct. The height of each bar in the bar graph given represents the number of students that voted for the activity specified at the bottom of the bar. The bar for activity 3 has a height that is between 35 and 40. In other words, the number of students that chose activity 3 is between 35 students and 40 students. Of the given choices, 39 is the only value between 35 and 40. Therefore, 39 students chose activity 3.

Choice A is incorrect and may result from conceptual errors. *Choice C* is incorrect. This is the number of students that chose activity 5, not activity 3. *Choice D* is incorrect and may result from conceptual errors.

QUESTION 2

Choice A is correct. Let x represent the percentage of 300 that is 75. This can be written as $\frac{x}{100}(300)=75$, or $3x=75$. Dividing both sides of this equation by 3 yields $x=25$. Therefore, 25% of 300 is 75.

Choice B is incorrect. 50% of 300 is 150, not 75. *Choice C* is incorrect. 75% of 300 is 225, not 75. *Choice D* is incorrect. 225% of 300 is 675, not 75.

QUESTION 3

Choice B is correct. Multiplying the left- and right-hand sides of the given equation by 25 yields $x^2=900$. Taking the square root of the left- and right-hand sides of this equation yields $x=30$ or $x=-30$. Of these two solutions, only 30 is given as a choice.

Choice A is incorrect. This is a solution to the equation $x^2=36$. *Choice C* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect and may result from conceptual or calculation errors.

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QUESTION 4

Choice D is correct. The given phrase "8 times a number x" can be represented by the expression $8x$. The given phrase "3 more than" indicates an increase of 3 to a quantity. Therefore "3 more than 8 times a number x" can be represented by the expression $8x+3$. Since it's given that 3 more than 8 times a number x is equal to 83, it follows that $8x+3$ is equal to 83, or $8x+3=83$. Therefore, the equation that represents this situation is $8x+3=83$.

Choice A is incorrect. This equation represents 3 times the quantity 8 times a number x is equal to 83. *Choice B* is incorrect. This equation represents 8 times a number x is equal to 3 more than 83. *Choice C* is incorrect. This equation represents 8 more than 3 times a number x is equal to 83.

QUESTION 5

Choice A is correct. It's given that t represents the number of monthly deposits. In the given function $f(t)=100+25t$, the coefficient of t is 25. This means that for every increase in the value of t by 1, the value of $f(t)$ increases by 25. It follows that with each monthly deposit, the amount in Hana's bank account increased by \$25.

Choice B is incorrect. Before Hana made any monthly deposits, the amount in her bank account was \$100. *Choice C* is incorrect. After 1 monthly deposit, the amount in Hana's bank account was \$125. *Choice D* is incorrect and may result from conceptual errors.

QUESTION 6

The correct answer is 9. It's given that the customer spent \$27 to purchase oranges at \$3 per pound. Therefore, the number of pounds of oranges the customer purchased is $\$27\left(\frac{1\text{ pound}}{\$3}\right)$, or 9 pounds.

QUESTION 7

The correct answer is 10. It's given that the cost for the entire purchase was \$27 after a coupon was used for \$63 off the entire purchase. Adding the amount of the coupon to the purchase price yields $27+63=90$. Thus, the cost for the entire purchase before using the coupon was \$90. It's given that Nasir bought 9 storage bins. The original price for 1 storage bin can be found by dividing the total cost by 9. Therefore, the original price, in dollars, for 1 storage bin is $\frac{90}{9}$, or 10.

QUESTION 8

Choice A is correct. An equation that defines a linear function f can be written in the form $f(x)=mx+b$, where m and b are constants. It's given in the table that when $x=0$, $f(x)=29$. Substituting 0 for x and 29 for $f(x)$ in the equation $f(x)=mx+b$ yields $29=m(0)+b$, or $29=b$. Substituting 29 for b in the equation $f(x)=mx+b$ yields $f(x)=mx+29$. It's also given in the table that when $x=1$, $f(x)=32$. Substituting 1 for x and 32 for $f(x)$ in the equation

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$f(x)=mx+29$ yields $32=m(1)+29$, or $32=m+29$. Subtracting 29 from both sides of this equation yields $3=m$. Substituting 3 for m in the equation $f(x)=mx+29$ yields $f(x)=3x+29$.

Choice B is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 9

Choice B is correct. In similar triangles, corresponding angles are congruent. It's given that right triangles PQR and STU are similar, where angle P corresponds to angle S. It follows that angle P is congruent to angle S. In the triangles shown, angle R and angle U are both marked as right angles, so angle R and angle U are corresponding angles. It follows that angle Q and angle T are corresponding angles, and thus, angle Q is congruent to angle T. It's given that the measure of angle Q is $18°$, so the measure of angle T is also $18°$. Angle U is a right angle, so the measure of angle U is $90°$. The sum of the measures of the interior angles of a triangle is $180°$. Thus, the sum of the measures of the interior angles of triangle STU is 180 degrees. Let s represent the measure, in degrees, of angle S. It follows that $s+18+90=180$, or $s+108=180$. Subtracting 108 from both sides of this equation yields $s=72$. Therefore, if the measure of angle Q is 18 degrees, then the measure of angle S is 72 degrees.

Choice A is incorrect. This is the measure of angle T. *Choice C* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect. This is the sum of the measures of angle S and angle U.

QUESTION 10

Choice D is correct. The data points suggest that as the variable x increases, the variable y decreases, which implies that an appropriate linear model for the data has a negative slope. The data points also show that when x is close to 0, y is greater than 9. Therefore, the y-intercept of the graph of an appropriate linear model has a y-coordinate greater than 9. The graph of an equation of the form $y=a+bx$, where a and b are constants, has a y-intercept with a y-coordinate of a and has a slope of b. Of the given choices, only choice D represents a graph that has a negative slope, -0.9, and a y-intercept with a y-coordinate greater than 9, 9.4.

Choice A is incorrect. The graph of this equation has a positive slope, not a negative slope, and a y-intercept with a y-coordinate less than 1, not greater than 9. *Choice B* is incorrect. The graph of this equation has a y-intercept with a y-coordinate less than 1, not greater than 9. *Choice C* is incorrect. The graph of this equation has a positive slope, not a negative slope.

QUESTION 11

Choice A is correct. The number of birds can be found by calculating the value of b when $r=16$ in the given equation. Substituting 16 for r in the given equation

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yields $2.5b+5(16)=80$, or $2.5b+80=80$. Subtracting 80 from both sides of this equation yields $2.5b=0$. Dividing both sides of this equation by 2.5 yields $b=0$. Therefore, if the business cares for 16 reptiles on a given day, it can care for 0 birds on this day.

Choice B is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 12

Choice C is correct. An equation of a line can be written in the form $y=mx+b$, where m is the slope of the line and $(0, b)$ is the y-intercept of the line. The line shown passes through the point $(0, -8)$, so $b=-8$. The line shown also passes through the point $(-8, 0)$. The slope, m, of a line passing through two points (x_1, y_1) and (x_2, y_2) can be calculated using the equation $m=\frac{y_2-y_1}{x_2-x_1}$. For the points $(0, -8)$ and $(-8, 0)$, this gives $m=\frac{(-8)-0}{0-(-8)}$, or $m=-1$. Substituting -8 for b and -1 for m in $y=mx+b$ yields $y=(-1)x+(-8)$, or $y=-x-8$. Therefore, an equation of the graph shown is $y=-x-8$.

Choice A is incorrect. This is an equation of a line with a slope of -2, not -1.
Choice B is incorrect. This is an equation of a line with a slope of 1, not -1.
Choice D is incorrect. This is an equation of a line with a slope of 2, not -1.

QUESTION 13

The correct answer is $\frac{1}{5}$. Since the number 5 can also be written as $\frac{5}{1}$, the given equation can also be written as $\frac{x}{8}=\frac{5}{1}$. This equation is equivalent to $\frac{8}{x}=\frac{1}{5}$. Therefore, the value of $\frac{8}{x}$ is $\frac{1}{5}$. Note that 1/5 and .2 are examples of ways to enter a correct answer.

Alternate approach: Multiplying both sides of the equation $\frac{x}{8}=5$ by 8 yields $x=40$. Substituting 40 for x into the expression $\frac{8}{x}$ yields $\frac{8}{40}$, or $\frac{1}{5}$.

QUESTION 14

The correct answer is 80. Subtracting the second equation in the given system from the first equation yields $(24x+y)-(6x+y)=48-72$, which is equivalent to $24x-6x+y-y=-24$, or $18x=-24$. Dividing each side of this equation by 3 yields $6x=-8$. Substituting -8 for $6x$ in the second equation yields $-8+y=72$. Adding 8 to both sides of this equation yields $y=80$.

Alternate approach: Multiplying each side of the second equation in the given system by 4 yields $24x+4y=288$. Subtracting the first equation in the given system from this equation yields $(24x+4y)-(24x+y)=288-48$, which is

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equivalent to $24x-24x+4y-y=240$, or $3y=240$. Dividing each side of this equation by 3 yields $y=80$.

QUESTION 15

Choice D is correct. The equation that defines line t in the *xy*-plane can be written in slope-intercept form $y=mx+b$, where m is the slope of line t and $(0, b)$ is its *y*-intercept. It's given that line t has a slope of $-\frac{1}{3}$. Therefore, $m=-\frac{1}{3}$. Substituting $-\frac{1}{3}$ for m in the equation $y=mx+b$ yields $y=-\frac{1}{3}x+b$, or $y=-\frac{x}{3}+b$. It's also given that line t passes through the point $(9, 10)$. Substituting 9 for x and 10 for y in the equation $y=-\frac{x}{3}+b$ yields $10=-\frac{9}{3}+b$, or $10=-3+b$. Adding 3 to both sides of this equation yields $13=b$. Substituting 13 for b in the equation $y=-\frac{x}{3}+b$ yields $y=-\frac{x}{3}+13$.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice B* is incorrect. This equation defines a line that has a slope of 9, not $-\frac{1}{3}$, and passes through the point $(0, 10)$, not $(9, 10)$. *Choice C* is incorrect. This equation defines a line that passes through the point $(0, 10)$, not $(9, 10)$.

QUESTION 16

Choice B is correct. It's given that the function $f(x)=206(1.034)^x$ models the value, in dollars, of a certain bank account by the end of each year from 1957 through 1972, where x is the number of years after 1957. It follows that $f(x)$ represents the estimated value, in dollars, of the bank account x years after 1957. Since the value of $f(5)$ is the value of $f(x)$ when $x=5$, it follows that "$f(5)$ is approximately equal to 243" means that $f(x)$ is approximately equal to 243 when $x=5$. In the given context, this means that the value of the bank account is estimated to be approximately 243 dollars 5 years after 1957. Therefore, the best interpretation of the statement "$f(5)$ is approximately equal to 243" in this context is the value of the bank account is estimated to be approximately 243 dollars in 1962.

Choice A is incorrect and may result from conceptual errors. *Choice C* is incorrect and may result from conceptual errors. *Choice D* is incorrect and may result from conceptual errors.

QUESTION 17

Choice B is correct. It's given that the ratio of the rectangular region's length to its width is 35 to 10. This can be written as a proportion: $\frac{\text{length}}{\text{width}}=\frac{35}{10}$, or $\frac{\ell}{w}=\frac{35}{10}$. This proportion can be rewritten as $10\ell=35w$, or $\ell=3.5w$. If the width of the rectangular region increases by 7, then the length will increase by some number x in order to maintain this ratio. The value of x can be found by replacing ℓ with $\ell+x$ and w with $w+7$ in the equation, which gives $\ell+x=3.5(w+7)$. This equation can be rewritten using the distributive property as $\ell+x=3.5w+24.5$.

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Since $\ell = 3.5w$, the right-hand side of this equation can be rewritten by substituting ℓ for $3.5w$, which gives $\ell + x = \ell + 24.5$, or $x = 24.5$. Therefore, if the width of the rectangular region increases by 7 units, the length must increase by 24.5 units in order to maintain the given ratio.

Choice A is incorrect. If the width of the rectangular region increases, the length must also increase, not decrease. *Choice C* is incorrect. If the width of the rectangular region increases, the length must also increase, not decrease. *Choice D* is incorrect. Since the ratio of the length to the width of the rectangular region is 35 to 10, if the width of the rectangular region increases by 7 units, the length would have to increase by a proportional amount, which would have to be greater than 7 units.

QUESTION 18

Choice A is correct. Let x represent the side length, in inches, of square P. It follows that the perimeter of square P is $4x$ inches. It's given that square Q has a perimeter that is 176 inches greater than the perimeter of square P. Thus, the perimeter of square Q is 176 inches greater than $4x$ inches, or $4x + 176$ inches. Since the perimeter of a square is 4 times the side length of the square, each side length of Q is $\frac{4x+176}{4}$, or $x + 44$ inches. Since the area of a square is calculated by multiplying the length of two sides, the area of square Q is $(x + 44)(x + 44)$, or $(x + 44)^2$ square inches. It follows that function f is defined by $f(x) = (x + 44)^2$.

Choice B is incorrect. This function represents a square with side lengths $(x + 176)$ inches. *Choice C* is incorrect. This function represents a square with side lengths $(176x + 44)$ inches. *Choice D* is incorrect. This function represents a square with side lengths $(176x + 176)$ inches.

QUESTION 19

Choice C is correct. Dividing each side of the given equation by 2 yields $\frac{14x}{14y} = \frac{2\sqrt{w+19}}{2}$, or $\frac{x}{y} = \sqrt{w + 19}$. Because it's given that each of the variables is positive, squaring each side of this equation yields the equivalent equation $\left(\frac{x}{y}\right)^2 = w + 19$. Subtracting 19 from each side of this equation yields $\left(\frac{x}{y}\right)^2 - 19 = w$, or $w = \left(\frac{x}{y}\right)^2 - 19$.

Choice A is incorrect. This equation isn't equivalent to the given equation. *Choice B* is incorrect. This equation isn't equivalent to the given equation. *Choice D* is incorrect. This equation isn't equivalent to the given equation.

QUESTION 20

The correct answer is 100. It's given that point O is the center of a circle and the measure of arc RS on the circle is $100°$. It follows that points R and S lie on the circle. Therefore, $\overline{OR}$ and $\overline{OS}$ are radii of the circle. A central angle is an angle

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formed by two radii of a circle, with its vertex at the center of the circle. Therefore, $\angle ROS$ is a central angle. Because the degree measure of an arc is equal to the measure of its associated central angle, it follows that the measure, in degrees, of $\angle ROS$ is 100.

QUESTION 21

The correct answer is $\frac{361}{8}$. The rational exponent property is $\sqrt[n]{y^m} = y^{\frac{m}{n}}$, where $y > 0$, m and n are integers, and $n > 0$. This property can be applied to rewrite the given expression $6\sqrt[5]{3^5 x^{45}} \cdot \sqrt[8]{2^8 x}$ as $6\left(3^{\frac{5}{5}}\right)\left(x^{\frac{45}{5}}\right)\left(2^{\frac{8}{8}}\right)\left(x^{\frac{1}{8}}\right)$, or $6(3)(x^9)(2)\left(x^{\frac{1}{8}}\right)$. This expression can be rewritten by multiplying the constants, which gives $36(x^9)\left(x^{\frac{1}{8}}\right)$. The multiplication exponent property is $y^n \cdot y^m = y^{n+m}$, where $y > 0$. This property can be applied to rewrite the expression $36(x^9)\left(x^{\frac{1}{8}}\right)$ as $36x^{9+\frac{1}{8}}$, or $36x^{\frac{73}{8}}$. Therefore, $6\sqrt[5]{3^5 x^{45}} \cdot \sqrt[8]{2^8 x} = 36x^{\frac{73}{8}}$. It's given that $6\sqrt[5]{3^5 x^{45}} \cdot \sqrt[8]{2^8 x}$ is equivalent to ax^b; therefore, $a = 36$ and $b = \frac{73}{8}$. It follows that $a + b = 36 + \frac{73}{8}$. Finding a common denominator on the right-hand side of this equation gives $a + b = \frac{288}{8} + \frac{73}{8}$, or $a + b = \frac{361}{8}$. Note that 361/8, 45.12, and 45.13 are examples of ways to enter a correct answer.

QUESTION 22

Choice B is correct. The area, A, of a triangle can be found using the formula $A = \frac{1}{2}bh$, where b is the length of the base of the triangle and h is the height of the triangle. It's given that the triangle is a right triangle. Therefore, its base and height can be represented by the two legs. It's also given that the triangle has sides of length $2\sqrt{2}$, $6\sqrt{2}$, and $\sqrt{80}$ units. Since $\sqrt{80}$ units is the greatest of these lengths, it's the length of the hypotenuse. Therefore, the two legs have lengths $2\sqrt{2}$ and $6\sqrt{2}$ units. Substituting these values for b and h in the formula $A = \frac{1}{2}bh$ gives $A = \frac{1}{2}\left(2\sqrt{2}\right)\left(6\sqrt{2}\right)$, which is equivalent to $A = 6\sqrt{4}$ square units, or $A = 12$ square units.

Choice A is incorrect. This expression represents the perimeter, rather than the area, of the triangle. *Choice C* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 23

Choice D is correct. It's given that $4x^2 + bx - 45$ can be rewritten as $(hx + k)(x + j)$. The expression $(hx + k)(x + j)$ can be rewritten as $hx^2 + jhx + kx + kj$, or $hx^2 + (jh + k)x + kj$. Therefore, $hx^2 + (jh + k)x + kj$ is equivalent to $4x^2 + bx - 45$. It follows that $kj = -45$. Dividing each side of this equation by k yields $j = \frac{-45}{k}$. Since j is an integer, $-\frac{45}{k}$ must be an integer. Therefore, $\frac{45}{k}$ must also be an integer.

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Choice A is incorrect and may result from conceptual or calculation errors.
Choice B is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.

QUESTION 24

Choice C is correct. It's given that the graphs of the equations in the given system intersect at exactly one point, (x, y), in the *xy*-plane. Therefore, (x, y) is the only solution to the given system of equations. The given system of equations can be solved by subtracting the second equation, $y = 3x + a$, from the first equation, $y = 2x^2 - 21x + 64$. This yields $y - y = (2x^2 - 21x + 64) - (3x + a)$, or $0 = 2x^2 - 24x + 64 - a$. Since the given system has only one solution, this equation has only one solution. A quadratic equation in the form $rx^2 + sx + t = 0$, where r, s, and t are constants, has one solution if and only if the discriminant, $s^2 - 4rt$, is equal to zero. Substituting 2 for r, -24 for s, and $-a + 64$ for t in the expression $s^2 - 4rt$ yields $(-24)^2 - (4)(2)(64 - a)$. Setting this expression equal to zero yields $(-24)^2 - (4)(2)(64 - a) = 0$, or $8a + 64 = 0$. Subtracting 64 from both sides of this equation yields $8a = -64$. Dividing both sides of this equation by 8 yields $a = -8$. Substituting -8 for a in the equation $0 = 2x^2 - 24x + 64 - a$ yields $0 = 2x^2 - 24x + 64 + 8$, or $0 = 2x^2 - 24x + 72$. Factoring 2 from the right-hand side of this equation yields $0 = 2(x^2 - 12x + 36)$. Dividing both sides of this equation by 2 yields $0 = x^2 - 12x + 36$, which is equivalent to $0 = (x - 6)(x - 6)$, or $0 = (x - 6)^2$. Taking the square root of both sides of this equation yields $0 = x - 6$. Adding 6 to both sides of this equation yields $x = 6$.

Choice A is incorrect. This is the value of a, not x. *Choice B* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 25

Choice C is correct. Since the triangle is an isosceles right triangle, the two sides that form the right angle must be the same length. Let x be the length, in inches, of each of those sides. The Pythagorean theorem states that in a right triangle, $a^2 + b^2 = c^2$, where c is the length of the hypotenuse and a and b are the lengths of the other two sides. Substituting x for a, x for b, and 58 for c in this equation yields $x^2 + x^2 = 58^2$, or $2x^2 = 58^2$. Dividing each side of this equation by 2 yields $x^2 = \frac{58^2}{2}$, or $x^2 = \frac{2 \cdot 58^2}{4}$. Taking the square root of each side of this equation yields two solutions: $x = \frac{58\sqrt{2}}{2}$ and $x = -\frac{58\sqrt{2}}{2}$. The value of x must be positive because it represents a side length. Therefore, $x = \frac{58\sqrt{2}}{2}$, or $x = 29\sqrt{2}$. The perimeter, in inches, of the triangle is $58 + x + x$, or $58 + 2x$. Substituting $29\sqrt{2}$ for x in this expression gives a perimeter, in inches, of $58 + 2(29\sqrt{2})$, or $58 + 58\sqrt{2}$.

Choice A is incorrect. This is the length, in inches, of each of the congruent sides of the triangle, not the perimeter, in inches, of the triangle. *Choice B* is incorrect. This is the sum of the lengths, in inches, of the congruent sides of the triangle, not

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the perimeter, in inches, of the triangle. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 26

Choice D is correct. The equation of a parabola in the *xy*-plane can be written in the form $y=a(x-h)^2+k$, where a is a constant and (h, k) is the vertex of the parabola. If a is positive, the parabola will open upward, and if a is negative, the parabola will open downward. It's given that the parabola has vertex $(9, -14)$. Substituting 9 for h and -14 for k in the equation $y=a(x-h)^2+k$ gives $y=a(x-9)^2-14$, which can be rewritten as $y=a(x-9)(x-9)-14$, or $y=a(x^2-18x+81)-14$. Distributing the factor of a on the right-hand side of this equation yields $y=ax^2-18ax+81a-14$. Therefore, the equation of the parabola, $y=ax^2-18ax+81a-14$, can be written in the form $y=ax^2+bx+c$, where $a=a$, $b=-18a$, and $c=81a-14$. Substituting $-18a$ for b and $81a-14$ for c in the expression $a+b+c$ yields $(a)+(-18a)+(81a-14)$, or $64a-14$. Since the vertex of the parabola, $(9, -14)$, is below the *x*-axis, and it's given that the parabola intersects the *x*-axis at two points, the parabola must open upward. Therefore, the constant a must have a positive value. Setting the expression $64a-14$ equal to the value in choice D yields $64a-14=-12$. Adding 14 to both sides of this equation yields $64a=2$. Dividing both sides of this equation by 64 yields $a=\frac{2}{64}$, which is a positive value. Therefore, if the equation of the parabola is written in the form $y=ax^2+bx+c$, where a, b, and c are constants, the value of $a+b+c$ could be -12.

Choice A is incorrect. If the equation of a parabola with a vertex at $(9, -14)$ is written in the form $y=ax^2+bx+c$, where a, b, and c are constants and $a+b+c=-23$, then the value of a will be negative, which means the parabola will open downward, not upward, and will intersect the *x*-axis at zero points, not two points. *Choice B* is incorrect. If the equation of a parabola with a vertex at $(9, -14)$ is written in the form $y=ax^2+bx+c$, where a, b, and c are constants and $a+b+c=-19$, then the value of a will be negative, which means the parabola will open downward, not upward, and will intersect the *x*-axis at zero points, not two points. *Choice C* is incorrect. If the equation of a parabola with a vertex at $(9, -14)$ is written in the form $y=ax^2+bx+c$, where a, b, and c are constants and $a+b+c=-14$, then the value of a will be 0, which is inconsistent with the equation of a parabola.

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QUESTION 27

The correct answer is 5. It's given that $f(x)=-a^x+b$. Substituting $-a^x+b$ for $f(x)$ in the equation $y=f(x)-15$ yields $y=-a^x+b-15$. It's given that the y-intercept of the graph of $y=f(x)-15$ is $\left(0, -\frac{99}{7}\right)$. Substituting 0 for x and $-\frac{99}{7}$ for y in the equation $y=-a^x+b-15$ yields $-\frac{99}{7}=-a^0+b-15$, which is equivalent to $-\frac{99}{7}=-1+b-15$, or $-\frac{99}{7}=b-16$. Adding 16 to both sides of this equation yields $\frac{13}{7}=b$. It's given that the product of a and b is $\frac{65}{7}$, or $ab=\frac{65}{7}$. Substituting $\frac{13}{7}$ for b in this equation yields $(a)\left(\frac{13}{7}\right)=\frac{65}{7}$. Dividing both sides of this equation by $\frac{13}{7}$ yields $a=5$.

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Math

Module 2
(27 questions)

QUESTION 1

Choice B is correct. For the given line graph, the estimated number of chipmunks is represented on the vertical axis. The greatest estimated number of chipmunks in the state park is indicated by the greatest height in the line graph. This height is achieved when the year is 1994.

Choice A is incorrect and may result from conceptual errors. *Choice C* is incorrect and may result from conceptual errors. *Choice D* is incorrect and may result from conceptual errors.

QUESTION 2

Choice B is correct. It's given that the fish swam 5,104 yards and that 1 mile is equal to 1,760 yards. Therefore, the fish swam $5{,}104 \text{ yards}\left(\frac{1 \text{ mile}}{1{,}760 \text{ yards}}\right)$, which is equivalent to $\frac{5{,}104}{1{,}760}$ miles, or 2.9 miles.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 3

Choice C is correct. The given expression shows subtraction of two like terms. The two terms can be subtracted as follows: $12x^3 - 5x^3 = (12 - 5)x^3$, or $7x^3$.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice B is incorrect. This is the result of adding, not subtracting, the two like terms. *Choice D* is incorrect and may result from conceptual or calculation errors.

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QUESTION 4

Choice A is correct. The second equation in the given system defines the value of x as $5y$. Substituting $5y$ for x into the first equation yields $5y+y=18$ or $6y=18$. Dividing each side of this equation by 6 yields $y=3$. Substituting 3 for y in the second equation yields $5(3)=x$ or $x=15$. Therefore, the solution (x, y) to the given system of equations is $(15, 3)$.

Choice B is incorrect. Substituting 16 for x and 2 for y in the second equation yields $5(2)=16$, which is not true. Therefore, $(16, 2)$ is not a solution to the given system of equations. *Choice C* is incorrect. Substituting 17 for x and 1 for y in the second equation yields $5(1)=17$, which is not true. Therefore, $(17, 1)$ is not a solution to the given system of equations. *Choice D* is incorrect. Substituting 18 for x and 0 for y in the second equation yields $5(0)=18$, which is not true. Therefore, $(18, 0)$ is not a solution to the given system of equations.

QUESTION 5

Choice A is correct. The given point, $(8, 2)$, is located in the first quadrant in the *xy*-plane. The system of inequalities in choice A represents all the points in the first quadrant in the *xy*-plane. Therefore, $(8, 2)$ is a solution to the system of inequalities in choice A.

Alternate approach: Substituting 8 for x in the first inequality in choice A, $x>0$, yields $8>0$, which is true. Substituting 2 for y in the second inequality in choice A, $y>0$, yields $2>0$, which is true. Since the coordinates of the point $(8, 2)$ make the inequalities $x>0$ and $y>0$ true, the point $(8, 2)$ is a solution to the system of inequalities consisting of $x>0$ and $y>0$.

Choice B is incorrect. This system of inequalities represents all the points in the fourth quadrant, not the first quadrant, in the *xy*-plane. *Choice C* is incorrect. This system of inequalities represents all the points in the second quadrant, not the first quadrant, in the *xy*-plane. *Choice D* is incorrect. This system of inequalities represents all the points in the third quadrant, not the first quadrant, in the *xy*-plane.

QUESTION 6

The correct answer is 15 or -5. By the definition of absolute value, if $|x-5|=10$, then $x-5=10$ or $x-5=-10$. Adding 5 to both sides of the first equation yields $x=15$. Adding 5 to both sides of the second equation yields $x=-5$. Thus, the given equation has two possible solutions, 15 and -5. Note that 15 and –5 are examples of ways to enter a correct answer.

QUESTION 7

The correct answer is 50. It's given that the function f gives the total number of people on a company retreat with x managers. It's also given that 7 managers are on the company retreat. Substituting 7 for x in the given function yields $f(7)=7(7)+1$, or $f(7)=50$. Therefore, there are a total of 50 people on a company retreat with 7 managers.

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QUESTION 8

Choice B is correct. It's given that $h(x)=x^2-3$. Each table gives 1, 2, and 3 as the three given values of x. Substituting 1 for x in the equation $h(x)=x^2-3$ yields $h(1)=(1)^2-3$, or $h(1)=-2$. Substituting 2 for x in the equation $h(x)=x^2-3$ yields $h(2)=(2)^2-3$, or $h(2)=1$. Finally, substituting 3 for x in the equation $h(x)=x^2-3$ yields $h(3)=(3)^2-3$, or $h(3)=6$. Therefore, $h(x)$ is -2 when x is 1, $h(x)$ is 1 when x is 2, and $h(x)$ is 6 when x is 3. Choice B is a table with these values of x and their corresponding values of $h(x)$.

Choice A is incorrect. This is a table of values for the function $h(x)=x+3$, not $h(x)=x^2-3$. *Choice C* is incorrect. This is a table of values for the function $h(x)=2x-3$, not $h(x)=x^2-3$. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 9

Choice D is correct. The value of $f(0)$ is the value of $f(x)$ when $x=0$. Substituting 0 for x in the given function yields $f(0)=270(0.1)^0$, or $f(0)=270(1)$, which is equivalent to $f(0)=270$. Therefore, the value of $f(0)$ is 270.

Choice A is incorrect. This is the value of x, not $f(x)$. *Choice B* is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect. This is the value of $f(1)$, not $f(0)$.

QUESTION 10

Choice A is correct. It's given that the estimate for the proportion of the population that has the characteristic is 0.49 with an associated margin of error of 0.04. Subtracting the margin of error from the estimate and adding the margin of error to the estimate gives an interval of plausible values for the true proportion of the population that has the characteristic. Therefore, it's plausible that the proportion of the population that has this characteristic is between 0.45 and 0.53.

Choice B is incorrect. A value less than 0.45 is outside the interval of plausible values for the proportion of the population that has the characteristic. *Choice C* is incorrect. The value 0.49 is an estimate for the proportion based on this sample. However, since the margin of error for this estimate is known, the most appropriate conclusion is not that the proportion is exactly one value but instead lies in an interval of plausible values. *Choice D* is incorrect. A value greater than 0.53 is outside the interval of plausible values for the proportion of the population that has the characteristic.

QUESTION 11

Choice A is correct. It's given that the truck can tow a trailer if the combined weight of the trailer and the boxes it contains is no more than 4,600 pounds. If the trailer has a weight of 500 pounds and each box weighs 120 pounds, the

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expression $500+120b$, where b is the number of boxes, gives the combined weight of the trailer and the boxes. Since the combined weight must be no more than $4{,}600$ pounds, the possible numbers of boxes the truck can tow are given by the inequality $500+120b \leq 4{,}600$. Subtracting 500 from both sides of this inequality yields $120b \leq 4{,}100$. Dividing both sides of this inequality by 120 yields $b \leq \frac{205}{6}$, or b is less than or equal to approximately 34.17. Since the number of boxes, b, must be a whole number, the maximum number of boxes the truck can tow is the greatest whole number less than 34.17, which is 34.

Choice B is incorrect. Towing the trailer and 35 boxes would yield a combined weight of $4{,}700$ pounds, which is greater than $4{,}600$ pounds. *Choice C* is incorrect. Towing the trailer and 38 boxes would yield a combined weight of $5{,}060$ pounds, which is greater than $4{,}600$ pounds. *Choice D* is incorrect. Towing the trailer and 39 boxes would yield a combined weight of $5{,}180$ pounds, which is greater than $4{,}600$ pounds.

QUESTION 12

Choice B is correct. Multiplying each side of the given equation by -16 yields $64x^2+112x=576$. To complete the square, adding 49 to each side of this equation yields $64x^2+112x+49=576+49$, or $(8x+7)^2=625$. Taking the square root of each side of this equation yields two equations: $8x+7=25$ and $8x+7=-25$. Subtracting 7 from each side of the equation $8x+7=25$ yields $8x=18$. Dividing each side of this equation by 8 yields $x=\frac{18}{8}$, or $x=\frac{9}{4}$. Therefore, $\frac{9}{4}$ is a solution to the given equation. Subtracting 7 from each side of the equation $8x+7=-25$ yields $8x=-32$. Dividing each side of this equation by 8 yields $x=-4$. Therefore, the given equation has two solutions, $\frac{9}{4}$ and -4. Since $\frac{9}{4}$ is positive, it follows that $\frac{9}{4}$ is the positive solution to the given equation.

Alternate approach: Adding $4x^2$ and $7x$ to each side of the given equation yields $0=4x^2+7x-36$. The right-hand side of this equation can be rewritten as $4x^2+16x-9x-36$. Factoring out the common factor of $4x$ from the first two terms of this expression and the common factor of -9 from the second two terms yields $4x(x+4)-9(x+4)$. Factoring out the common factor of $(x+4)$ from these two terms yields the expression $(4x-9)(x+4)$. Since this expression is equal to 0, it follows that either $4x-9=0$ or $x+4=0$. Adding 9 to each side of the equation $4x-9=0$ yields $4x=9$. Dividing each side of this equation by 4 yields $x=\frac{9}{4}$. Therefore, $\frac{9}{4}$ is a positive solution to the given equation. Subtracting 4 from each side of the equation $x+4=0$ yields $x=-4$. Therefore, the given equation has two solutions, $\frac{9}{4}$ and -4. Since $\frac{9}{4}$ is positive, it follows that $\frac{9}{4}$ is the positive solution to the given equation.

Choice A is incorrect. Substituting $\frac{7}{4}$ for x in the given equation yields $-\frac{49}{2}=-36$, which is false. *Choice C* is incorrect. Substituting 4 for x in the given equation yields $-92=-36$, which is false. *Choice D* is incorrect. Substituting 7 for x in the given equation yields $-245=-36$, which is false.

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QUESTION 13

The correct answer is $\frac{3}{10}$. It's given that there are a total of 100 tiles of equal area, which is the total number of possible outcomes. According to the table, there are a total of 30 red tiles. The probability of an event occurring is the ratio of the number of favorable outcomes to the total number of possible outcomes. By definition, the probability of selecting a red tile is given by $\frac{30}{100}$, or $\frac{3}{10}$. Note that 3/10 and .3 are examples of ways to enter a correct answer.

QUESTION 14

The correct answer is 2. It's given that function f is defined by $f(x)=2x+3$. Therefore, the equation representing the graph of $y=f(x)$ in the xy-plane is $y=2x+3$, and the graph is a line. For a linear equation in the form $y=mx+b$, m represents the slope of the line. Since the value of m in the equation $y=2x+3$ is 2, the slope of the line defined by function f is 2. It's given that line j is parallel to the line defined by function f. The slopes of parallel lines are equal. Therefore, the slope of line j is also 2.

QUESTION 15

Choice A is correct. It's given that a radio show stated that 3 times as many people voted in favor of the proposal as people who voted against it. Let x represent the number of people who voted against the proposal. It follows that $3x$ is the number of people who voted in favor of the proposal and $3x-x$, or $2x$, is how many more people voted in favor of the proposal than voted against it. It's also given that a social media post reported that $15{,}000$ more people voted in favor of the proposal than voted against it. Thus, $2x=15{,}000$. Since $2x=15{,}000$, the value of x must be half of $15{,}000$, or $7{,}500$. Therefore, $7{,}500$ people voted against the proposal.

Choice B is incorrect. This is how many more people voted in favor of the proposal than voted against it, not the number of people who voted against the proposal. *Choice C* is incorrect. This is the number of people who voted in favor of the proposal, not the number of people who voted against the proposal. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 16

Choice C is correct. Vertical angles, which are angles that are opposite each other when two lines intersect, are congruent. The figure shows that lines t and m intersect. It follows that the angle with measure x° and the angle with measure y° are vertical angles, so $x=y$. It's given that $x=6k+13$ and $y=8k-29$. Substituting $6k+13$ for x and $8k-29$ for y in the equation $x=y$ yields $6k+13=8k-29$. Subtracting $6k$ from both sides of this equation yields $13=2k-29$. Adding 29 to both sides of this equation yields $42=2k$, or $2k=42$. Dividing both sides of this equation by 2 yields $k=21$. It's given that lines m and n are parallel, and the figure shows that lines m and n are intersected by a transversal, line t. If two parallel lines are intersected by a transversal, then the same-side interior angles are supplementary. It follows that the same-side interior

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angles with measures $y°$ and $z°$ are supplementary, so $y+z=180$. Substituting $8k-29$ for y in this equation yields $8k-29+z=180$. Substituting 21 for k in this equation yields $8(21)-29+z=180$, or $139+z=180$. Subtracting 139 from both sides of this equation yields $z=41$. Therefore, the value of z is 41.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice B* is incorrect. This is the value of k, not z. *Choice D* is incorrect. This is the value of x or y, not z.

QUESTION 17

Choice B is correct. A linear equation in one variable has no solution if and only if the equation is false; that is, when there is no value of x that produces a true statement. It's given that in the equation $-3x+21px=84$, p is a constant and the equation has no solution for x. Therefore, the value of the constant p is one that results in a false equation. Factoring out the common factor of $-3x$ on the left-hand side of the given equation yields $-3x(1-7p)=84$. Dividing both sides of this equation by -3 yields $x(1-7p)=-28$. Dividing both sides of this equation by $(1-7p)$ yields $x=\frac{-28}{1-7p}$. This equation is false if and only if $1-7p=0$. Adding $7p$ to both sides of $1-7p=0$ yields $1=7p$. Dividing both sides of this equation by 7 yields $\frac{1}{7}=p$. It follows that the equation $x=\frac{-28}{1-7p}$ is false if and only if $p=\frac{1}{7}$. Therefore, the given equation has no solution if and only if the value of p is $\frac{1}{7}$.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice C* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 18

Choice D is correct. It's given that $f(x)=(x-10)(x+13)$, which can be rewritten as $f(x)=x^2+3x-130$. Since the coefficient of the x^2-term is positive, the graph of $y=f(x)$ in the xy-plane opens upward and reaches its minimum value at its vertex. The x-coordinate of the vertex is the value of x such that $f(x)$ reaches its minimum. For an equation in the form $f(x)=ax^2+bx+c$, where a, b, and c are constants, the x-coordinate of the vertex is $-\frac{b}{2a}$. For the equation $f(x)=x^2+3x-130$, $a=1$, $b=3$, and $c=-130$. It follows that the x-coordinate of the vertex is $-\frac{3}{2(1)}$, or $-\frac{3}{2}$. Therefore, $f(x)$ reaches its minimum when the value of x is $-\frac{3}{2}$.

Alternate approach: The value of x for the vertex of a parabola is the x-value of the midpoint between the two x-intercepts of the parabola. Since it's given that $f(x)=(x-10)(x+13)$, it follows that the two x-intercepts of the graph of $y=f(x)$ in the xy-plane occur when $x=10$ and $x=-13$, or at the points $(10, 0)$ and $(-13, 0)$. The midpoint between two points, (x_1, y_1) and (x_2, y_2), is $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$. Therefore, the midpoint between $(10, 0)$ and $(-13, 0)$ is $\left(\frac{10+(-13)}{2}, \frac{0+0}{2}\right)$, or $\left(-\frac{3}{2}, 0\right)$. It follows that $f(x)$ reaches its minimum when the value of x is $-\frac{3}{2}$.

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Choice A is incorrect. This is the *y*-coordinate of the *y*-intercept of the graph of $y=f(x)$ in the *xy*-plane. *Choice B* is incorrect. This is one of the *x*-coordinates of the *x*-intercepts of the graph of $y=f(x)$ in the *xy*-plane. *Choice C* is incorrect and may result from conceptual or calculation errors.

QUESTION 19

Choice A is correct. The graph of a quadratic equation in the form $y=a(x-h)^2+k$, where a, h, and k are positive constants, is a parabola that opens upward with vertex (h, k). The given function $f(x)=\frac{1}{9}(x-7)^2+3$ is in the form $y=a(x-h)^2+k$, where $y=f(x)$, $a=\frac{1}{9}$, $h=7$, and $k=3$. Therefore, the graph of $y=f(x)$ is a parabola that opens upward with vertex $(7, 3)$. Since the parabola opens upward, the vertex is the lowest point on the graph. It follows that the *y*-coordinate of the vertex of the graph of $y=f(x)$ is the minimum value of $f(x)$. Therefore, the minimum value of $f(x)$ is 3. It's given that $f(x)=\frac{1}{9}(x-7)^2+3$ represents the metal ball's height above the ground, in inches, x seconds after it started moving on a track. Therefore, the best interpretation of the vertex of the graph of $y=f(x)$ is that the metal ball's minimum height was 3 inches above the ground.

Choice B is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 20

The correct answer is $\frac{15}{17}$. It's given that angle J is the right angle in triangle JKL. Therefore, the acute angles of triangle JKL are angle K and angle L. The hypotenuse of a right triangle is the side opposite its right angle. Therefore, the hypotenuse of triangle JKL is side KL. The cosine of an acute angle in a right triangle is the ratio of the length of the side adjacent to the angle to the length of the hypotenuse. It's given that $\cos(K)=\frac{24}{51}$. This can be written as $\cos(K)=\frac{8}{17}$. Since the cosine of angle K is a ratio, it follows that the length of the side adjacent to angle K is $8n$ and the length of the hypotenuse is $17n$, where n is a constant. Therefore, $JK=8n$ and $KL=17n$. The Pythagorean theorem states that in a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the other two sides. For triangle JKL, it follows that $(JK)^2+(JL)^2=(KL)^2$. Substituting $8n$ for JK and $17n$ for KL yields $(8n)^2+(JL)^2=(17n)^2$. This is equivalent to $64n^2+(JL)^2=289n^2$. Subtracting $64n^2$ from each side of this equation yields $(JL)^2=225n^2$. Taking the square root of each side of this equation yields $JL=15n$. Since $\cos(L)=\frac{JL}{KL}$, it follows that $\cos(L)=\frac{15n}{17n}$, which can be rewritten as $\cos(L)=\frac{15}{17}$. Note that 15/17, .8824, .8823, and 0.882 are examples of ways to enter a correct answer.

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QUESTION 21

The correct answer is 51. A quadratic equation of the form $ax^2+bx+c=0$, where a, b, and c are constants, has no real solution if and only if its discriminant, $-4ac+b^2$, is negative. In the given equation, $a=-1$ and $c=-676$. Substituting -1 for a and -676 for c in this expression yields a discriminant of $b^2-4(-1)(-676)$, or $b^2-2{,}704$. Since this value must be negative, $b^2-2{,}704<0$, or $b^2<2{,}704$. Taking the positive square root of each side of this inequality yields $b<52$. Since b is a positive integer, and the greatest integer less than 52 is 51, the greatest possible value of b is 51.

QUESTION 22

Choice A is correct. A solution to a system of equations must satisfy each equation in the system. It follows that if an ordered pair (x, y) is a solution to the system, the point (x, y) lies on the graph in the *xy*-plane of each equation in the system. The only point that lies on each graph of the system of two linear equations shown is their intersection point $(8, 2)$. It follows that if a new graph of three linear equations is created using the system of equations shown and the graph of $x+4y=-16$, this system has either zero solutions or one solution, the point $(8, 2)$. Substituting 8 for x and 2 for y in the equation $x+4y=-16$ yields $8+4(2)=-16$, or $16=-16$. Since this equation is not true, the point $(8, 2)$ does not lie on the graph of $x+4y=-16$. Therefore, $(8, 2)$ is not a solution to the system of three equations. It follows that there are zero solutions to this system.

Choice B is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 23

Choice C is correct. For a function of the form $f(x)=a(r)^{\frac{x}{k}}$, where a, r, and k are constants and $r<1$, the value of $f(x)$ decreases by $100(1-r)\%$ for every increase of x by k. In the given function, $a=5{,}470$, $r=0.64$, and $k=12$. Therefore, for the given function, the value of $f(x)$ decreases by $100(1-0.64)\%$, or 36%, for every increase of x by 12. Since $f(x)$ represents the value, in dollars, of the equipment after x months of use, it follows that the value of the equipment decreases every 12 months by 36% of its value the preceding 12 months. Since there are 12 months in a year, the value of the equipment decreases each year by 36% of its value the preceding year. Thus, the value of p is 36.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice B is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 24

Choice C is correct. The median of a data set with an odd number of values, in ascending or descending order, is the middle value of the data set, and the range of a data set is the positive difference between the maximum and minimum values

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in the data set. Since the dot plot shown gives the values in data set A in ascending order and there are 15 values in the data set, the eighth value in data set A, 23, is the median. The maximum value in data set A is 26 and the minimum value is 22, so the range of data set A is $26-22$, or 4. It's given that data set B is created by adding 56 to each of the values in data set A. Increasing each of the 15 values in data set A by 56 will also increase its median value by 56 making the median of data set B 79. Increasing each value of data set A by 56 does not change the range, since the maximum value of data set B is $26+56$, or 82, and the minimum value is $22+56$, or 78, making the range of data set B $82-78$, or 4. Therefore, the median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.

Choice A is incorrect and may result from conceptual or calculation errors. *Choice B* is incorrect and may result from conceptual or calculation errors. *Choice D* is incorrect and may result from conceptual or calculation errors.

QUESTION 25

Choice D is correct. The graph in the *xy*-plane of an equation of the form $(x-h)^2+(y-k)^2=r^2$ is a circle with center (h, k) and a radius of length r. It's given that circle A is represented by $x^2+(y-1)^2=49$, which can be rewritten as $x^2+(y-1)^2=7^2$. Therefore, circle A has center $(0, 1)$ and a radius of length 7. Shifting circle A down two units is a rigid vertical translation of circle A that does not change its size or shape. Since circle B is obtained by shifting circle A down two units, it follows that circle B has the same radius as circle A, and for each point (x, y) on circle A, the point $(x, y-2)$ lies on circle B. Moreover, if (h, k) is the center of circle A, then $(h, k-2)$ is the center of circle B. Therefore, circle B has a radius of 7 and the center of circle B is $(0, 1-2)$, or $(0, -1)$. Thus, circle B can be represented by the equation $x^2+(y+1)^2=7^2$, or $x^2+(y+1)^2=49$.

Choice A is incorrect. This is the equation of a circle obtained by shifting circle A right 2 units. *Choice B* is incorrect. This is the equation of a circle obtained by shifting circle A up 2 units. *Choice C* is incorrect. This is the equation of a circle obtained by shifting circle A left 2 units.

QUESTION 26

Choice B is correct. Let x represent the side length, in cm, of each square base. If the two prisms are glued together along a square base, the resulting prism has a surface area equal to twice the surface area of one of the prisms, minus the area of the two square bases that are being glued together, which yields $2K-2x^2$ cm^2. It's given that this resulting surface area is equal to $\frac{92}{47}K$ cm^2, so $2K-2x^2=\frac{92}{47}K$. Subtracting $\frac{92}{47}K$ from both sides of this equation yields $2K-\frac{92}{47}K-2x^2=0$. This equation can be rewritten by multiplying $2K$ on the left-hand side by $\frac{47}{47}$, which yields $\frac{94}{47}K-\frac{92}{47}K-2x^2=0$, or $\frac{2}{47}K-2x^2=0$. Adding $2x^2$ to both sides of this equation yields $\frac{2}{47}K=2x^2$. Multiplying both sides of this equation by $\frac{47}{2}$ yields $K=47x^2$. The surface area K, in cm^2, of each rectangular prism is equivalent to the sum of the areas of the two square bases and the areas of the four lateral

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faces. Since the height of each rectangular prism is 90 cm and the side length of each square base is x cm, it follows that the area of each square base is x^2 cm^2 and the area of each lateral face is $90x$ cm^2. Therefore, the surface area of each rectangular prism can be represented by the expression $2x^2+4(90x)$, or $2x^2+360x$. Substituting this expression for K in the equation $K=47x^2$ yields $2x^2+360x=47x^2$. Subtracting $2x^2$ and $360x$ from both sides of this equation yields $0=45x^2-360x$. Factoring x from the right-hand side of this equation yields $0=x(45x-360)$. Applying the zero product property, it follows that $x=0$ and $45x-360=0$. Adding 360 to both sides of the equation $45x-360=0$ yields $45x=360$. Dividing both sides of this equation by 45 yields $x=8$. Since a side length of a rectangular prism can't be 0, the length of each square base is 8 cm.

Choice A is incorrect and may result from conceptual or calculation errors.
Choice C is incorrect and may result from conceptual or calculation errors.
Choice D is incorrect and may result from conceptual or calculation errors.

QUESTION 27

The correct answer is 600. It's given that 210 is p% greater than 30. It follows that $210=\left(1+\frac{p}{100}\right)(30)$. Dividing both sides of this equation by 30 yields $7=1+\frac{p}{100}$. Subtracting 1 from both sides of this equation yields $6=\frac{p}{100}$. Multiplying both sides of this equation by 100 yields $p=600$. Therefore, the value of p is 600.

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